D.C. CIRCUITS

10 D.C. circuits

10.1 Practical circuits

Candidates should be able to:

1 recall and use the circuit symbols shown in section 6 of this syllabus

2 draw and interpret circuit diagrams containing the circuit symbols shown in section 6 of this syllabus

3 define and use the electromotive force (e.m.f.) of a source as energy transferred per unit charge in driving charge around a complete circuit

4 distinguish between e.m.f. and potential difference (p.d.) in terms of energy considerations

5 understand the effects of the internal resistance of a source of e.m.f. on the terminal potential difference

10.2 Kirchhoff's laws

Candidates should be able to:

1 recall Kirchhoff's first law and understand that it is a consequence of conservation of charge

2 recall Kirchhoff's second law and understand that it is a consequence of conservation of energy

3 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in series

4 use the formula for the combined resistance of two or more resistors in series

5 derive, using Kirchhoff's laws, a formula for the combined resistance of two or more resistors in parallel

6 use the formula for the combined resistance of two or more resistors in parallel

7 use Kirchhoff's laws to solve simple circuit problems

10.3 Potential dividers

Candidates should be able to:

1 understand the principle of a potential divider circuit

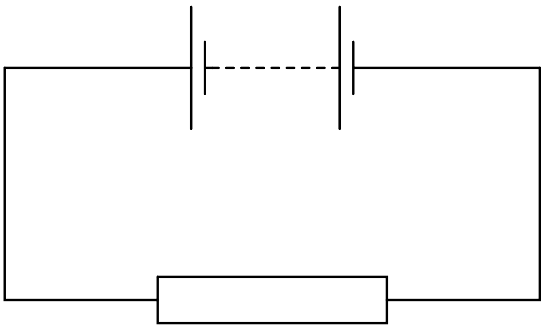
2 recall and use the principle of the potentiometer as a means of comparing potential differences

3 understand the use of a galvanometer in null methods

4 explain the use of thermistors and light-dependent resistors in potential dividers to provide a potential difference that is dependent on temperature and light intensity

1 9702/1/M/J/02/Q31*

In the circuit below, the battery converts an amount E of chemical energy to electrical energy when charge Q passes through the resistor in time t.



Which expressions give the e.m.f. of the battery and the current in the resistor?

	e.m.f.	current
A	EQ	Q/t
B	EQ	Qt
C	E/Q	Q/t
D	E/Q	Qt

2 9702/1/M/J/02/Q32

The filament of a 240 V, 100 W electric lamp heats up from room temperature to its operating temperature. As it heats up, its resistance increases by a factor of 16.

What is the resistance of this lamp at room temperature?

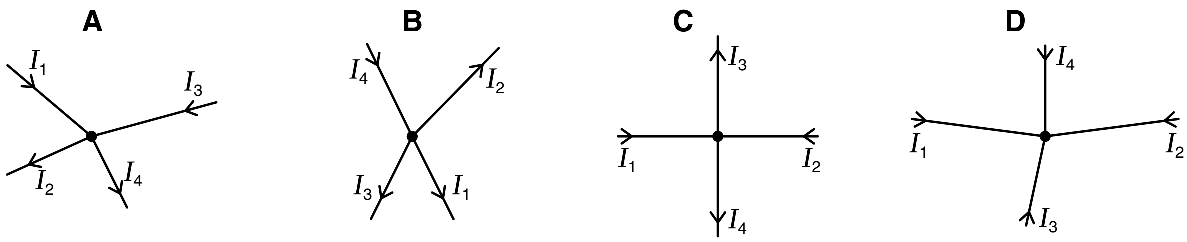
A 36 Ω **B** 580 Ω **C** 1.5 kΩ **D** 9.2 kΩ

3 9702/1/M/J/02/Q33

The diagrams show connected wires which carry currents I_1, I_2, I_3 and I_4.

The currents are related by the equation $I_1 + I_2 = I_3 + I_4$.

To which diagram does this equation apply?



4 9702/1/M/J/02/Q34

When four identical lamps P, Q, R and S are connected as shown in diagram 1, they have normal brightness.

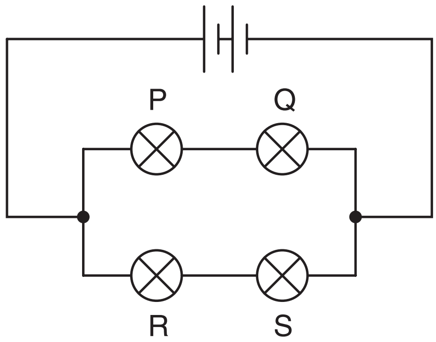


diagram 1

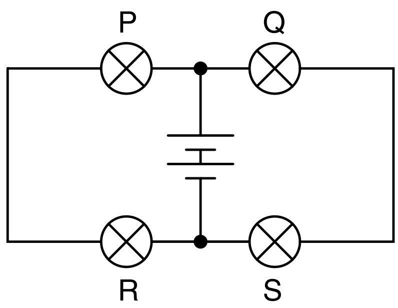


diagram 2

When the four lamps are connected as shown in diagram 2, which statement is correct?

A The lamps do not light.

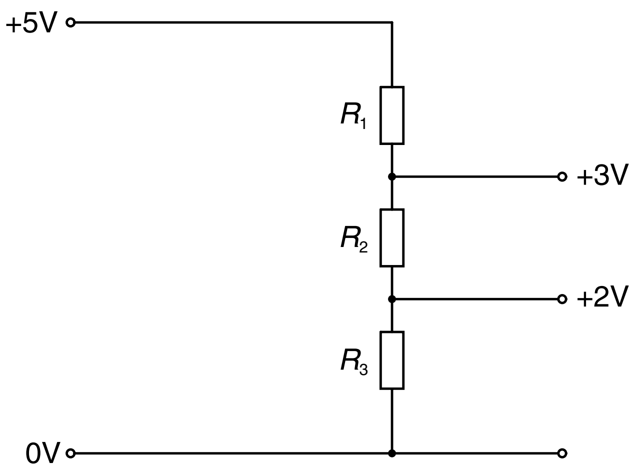
B The lamps are less bright than normal.

C The lamps have normal brightness.

D The lamps are brighter than normal.

5 9702/1/M/J/02/Q35

A potential divider is used to give outputs of 2 V and 3 V from a 5 V source, as shown.

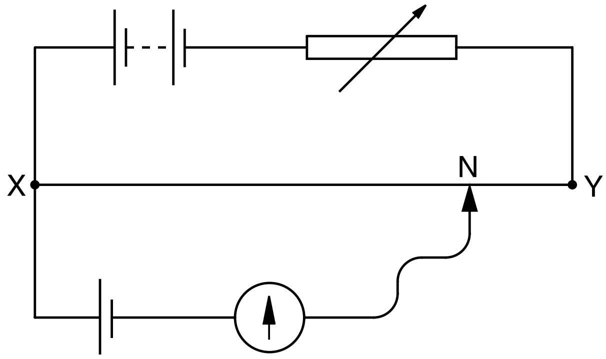


What are possible values for the resistances R_1, R_2 and R_3?

	R_1/kΩ	R_2/kΩ	R_3/kΩ
A	2	1	5
B	3	2	2
C	4	2	4
D	4	6	10

6 **9702/1/O/N/02/Q35**

In the potentiometer circuit below, the moveable contact is placed at N on the bare wire XY, such that the galvanometer shows zero deflection.



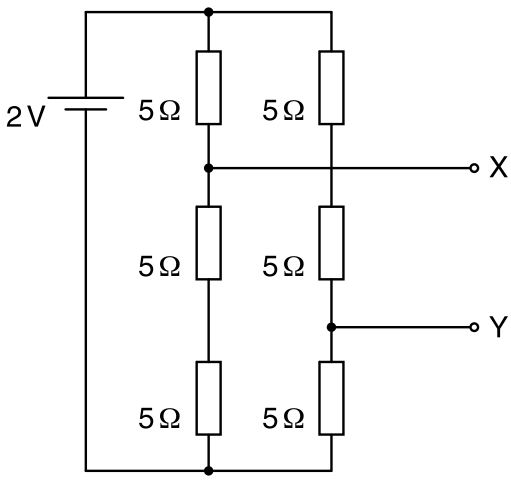
The resistance of the variable resistor is now increased.

What is the effect of this increase on the potential difference across the wire XY and on the position of the moveable contact for zero deflection?

	potential difference across XY	position of moveable contact
A	increases	nearer to X
B	increases	nearer to Y
C	decreases	nearer to X
D	decreases	nearer to Y

7 **9702/1/O/N/02/Q36**

Six resistors, each of resistance 5 Ω, are connected to a 2 V cell of negligible internal resistance.



What is the potential difference between terminals X and Y?

A $\frac{2}{3}$ V **B** $\frac{8}{9}$ V **C** $\frac{4}{3}$ V **D** 2 V

8 **9702/01/M/J/03/Q31**

The sum of the electrical currents into a point in a circuit is equal to the sum of the currents out of the point.

Which of the following is correct?

A This is Kirchhoff's first law, which results from the conservation of charge.

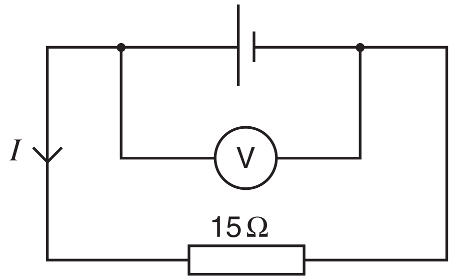
B This is Kirchhoff's first law, which results from the conservation of energy.

C This is Kirchhoff's second law, which results from the conservation of charge.

D This is Kirchhoff's second law, which results from the conservation of energy.

9 **9702/01/M/J/03/Q32**

The e.m.f. of the cell in the following circuit is 9.0 V. The reading on the high-resistance voltmeter is 7.5 V.

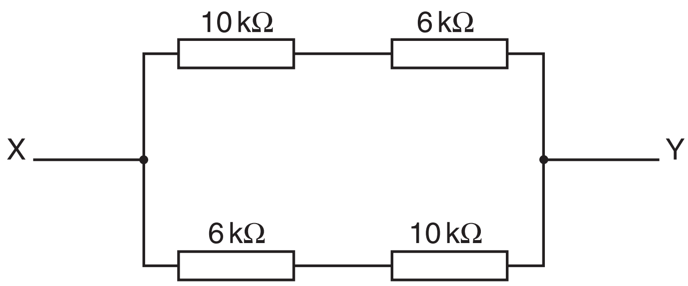


What is the current I?

A 0.1 A **B** 0.5 A **C** 0.6 A **D** 2.0 A

10 **9702/01/M/J/03/Q33**

The diagram shows an arrangement of four resistors.



What is the resistance between X and Y?

A 4 kΩ

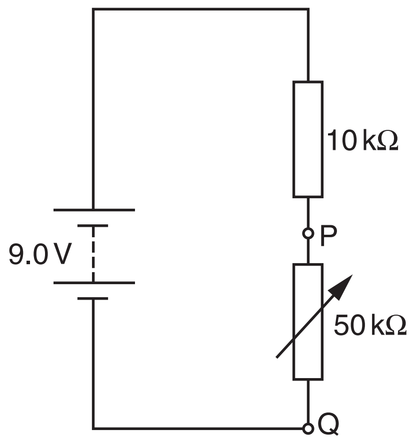
B 8 kΩ

C 16 kΩ

D 32 kΩ

11 **9702/01/M/J/03/Q34**

The diagram shows a potential divider connected to a 9.0 V supply of negligible internal resistance.



What range of voltages can be obtained between P and Q?

A zero to 1.5 V

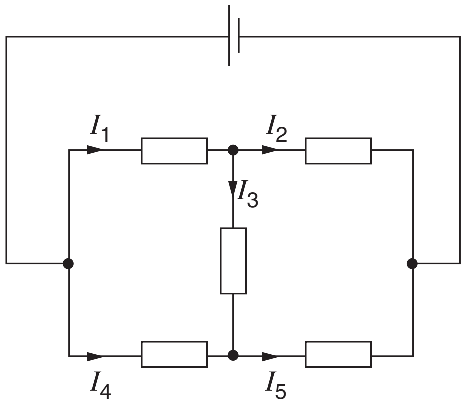
B zero to 7.5 V

C 1.5 V to 7.5 V

D 1.5 V to 9.0 V

12 **9702/11/O/N/16/Q37**

The diagram shows currents I_1, I_2, I_3, I_4 and I_5 in different branches of a circuit.



Which one of the following is correct?

A $I_1 = I_2 + I_3$

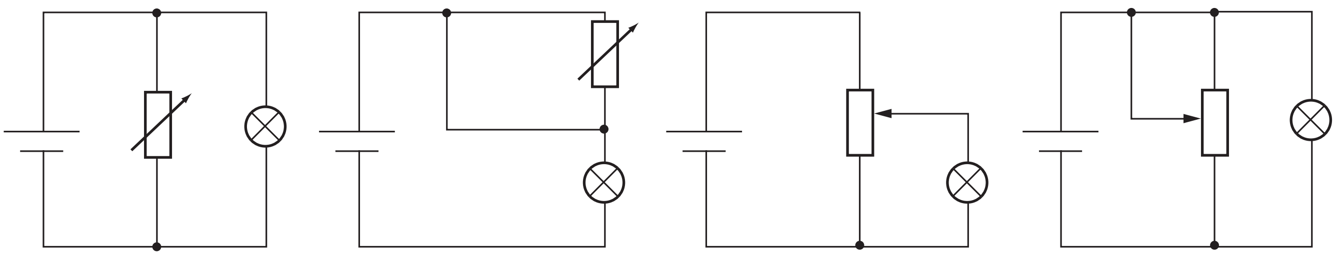
B $I_2 = I_1 + I_3$

C $I_3 = I_4 + I_5$

D $I_4 = I_5 + I_3$

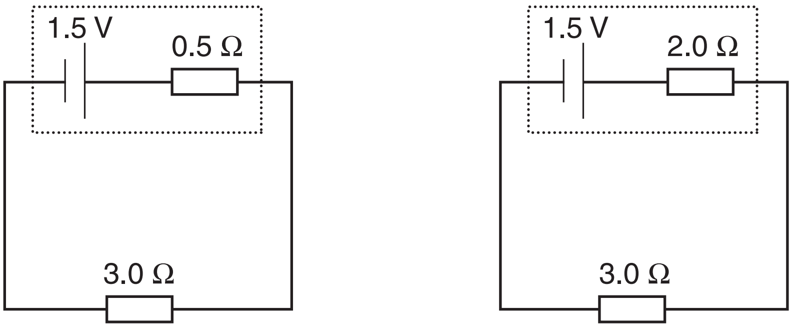
13 **9702/01/O/N/03/Q33**

Which diagram shows a potential divider circuit that can vary the voltage across the lamp?



14 **9702/01/O/N/03/Q34**

The diagram shows two circuits. In these circuits, only the internal resistances differ.



circuit **X** circuit **Y**

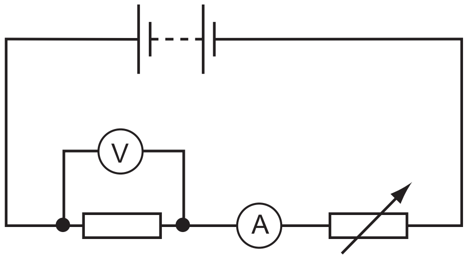
Which line in the table is correct?

	potential difference across 3.0 Ω resistor	power dissipated in 3.0 Ω resistor
A	greater in X than in Y	less in X than in Y
B	greater in X than in Y	greater in X than in Y
C	less in X than in Y	less in X than in Y
D	less in X than in Y	greater in X than in Y

15 **9702/01/M/J/04/Q35***

The diagram shows a battery, a fixed resistor, an ammeter and a variable resistor connected in series.

A voltmeter is connected across the fixed resistor.



The value of the variable resistor is reduced.

Which correctly describes the changes in the readings of the ammeter and of the voltmeter?

	ammeter	voltmeter
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

16 **9702/01/M/J/04/Q36**

Kirchhoff's two laws for electric circuits can be derived by using conservation laws.

On which conservation laws do Kirchhoff's laws depend?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	current
B	charge	energy
C	current	mass
D	energy	current

17 **9702/01/M/J/04/Q37**

The diagram shows a parallel combination of three resistors. The total resistance of the combination is $3\,\Omega$.

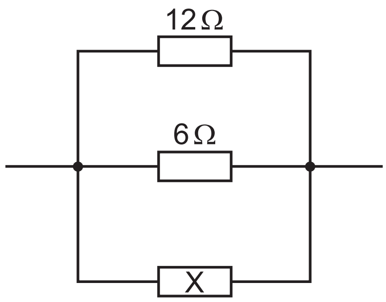
What is the resistance of resistor X?

A $2\,\Omega$

B $3\,\Omega$

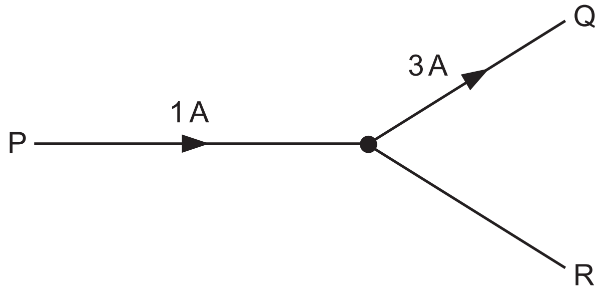
C $6\,\Omega$

D $12\,\Omega$



18 **9702/01/O/N/04/Q35**

The diagram shows a junction in a circuit where three wires P, Q and R meet. The currents in P and Q are 1 A and 3 A respectively, in the directions shown.



How many coulombs of charge pass a given point in wire R in 5 seconds?

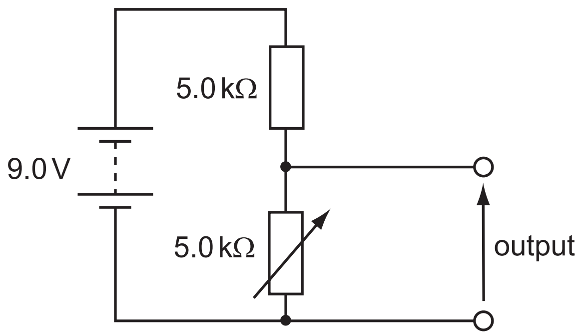
A 0.4 **B** 0.8 **C** 2 **D** 10

19 **9702/01/O/N/04/Q36**

The diagram shows a potential divider circuit designed to provide a variable output p.d.

Which gives the available range of output p.d?

	maximum output	minimum output
A	3.0 V	0
B	4.5 V	0
C	9.0 V	0
D	9.0 V	4.5 V



20 **9702/01/O/N/04/Q37***

In the circuit shown, the ammeters have negligible resistance and the voltmeters have infinite resistance.

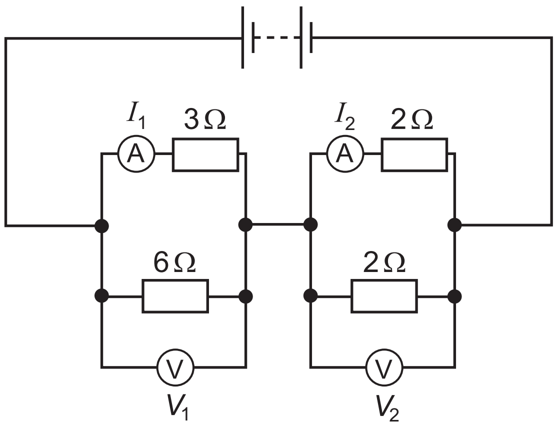
The readings on the meters are I_1, I_2, V_1 and V_2, as labelled on the diagram.
Which is correct?

A $I_1 > I_2$ and $V_1 > V_2$

C $I_1 < I_2$ and $V_1 > V_2$

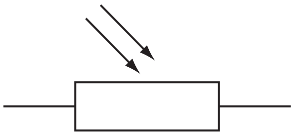
B $I_1 > I_2$ and $V_1 < V_2$

D $I_1 < I_2$ and $V_1 < V_2$



21 9702/01/M/J/05/Q35

Which electrical component is represented by the following symbol?



A a diode

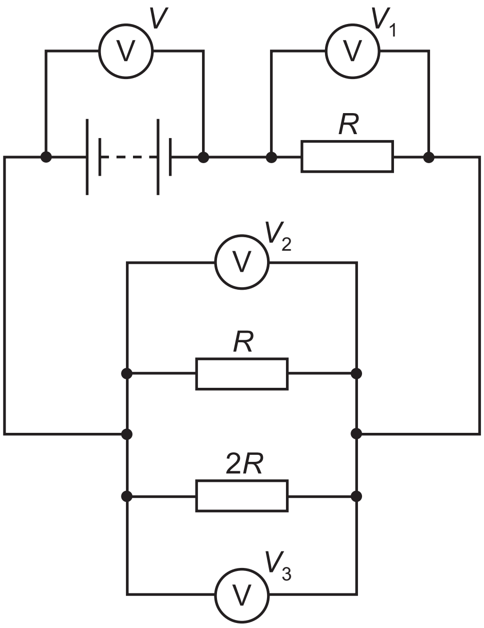
B a light-dependent resistor

C a resistor

D a thermistor

22 9702/01/M/J/05/Q36

The diagram shows a circuit with four voltmeter readings V, V_1, V_2 and V_3.



Which equation relating the voltmeter readings must be true?

A $V = V_1 + V_2 + V_3$

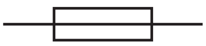
B $V + V_1 = V_2 + V_3$

C $V_3 = 2(V_2)$

D $V - V_1 = V_3$

23 9702/01/O/N/05/Q35

An electrical component has the following circuit symbol.



What does this symbol represent?

A variable resistor (rheostat)

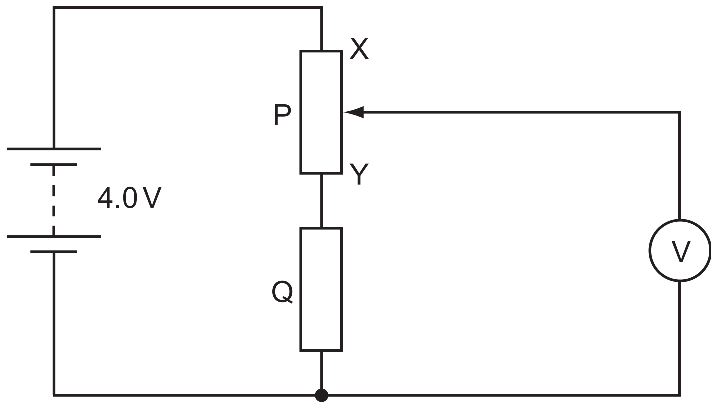
B fuse

C light-dependent resistor

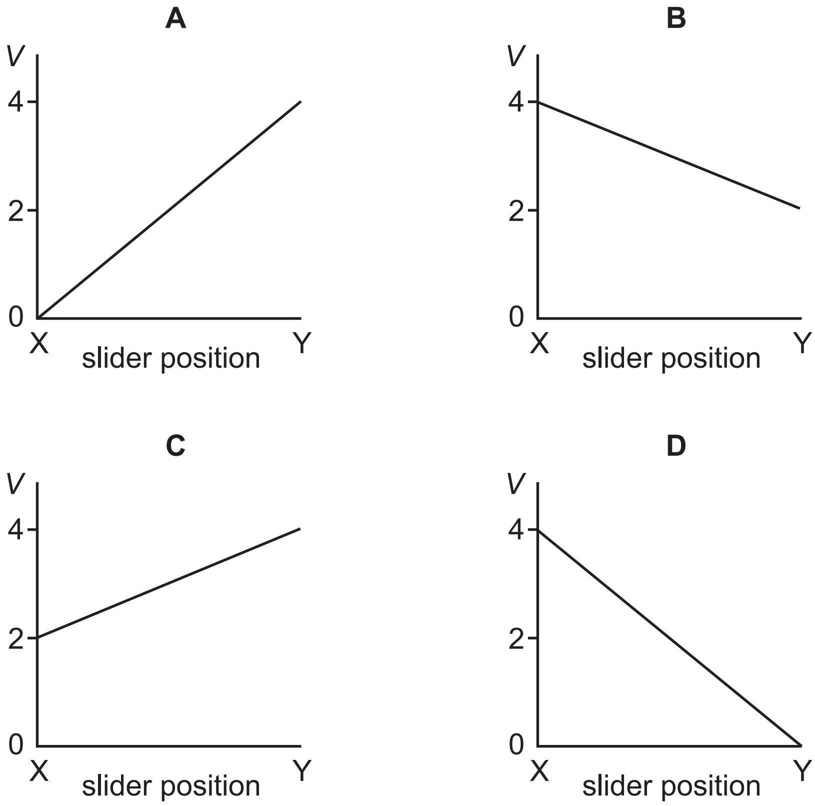
D thermistor

24 9702/13/O/N/12/Q36**

In the circuit below, P is a potentiometer of total resistance 10 Ω and Q is a fixed resistor of resistance 10 Ω. The battery has an e.m.f. of 4.0 V and negligible internal resistance. The voltmeter has a very high resistance. The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading *V* is plotted against slider position.



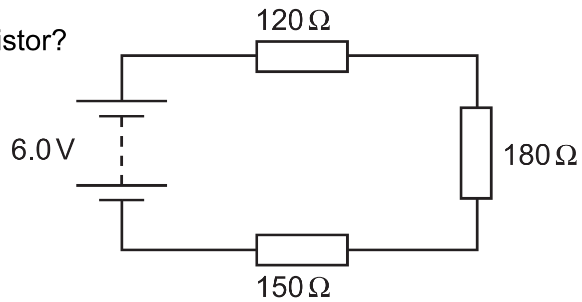
Which graph is obtained?



25 **9702/01/O/N/05/Q36**

Three resistors are connected in series with a battery as shown in the diagram. The battery has negligible internal resistance.

What is the potential difference across the 180 Ω resistor?



A 1.6 V

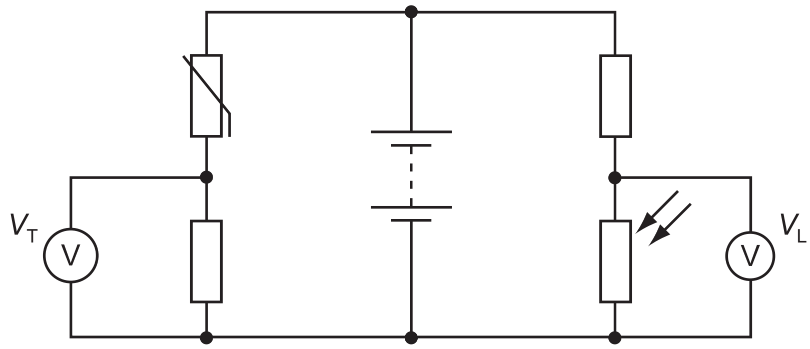
B 2.4 V

C 3.6 V

D 6.0 V

26 **9702/12/M/J/13/Q38**

In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



The readings on V_T and V_L are both high.

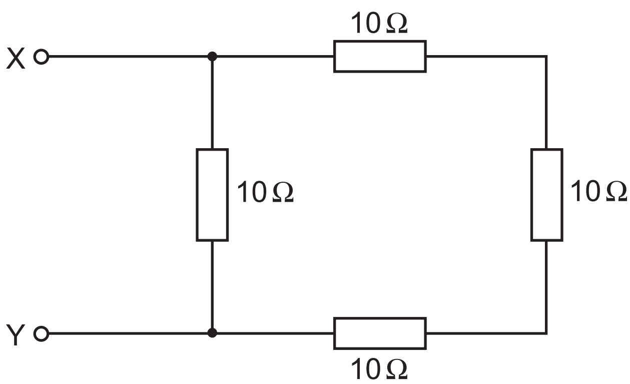
What are the conditions of temperature and light level?

	temperature	light level
A	low	low
B	low	high
C	high	low
D	high	high

27 **9702/12/O/N/15/Q36**

The diagram shows an arrangement of resistors.

What is the total electrical resistance between X and Y?



A less than 1 Ω

B between 1 Ω and 10 Ω

C between 10 Ω and 30 Ω

D 40 Ω

28 **9702/01/M/J/06/Q36**

When four identical lamps P, Q, R and S are connected as shown in diagram 1, they have normal brightness.

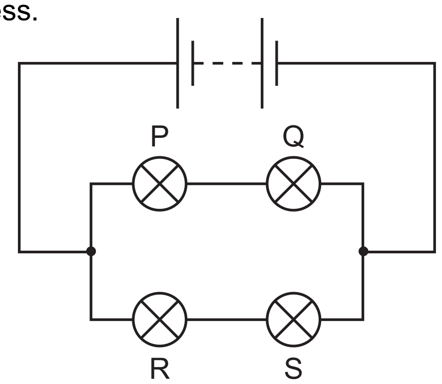


diagram 1

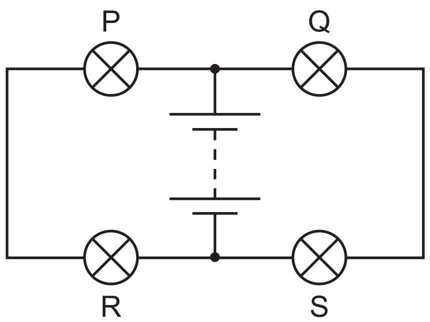


diagram 2

The four lamps and the battery are then connected as shown in diagram 2.

Which statement is correct?

A The lamps do not light.

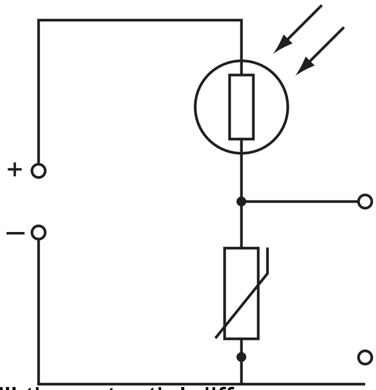
B The lamps are less bright than normal.

C The lamps have normal brightness.

D The lamps are brighter than normal.

29 **9702/01/M/J/06/Q37**

The diagram shows a light-dependent resistor (LDR) and a thermistor forming a potential divider.



Under which set of conditions will the potential difference across the thermistor have the greatest value?

	illumination	temperature
A	low	low
B	high	low
C	low	high
D	high	high

30 **9702/01/O/N/06/Q35**

The resistance of a device is designed to change with temperature.

What is the device?

A a light-dependent resistor

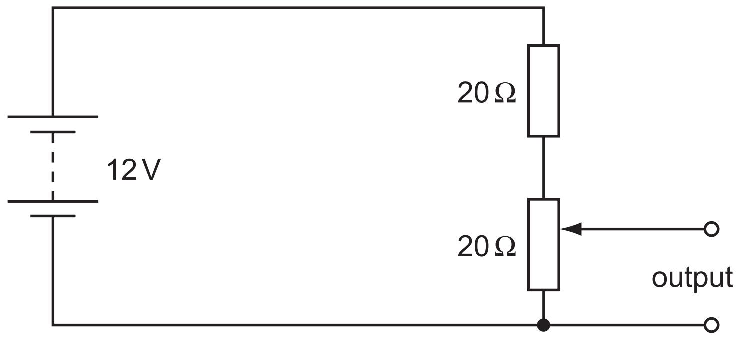
B a potential divider

C a semiconductor diode

D a thermistor

31 **9702/01/O/N/06/Q34**

The diagram shows a potentiometer and a fixed resistor connected across a 12 V battery of negligible internal resistance.



The fixed resistor and the potentiometer each have resistance 20 Ω. The circuit is designed to provide a variable output voltage.

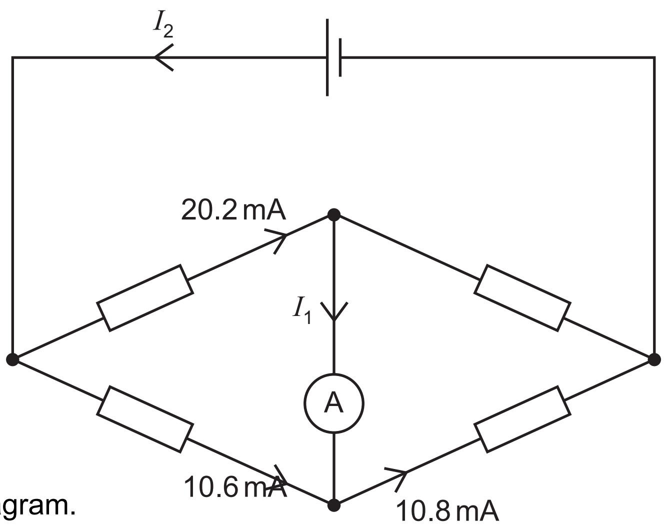
What is the range of output voltages?

A 0 – 6 V **B** 0 – 12 V

C 6 – 12 V **D** 12 – 20 V

32 **9702/01/O/N/06/Q36**

The diagram represents a circuit.



Some currents have been shown on the diagram.

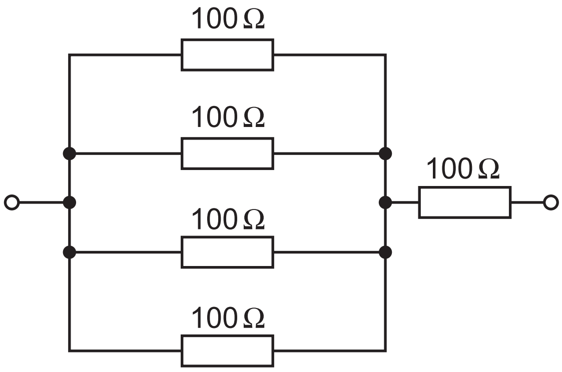
What are the currents I_1 and I_2?

	I_1	I_2
A	0.2 mA	10.8 mA
B	0.2 mA	30.8 mA
C	–0.2 mA	20.0 mA
D	–0.2 mA	30.8 mA

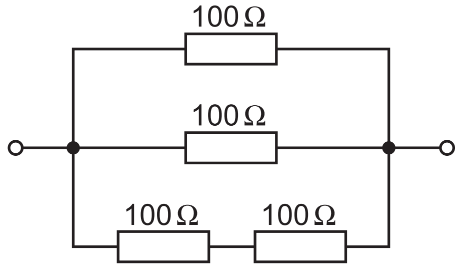
33 9702/01/O/N/06/Q37

Which circuit has a resistance of 40 Ω between the terminals?

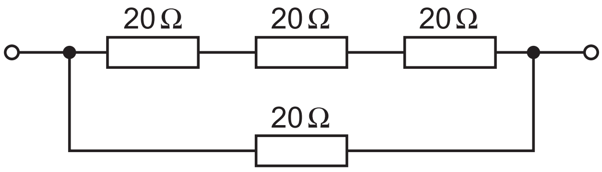
A



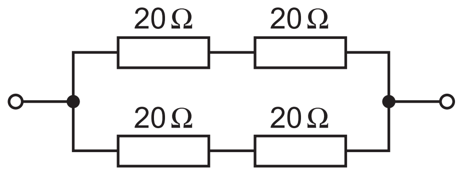
B



C

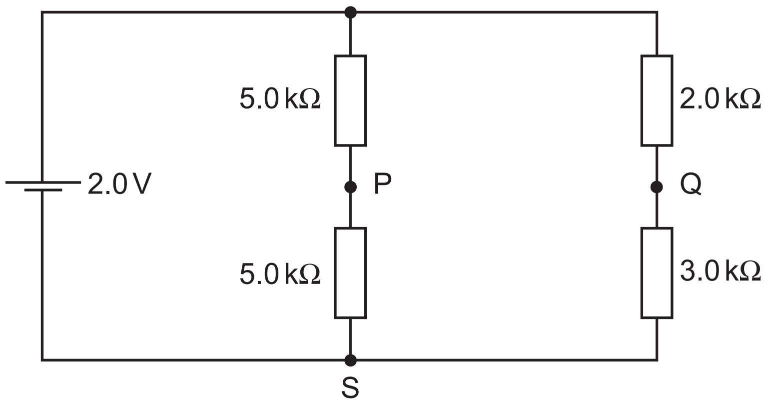


D



34 9702/01/M/J/07/Q33

A cell of e.m.f. 2.0 V and negligible internal resistance is connected to the network of resistors shown.



V_1 is the potential difference between S and P. V_2 is the potential difference between S and Q.

What is the value of $V_1 - V_2$?

A +0.50 V **B** +0.20 V **C** –0.20 V **D** –0.50 V

35 9702/01/M/J/07/Q34

A circuit is set up with an LDR and a fixed resistor as shown.

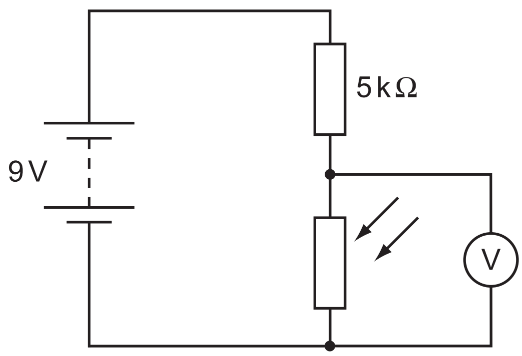
The voltmeter reads 4 V.

The light intensity is increased.

What is a possible voltmeter reading?

A 3 V **B** 4 V

C 6 V **D** 8 V

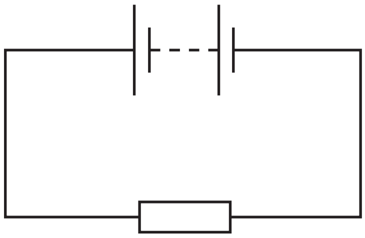


36 9702/01/M/J/07/Q35

In the circuit below, the battery converts an amount E of chemical energy to electrical energy when charge Q passes through the resistor in time t.

Which expressions give the e.m.f. of the battery and the current in the resistor?

	e.m.f.	current
A	EQ	Q/t
B	EQ	Qt
C	E/Q	Q/t
D	E/Q	Qt



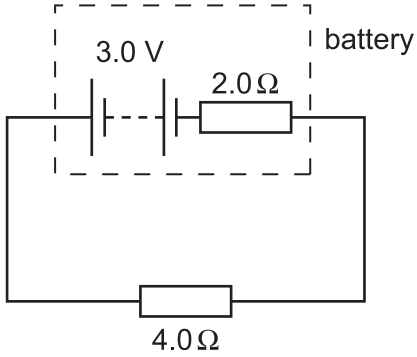
37 9702/01/M/J/07/Q36

A battery has an e.m.f. of 3.0 V and an internal resistance of 2.0 Ω.

The battery is connected to a load of 4.0 Ω.

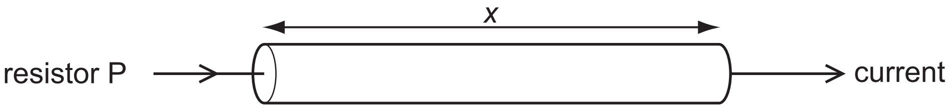
What are the terminal potential difference V and output power P?

	V/V	P/W
A	1.0	0.50
B	1.0	1.5
C	2.0	1.0
D	2.0	1.5

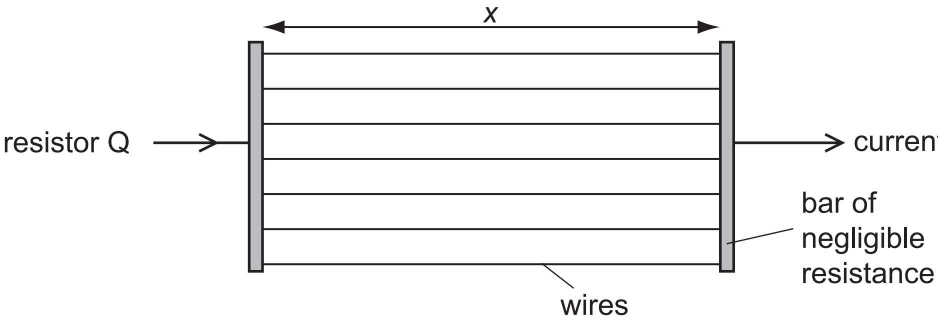


38 9702/01/M/J/07/Q37

A researcher has two pieces of copper of the same volume. All of the first piece is made into a cylindrical resistor P of length x.



All of the second piece is made into uniform wires each of the same length x which he connects between two bars of negligible resistance to form a resistor Q.



How do the electrical resistances of P and Q compare?

A P has a larger resistance than Q.

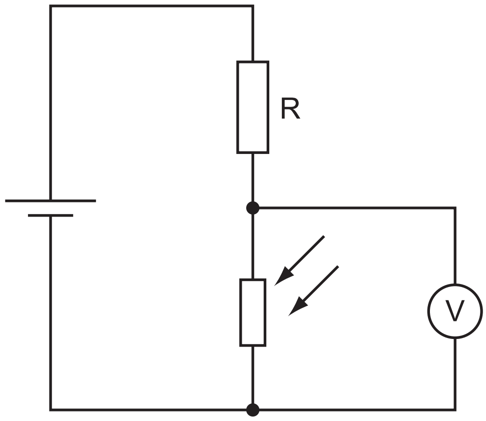
B Q has a larger resistance than P.

C P and Q have equal resistance.

D Q may have a larger or smaller resistance than P, depending on the number of wires made.

39 **9702/01/O/N/07/Q32**

A potential divider consists of a fixed resistor R and a light-dependent resistor (LDR).

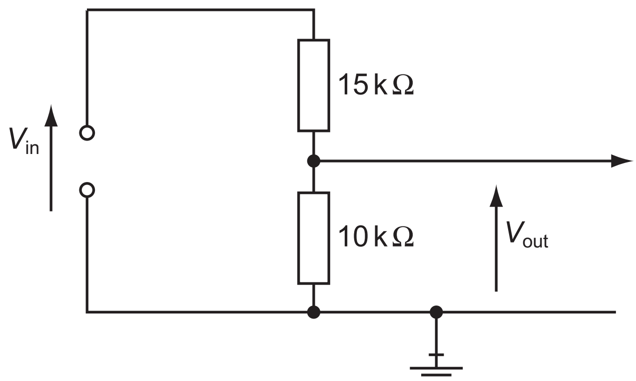


What happens to the voltmeter reading, and why does it happen, when the intensity of light on the LDR increases?

A The voltmeter reading decreases because the LDR resistance decreases.
B The voltmeter reading decreases because the LDR resistance increases.
C The voltmeter reading increases because the LDR resistance decreases.
D The voltmeter reading increases because the LDR resistance increases.

40 **9702/01/O/N/07/Q33**

The circuit is designed to trigger an alarm system when the input voltage exceeds some preset value. It does this by comparing V_{out} with a fixed reference voltage, which is set at 4.8 V.



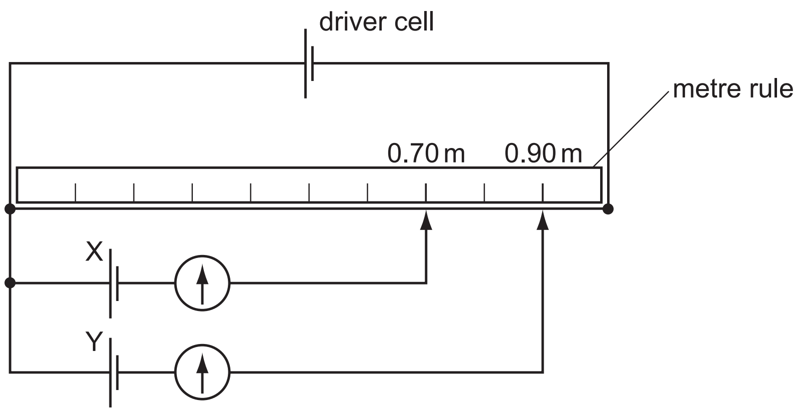
V_{out} is equal to 4.8 V.

What is the input voltage V_{in}?

A 4.8 V
B 7.2 V
C 9.6 V
D 12 V

41 **9702/01/O/N/07/Q34**

A potentiometer is used as shown to compare the e.m.f.s of two cells.



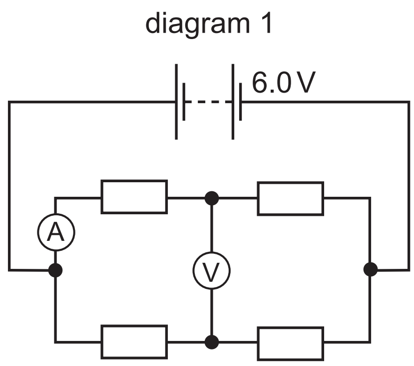
The balance points for cells X and Y are 0.70 m and 0.90 m respectively.

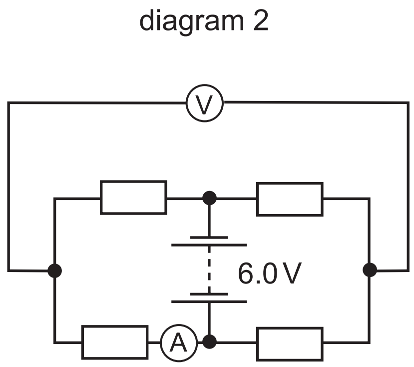
If the e.m.f. of cell X is 1.1 V, what is the e.m.f. of cell Y?

A 0.69 V **B** 0.86 V **C** 0.99 V **D** 1.4 V

42 9702/01/O/N/07/Q35

When four identical resistors are connected as shown in diagram 1, the ammeter reads 1.0 A and the voltmeter reads zero.





The resistors and meters are reconnected to the supply as shown in diagram 2.

What are the meter readings in diagram 2?

	voltmeter reading / V	ammeter reading / A
A	0	1.0
B	3.0	0.5
C	3.0	1.0
D	6.0	0

43 9702/01/O/N/07/Q36

How is it possible to distinguish between the isotopes of uranium?

A Their nuclei have different charge and different mass, and they emit different particles when they decay.

B Their nuclei have different charge but the same mass.

C Their nuclei have the same charge but different mass.

D Their nuclei have the same charge and mass, but they emit different particles when they decay.

44 9702/01/O/N/07/Q37

What is **not** conserved in nuclear processes?

A energy and mass together

B nucleon number

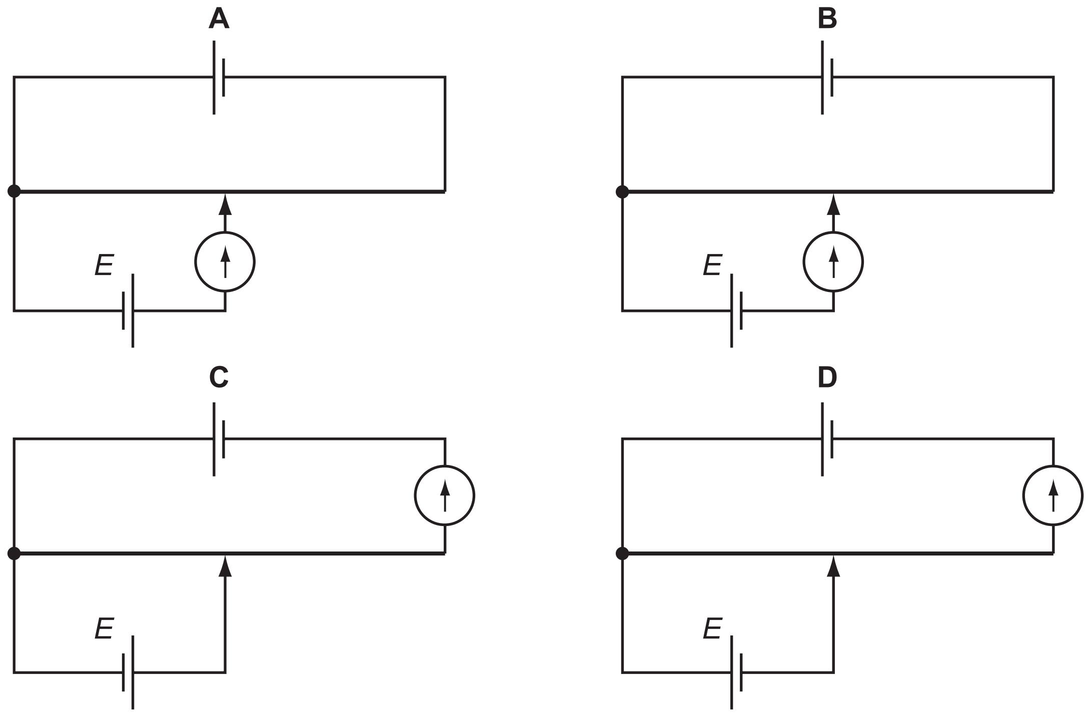
C neutron number

D charge

45 9702/01/M/J/08/Q38

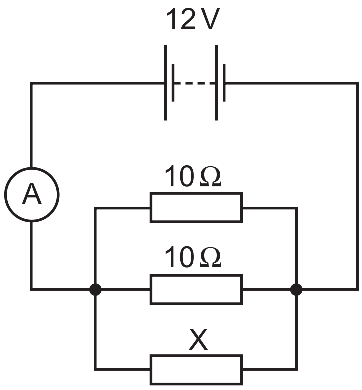
The unknown e.m.f. E of a cell is to be determined using a potentiometer circuit. The balance length is to be measured when the galvanometer records a null reading.

What is the correct circuit to use?



46 9702/01/O/N/08/Q35

The diagram shows a circuit containing three resistors in parallel.



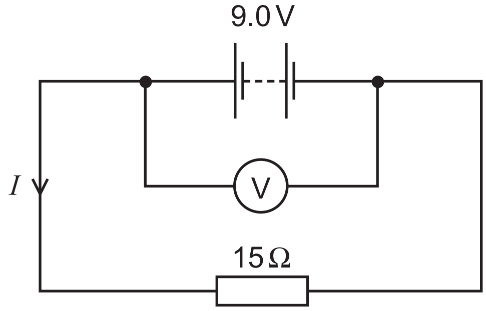
The battery has e.m.f. 12 V and negligible internal resistance. The ammeter reading is 3.2 A.

What is the resistance of X?

A 2.1 Ω **B** 4.6 Ω **C** 6.0 Ω **D** 15 Ω

47 **9702/01/O/N/08/Q36**

The e.m.f. of the battery is 9.0 V. The reading on the high-resistance voltmeter is 7.5 V.

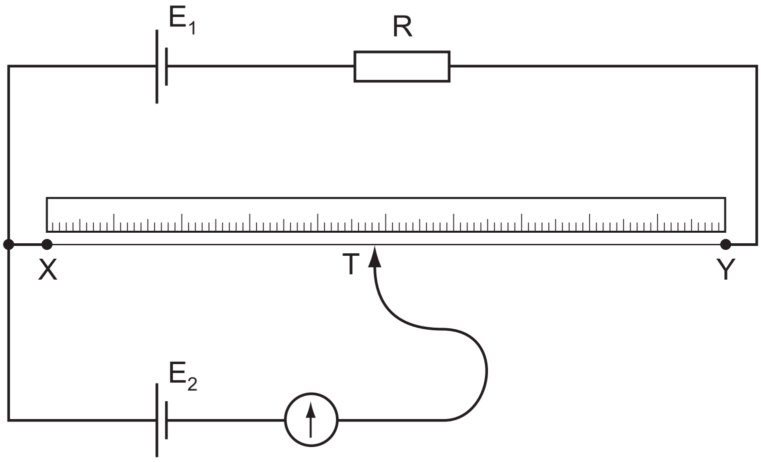


What is the current I?

A 0.10 A **B** 0.50 A **C** 0.60 A **D** 2.0 A

48 **9702/01/O/N/08/Q37**

The diagram shows a potentiometer circuit.



The contact T is placed on the wire and moved along the wire until the galvanometer reading is zero. The length XT is then noted.

In order to calculate the potential difference per unit length on the wire XY, which value must also be known?

A the e.m.f. of the cell E_1

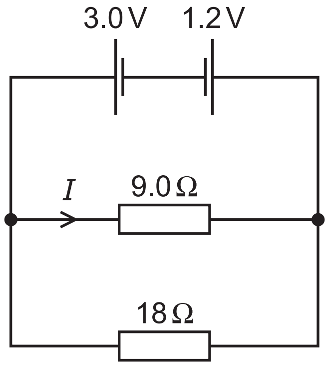
B the e.m.f. of the cell E_2

C the resistance of resistor R

D the resistance of the wire XY

49 **9702/01/M/J/09/Q33**

Two cells of e.m.f. 3.0 V and 1.2 V and negligible internal resistance are connected to resistors of resistance 9.0 Ω and 18 Ω as shown.

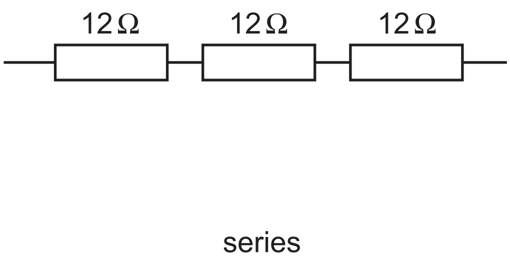


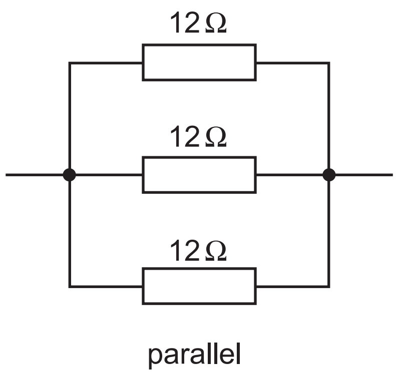
What is the value of the current I in the 9.0 Ω resistor?

A 0.10 A **B** 0.20 A **C** 0.30 A **D** 0.47 A

50 **9702/01/M/J/09/Q34**

Six identical 12 Ω resistors are arranged in two groups, one with three in series and the other with three in parallel.



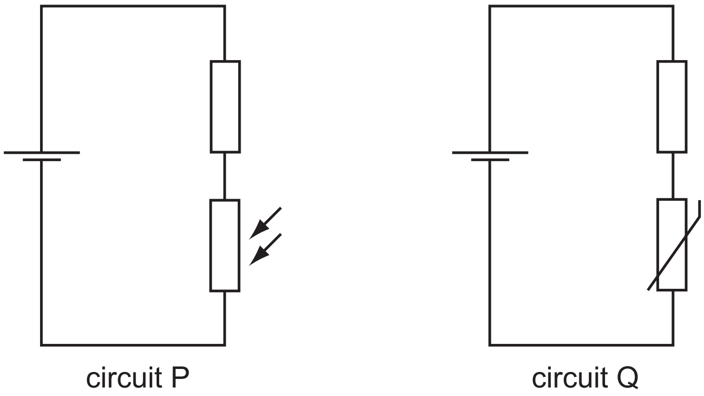


What are the combined resistances of each of these two arrangements?

	series	parallel
A	4.0 Ω	0.25 Ω
B	4.0 Ω	36 Ω
C	36 Ω	0.25 Ω
D	36 Ω	4.0 Ω

51 9702/01/M/J/09/Q35

The diagrams show a light-dependent resistor in circuit P, and a thermistor in circuit Q.



How does the potential difference across the fixed resistor in each circuit change when both the brightness of the light on the light-dependent resistor and the temperature of the thermistor are increased?

	circuit P	circuit Q
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase

52 9702/12/O/N/09/Q32

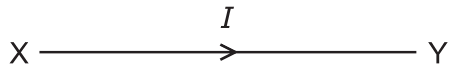
Each of Kirchhoff's two laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	the algebraic sum of currents into a junction is zero	charge
B	the algebraic sum of currents into a junction is zero	energy
C	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	charge
D	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	energy

53 **9702/12/O/N/09/Q33**

The diagram shows the symbol for a wire carrying a current I.



What does this current represent?

A the amount of charge flowing past a point in XY per second

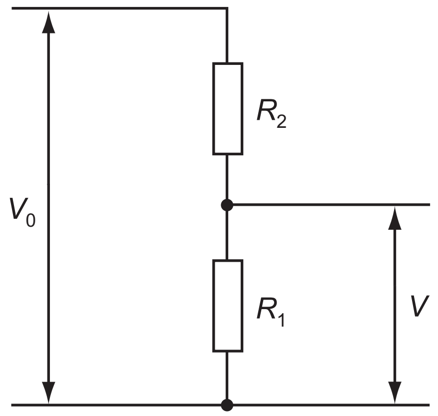
B the number of electrons flowing past a point in XY per second

C the number of positive ions flowing past a point in XY per second

D the number of protons flowing past a point in XY per second

54 **9702/12/O/N/09/Q34**

A potential divider consisting of resistors of resistance R_1 and R_2 is connected to an input potential difference of V_0 and gives an output p.d. of V.

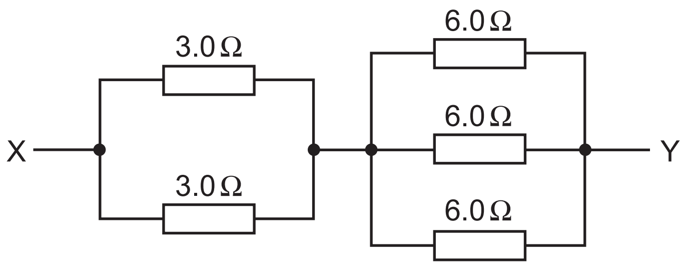


What is the value of V?

A $\dfrac{V_0R_1}{R_2}$ **B** $\dfrac{V_0R_1}{R_1+R_2}$ **C** $\dfrac{V_0R_2}{R_1+R_2}$ **D** $\dfrac{V_0(R_1+R_2)}{R_1}$

55 **9702/12/O/N/09/Q35**

A network of resistors consists of two 3.0 Ω resistors and three 6.0 Ω resistors.



What is the combined resistance of this network between points X and Y?

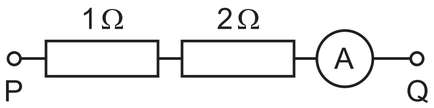
A 0.86 Ω **B** 1.2 Ω **C** 3.5 Ω **D** 24 Ω

56 **9702/12/M/J/10/Q32***

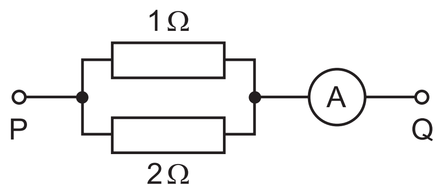
In each arrangement of resistors, the ammeter has a resistance of $2\,\Omega$.

Which arrangement gives the largest reading on the ammeter when the same potential difference is applied between points P and Q?

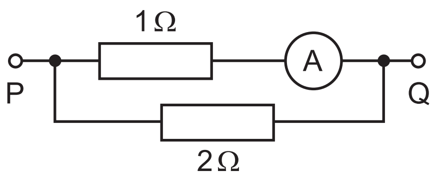
A



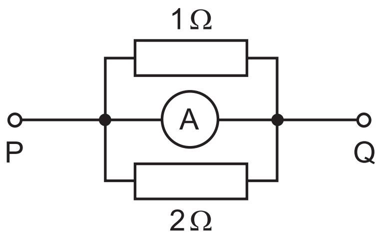
B



C



D



57 **9702/12/M/J/10/Q33**

A source of e.m.f. of 9.0 mV has an internal resistance of $6.0\,\Omega$.

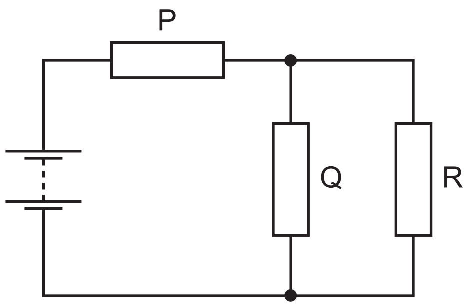
It is connected across a galvanometer of resistance $30\,\Omega$.

What will be the current in the galvanometer?

A $250\,\mu$A **B** $300\,\mu$A **C** 1.5 mA **D** 2.5 mA

58 **9702/12/M/J/10/Q34**

The resistors P, Q and R in the circuit have equal resistance.



The battery, of negligible internal resistance, supplies a total power of 12 W.

What is the power dissipated by heating in resistor R?

A 2 W **B** 3 W **C** 4 W **D** 6 W

59 **9702/12/M/J/10/Q35**

In deriving a formula for the combined resistance of three different resistors in series, Kirchhoff's laws are used.

Which physics principle is involved in this derivation?

A the conservation of charge

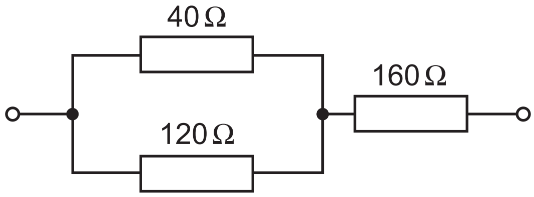
B the direction of the flow of charge is from negative to positive

C the potential difference across each resistor is the same

D the current varies in each resistor, in proportion to the resistor value

60 **9702/11/O/N/10/Q35**

The diagram shows part of a circuit.

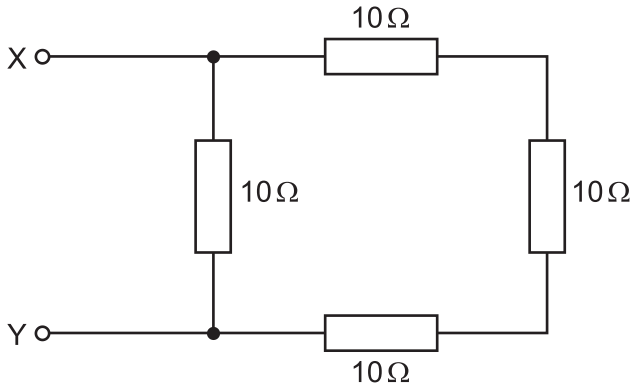


What is the total resistance of the combination of the three resistors?

A 320 Ω **B** 240 Ω **C** 190 Ω **D** 80 Ω

61 **9702/11/O/N/10/Q36**

The diagram shows an arrangement of resistors.



What is the total electrical resistance between X and Y?

A less than 1 Ω

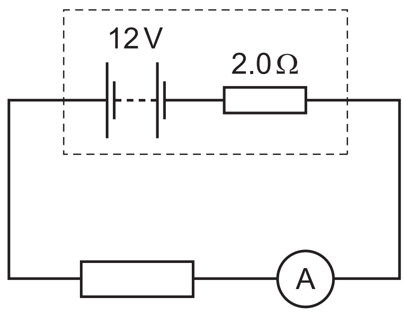
B between 1 Ω and 10 Ω

C between 10 Ω and 30 Ω

D 40 Ω

62 **9702/12/O/N/10/Q33**

A battery of e.m.f. 12 V and internal resistance 2.0 Ω is connected in series with an ammeter of negligible resistance and an external resistor. External resistors of various different values are used.

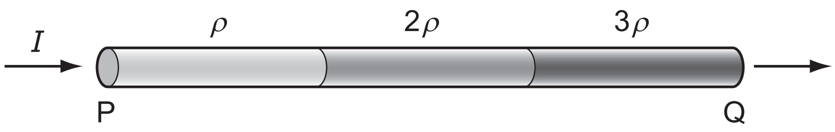


Which combination of current and resistor value is **not** correct?

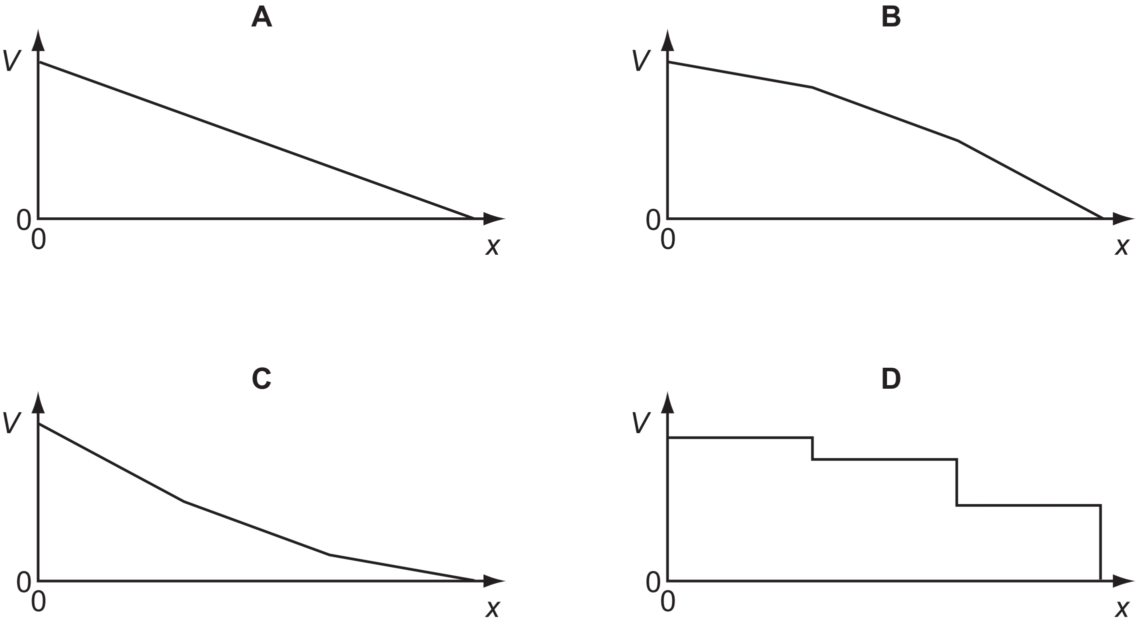
	current / A	external resistor value / Ω
A	1.0	10
B	1.2	8
C	1.5	6
D	1.8	4

63 9702/12/O/N/10/Q34

A wire PQ is made of three different materials, with resistivities ρ, 2ρ and 3ρ. There is a current I in this composite wire, as shown.

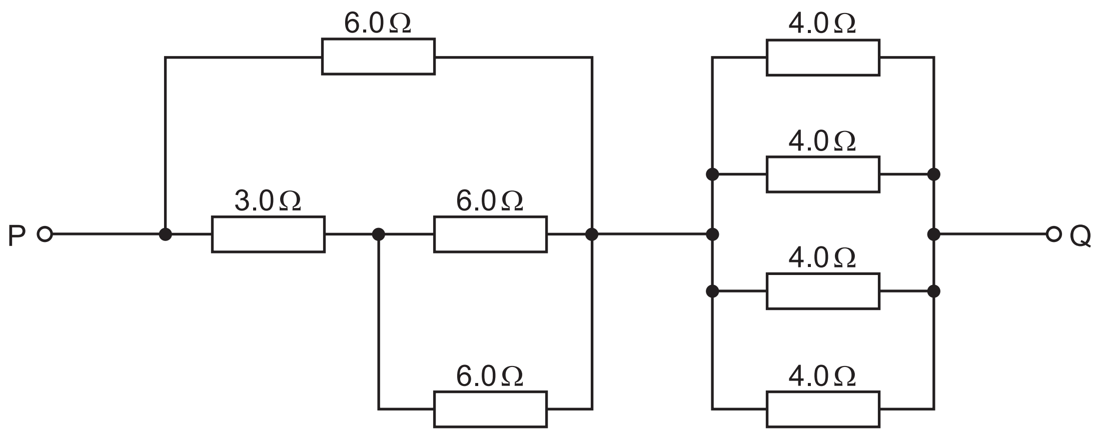


Which graph best shows how the potential V along the wire varies with distance x from P?



64 9702/12/O/N/10/Q35

The diagram shows part of a circuit.

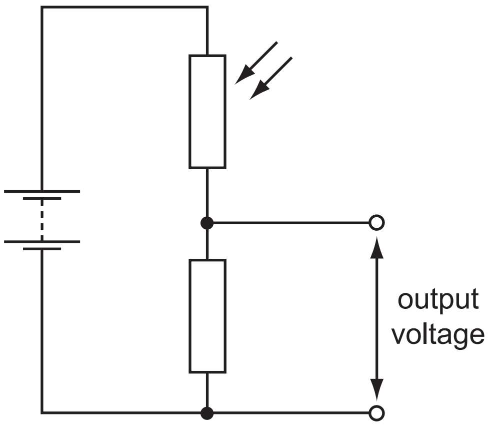


What is the resistance between the points P and Q due to the resistance network?

A 1.3Ω **B** 4.0Ω **C** 10Ω **D** 37Ω

65 9702/12/O/N/10/Q36

The diagram shows a potential divider circuit.



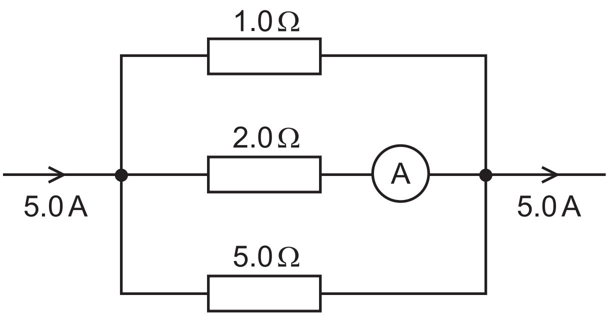
The light level increases.

What is the effect on the resistance of the light-dependent resistor (LDR) and on the output voltage?

	resistance of the LDR	output voltage
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

66 9702/11/M/J/11/Q35

The diagram shows part of a current-carrying circuit. The ammeter has negligible internal resistance.

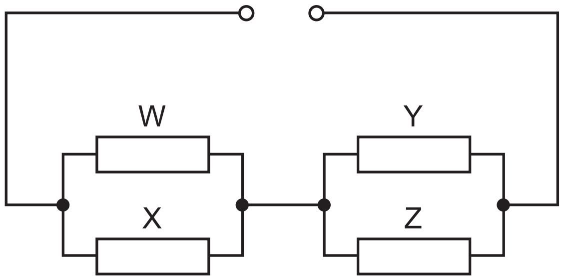


What is the reading on the ammeter?

A 0.7 A **B** 1.3 A **C** 1.5 A **D** 1.7 A

67 9702/11/M/J/11/Q36

Four resistors of equal value are connected as shown.



How will the powers to the resistors change when resistor W is removed?

A The powers to X, Y and Z will all increase.

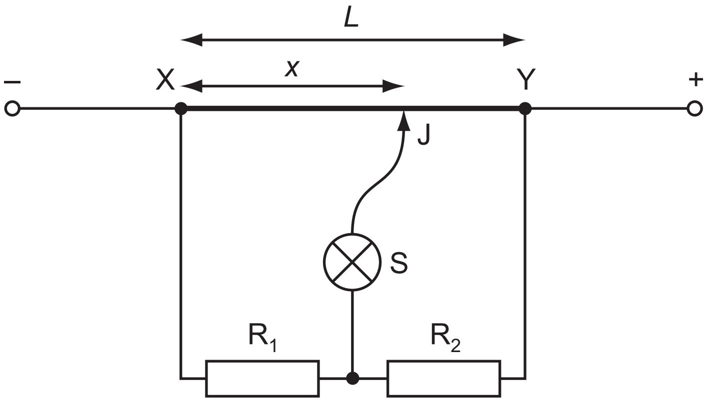
B The power to X will decrease and the powers to Y and Z will increase.

C The power to X will increase and the powers to Y and Z will decrease.

D The power to X will increase and the powers to Y and Z will remain unaltered.

68 9702/11/M/J/11/Q37

In the circuit shown, XY is a length *L* of uniform resistance wire. R_1 and R_2 are unknown resistors. J is a sliding contact that joins the junction of R_1 and R_2 to points on XY through a small signal lamp S.



To determine the ratio $\frac{V_1}{V_2}$ of the potential differences across R_1 and R_2, a point is found on XY at which the lamp is off. This point is at a distance *x* from X.

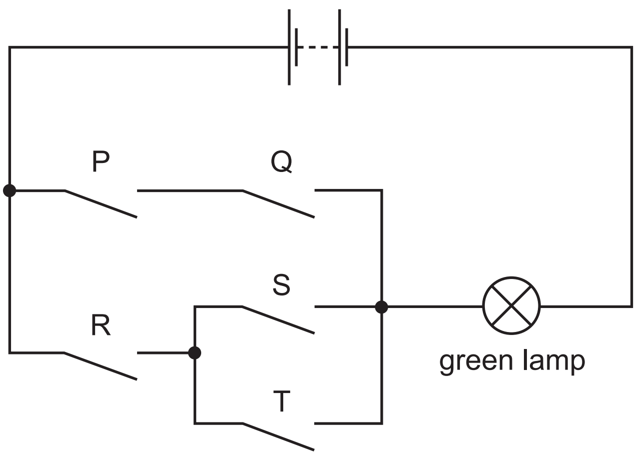
What is the value of the ratio $\frac{V_1}{V_2}$?

A $\frac{L}{x}$ **B** $\frac{x}{L}$ **C** $\frac{L-x}{x}$ **D** $\frac{x}{L-x}$

69 9702/12/M/J/11/Q36

Safety on railways is increased by using several electrical switches.

In the diagram, switches P, Q, R, S and T control the current through a green lamp.

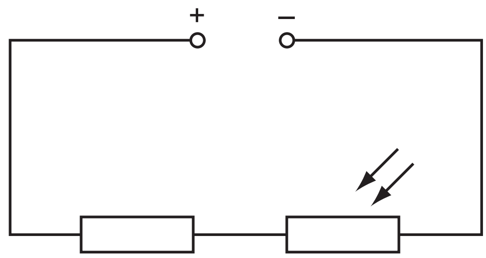


Which row does **not** allow the green lamp to light?

	P	Q	R	S	T
A	closed	closed	closed	open	closed
B	closed	open	closed	closed	open
C	closed	open	open	closed	closed
D	open	open	closed	open	closed

70 9702/12/M/J/11/Q37

The diagram shows a fixed resistor and a light-dependent resistor (LDR) in series with a constant low-voltage supply.



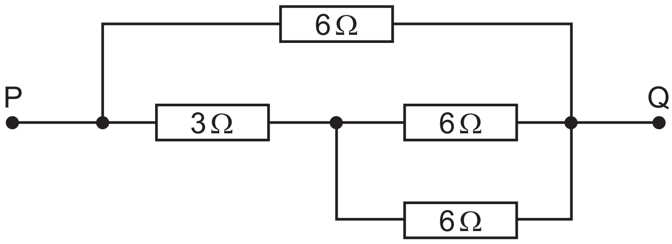
When the LDR is in the dark, the fixed resistor and the LDR have the same value of resistance.

Light is shone on the LDR.
What happens to the potential differences across the two components?

	p.d. across resistor	p.d. across LDR
A	decreased	increased
B	increased	decreased
C	no change	increased
D	no change	decreased

71 9702/12/M/J/11/Q38

The diagram shows a d.c. circuit.

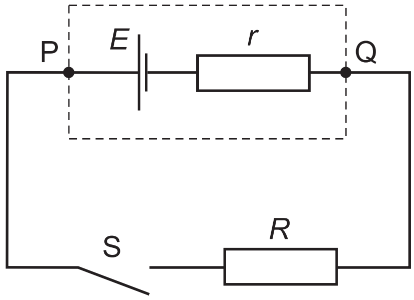


What is the resistance between the points P and Q due to the resistance network?

A 0.47 Ω **B** 2.1 Ω **C** 3.0 Ω **D** 21 Ω

72 9702/11/O/N/11/Q36

A cell of e.m.f. *E* and internal resistance *r* is connected in series with a switch S and an external resistor of resistance *R*.



The p.d. between P and Q is *V*.

When S is closed,

A *V* decreases because there is a p.d. across *R*.

B *V* decreases because there is a p.d. across *r*.

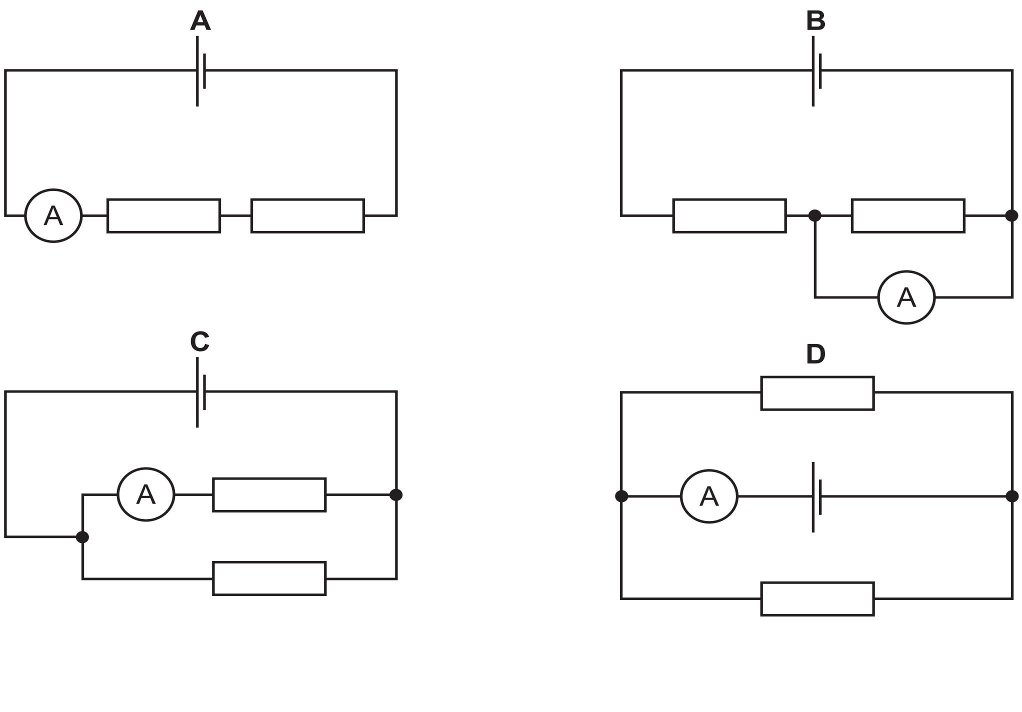
C *V* remains the same because the decrease of p.d. across *r* is balanced by the increase of p.d. across *R*.

D *V* remains the same because the sum of the p.d.s across *r* and *R* is still equal to *E*.

73 9702/11/O/N/11/Q37

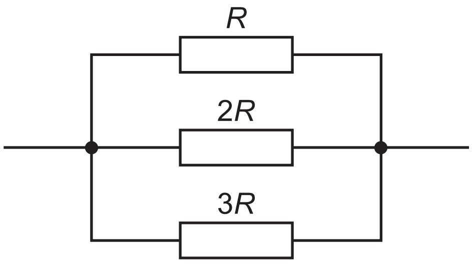
A cell, two resistors of equal resistance and an ammeter are used to construct four circuits. The resistors are the only parts of the circuits that have resistance.

In which circuit will the ammeter show the greatest reading?



74 9702/11/O/N/11/Q38

Three resistors of resistance *R*, 2*R* and 3*R* are connected in parallel.

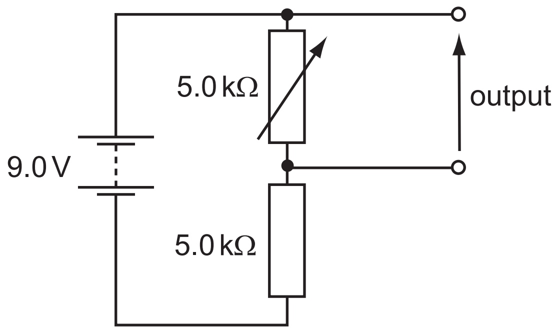


Using I to represent the current through the resistor of resistance *R*, which row represents the relationships between the currents through the resistors?

	resistor resistance		
	R	2*R*	3*R*
A	I	$\frac{1}{3}I$	$\frac{1}{2}I$
B	I	$\frac{1}{2}I$	$\frac{1}{3}I$
C	I	$\frac{2}{3}I$	$\frac{1}{3}I$
D	I	$2I$	$3I$

75 9702/11/O/N/11/Q39

The diagram shows a potential divider circuit designed to provide a variable output p.d.



Which row gives the available range of output p.d.?

	maximum output	minimum output
A	3.0 V	0
B	4.5 V	0
C	9.0 V	0
D	9.0 V	4.5 V

76 9702/12/O/N/11/Q36

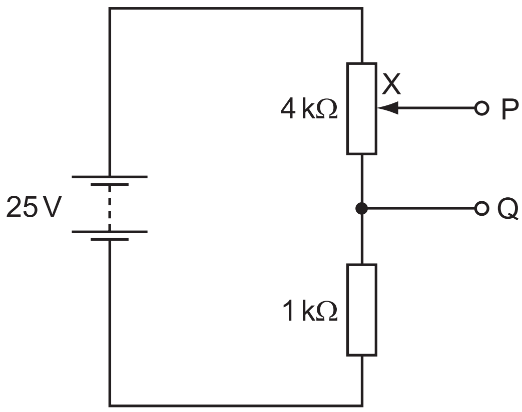
Each of Kirchhoff's laws is linked to the conservation of a physical quantity.

Which physical quantities are assumed to be conserved in the formulation of Kirchhoff's first law and of Kirchhoff's second law?

	Kirchhoff's first law	Kirchhoff's second law
A	energy	charge
B	energy	momentum
C	charge	energy
D	momentum	energy

77 9702/12/O/N/11/Q37

The diagram shows a potential divider circuit which, by adjustment of the contact X, can be used to provide a variable potential difference between the terminals P and Q.

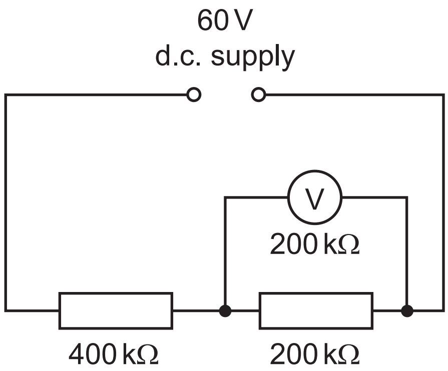


What are the limits of this potential difference?

A 0 and 5 V **B** 0 and 20 V **C** 0 and 25 V **D** 5 V and 25 V

78 9702/12/O/N/11/Q38

A constant 60 V d.c. supply is connected across two resistors of resistance 400 kΩ and 200 kΩ.

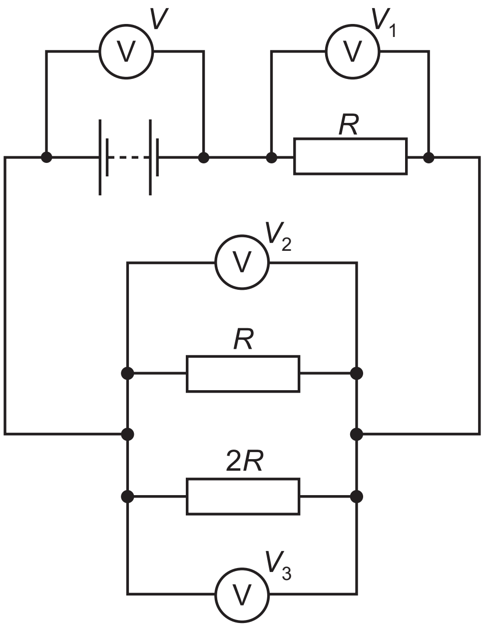


What is the reading on a voltmeter, also of resistance 200 kΩ, when connected across the 200 kΩ resistor as shown in the diagram?

A 12 V **B** 15 V **C** 20 V **D** 30 V

79 **9702/01/M/J/05/Q36**

The diagram shows a circuit with four voltmeter readings V, V_1, V_2 and V_3.



Which equation relating the voltmeter readings must be true?

A $V = V_1 + V_2 + V_3$

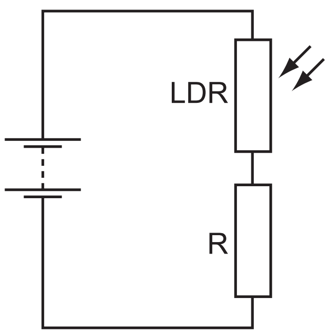
B $V + V_1 = V_2 + V_3$

C $V_3 = 2(V_2)$

D $V - V_1 = V_3$

80 **9702/12/M/J/12/Q36**

A light-dependent resistor (LDR) is connected in series with a resistor R and a battery.



The resistance of the LDR is equal to the resistance of R when no light falls on the LDR.

When the light intensity falling on the LDR increases, which statement is correct?

A The current in R decreases.

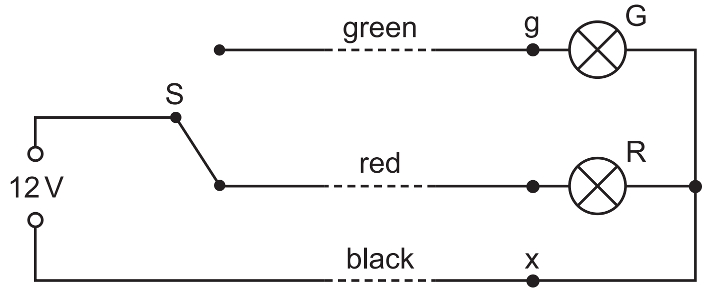
B The current in the LDR decreases.

C The p.d. across R decreases.

D The p.d. across the LDR decreases.

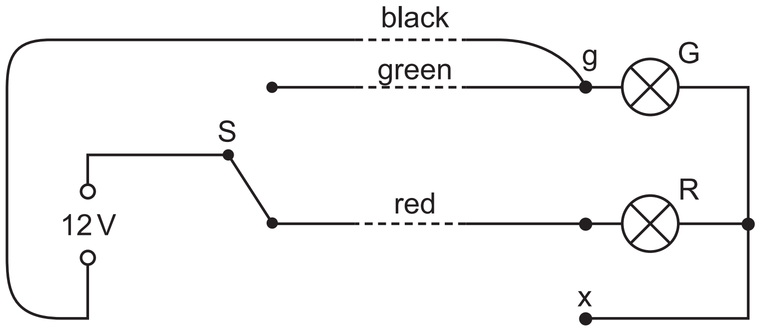
81 9702/12/M/J/12/Q37

The diagram shows the circuit for a signal to display a green or a red light. It is controlled by the switch S.



The signal is some way from S to which it is connected by a cable with green, red and black wires. At the signal, the green and red wires are connected to the corresponding lamp and the black wire is connected to a terminal x to provide a common return. The arrangement is shown correctly connected and with the switch set to illuminate the red lamp.

During maintenance, the wires at the signal are disconnected and, when reconnected, the black wire is connected in error to the green lamp (terminal g) instead of terminal x. The red wire is connected correctly to its lamp and connections at S remain as in the diagram.



When the system is tested with the switch connection to the red wire, what does the signal show?

A the green lamp illuminated normally

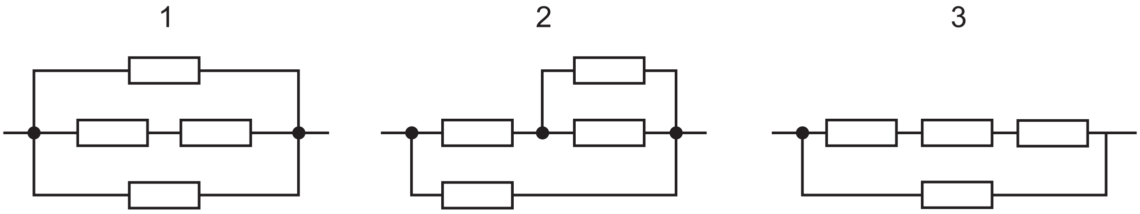
B the red lamp illuminated normally

C the red and green lamps both illuminated normally

D the red and green lamps both illuminated dimly

82 9702/12/M/J/12/Q38

Four identical resistors are connected in the three networks below.

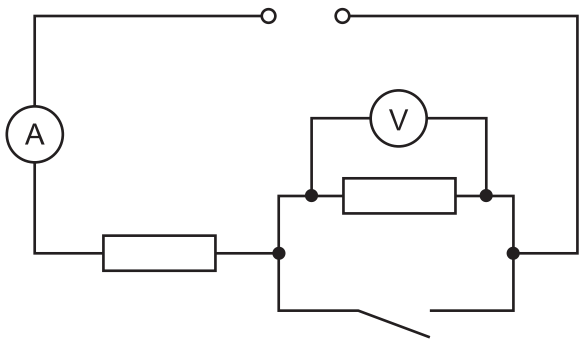


Which arrangement has the highest total resistance and which has the lowest?

	highest	lowest
A	1	2
B	1	3
C	3	1
D	3	2

83 9702/13/M/J/12/Q35

In the circuit below, the ammeter reading is I and the voltmeter reading is V.

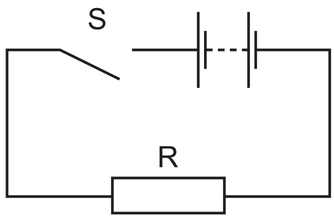


When the switch is closed, which row describes what happens to I and V?

	I	V
A	decreases	decreases to zero
B	increases	decreases to zero
C	increases	stays the same
D	stays the same	increases

84 9702/13/M/J/12/Q36

The diagram shows a simple circuit.



Which statement is correct?

A When switch S is closed, the electromotive force (e.m.f.) of the battery falls because work is done against the internal resistance of the battery.

B When switch S is closed, the e.m.f. of the battery falls because work is done against the resistance R.

C When switch S is closed, the potential difference across the battery falls because work is done against the internal resistance of the battery.

D When switch S is closed, the potential difference across the battery falls because work is done against the resistance R.

85 9702/13/M/J/12/Q37

The circuit below has a current I in the resistor R.

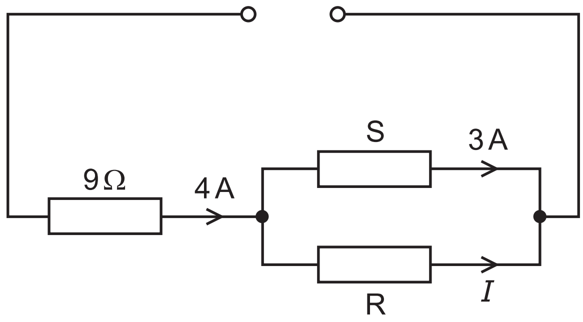
What must be known in order to determine the value of I?

A e.m.f. of the power supply

B resistance of resistor S

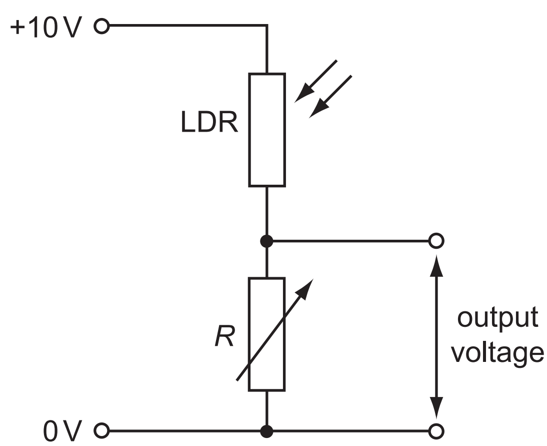
C Kirchhoff's first law

D Kirchhoff's second law



86 9702/13/M/J/12/Q38

A potential divider consists of a light-dependent resistor (LDR) in series with a variable resistor of resistance R. The resistance of the LDR decreases when the light level increases. The variable resistor can be set at either high resistance or low resistance.

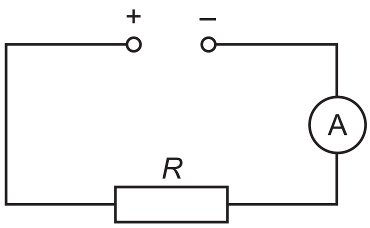


Which situation gives the largest output voltage?

	light level at LDR	R
A	high	high
B	high	low
C	low	high
D	low	low

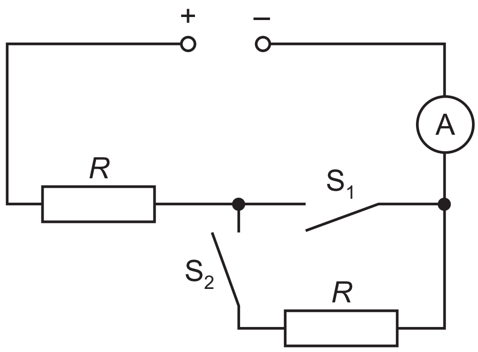
87 9702/12/O/N/12/Q37

The ammeter reading in the circuit below is I.



Another circuit containing the same voltage supply, two switches, an ammeter and two resistors each of resistance R, is shown.

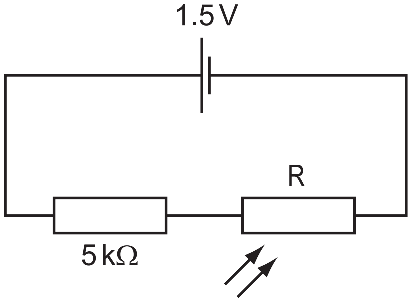
Which row is **not** correct?



	S_1	S_2	ammeter reading
A	closed	closed	I
B	closed	open	I
C	open	closed	I
D	open	open	0

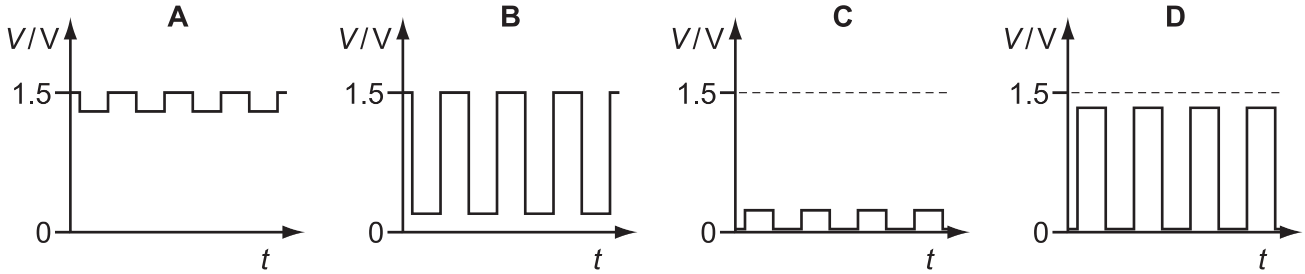
88 9702/12/O/N/12/Q38

A light-dependent resistor R has resistance of about 1 MΩ in the dark and about 1 kΩ when illuminated. It is connected in series with a 5 kΩ resistor to a 1.5 V cell of negligible internal resistance.



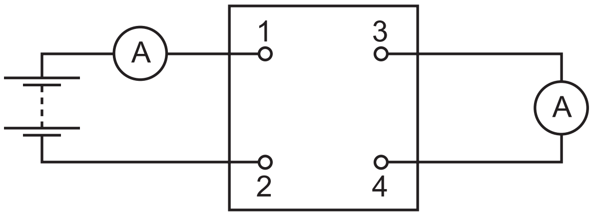
The light-dependent resistor is illuminated (in an otherwise dark room) by a flashing light.

Which graph best shows the variation with time t of potential difference V across R?



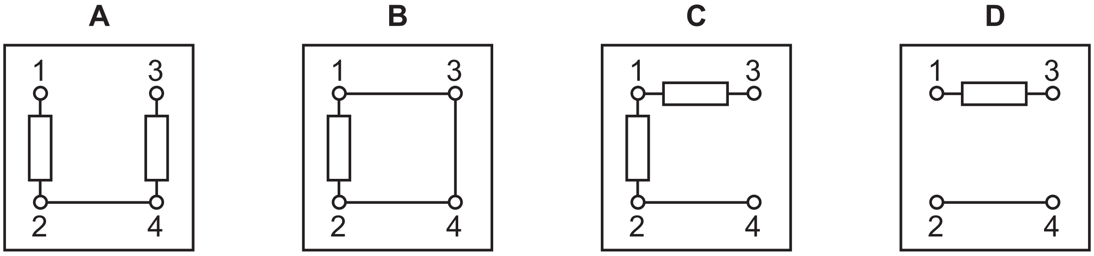
89 **9702/13/O/N/12/Q35**

The diagram shows a four-terminal box connected to a battery and two ammeters.



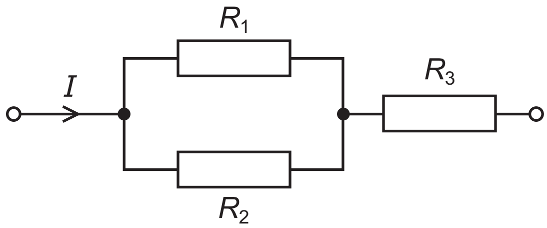
The currents in the two meters are identical.

Which circuit, within the box, will give this result?



90 **9702/13/O/N/12/Q37**

The diagram shows a resistor network. The potential difference across the network is V.



Is the equation shown below correct for the network? $V = I(1/R_1 + 1/R_2 + R_3)$

A Yes, it correctly combines two series resistors with one parallel resistor, and correctly uses Ohm's Law.

B Yes, it correctly combines two parallel resistors with one series resistor, and correctly uses Ohm's Law.

C No, because it should read $V = I \div (1/R_1 + 1/R_2 + R_3)$.

D No, because the terms $1/R_2$ and R_3 have different units and cannot be added.

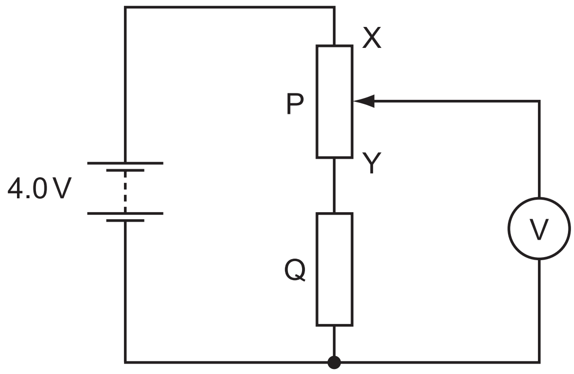
91 **9702/11/M/J/13/Q34**

The principles of conservation of which two quantities are associated with Kirchhoff's first and second laws?

	first law	second law
A	charge	energy
B	charge	voltage
C	energy	charge
D	voltage	charge

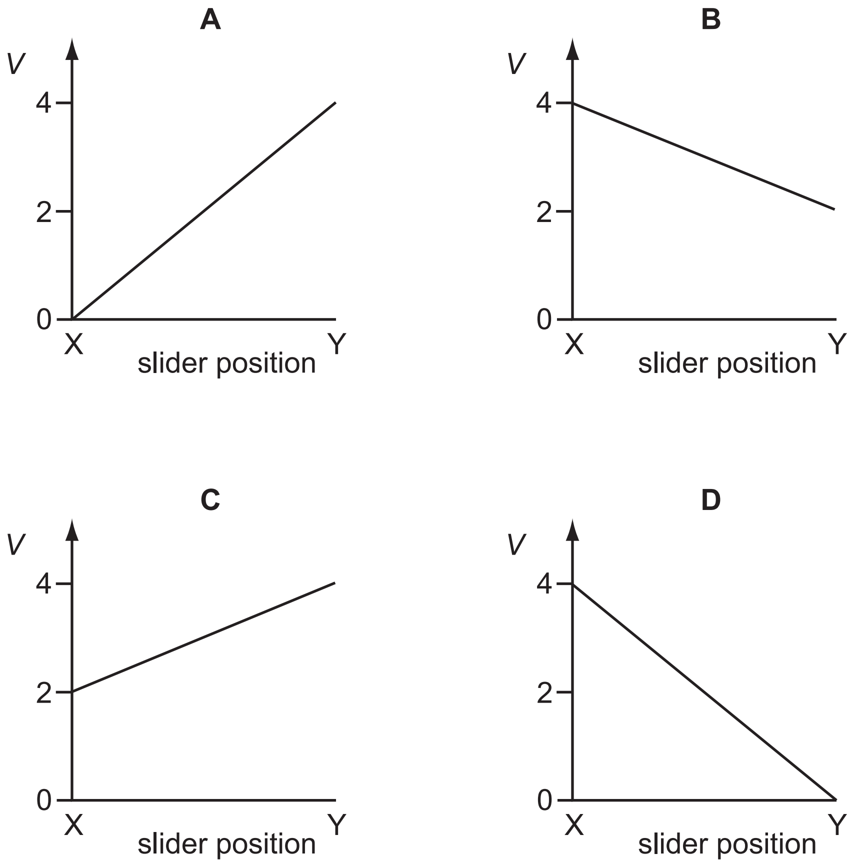
92 9702/01/M/J/05/Q37

In the circuit below, P is a potentiometer of total resistance 10 Ω and Q is a fixed resistor of resistance 10 Ω. The battery has an e.m.f. of 4.0 V and negligible internal resistance. The voltmeter has a very high resistance.



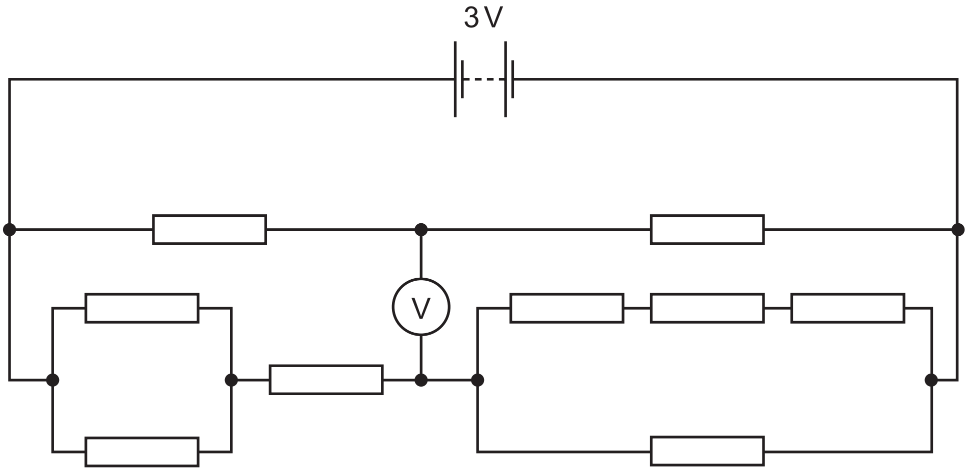
The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading *V* is plotted against slider position.

Which graph is obtained?



93 **9702/11/M/J/13/Q35**

A circuit is set up as shown, supplied by a 3 V battery. All resistances are 1 kΩ.

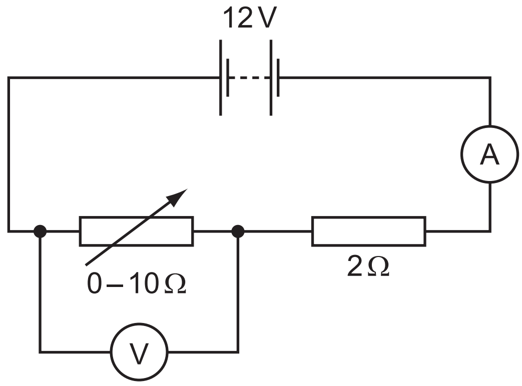


What will be the reading on the voltmeter?

A 0 **B** 0.5 V **C** 1.0 V **D** 1.5 V

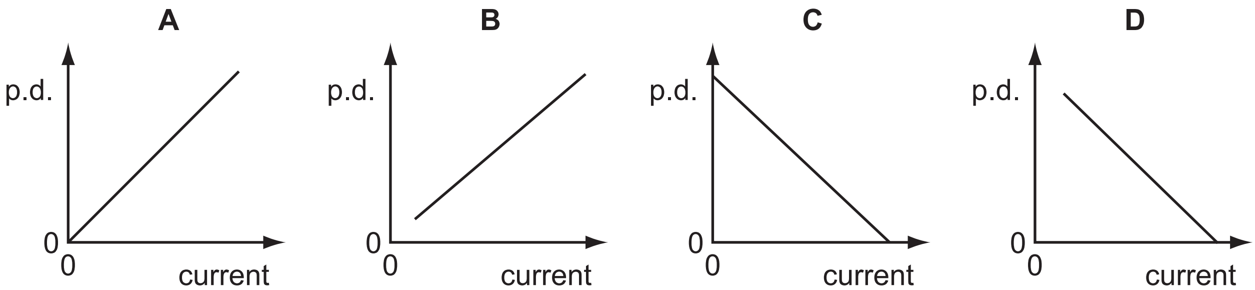
94 **9702/11/M/J/13/Q37***

A 12 V battery is in series with an ammeter, a 2 Ω fixed resistor and a 0 – 10 Ω variable resistor. A high-resistance voltmeter is connected across the variable resistor.



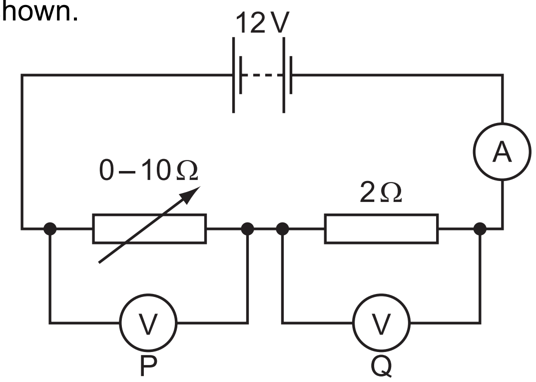
The resistance of the variable resistor is changed from zero to its maximum value.

Which graph shows how the potential difference (p.d.) measured by the voltmeter varies with the current measured by the ammeter?



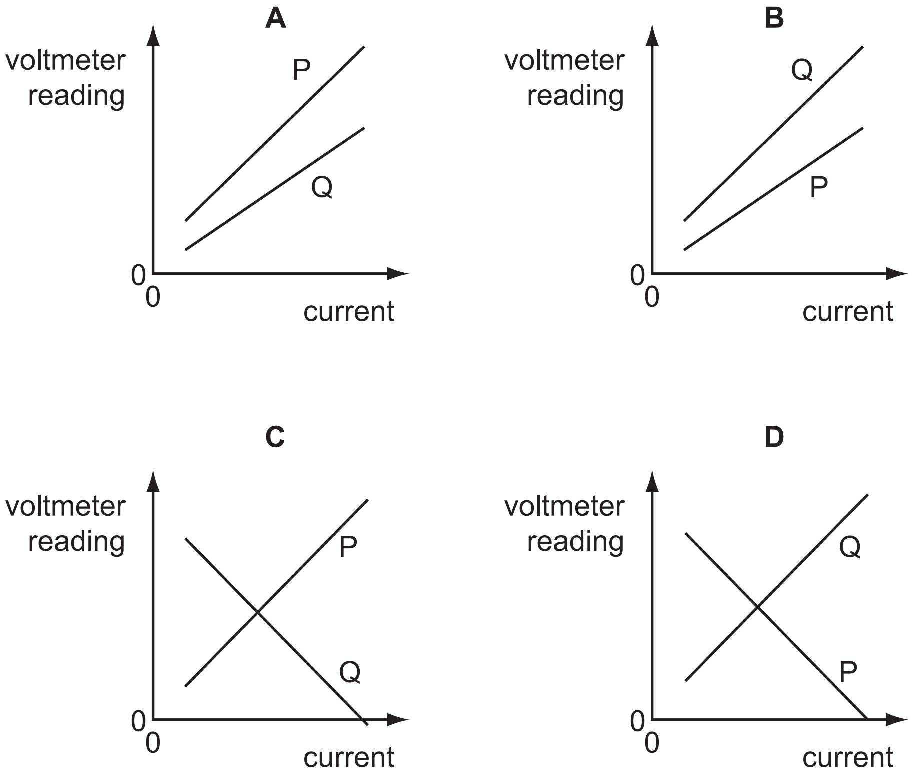
95 9702/12/M/J/13/Q36

A 12 V battery is in series with an ammeter, a 2 Ω fixed resistor and a 0 – 10 Ω variable resistor. High-resistance voltmeters P and Q are connected across the variable resistor and the fixed resistor respectively, as shown.



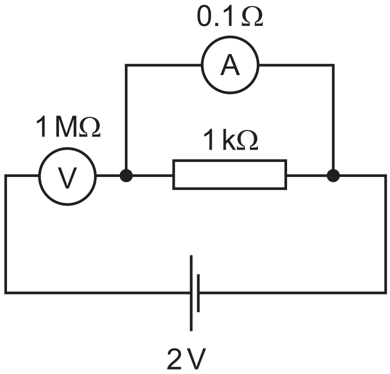
The resistance of the variable resistor is changed from its maximum value to zero.

Which graph shows the variation with current of the voltmeter readings?



96 9702/11/M/J/13/Q36

The diagram shows an incorrectly connected circuit. The ammeter has a resistance of 0.1 Ω and the voltmeter has a resistance of 1 MΩ.



Which statement is correct?

A The ammeter reads 2 mA.

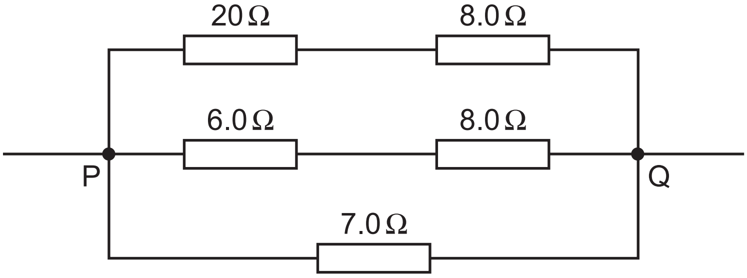
B The ammeter reads 20 A.

C The voltmeter reads zero.

D The voltmeter reads 2 V.

97 9702/12/M/J/13/Q37*

Five resistors are connected as shown.

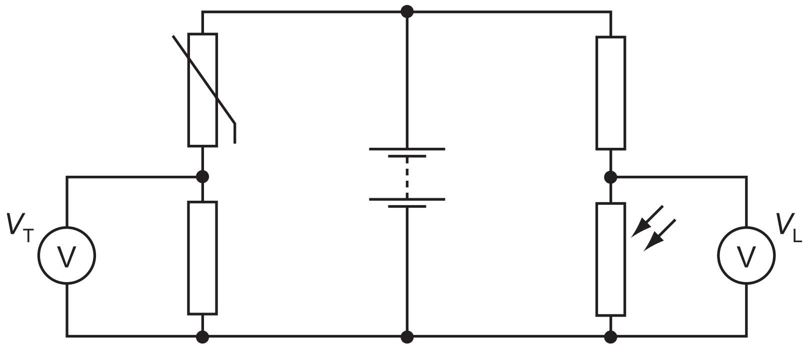


What is the total resistance between P and Q?

A 0.25 Ω **B** 0.61 Ω **C** 4.0 Ω **D** 16 Ω

98 9702/12/M/J/13/Q38

In the circuit below, the reading V_T on the voltmeter changes from high to low as the temperature of the thermistor changes. The reading V_L on the voltmeter changes from high to low as the level of light on the light-dependent resistor (LDR) changes.



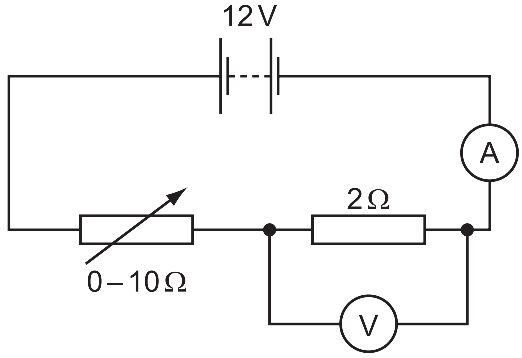
The readings V_T and V_L are both high.

What are the conditions of temperature and light level?

	temperature	light level
A	low	low
B	low	high
C	high	low
D	high	high

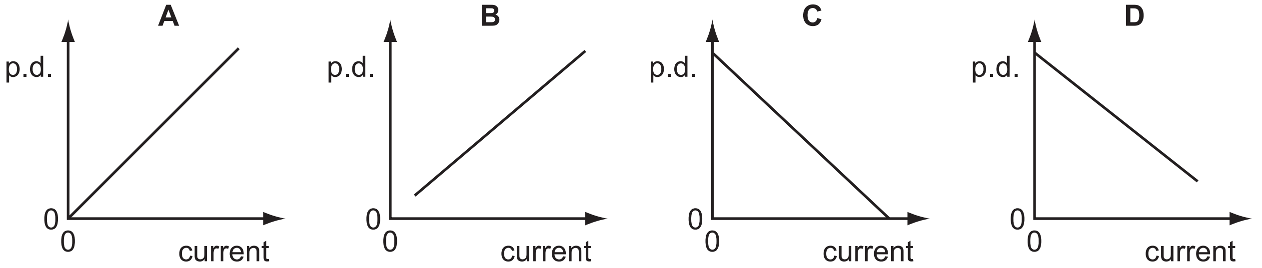
99 **9702/13/M/J/13/Q36**

A 12 V battery is in series with an ammeter, a 2 Ω fixed resistor and a 0 – 10 Ω variable resistor. A high-resistance voltmeter is connected across the fixed resistor.



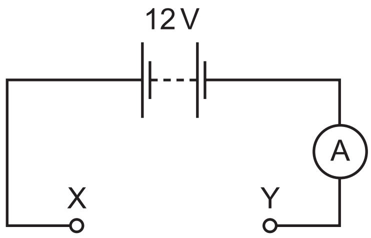
The resistance of the variable resistor is changed from zero to its maximum value.

Which graph shows how the potential difference (p.d.) measured by the voltmeter varies with the current measured by the ammeter?



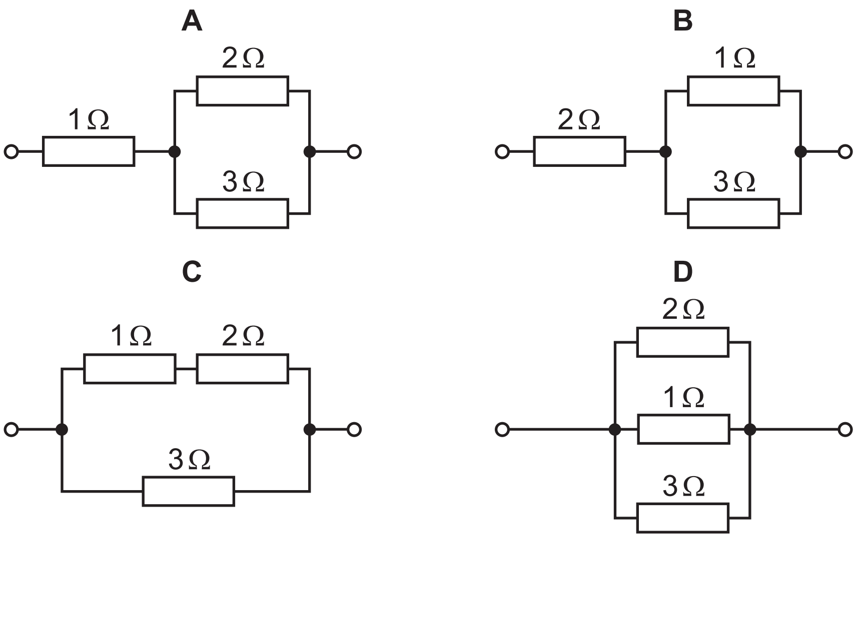
100 **9702/13/M/J/13/Q37**

In the circuit shown, the battery and ammeter each have negligible resistance.



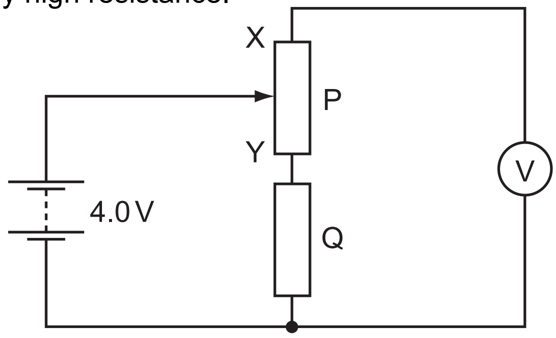
The following combinations of resistors are placed in turn between the terminals X and Y of the circuit.

Which combination would give an ammeter reading of 8 A?



101 **9702/12/O/N/13/Q36**

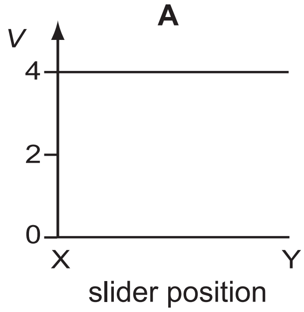
In the circuit below, P is a potentiometer of total resistance 10 Ω and Q is a fixed resistor of resistance 10 Ω. The battery has an electromotive force (e.m.f.) of 4.0 V and negligible internal resistance. The voltmeter has a very high resistance.



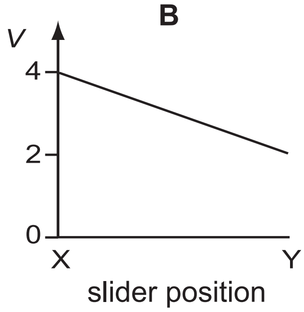
The slider on the potentiometer is moved from X to Y and a graph of voltmeter reading V is plotted against slider position.

Which graph would be obtained?

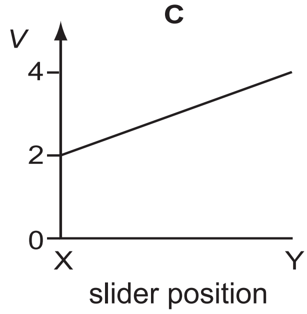
A



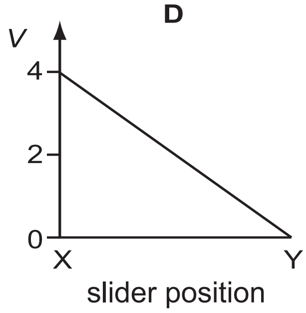
B



C

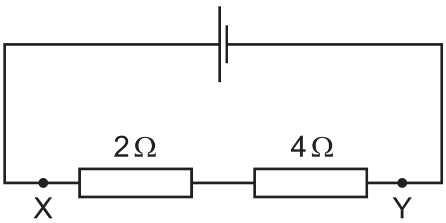


D



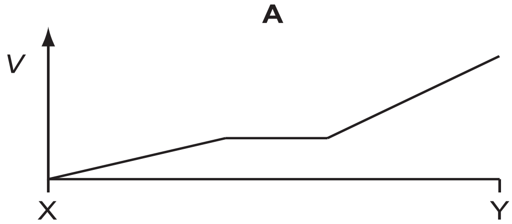
102 **9702/12/O/N/13/Q37**

A 2 Ω resistor and a 4 Ω resistor are connected to a cell.

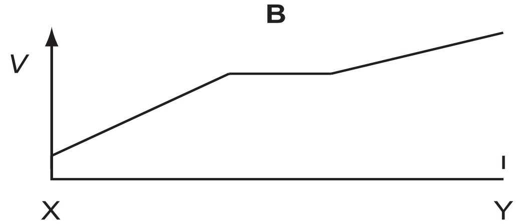


Which graph shows how the potential V varies with distance between X and Y?

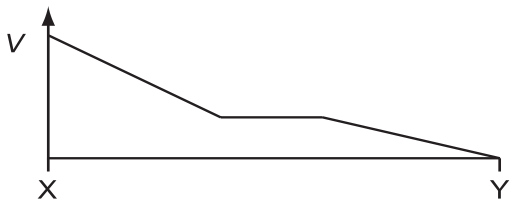
A



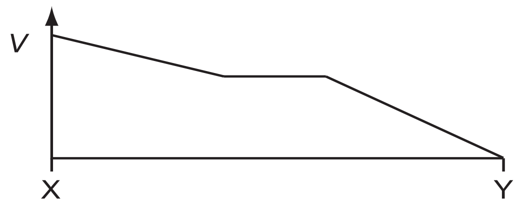
B



C

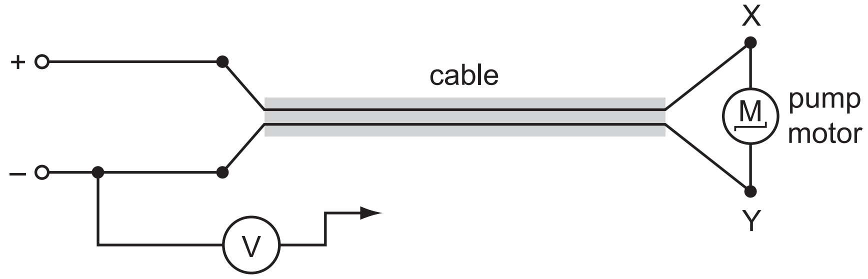


D



103 9702/13/O/N/13/Q36

The diagram shows the electric motor for a garden pump connected to a 24 V power supply by an insulated two-core cable.



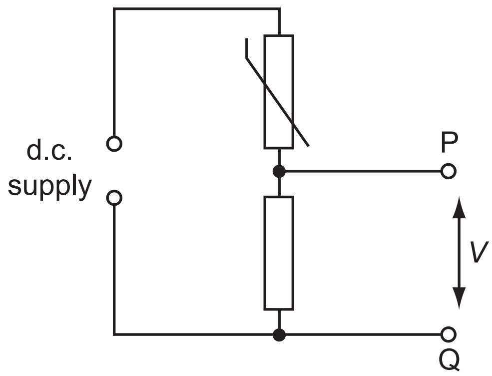
The motor does not work so, to find the fault, the negative terminal of a voltmeter is connected to the negative terminal of the power supply and its other end is connected in turn to terminals X and Y at the motor.

Which row represents two readings and a correct conclusion?

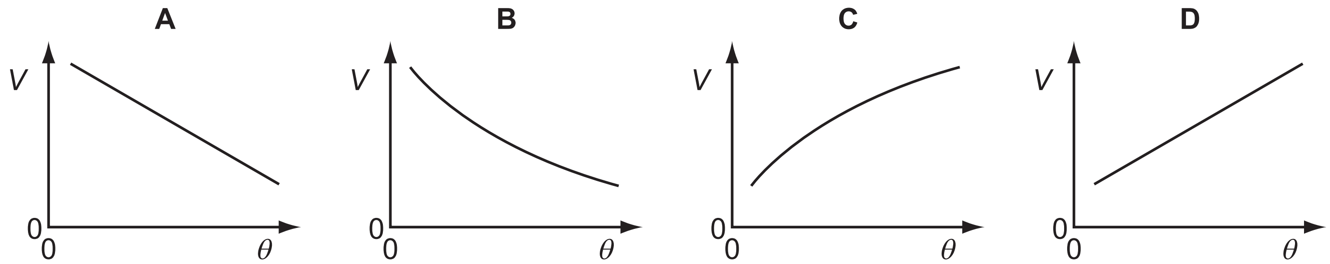
	voltmeter reading when connected to X / V	voltmeter reading when connected to Y / V	conclusion
A	24	0	break in positive wire of cable
B	24	12	break in negative wire of cable
C	24	24	break in connection within the motor
D	24	24	break in negative wire of cable

104 9702/13/O/N/13/Q37

In the circuit shown, the resistance of the thermistor decreases as temperature increases.

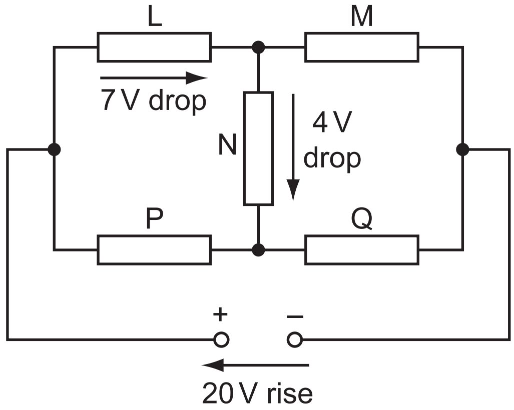


Which graph shows the variation with Celsius temperature θ of potential difference V between points P and Q?



105 9702/13/O/N/13/Q38

A 20 V d.c. supply is connected to a circuit consisting of five resistors L, M, N, P and Q.



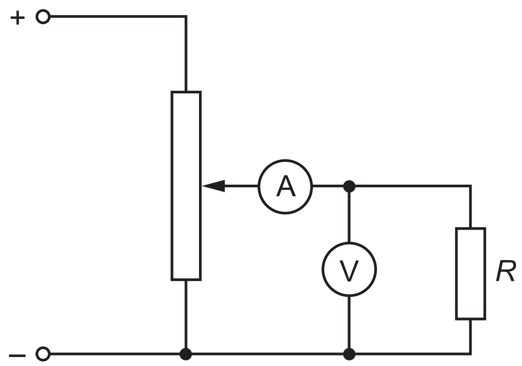
There is a potential drop of 7 V across L and a further 4 V potential drop across N.

What are the potential drops across M, P and Q?

	potential drop across M / V	potential drop across P / V	potential drop across Q / V
A	9	7	13
B	13	7	13
C	13	11	9
D	17	3	17

106 9702/11/M/J/14/Q33

In the circuit below, a voltmeter of resistance R_V and an ammeter of resistance R_A are used to measure the resistance R of the fixed resistor.



Which condition is necessary for an accurate value to be obtained for R?

A R is much smaller than R_V.

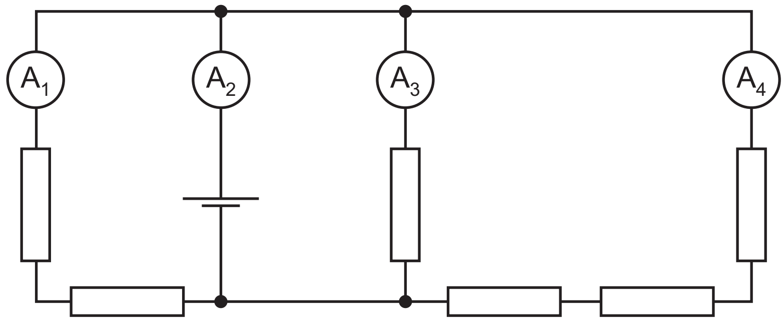
B R is much smaller than R_A.

C R is much greater than R_V.

D R is much greater than R_A.

107 **9702/11/M/J/14/Q34**

In the circuit shown, all the resistors are identical and all the ammeters have negligible resistance.



The reading on ammeter A_1 is 0.6 A.

What are the readings on the other ammeters?

	reading on ammeter A_2 / A	reading on ammeter A_3 / A	reading on ammeter A_4 / A
A	1.0	0.3	0.1
B	1.4	0.6	0.2
C	1.8	0.9	0.3
D	2.2	1.2	0.4

108 **9702/11/M/J/14/Q36**

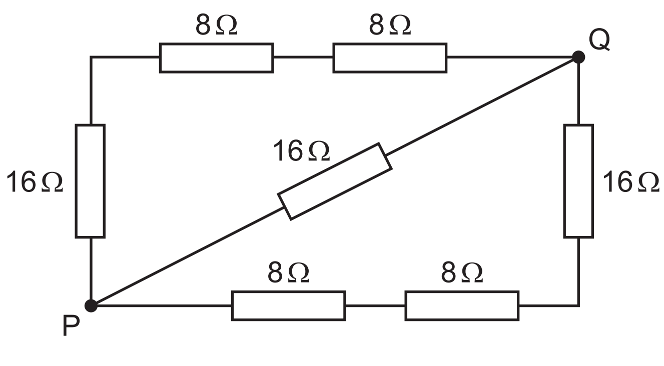
What is the total resistance between points P and Q in this network of resistors?

A 8 Ω

B 16 Ω

C 24 Ω

D 32 Ω



109 **9702/13/M/J/15/Q37**

The diagram shows part of a current-carrying circuit. The ammeter has negligible resistance.

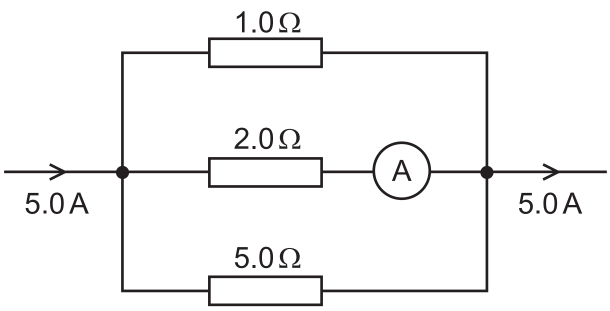
What is the reading on the ammeter?

A 0.7 A

B 1.3 A

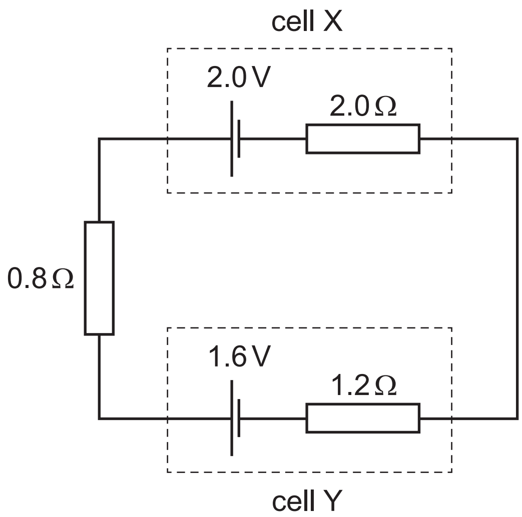
C 1.5 A

D 1.7 A



110 **9702/12/M/J/14/Q33**

Cell X has an e.m.f. of 2.0 V and an internal resistance of 2.0 Ω. Cell Y has an e.m.f. of 1.6 V and an internal resistance of 1.2 Ω. These two cells are connected to a resistor of resistance 0.8 Ω, as shown.

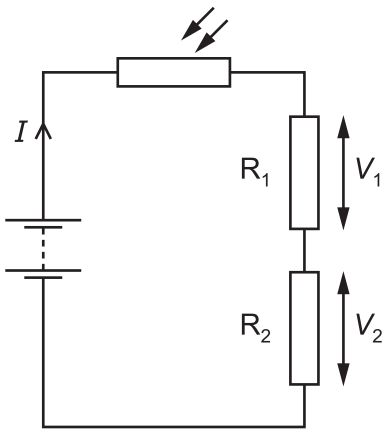


What is the current in cell X?

A 0.10 A **B** 0.50 A **C** 0.90 A **D** 1.0 A

111 **9702/12/M/J/14/Q34**

In the circuit shown, a light-dependent resistor (LDR) is connected to two resistors R_1 and R_2. The potential difference (p.d.) across R_1 is V_1 and the p.d. across R_2 is V_2. The current in the circuit is I.



Which statement about this circuit is correct?

A The current I increases when the light intensity decreases.

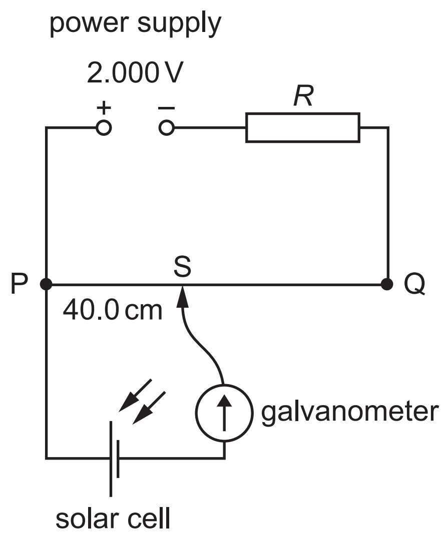
B The LDR is an ohmic conductor.

C The p.d. V_2 increases when the light intensity decreases.

D The ratio $\frac{V_1}{V_2}$ is independent of light intensity.

112 9702/12/M/J/14/Q35

A power supply and a solar cell are compared using the potentiometer circuit shown.



The e.m.f. produced by the solar cell is measured on the potentiometer.

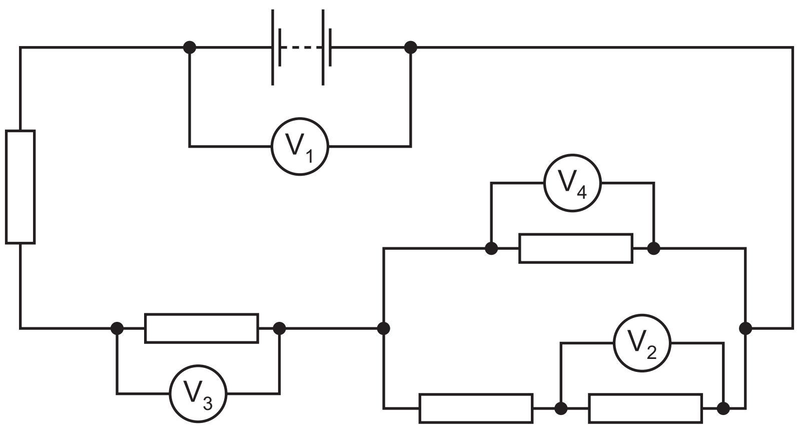
The potentiometer wire PQ is 100.0 cm long and has a resistance of 5.00 Ω. The power supply has an e.m.f. of 2.000 V and the solar cell has an e.m.f. of 5.00 mV.

Which resistance R must be used so that the galvanometer reads zero when PS = 40.0 cm?

A 395 Ω **B** 795 Ω **C** 995 Ω **D** 1055 Ω

113 9702/12/M/J/14/Q36

In the circuit shown, all the resistors are identical.



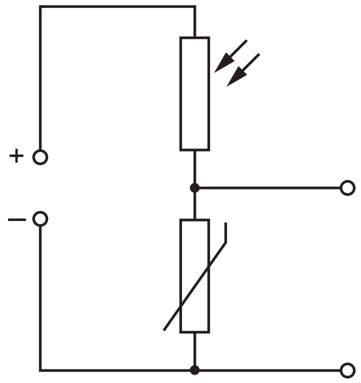
The reading on voltmeter V_1 is 8.0 V and the reading on voltmeter V_2 is 1.0 V.

What are the readings on the other voltmeters?

	reading on voltmeter V_3 / V	reading on voltmeter V_4 / V
A	1.5	1.0
B	3.0	2.0
C	4.5	3.0
D	6.0	4.0

114 **9702/12/M/J/14/Q37**

The diagram shows a light-dependent resistor (LDR) and a thermistor forming a potential divider.

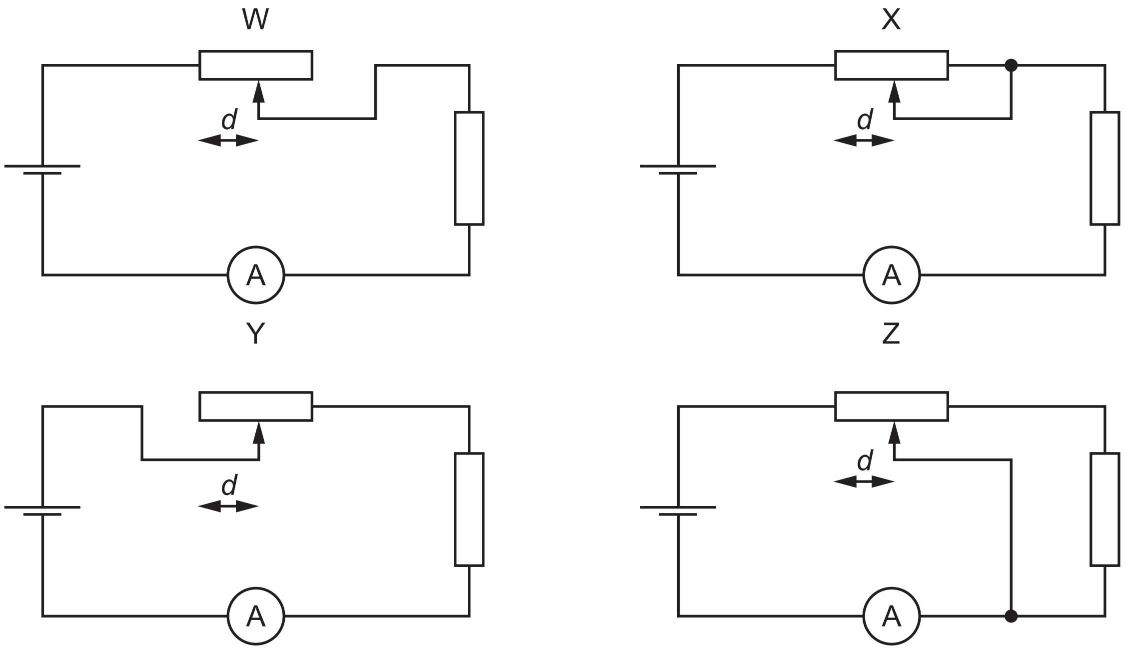


Under which set of conditions will the potential difference across the thermistor have the greatest value?

	illumination	temperature
A	low	low
B	high	low
C	low	high
D	high	high

115 **9702/13/M/J/14/Q36**

The diagrams show the same cell, ammeter, potentiometer and fixed resistor connected in different ways.



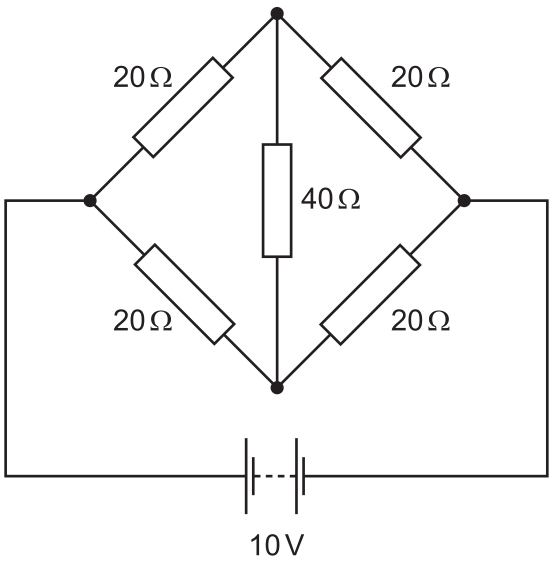
The distance *d* between the sliding contact and a particular end of the potentiometer is varied. The current measured is then plotted against the distance *d*.

For which two circuits will the graphs be identical?

A W and X **B** W and Y **C** X and Y **D** Y and Z

116 9702/13/M/J/14/Q37

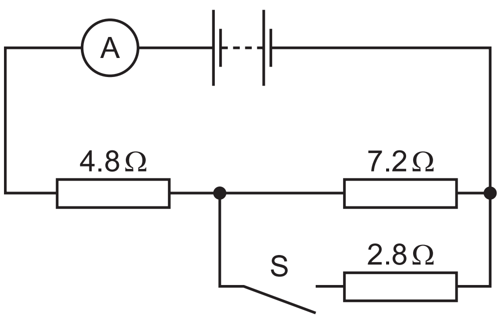
What is the current in the $40\,\Omega$ resistor of the circuit shown?



A zero **B** 0.13 A **C** 0.25 A **D** 0.50 A

117 9702/13/M/J/14/Q38

A battery of negligible internal resistance is connected to a resistor network, an ammeter and a switch S, as shown.



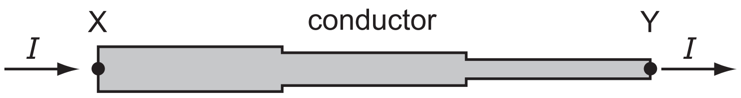
When S is open, the reading on the ammeter is 250 mA.

When S is closed, what is the **change** in the reading on the ammeter?

A 1.07 A **B** 1.32 A **C** 190 mA **D** 440 mA

118 **9702/11/O/N/14/Q33**

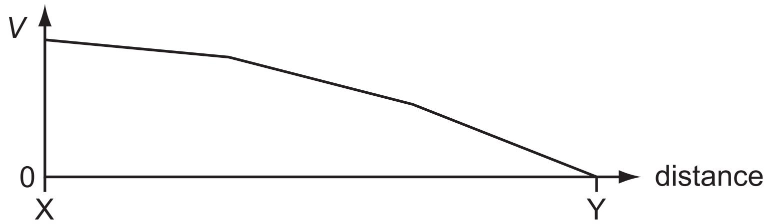
A conductor consists of three wires connected in series. The wires are all made of the same metal but have different cross-sectional areas. There is a current I in the conductor.



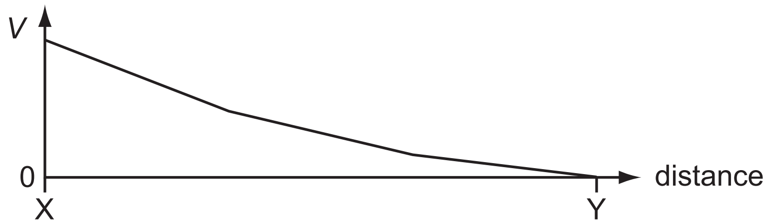
Point Y on the conductor is at zero potential.

Which graph best shows the variation of potential V with distance along the conductor?

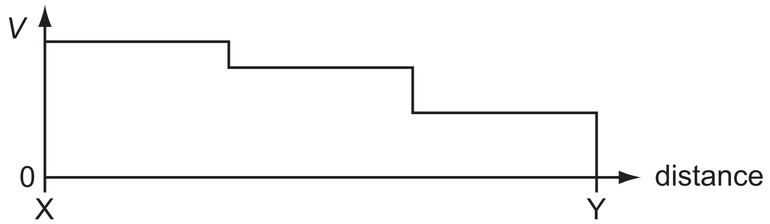
A



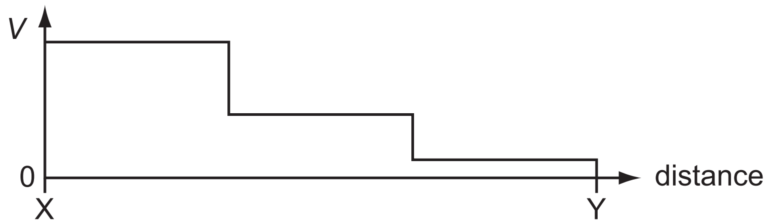
B



C

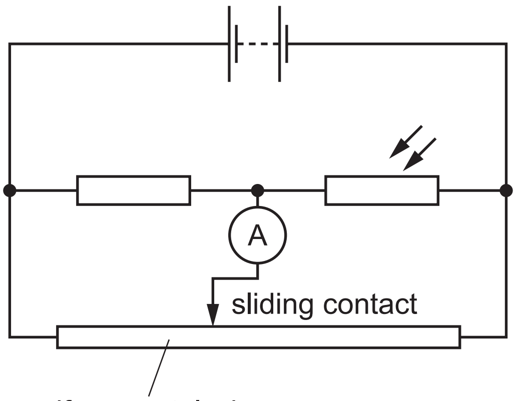


D



119 9702/11/O/N/14/Q36

In the potentiometer circuit shown, the reading on the ammeter is zero.



The light-dependent resistor (LDR) is then covered up and the ammeter gives a non-zero reading.

Which change could return the ammeter reading to zero?

A Decrease the supply voltage.

B Increase the supply voltage.

C Move the sliding contact to the left.

D Move the sliding contact to the right.

120 9702/11/O/N/14/Q37

Six resistors, each of resistance *R*, are connected as shown.

The combined resistance is 66 kΩ.

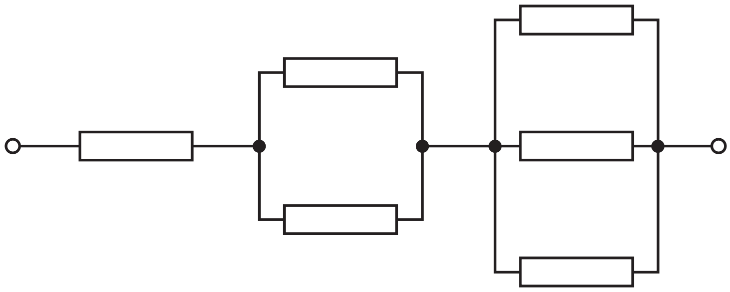
What is the value of *R*?

A 11 kΩ

B 18 kΩ

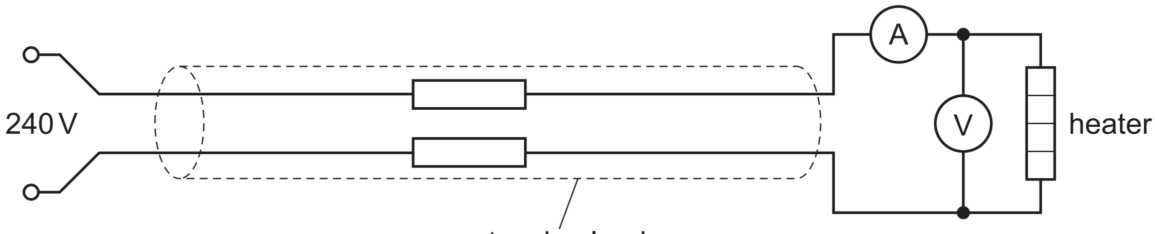
C 22 kΩ

D 36 kΩ



121 9702/13/O/N/14/Q36

An extension lead is used to connect a 240 V electrical supply to a heater as shown.



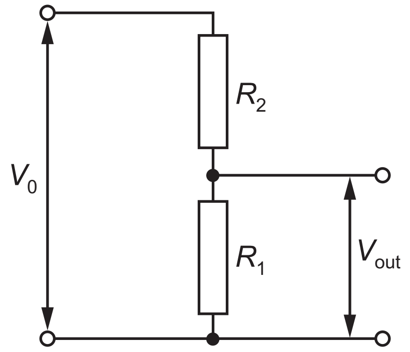
A voltmeter measures the potential difference (p.d.) across the heater as 216 V and an ammeter measures the current through the heater as 7.7 A.

What is the total resistance of the extension lead?

A 3.1 Ω **B** 6.2 Ω **C** 28 Ω **D** 31 Ω

122 **9702/13/O/N/14/Q35**

A potential divider consists of resistors of resistance R_1 and R_2 connected in series across a source of potential difference V_0. The potential difference across R_1 is V_{out}.

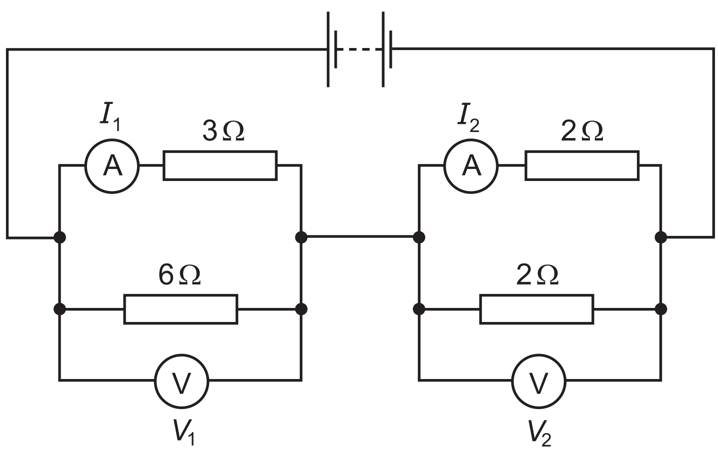


Which changes to R_1 and R_2 will increase the value of V_{out}?

	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

123 **9702/13/O/N/14/Q37**

In the circuit shown, the ammeters have negligible resistance and the voltmeters have infinite resistance.



The readings on the meters are I_1, I_2, V_1 and V_2, as labelled on the diagram.

Which statement is correct?

A $I_1 > I_2$ and $V_1 > V_2$

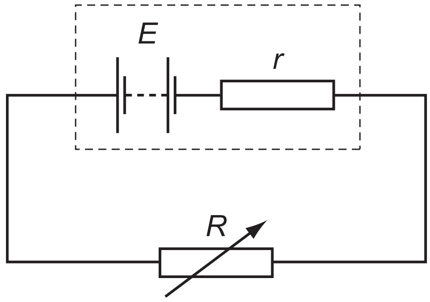
B $I_1 > I_2$ and $V_1 < V_2$

C $I_1 < I_2$ and $V_1 > V_2$

D $I_1 < I_2$ and $V_1 < V_2$

124 9702/11/M/J/15/Q36

A battery with e.m.f. E and internal resistance r is connected in series with a variable external resistor.



The value of the external resistance R is slowly increased from zero.

Which statement is correct? (Ignore any temperature effects.)

A The potential difference across the external resistance decreases.

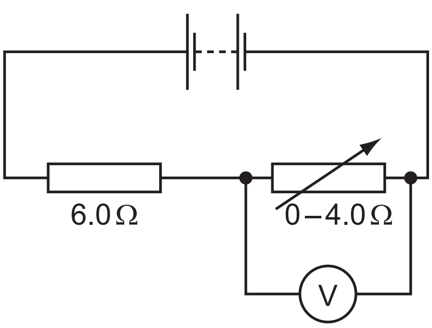
B The potential difference across the internal resistance increases.

C The power dissipated in r increases and then decreases.

D The power dissipated in R increases and then decreases.

125 9702/11/M/J/15/Q37

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected in series with a resistor of resistance 6.0 Ω and a variable resistor of resistance from zero to 4.0 Ω. A voltmeter is connected across the variable resistor. The resistance of the variable resistor is changed.



What is the range of the voltmeter reading?

A 0 V – 2.4 V

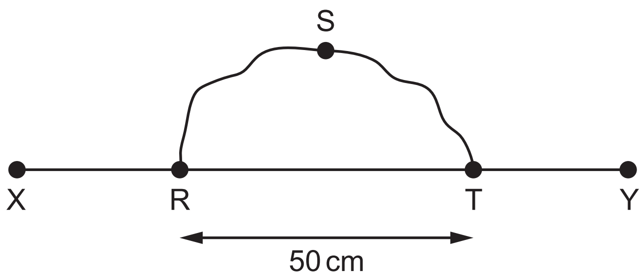
B 0 V – 3.6 V

C 2.4 V – 6.0 V

D 3.6 V – 6.0 V

126 9702/11/M/J/15/Q38

A wire RST is connected to another wire XY as shown.



Each wire is 100 cm long with a resistance per unit length of $10\,\Omega\,m^{-1}$.

What is the total resistance between X and Y?

A 3.3 Ω **B** 5.0 Ω **C** 8.3 Ω **D** 13.3 Ω

127 9702/12/M/J/15/Q35

Each of Kirchhoff's two laws presumes that some quantity is conserved.

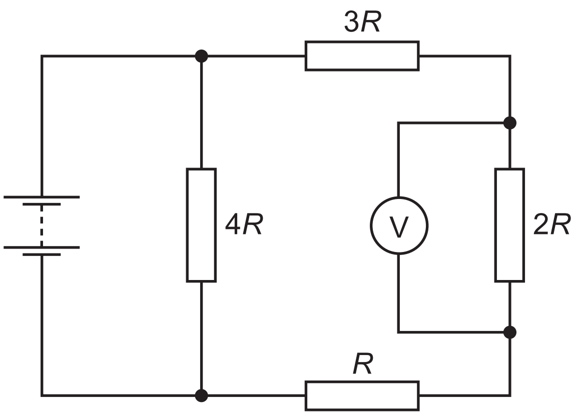
Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	the algebraic sum of currents into a junction is zero	charge
B	the algebraic sum of currents into a junction is zero	energy
C	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	charge
D	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	energy

128 9702/12/O/N/13/Q38

Four resistors of resistance R, $2R$, $3R$ and $4R$ are connected to form a network.

A battery of negligible internal resistance and a voltmeter are connected to the resistor network as shown.



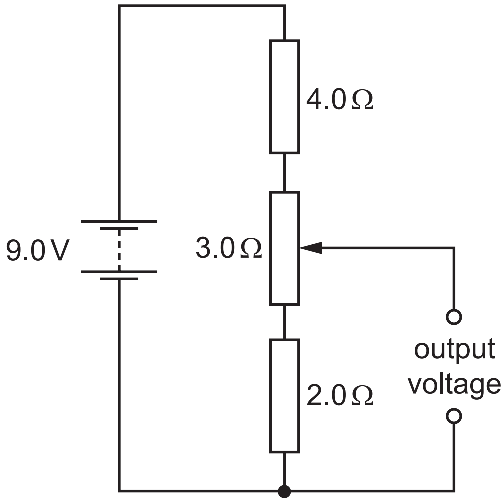
The voltmeter reading is 2 V.

What is the electromotive force (e.m.f.) of the battery?

A 2 V **B** 4 V **C** 6 V **D** 10 V

129 9702/12/M/J/15/Q36

A potential divider circuit consists of fixed resistors of resistance 2.0 Ω and 4.0 Ω connected in series with a 3.0 Ω resistor fitted with a sliding contact. These are connected across a battery of e.m.f. 9.0 V and zero internal resistance, as shown.

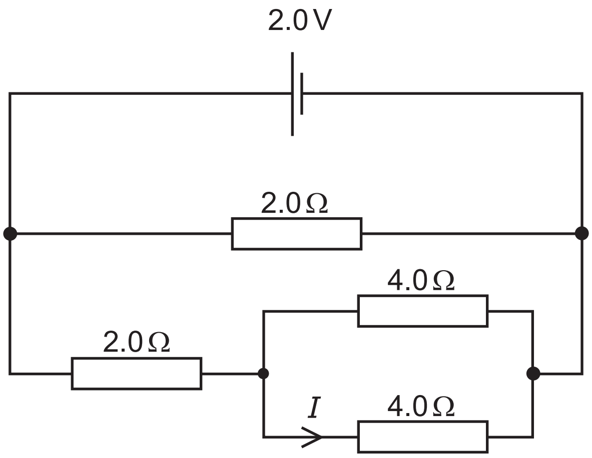


What are the maximum and the minimum output voltages of this potential divider circuit?

	maximum voltage / V	minimum voltage / V
A	4.0	2.0
B	5.0	2.0
C	9.0	0
D	9.0	2.0

130 9702/12/M/J/15/Q37

A cell of e.m.f. 2.0 V and negligible internal resistance is connected to a network of resistors as shown.

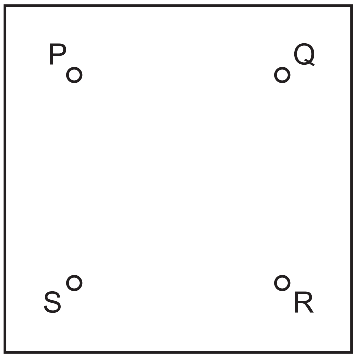


What is the current I?

A 0.25 A **B** 0.33 A **C** 0.50 A **D** 1.5 A

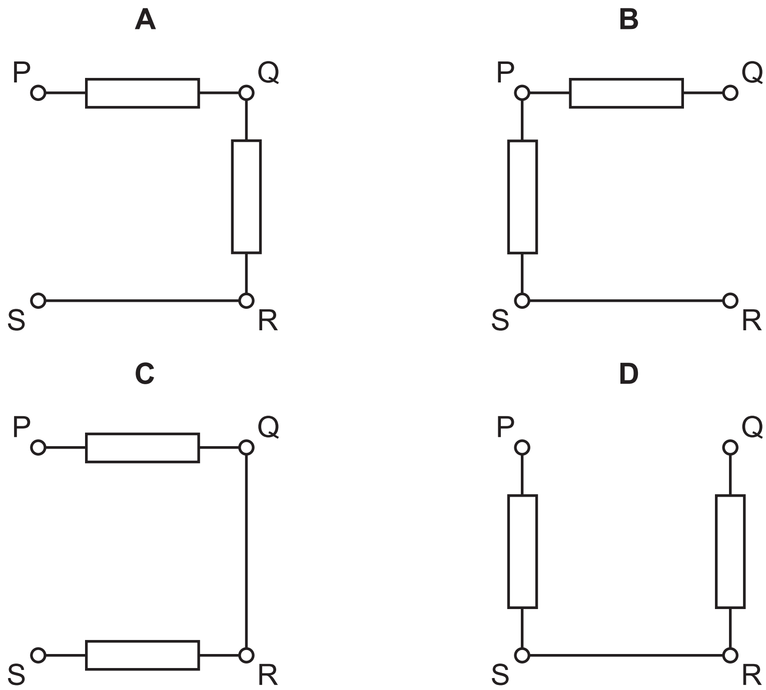
131 9702/13/M/J/15/Q36

A box with four terminals P, Q, R and S contains two identical resistors.



When a battery of electromotive force (e.m.f.) E and negligible internal resistance is connected across PS, a high-resistance voltmeter connected across QR reads $\frac{E}{2}$.

Which diagram shows the correct arrangement of the two resistors inside the box?



132 9702/11/M/J/14/Q32

Which symbol represents a component whose resistance is designed to change with temperature?

133 **9702/11/M/J/17/Q36**

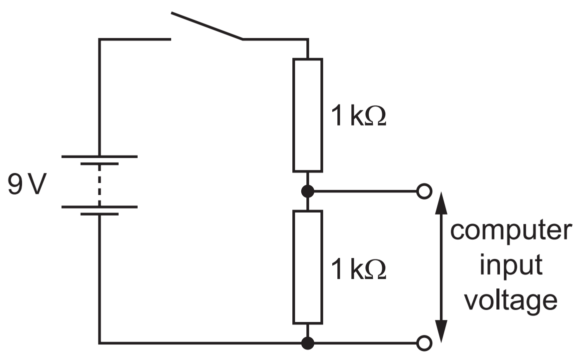
A computer is used to detect the change of position of a switch.

To detect the change of position, the computer requires a potential difference (p.d.) of 0 V to its input at one switch position and a p.d. of between 5 V and 7 V at the other switch position.

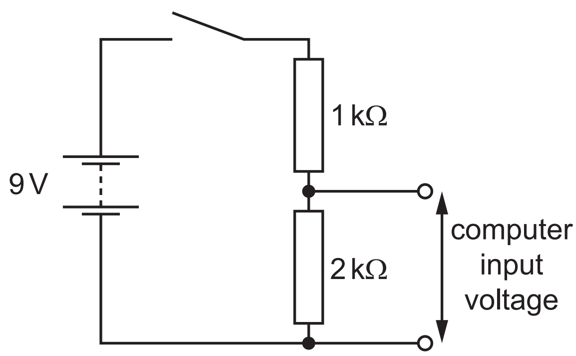
For each of the circuits, assume the battery has negligible internal resistance.

Which circuit provides an input voltage to the computer that enables it to detect the change of position of the switch?

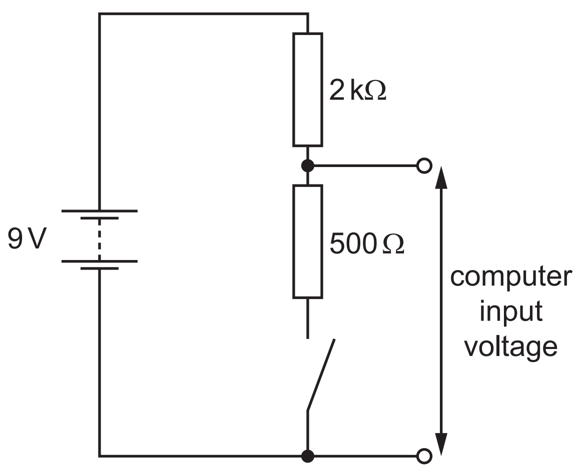
A



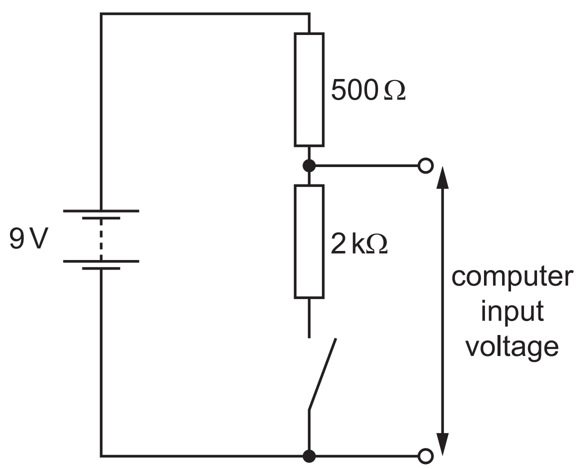
B



C

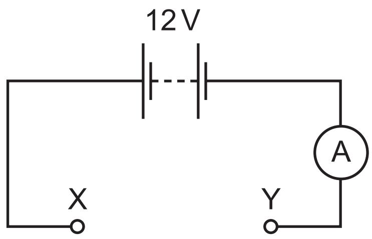


D



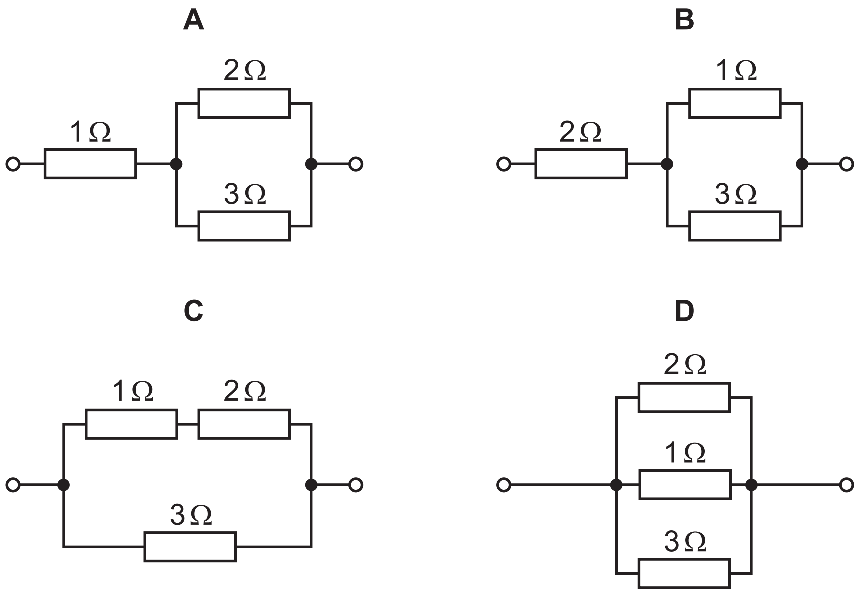
134 **9702/11/M/J/17/Q37**

In the circuit shown, the battery and ammeter have negligible resistance.



The following combinations of resistors are each separately placed between the terminals X and Y of the circuit.

Which combination would give an ammeter reading of 8 A?



135 **9702/12/M/J/17/Q35**

A cell has a constant electromotive force.

A variable resistor is connected between the terminals of the cell.

The resistance of the variable resistor is decreased.

Which statement about the change of the cell's terminal potential difference (p.d.) is correct?

A The terminal p.d. is decreased because more work is done moving unit charge through the internal resistance of the cell.

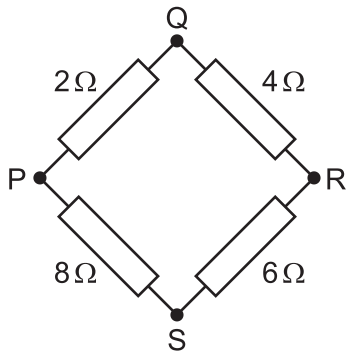
B The terminal p.d. is decreased because the current in the variable resistor is decreased.

C The terminal p.d. is increased because more work is done moving unit charge through the variable resistor.

D The terminal p.d. is increased because the current in the variable resistor is increased.

136 **9702/12/M/J/17/Q36**

Four resistors are connected in a square as shown.



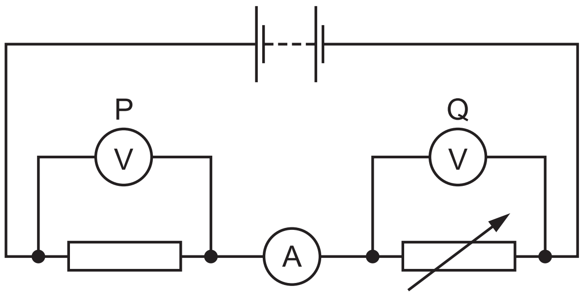
The resistance may be measured between any two junctions.

Between which two junctions is the measured resistance greatest?

A P and Q **B** Q and S **C** R and S **D** S and P

137 **9702/12/M/J/17/Q37**

A circuit is set up as shown.



The variable resistor is adjusted so that the ammeter reading decreases.

How do the readings of the voltmeters change?

	reading on voltmeter P	reading on voltmeter Q
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

138 **9702/11/O/N/17/Q34**

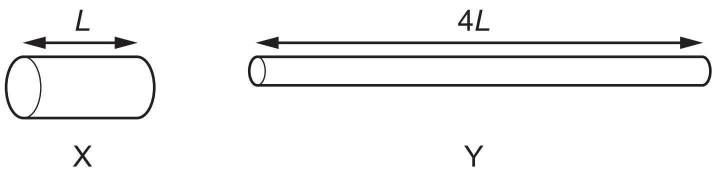
A filament lamp has a resistance of 180 Ω when the current in it is 500 mA.

What is the power dissipated in the lamp?

A 45 W **B** 90 W **C** 290 W **D** 360 W

139 **9702/11/O/N/17/Q35**

Two copper wires X and Y have the same volume. Wire Y is four times as long as wire X.



What is the ratio $\frac{\text{resistance of wire Y}}{\text{resistance of wire X}}$?

A 1 **B** 4 **C** 8 **D** 16

140 **9702/11/O/N/17/Q38**

The circuit diagram shows four resistors of different resistances *P*, *Q*, *R* and *S* connected to a battery.

The voltmeter reading is zero.

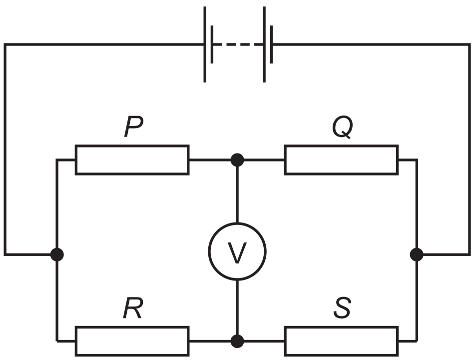
Which equation is correct?

A $P - Q = R - S$

B $P - S = Q - R$

C $PQ = RS$

D $PS = QR$



141 **9702/12/O/N/17/Q19**

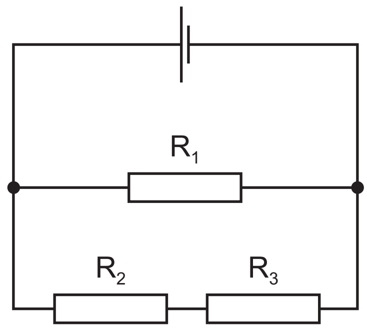
During refuelling, a petrol car receives 50 litres of fuel in 90 seconds. The petrol has 34 MJ of energy per litre.

For an electric car to receive the same amount of energy in the same time from a 230 V supply, what is the minimum current required?

A 2700 A **B** 8.2×10^4 A **C** 7.4×10^6 A **D** 6.6×10^8 A

142 9702/12/O/N/17/Q34

A cell of negligible internal resistance is connected to resistors R_1, R_2 and R_3, as shown. The cell provides power to the circuit and power is dissipated in the resistors.



Which word equation **must** be correct?

A power loss in R_1 = power loss in R_2 + power loss in R_3

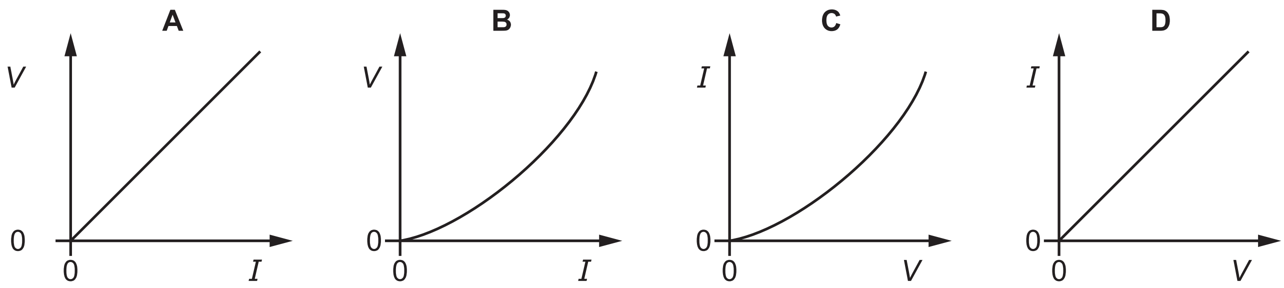
B power loss in R_2 = power loss in R_3

C power output of cell = power loss in R_1 + power loss in R_2 + power loss in R_3

D power output of cell = power loss in R_1

143 9702/12/O/N/17/Q35

Which graph represents a metallic conductor, where the resistance of the conductor is given by the gradient of the graph?



144 9702/12/O/N/17/Q36

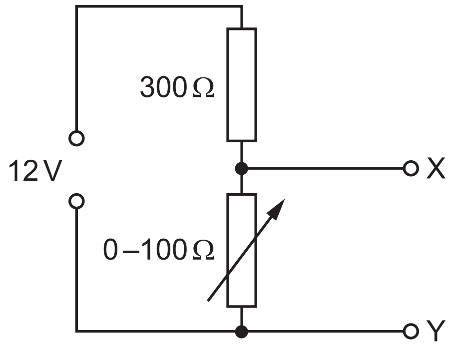
A typical mobile phone battery has an e.m.f. of 5.0 V and an internal resistance of 200 mΩ.

What is the terminal p.d. of the battery when it supplies a current of 500 mA?

A 4.8 V **B** 4.9 V **C** 5.0 V **D** 5.1 V

145 9702/12/O/N/17/Q38

The diagram shows a potential divider connected to a 12 V supply of negligible internal resistance.



Which range of voltages can be obtained between X and Y?

A 0 to 3 V **B** 0 to 4 V **C** 0 to 8 V **D** 0 to 9 V

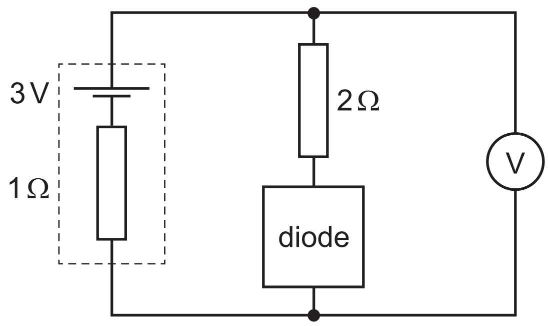
146 9702/13/O/N/17/Q35

An ideal diode has zero resistance when forward biased and infinite resistance when reverse biased. The diode is connected in series with a 2 Ω resistor across the terminals of a source having electromotive force (e.m.f.) 3 V and internal resistance 1 Ω, as shown.

A high-resistance voltmeter is connected across the diode and resistor.

Which row gives the readings of the voltmeter for the two ways of connecting the diode?

	forward biased	reverse biased
A	1 V	3 V
B	2 V	0 V
C	2 V	3 V
D	3 V	0 V



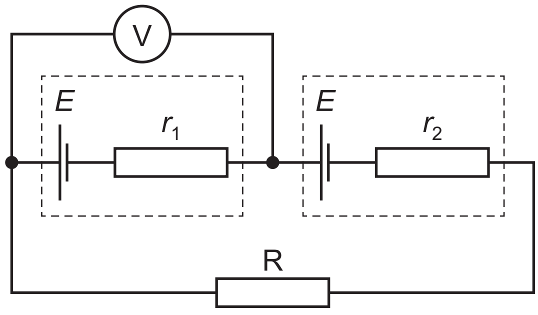
147 9702/13/O/N/17/Q36

Two cells, each with electromotive force (e.m.f.) E, but different internal resistances r_1 and r_2, are connected in series to a resistor R. The reading on the voltmeter is 0 V.

What is the resistance of R?

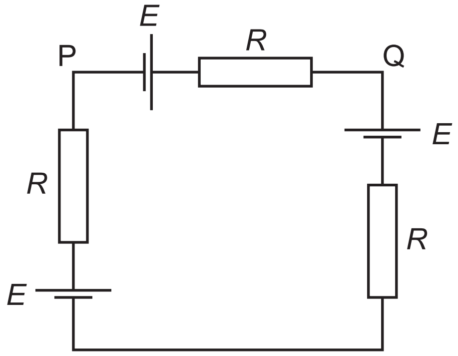
A 0 **B** $r_1 - r_2$

C $r_1 + r_2$ **D** $\dfrac{r_1 r_2}{r_1 + r_2}$



148 **9702/13/O/N/17/Q37**

Three identical cells each have electromotive force (e.m.f.) E and negligible internal resistance. The cells are connected to three identical resistors, each of resistance R, as shown.



What is the potential difference between P and Q?

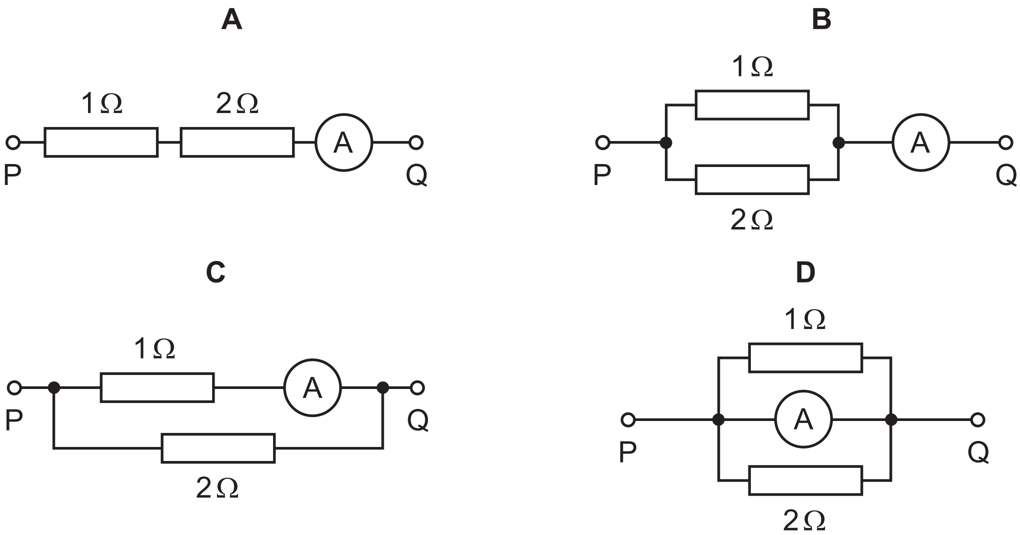
A 0 **B** $\frac{E}{3}$

C $\frac{2E}{3}$ **D** E

149 **9702/13/O/N/17/Q38**

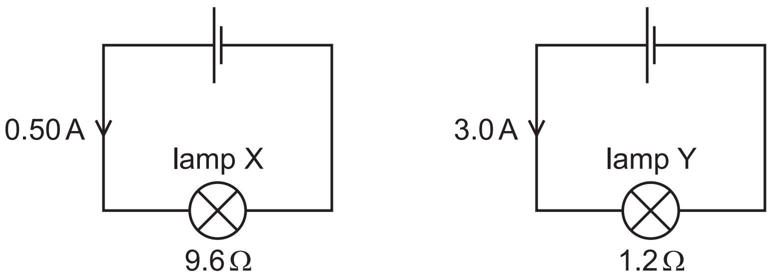
In each arrangement of resistors, the ammeter has a resistance of 2 Ω.

Which arrangement gives the largest reading on the ammeter when the same potential difference is applied between points P and Q?



150 **9702/11/M/J/18/Q31**

The circuit diagrams show two lamps X and Y each connected to a cell. The current in lamp X is 0.50 A and its resistance is 9.6 Ω. The current in lamp Y is 3.0 A and its resistance is 1.2 Ω.



What is the ratio $\frac{\text{power in lamp X}}{\text{power in lamp Y}}$?

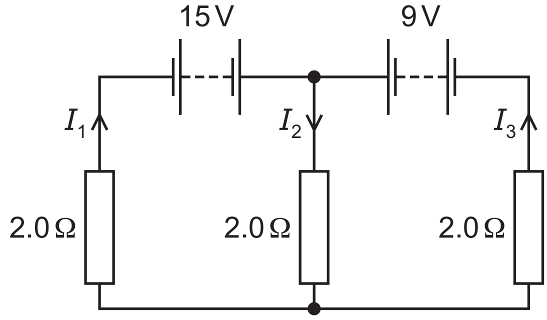
A 0.22 **B** 0.75 **C** 1.3 **D** 4.5

151 **9702/11/M/J/18/Q34**

In the circuit shown, the batteries have negligible internal resistance.

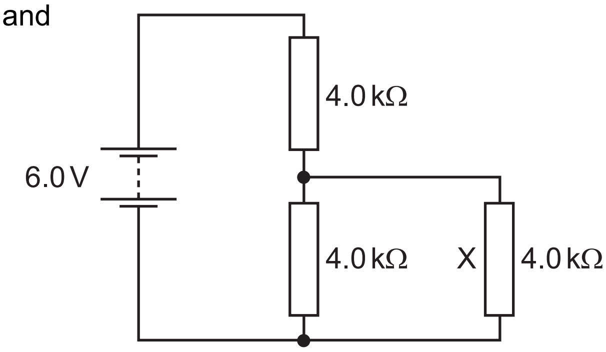
What are the values of the currents I_1, I_2 and I_3?

	I_1 / A	I_2 / A	I_3 / A
A	–5.5	1.0	6.5
B	0.5	4.0	3.5
C	3.5	4.0	0.5
D	6.5	1.0	–5.5



152 **9702/11/M/J/18/Q35**

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected to three resistors as shown.



Each resistor has a resistance of 4.0 kΩ.

What is the current in resistor X?

A 0.25 mA **B** 0.50 mA

C 0.75 mA **D** 1.0 mA

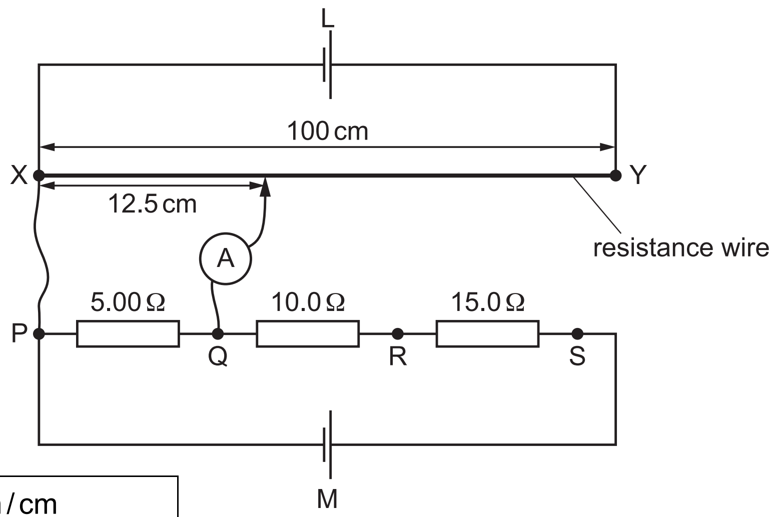
153 **9702/11/M/J/18/Q36**

A uniform resistance wire XY of length 100 cm is connected in series with a cell L. Another cell M is connected in series with resistors of resistances 5.00 Ω, 10.0 Ω and 15.0 Ω.

The potential difference (p.d.) between P and Q is balanced against 12.5 cm of the resistance wire, so that the ammeter reads zero.

The p.d. across the other resistors is then balanced against other lengths of the resistance wire.

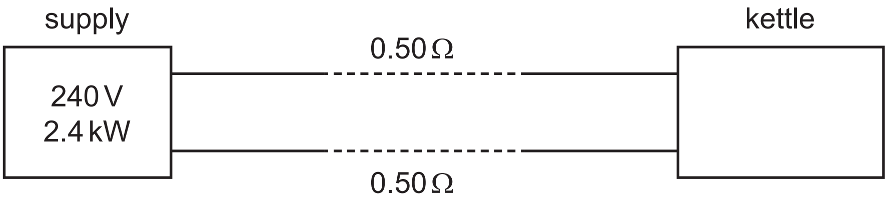
Which balanced lengths of resistance wire correspond to the connection points given in the table?



connection points	balanced length / cm			
	A	**B**	**C**	**D**
Q and R	12.5	25.0	25.0	25.0
Q and S	62.5	62.5	75.0	62.5
P and R	37.5	37.5	37.5	12.5

154 **9702/12/M/J/18/Q32**

The power output of an electrical supply is 2.4 kW at a potential difference (p.d.) of 240 V. The two wires between the supply and a kettle each have a resistance of 0.50 Ω, as shown.

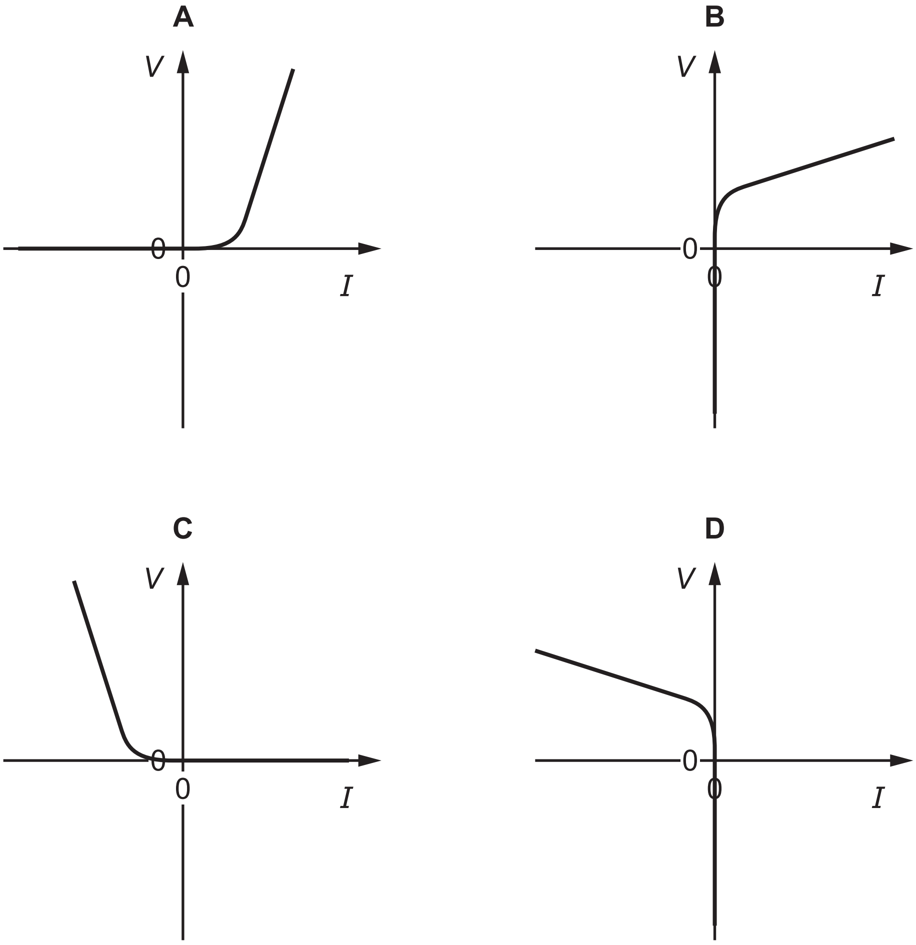


What is the power supplied to the kettle and what is the p.d. across the kettle?

	power / kW	p.d. / V
A	2.3	230
B	2.3	235
C	2.4	230
D	2.4	235

155 **9702/12/M/J/18/Q33**

Which graph shows the variation of voltage V with current I for a semiconductor diode?

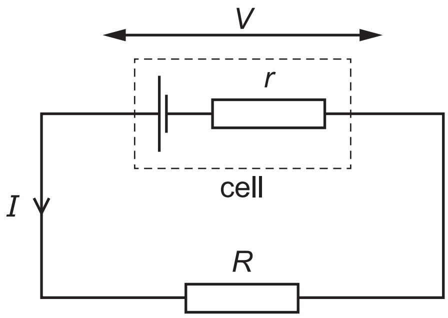


156 9702/12/M/J/18/Q35

A cell of constant electromotive force drives a current I through an external resistor of resistance R. The terminal potential difference (p.d.) across the cell is V.

When the internal resistance r of the cell increases, what is the effect on V and on I?

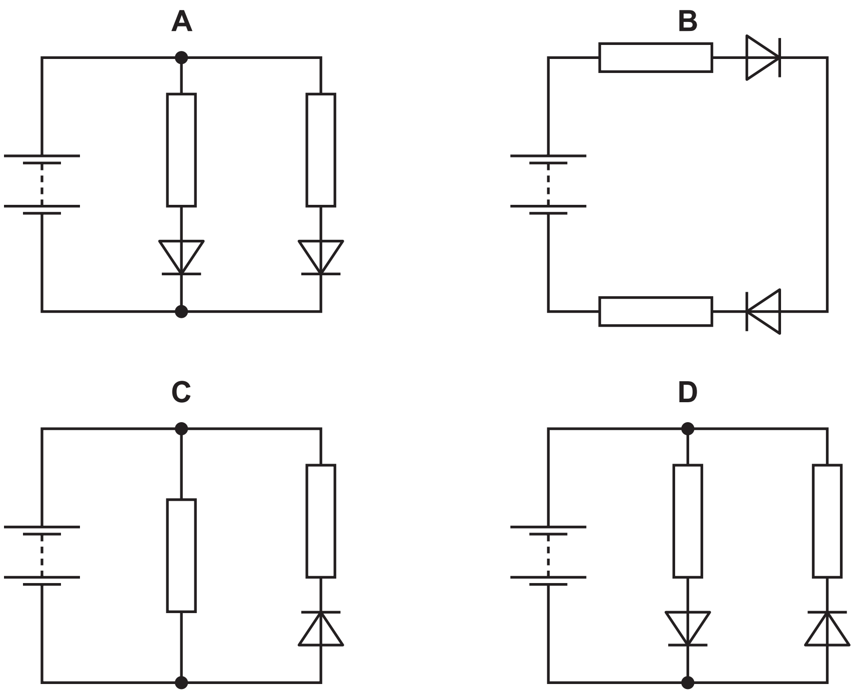
	V	I
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases



157 9702/12/M/J/18/Q36

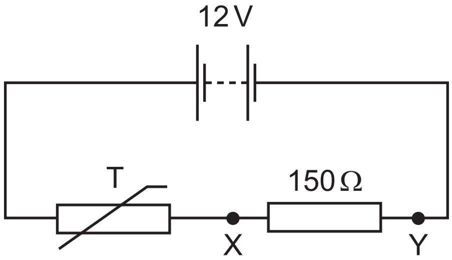
In the circuits shown, the batteries are identical and all have negligible internal resistance. All of the resistors have the same resistance. The diodes have zero resistance when conducting and infinite resistance when not conducting.

In which circuit is the current in the battery greatest?



158 **9702/12/M/J/18/Q37**

A thermistor is an electrical component with a resistance that varies with temperature. A thermistor T is used in a fire alarm system. The alarm is triggered when the potential difference between X and Y is 4.5 V.

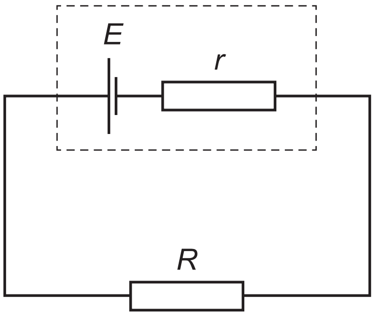


What is the resistance of T when the alarm is triggered?

A 90 Ω **B** 150 Ω **C** 250 Ω **D** 400 Ω

159 **9702/13/M/J/18/Q31**

A cell of electromotive force (e.m.f.) E and internal resistance r is connected to an external resistor of resistance R, as shown.

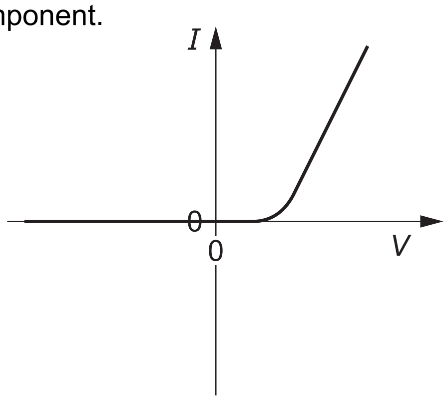


What is the power dissipated in the external resistor?

A $\dfrac{E^2(R+r)}{R^2}$ **B** $\dfrac{E^2R}{(R+r)^2}$ **C** $\dfrac{E^2(R+r)}{r^2}$ **D** $\dfrac{E^2r}{(R+r)^2}$

160 **9702/13/M/J/18/Q32**

The graph shows the I-V characteristic of an electrical component.



What is the component?

A a filament lamp

B a metallic conductor at constant temperature

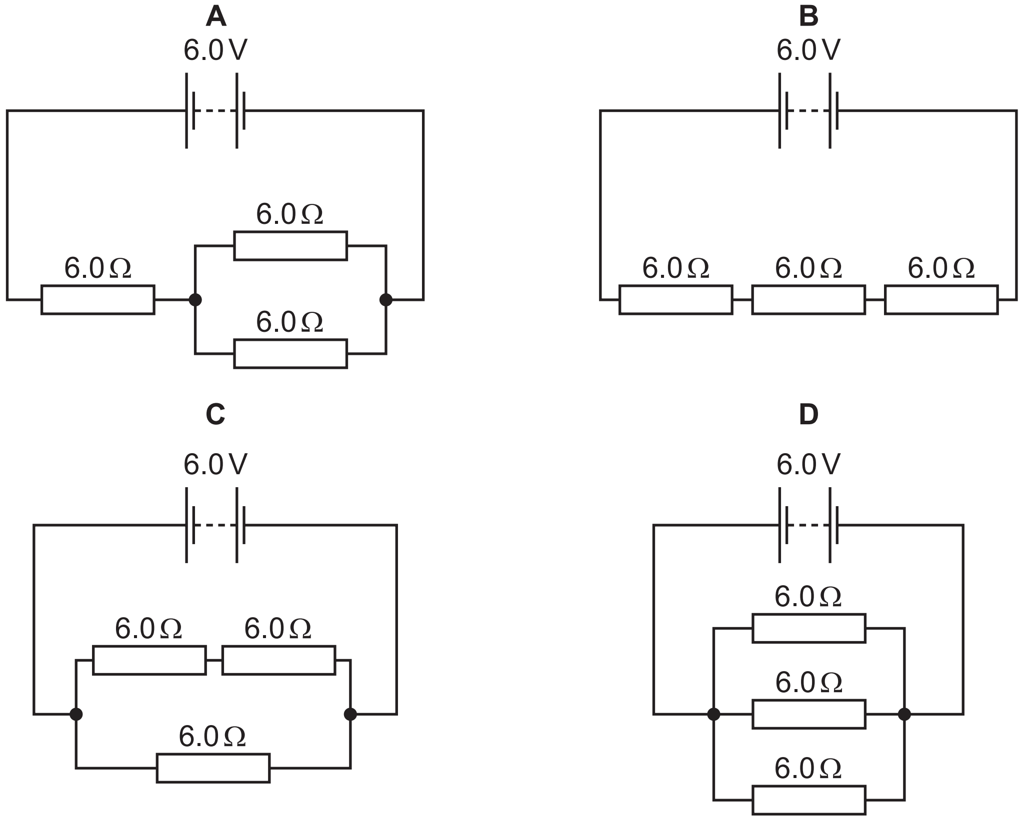
C a resistor

D a semiconductor diode

161 **9702/13/M/J/18/Q34**

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected to three resistors each of resistance 6.0 Ω.

Which circuit will produce a current through the battery of 0.67 A?



162 **9702/13/M/J/18/Q35**

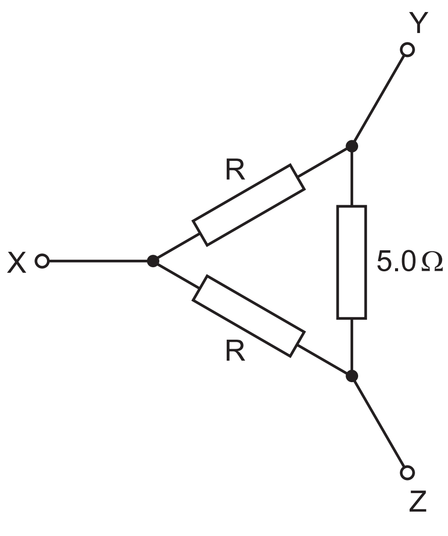
The diagram shows a network of three resistors. Two of these, marked R, are identical. The other resistor has a resistance of 5.0 Ω.

The resistance between Y and Z is found to be 2.5 Ω.

What is the resistance between X and Y?

A 0.30 Ω **B** 0.53 Ω

C 1.9 Ω **D** 3.3 Ω



163 9702/13/M/J/18/Q36

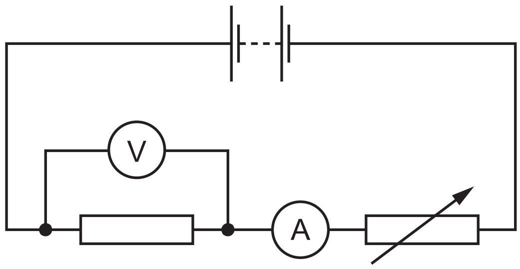
The diagram shows a battery, a fixed resistor, an ammeter and a variable resistor connected in series.

A voltmeter is connected across the fixed resistor.

The resistance of the variable resistor is reduced.

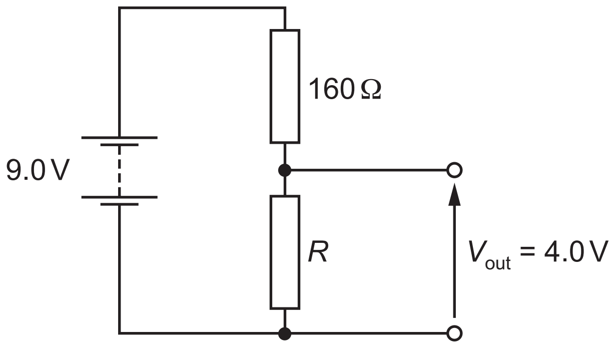
Which row describes the changes in the readings of the ammeter and of the voltmeter?

	ammeter	voltmeter
A	decrease	decrease
B	decrease	increase
C	increase	decrease
D	increase	increase



164 9702/13/M/J/18/Q37

The circuit diagram shows a battery of electromotive force (e.m.f.) 9.0 V and negligible internal resistance. It is connected to two resistors of resistances 160 Ω and R. The output potential difference V_{out} is 4.0 V.

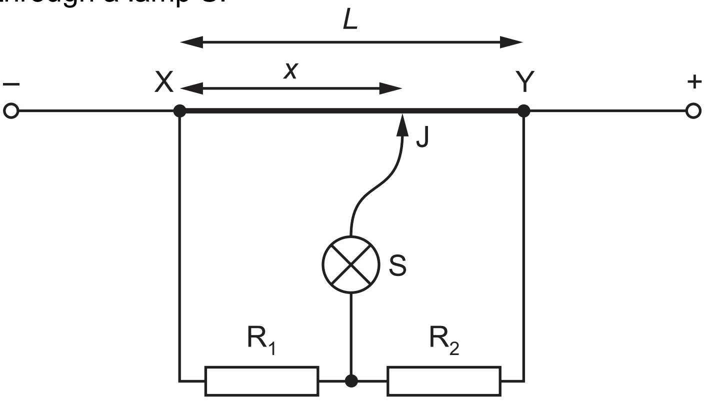


What is the resistance R?

A 32 Ω **B** 49 Ω **C** 71 Ω **D** 128 Ω

165 9702/13/M/J/18/Q38

In the circuit shown, XY is a length L of uniform resistance wire. A potential difference is applied across XY. R_1 and R_2 are unknown resistors. J is a sliding contact that joins the junction of R_1 and R_2 to points on XY through a lamp S.



J is moved along XY to a point at which the lamp is off. This point is at a distance x from X.

The potential difference across R_1 is V_1 and the potential difference across R_2 is V_2.

What is the value of the ratio $\frac{V_1}{V_2}$?

A $\frac{L}{x}$ **B** $\frac{x}{L}$ **C** $\frac{L-x}{x}$ **D** $\frac{x}{L-x}$

166 9702/12/F/M/18/Q33

A resistor has resistance R. When the potential difference across the resistor is V, the current in the resistor is I. The power dissipated in the resistor is P. Work W is done when charge Q flows through the resistor.

What is **not** a valid relationship between these variables?

A $I = \frac{P}{V}$ **B** $Q = \frac{W}{V}$ **C** $R = \frac{P}{I^2}$ **D** $R = \frac{V}{P}$

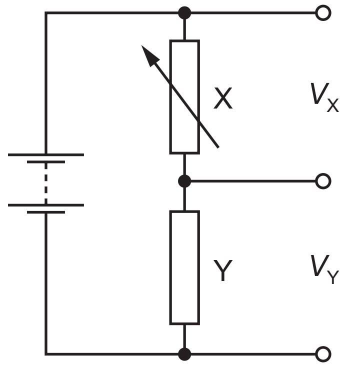
167 9702/12/F/M/18/Q36

A potential divider circuit is constructed with one variable resistor X and one fixed resistor Y, as shown.

The potential difference across resistor X is V_X and the potential difference of resistor Y is V_Y.

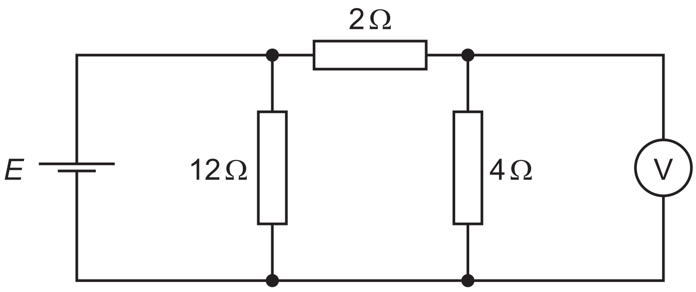
As the resistance of X is increased, what happens to V_X and to V_Y?

	V_X	V_Y
A	falls	rises
B	falls	stays the same
C	rises	falls
D	rises	stays the same



168 **9702/12/F/M/18/Q37**

A cell of electromotive force (e.m.f.) E and negligible internal resistance is connected into a circuit, as shown.



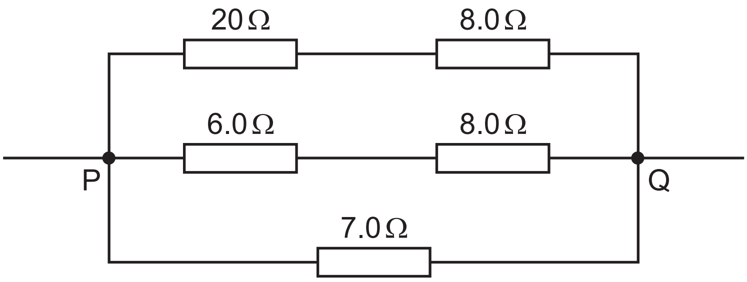
The voltmeter has a very high resistance and reads a potential difference V_{out}.

What is the ratio $\dfrac{V_{out}}{E}$?

A $\frac{1}{6}$ **B** $\frac{1}{3}$ **C** $\frac{1}{2}$ **D** $\frac{2}{3}$

169 **9702/12/F/M/18/Q38**

Five resistors are connected as shown.

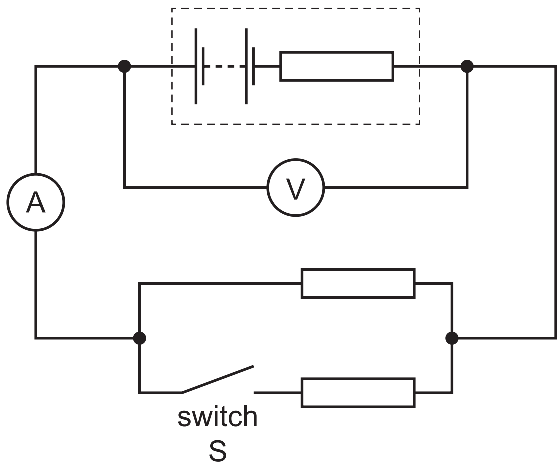


What is the total resistance between points P and Q?

A 0.25Ω **B** 0.61Ω **C** 4.0Ω **D** 16Ω

170 **9702/11/O/N/18/Q36**

A battery, with internal resistance, is connected to a parallel arrangement of two resistors and a switch S, as shown.



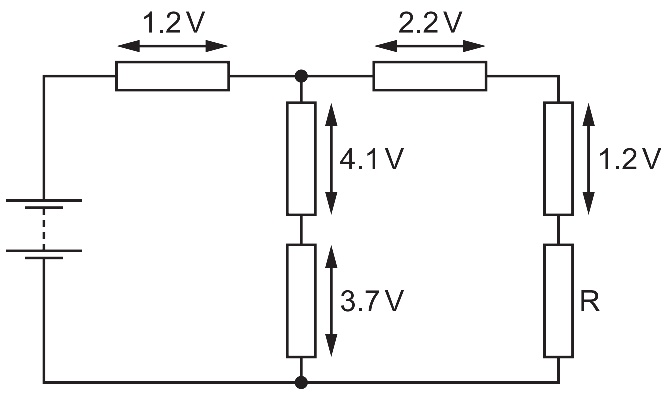
Initially switch S is open.

What happens to the voltmeter and ammeter readings when switch S is closed?

	voltmeter reading	ammeter reading
A	decreases	increases
B	decreases	decreases
C	increases	increases
D	increases	decreases

171 **9702/11/O/N/18/Q37**

A battery is connected to a network of six resistors, as shown.



The potential differences across five of the resistors are labelled on the diagram.

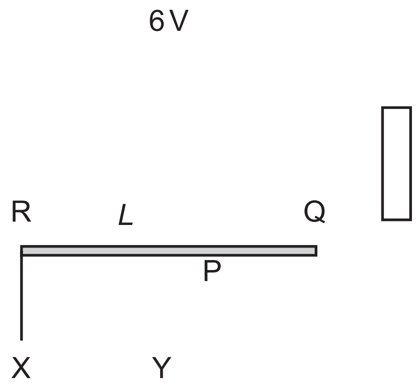
What is the potential difference across resistor R?

A 4.4 V **B** 4.6 V **C** 6.6 V **D** 11.2 V

172 9702/11/O/N/18/Q38

The diagram shows a battery of electromotive force (e.m.f.) 6 V, connected in series with a resistor and a uniform resistance wire RQ of length 60 cm.

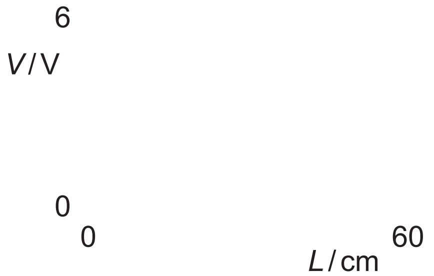
The resistance of RQ is equal to the resistance of the resistor.



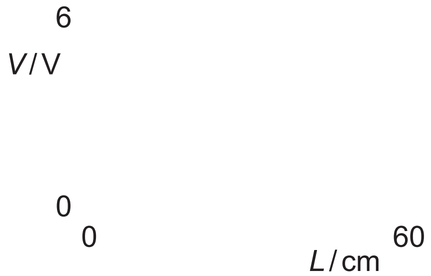
Terminal X is connected to fixed point R. Terminal Y is connected to point P, a connection that may be made at any position along the wire. *L* is the distance between R and P.

Which graph shows the variation with *L* of the potential difference (p.d.) *V* across XY?

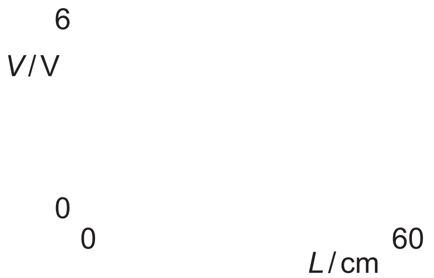
A



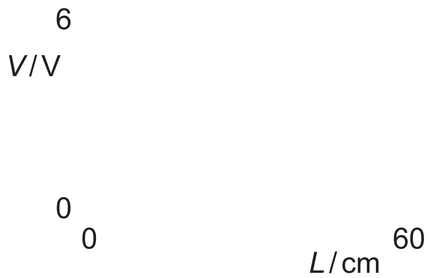
B



C

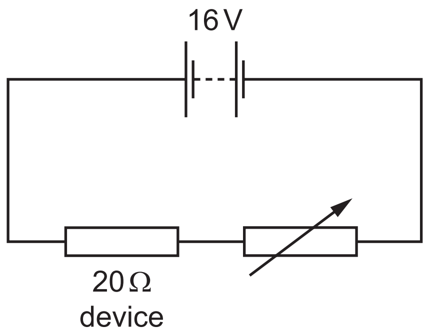


D



173 9702/12/O/N/18/Q34

An electrical device of fixed resistance 20 Ω is connected in series with a variable resistor and a battery of electromotive force (e.m.f.) 16 V and negligible internal resistance.

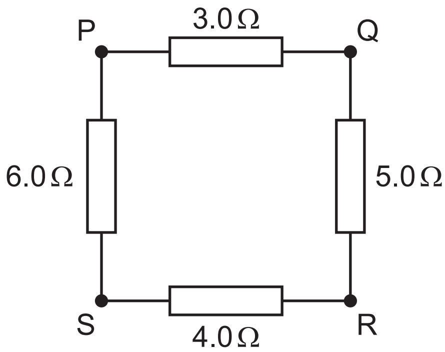


What is the resistance of the variable resistor when the power dissipated in the electrical device is 4.0 W?

A 16 Ω **B** 36 Ω **C** 44 Ω **D** 60 Ω

174 9702/12/O/N/18/Q36

A battery of negligible internal resistance may be connected between any two points P, Q, R and S of the network of resistors shown.



Which connections will give the largest current and the smallest current in the battery?

	largest current	smallest current
A	PQ	PR
B	PQ	QS
C	RS	PR
D	RS	QS

175 9702/12/O/N/18/Q37

Kirchhoff's second law is a consequence of a basic principle.

What is this principle?

A The charge flowing in an electric circuit is conserved.

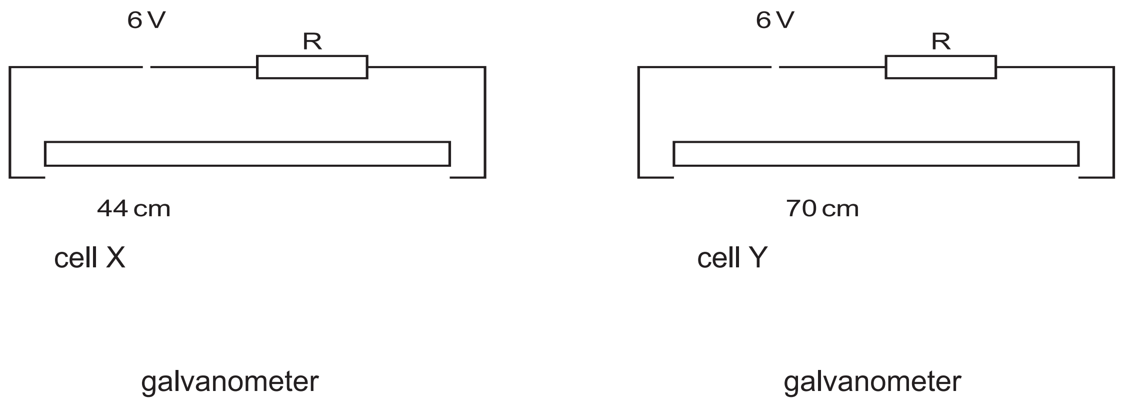
B The energy in an electric circuit is conserved.

C The sum of the electric currents entering a point in an electrical circuit is equal to the sum of the electric currents leaving that point.

D The sum of the potential differences in a circuit is equal to the sum of the products of the current and resistance.

176 9702/12/O/N/18/Q38

Two cells are investigated using a potentiometer. At the balance point, cell X gives a reading of 44 cm and cell Y gives a reading of 70 cm.



Which statement is **not** correct?

A A potentiometer balance point results in zero current through the galvanometer.

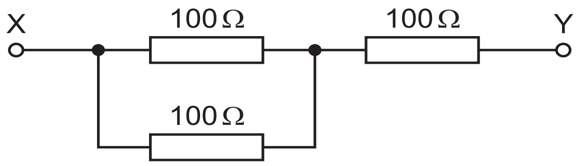
B At the balance point, the current through resistor R in both circuits is the same.

C The electromotive force (e.m.f.) of cell X is larger than that of cell Y.

D The value of the e.m.f. of each of the cells X and Y is less than 6 V.

177 9702/13/O/N/18/Q33

Three resistors are to be connected into a circuit with the arrangement shown.



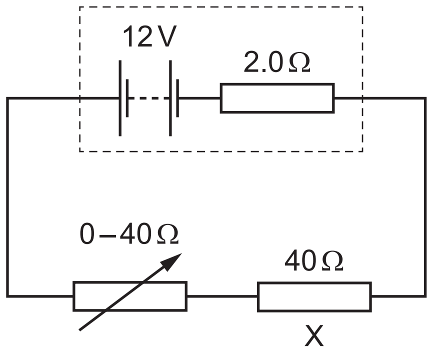
The power in any resistor must not be greater than 4.0 W.

What is the maximum voltage across XY?

A 24 V **B** 30 V **C** 40 V **D** 60 V

178 **9702/13/O/N/18/Q35**

A resistor X of resistance 40 Ω and a variable resistor are connected to a battery of electromotive force (e.m.f.) 12 V and internal resistance 2.0 Ω, as shown.



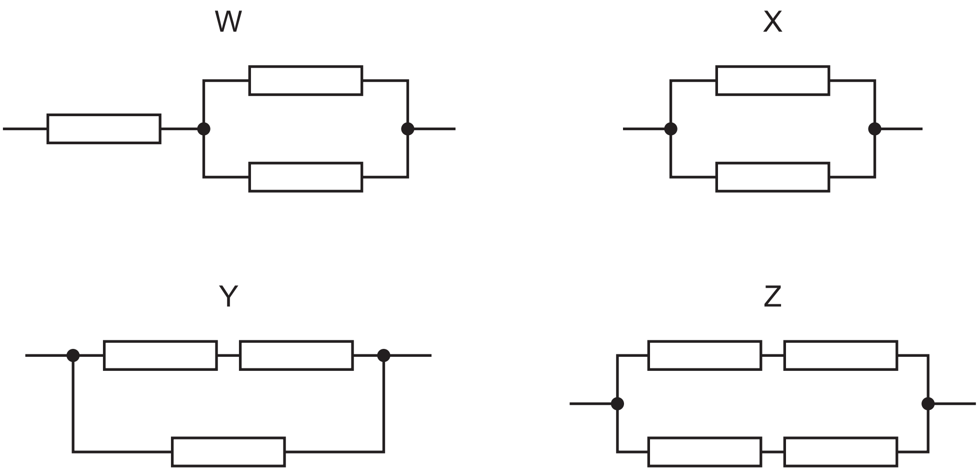
The resistance of the variable resistor is changed from 0 to 40 Ω.

What is the change in power dissipated in resistor X?

A 2.4 W **B** 2.7 W **C** 3.6 W **D** 5.6 W

179 **9702/13/O/N/18/Q36**

All the resistors shown in the resistor networks W, X, Y and Z have the same resistance.



Which list gives the networks in order of increasing total resistance?

A W → Z → Y → X

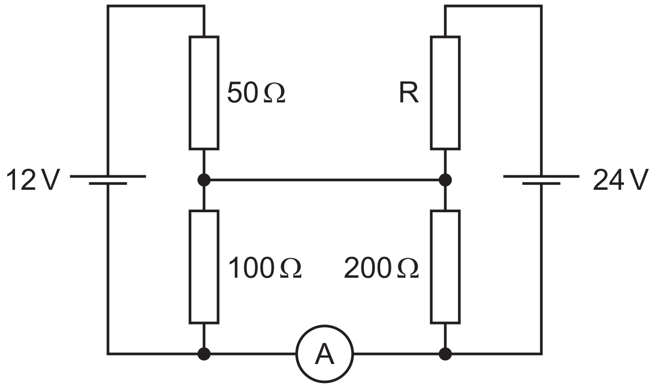
B X → W → Y → Z

C X → Y → W → Z

D X → Y → Z → W

180 9702/13/O/N/18/Q37

In the circuit shown, the ammeter reading is zero.



What is the resistance of resistor R?

A 100Ω **B** 200Ω **C** 400Ω **D** 600Ω

181 9702/11/M/J/19/Q35

When a battery is connected to a resistor, the battery gradually becomes warm. This causes the internal resistance of the battery to increase whilst its electromotive force (e.m.f.) stays unchanged.

As the internal resistance of the battery increases, how do the terminal potential difference and the output power change, if at all?

	terminal potential difference	output power
A	decreases	decreases
B	decreases	unchanged
C	unchanged	decreases
D	unchanged	unchanged

182 9702/11/M/J/19/Q36

A cell is connected to a resistor of resistance 3.00Ω. The current in the resistor is 1.00A.

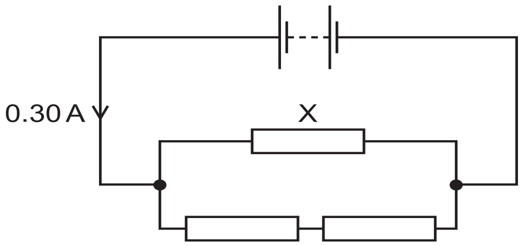
A second identical resistor is added in parallel. The current becomes 1.93A.

What are the e.m.f. E and internal resistance r of the cell?

	E/V	r/Ω
A	0.113	3.11
B	3.04	0.0358
C	3.11	0.113
D	9.34	6.34

183 9702/11/M/J/19/Q37

A battery with negligible internal resistance is connected to three resistors, as shown.



All three resistors have the same resistance.

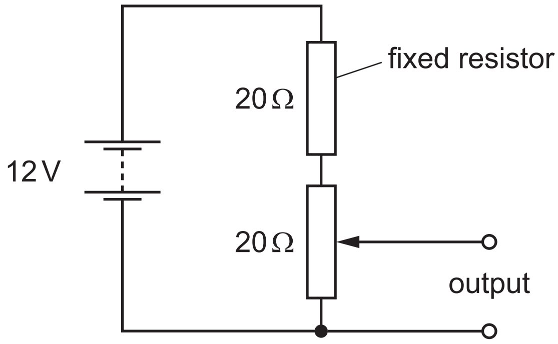
The current in the battery is 0.30 A.

What is the current in resistor X?

A 0.10 A **B** 0.15 A **C** 0.20 A **D** 0.30 A

184 9702/11/M/J/19/Q38

The diagram shows a potentiometer and a fixed resistor connected across a 12 V battery of negligible internal resistance.



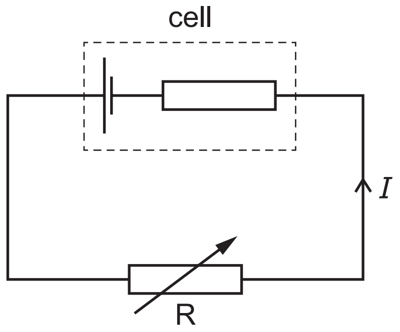
The fixed resistor and the potentiometer each have resistance 20 Ω. The circuit is designed to provide a variable output voltage.

What is the range of output voltages?

A 0 – 6 V **B** 0 – 12 V **C** 6 – 12 V **D** 12 – 20 V

185 9702/12/M/J/19/Q35

A cell with internal resistance is connected to a variable resistor R as shown.



The resistance of R is gradually decreased.

How do the current I and the terminal potential difference across the cell change?

	current I	terminal potential difference across cell
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

186 9702/12/M/J/19/Q36

Kirchhoff's first law states that the sum of the currents entering a junction in a circuit is equal to the sum of the currents leaving it.

The law is based on the conservation of a physical quantity.

What is this physical quantity?

A charge

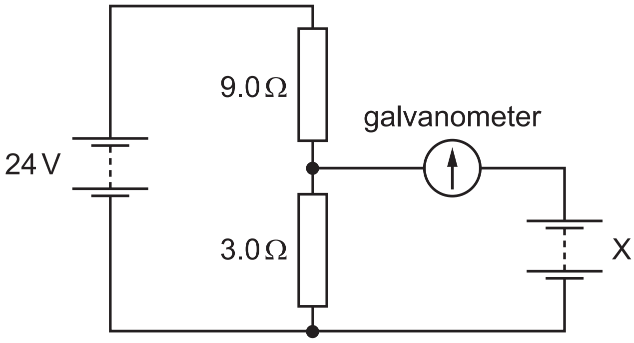
B energy

C mass

D momentum

187 **9702/12/M/J/19/Q37**

A circuit contains two batteries, each of negligible internal resistance, and two resistors as shown.



The galvanometer has a current reading of zero.

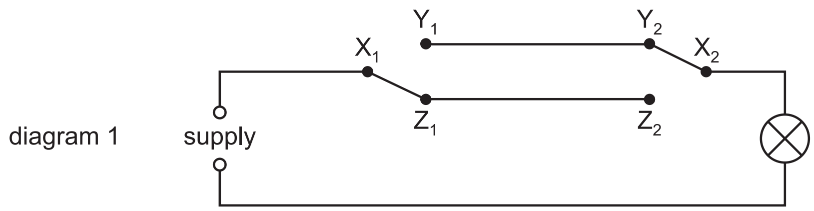
What is the electromotive force (e.m.f.) of battery X?

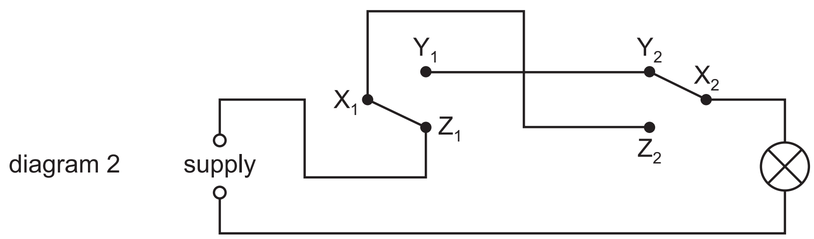
A 6.0 V **B** 8.0 V **C** 16.0 V **D** 18.0 V

188 **9702/13/M/J/19/Q34**

Diagram 1 shows a lamp connected to a supply through two switches.

During repairs, an electrician mistakenly reverses the connections X_1 and Z_1, so that Z_1 is connected to the supply and X_1 to the other switch at Z_2, as shown in diagram 2.





Which switch positions will now light the lamp?

A	X_1 to Y_1	X_2 to Y_2
B	X_1 to Y_1	X_2 to Z_2
C	X_1 to Z_1	X_2 to Y_2
D	X_1 to Z_1	X_2 to Z_2

189 **9702/13/M/J/19/Q35**

A wire supplying a shower heater with a current of 35 A has a resistance of 25 mΩ.

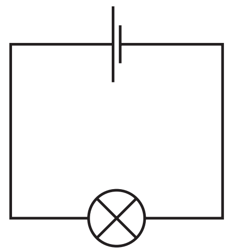
What is the power dissipated in the wire?

A 31 W **B** 49 W **C** 31 kW **D** 49 kW

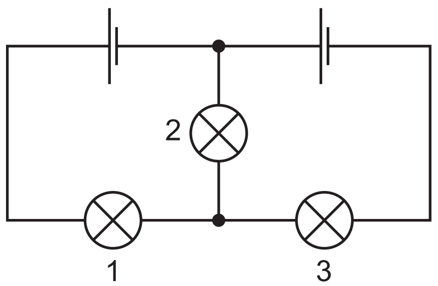
190 **9702/13/M/J/19/Q36**

A student has a set of identical cells and identical lamps. The cells have negligible internal resistance.

A lamp connected to a cell lights with normal brightness.



What happens when the student connects the lamps and the cells as shown?



A All three lamps light with normal brightness.

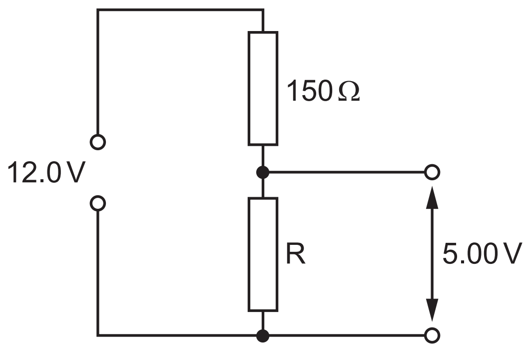
B Only lamp 2 lights with normal brightness.

C Only lamps 1 and 3 light with normal brightness.

D None of the lamps light with normal brightness.

191 9702/13/M/J/19/Q37

A potential divider circuit is shown.

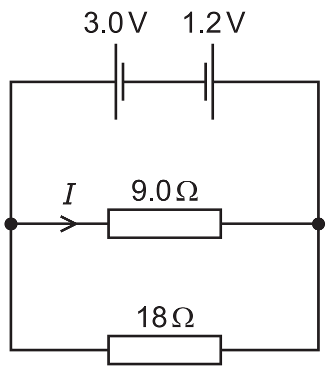


What is the resistance of resistor R in the potential divider circuit?

A 62.5Ω **B** 107Ω **C** 210Ω **D** 360Ω

192 9702/13/M/J/19/Q38

Two cells of electromotive force (e.m.f.) 3.0V and 1.2V and negligible internal resistance are connected to resistors of resistance 9.0Ω and 18Ω as shown.



What is the current I in the 9.0Ω resistor?

A 0.10A **B** 0.20A **C** 0.30A **D** 0.47A

193 9702/11/O/N/19/Q34

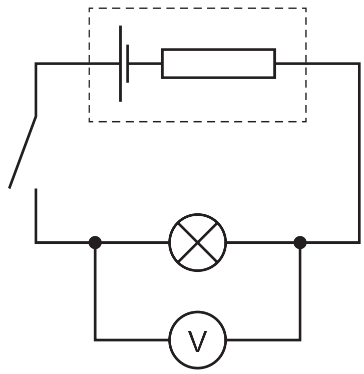
Kirchhoff's two laws for electric circuits can be derived by using conservation laws.

On which conservation laws do Kirchhoff's laws depend?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	current
B	charge	energy
C	current	mass
D	energy	current

194 9702/11/O/N/19/Q35

The diagram shows a circuit.



Which statement about the circuit is **not** correct?

A Electromotive force is the energy transferred per unit charge.

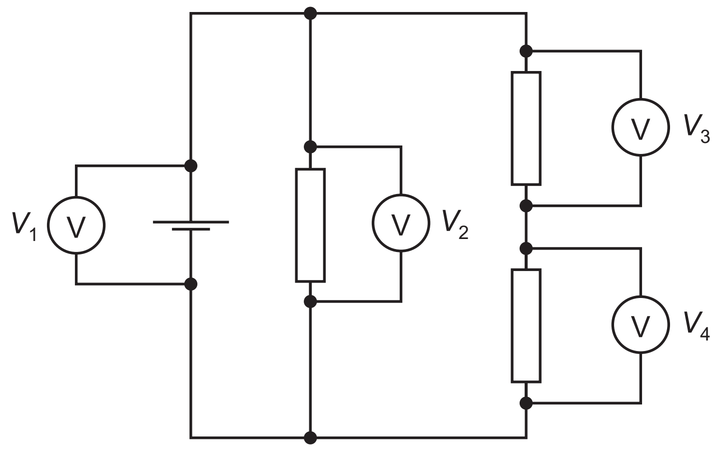
B Energy is transferred from chemical potential energy in the cell to other forms when the switch is closed.

C The electromotive force of the cell is greater than the terminal potential difference when the switch is closed.

D When the switch is open, the voltmeter measures the electromotive force of the cell.

195 **9702/11/O/N/19/Q36**

The diagram shows a circuit containing four voltmeters. The readings on the voltmeters are V_1, V_2, V_3 and V_4. All the readings are positive.



Which equation relating the voltmeter readings is correct?

A $V_1 = V_2 + V_4$

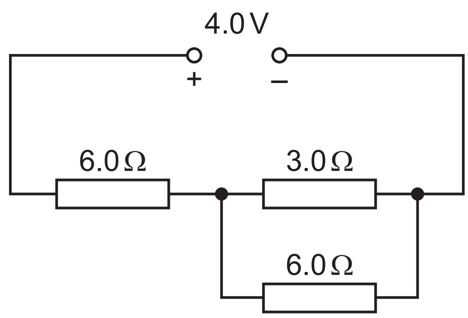
B $V_1 = V_2 + V_3 + V_4$

C $V_2 + V_3 = V_4$

D $V_3 + V_4 - V_2 = 0$

196 **9702/11/O/N/19/Q37**

A network consists of a 3.0 Ω resistor and two 6.0 Ω resistors, as shown.



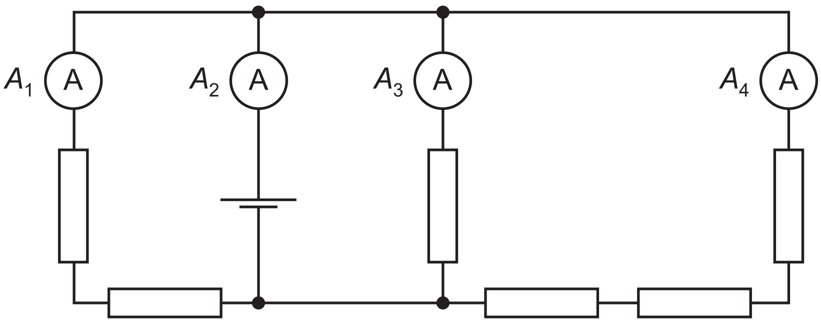
The potential difference (p.d.) across the network is 4.0 V.

What is the current through the 3.0 Ω resistor?

A 0.17 A **B** 0.25 A **C** 0.33 A **D** 1.3 A

197 9702/11/O/N/19/Q38

In the circuit shown, all the resistors are identical and all the ammeters have negligible resistance.



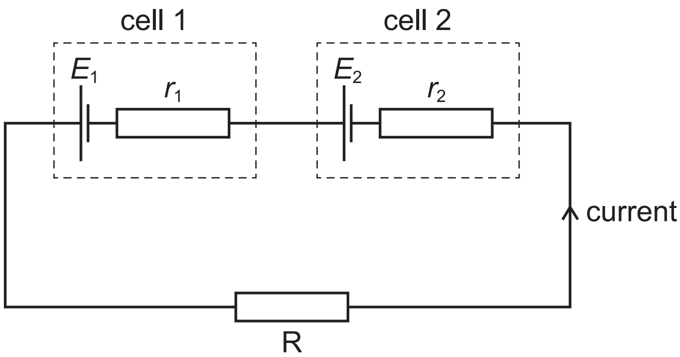
The reading A_1 is 0.6 A.

What are the readings on the other ammeters?

	A_2 / A	A_3 / A	A_4 / A
A	1.0	0.3	0.1
B	1.4	0.6	0.2
C	1.8	0.9	0.3
D	2.2	1.2	0.4

198 9702/12/O/N/19/Q34

Two cells with electromotive forces E_1 and E_2 and internal resistances r_1 and r_2 are connected to a resistor R as shown.



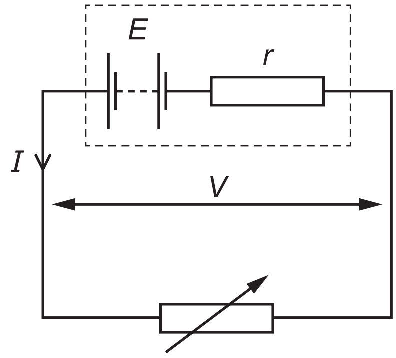
The terminal potential difference across cell 1 is zero.

Which expression gives the resistance of resistor R?

A $\dfrac{E_2r_1 - E_1r_2}{E_1}$ **B** $\dfrac{E_2r_1 - E_1r_2}{E_2}$ **C** $\dfrac{E_1r_2 - E_2r_1}{E_1}$ **D** $\dfrac{E_1r_2 - E_2r_1}{E_2}$

199 **9702/12/O/N/19/Q35**

A battery has an electromotive force (e.m.f.) E and internal resistance r. The battery delivers a current I to a variable resistor and the potential difference (p.d.) across its terminals is V.



The variable resistor is adjusted so that I increases.

Why does V decrease?

A The e.m.f. E decreases.

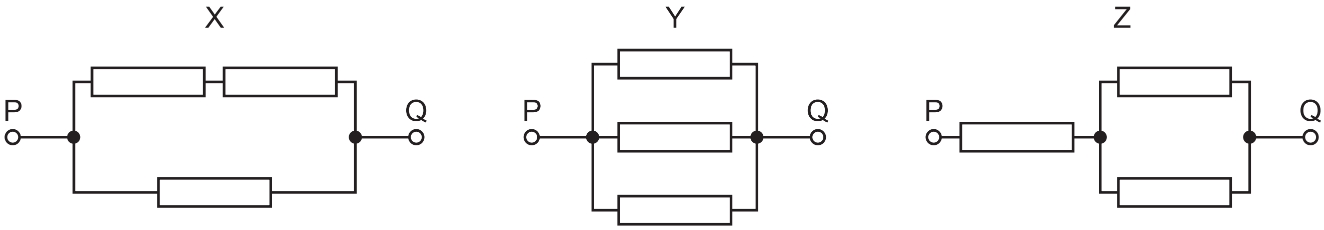
B The internal resistance r increases.

C The p.d. across r increases.

D The resistance of the variable resistor increases.

200 **9702/12/O/N/19/Q36**

Three identical resistors are connected between terminals P and Q in different networks X, Y and Z as shown.



What is the order of increasing combined resistance between P and Q (lowest first)?

A X → Y → Z

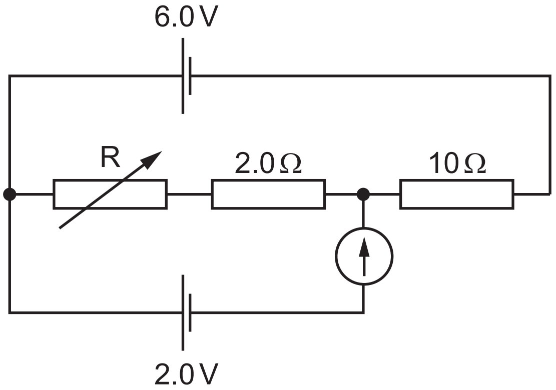
B X → Z → Y

C Y → X → Z

D Y → Z → X

201 **9702/12/O/N/19/Q37**

The diagram shows a variable resistor R and two fixed resistors connected in series in a circuit to act as a potential divider.



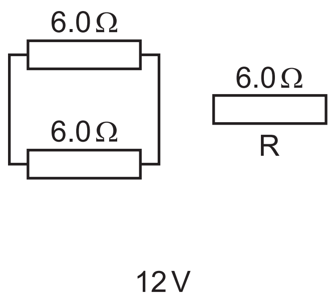
The cell of electromotive force (e.m.f.) 6.0 V has negligible internal resistance. A cell of e.m.f. 2.0 V and a galvanometer are connected into the potential divider. The resistance of R is varied until the galvanometer reads zero.

What is the resistance of resistor R?

A 3.0 Ω **B** 5.0 Ω **C** 8.0 Ω **D** 18 Ω

202 **9702/13/O/N/19/Q34**

A battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected to three resistors, each of resistance 6.0 Ω, as shown.

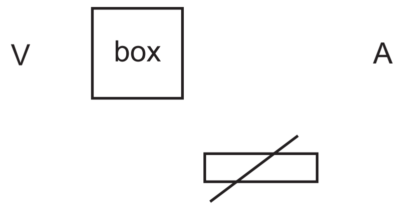


What is the power dissipated in resistor R?

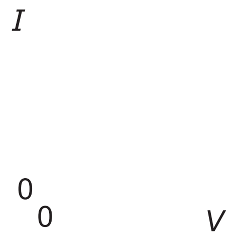
A 2.7 W **B** 6.0 W **C** 11 W **D** 24 W

203 9702/13/O/N/19/Q35

A box containing two electrical components is connected into a circuit.



The variable resistor is adjusted and measurements are taken to determine the I–V characteristic for the box, as shown.



Which arrangement of two electrical components in the box would create the best fit to the measured I–V characteristic?

A a filament lamp and a fixed resistor in parallel

B a filament lamp and a fixed resistor in series

C a semiconductor diode and a filament lamp in parallel

D a semiconductor diode and a filament lamp in series

204 9702/13/O/N/19/Q36

A cell of internal resistance $0.5\,\Omega$ is connected to a fixed resistor of resistance $10\,\Omega$.

The resistance of the resistor is changed to $20\,\Omega$.

Which statement is **not** correct?

A The current in the circuit will halve.

B The e.m.f. of the cell will remain constant.

C The power dissipated by the fixed resistor will decrease.

D The terminal p.d. of the cell will increase.

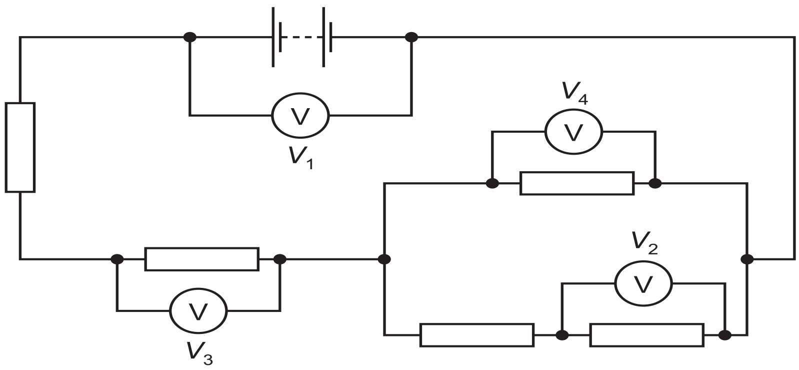
205 **9702/13/O/N/19/Q37**

Which row correctly describes Kirchhoff's laws?

	Kirchhoff's first law	physics principle applied for first law	Kirchhoff's second law	physics principle applied for second law
A	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy
B	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge
C	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge
D	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy

206 **9702/13/O/N/19/Q38**

In the circuit shown, all the resistors are identical.



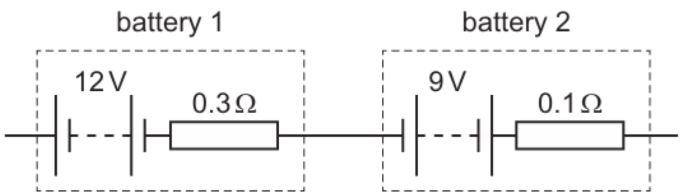
The reading V_1 is 8.0 V and the reading V_2 is 1.0 V.

What are the readings on the other voltmeters?

	V_3 / V	V_4 / V
A	1.5	1.0
B	3.0	2.0
C	4.5	3.0
D	6.0	4.0

207. 9702/12/F/M/20/Q35

Two batteries are connected together, as shown.



Battery 1 has electromotive force (e.m.f.) 12 V and internal resistance 0.3 Ω.

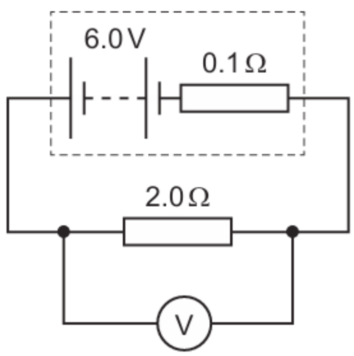
Battery 2 has e.m.f. 9 V and internal resistance 0.1 Ω.

What are the e.m.f. and the internal resistance of a single battery that has the same effect as the combination?

	e.m.f. / V	internal resistance / Ω
A	3	0.2
B	3	0.4
C	21	0.2
D	21	0.4

208. 9702/12/F/M/20/Q36

The diagram shows a circuit.



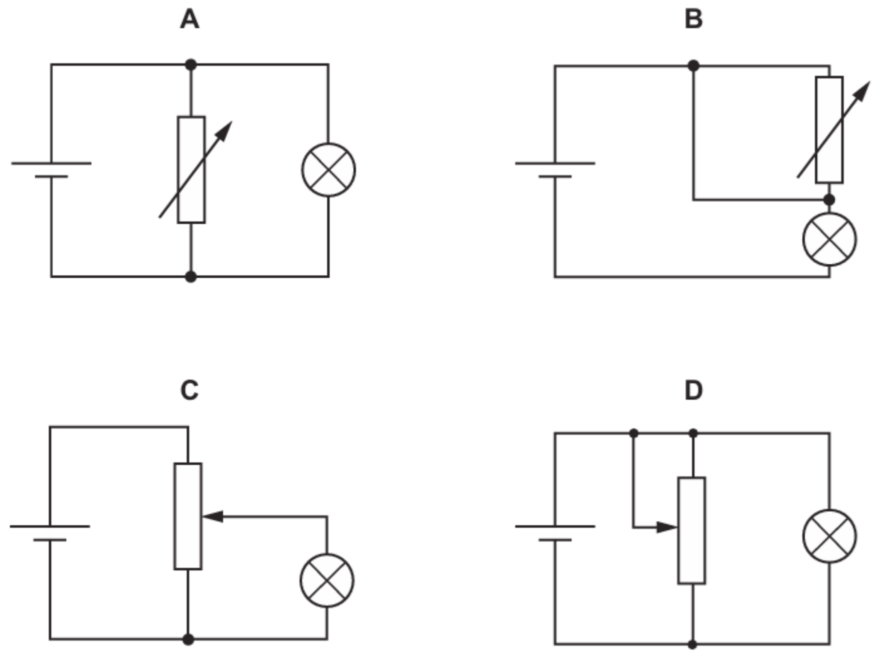
What is the reading on the voltmeter?

A 0.3 V **B** 5.7 V **C** 6.0 V **D** 6.3 V

209. 9702/12/F/M/20/Q37

In the circuits shown, the cell has negligible internal resistance.

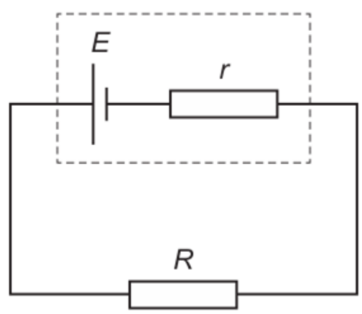
Which diagram shows a potential divider circuit that can vary the potential difference (p.d.) across the lamp?



210. 9702/11/M/J/20/Q36

A cell of electromotive force (e.m.f.) E and internal resistance r is connected to a resistor of resistance R.

A maximum power P can be dissipated by the resistor without overheating.



What is the maximum value of E if the resistor does not overheat?

A $R\sqrt{\frac{P}{(R-r)}}$ **B** $R\sqrt{\frac{P}{(R+r)}}$ **C** $(R-r)\sqrt{\frac{P}{R}}$ **D** $(R+r)\sqrt{\frac{P}{R}}$

211. 9702/11/M/J/20/Q37

Three identical resistors can be connected together in four different ways.

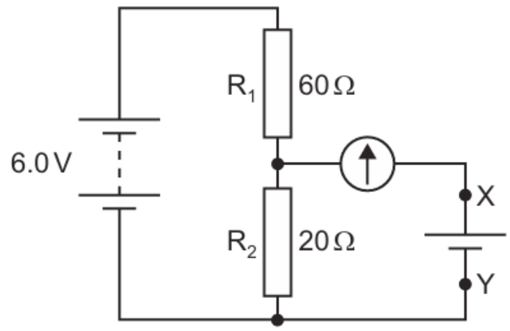
The resistances of two of these combinations are $4.0\,\Omega$ and $9.0\,\Omega$.

What is the resistance of each individual resistor?

A $3.0\,\Omega$ **B** $6.0\,\Omega$ **C** $12\,\Omega$ **D** $18\,\Omega$

212. 9702/11/M/J/20/Q38

In the circuit shown, a battery of negligible internal resistance is connected in series with a pair of fixed resistors R_1 and R_2.



The circuit is to be used to test whether the electromotive force (e.m.f.) of a particular cell is 1.5V. The cell is connected between terminals X and Y in parallel with R_2 and in series with a galvanometer.

Which statement about the test is correct?

A Any non-zero reading on the galvanometer means the cell has an e.m.f. of 1.5V.

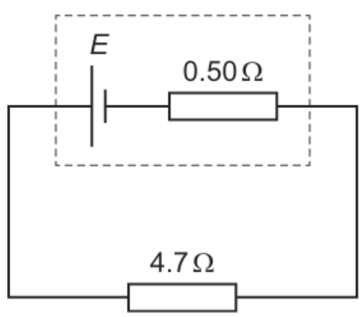
B The battery does not need to have an e.m.f. of 6.0V.

C The cell may be connected either way round between X and Y.

D The galvanometer does not need a scale calibrated in amperes.

213. 9702/12/M/J/20/Q36

A cell of electromotive force (e.m.f.) E and internal resistance 0.50Ω is connected to a resistor of resistance 4.7Ω.



The maximum power that can be dissipated by the resistor without overheating is 0.50W.

What is the maximum value of E for the resistor **not** to overheat?

A 1.4V **B** 1.5V **C** 1.7V **D** 2.9V

214. 9702/12/M/J/20/Q37

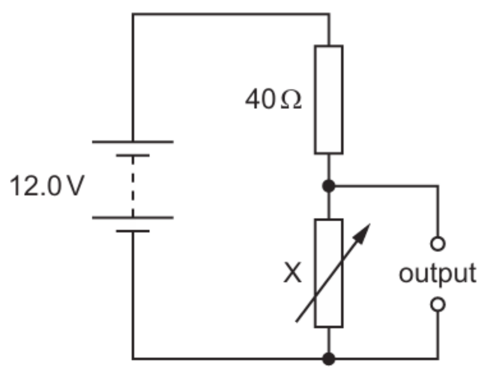
Kirchhoff's first and second laws link to the conservation of physical quantities.

Which quantities do they link to?

	first law	second law
A	charge	energy
B	charge	momentum
C	energy	charge
D	energy	momentum

215. 9702/12/M/J/20/Q38

In the circuit shown, X is a variable resistor whose resistance can be changed from 5.0 Ω to 500 Ω. The electromotive force (e.m.f.) of the battery is 12.0 V. It has negligible internal resistance.



What is the maximum range of values of potential difference across the output?

A 1.3 V to 11.1 V

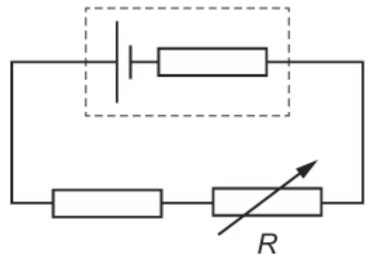
B 1.3 V to 12.0 V

C 1.5 V to 11.1 V

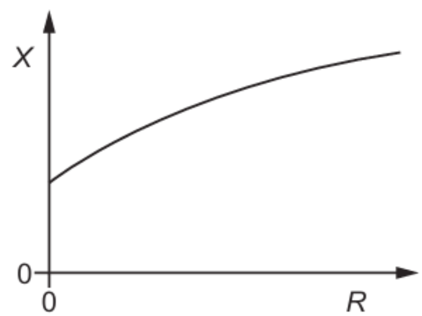
D 1.5 V to 12.0 V

216. 9702/13/M/J/20/Q37

A fixed resistor and a variable resistor are connected in series with a cell that has an internal resistance, as shown.



The graph shows the variation of a quantity X with the resistance R of the variable resistor as R is increased from zero to its maximum value.



What could X represent?

A the current in the circuit

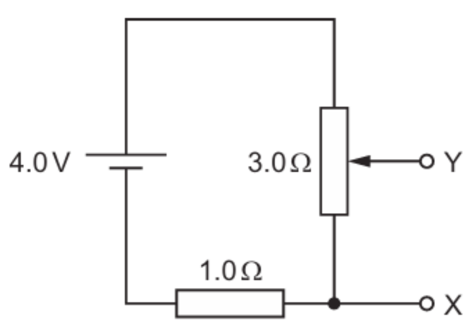
B the electromotive force of the cell

C the potential difference across the internal resistance

D the terminal potential difference across the cell

217. 9702/13/M/J/20/Q38

A cell of electromotive force (e.m.f.) 4.0 V and negligible internal resistance is connected to a fixed resistor of resistance 1.0 Ω and a potentiometer of maximum resistance 3.0 Ω, as shown.



Which range of potential differences can be obtained between the terminals X and Y?

A 0 V to 3.0 V

B 0 V to 4.0 V

C 1.0 V to 3.0 V

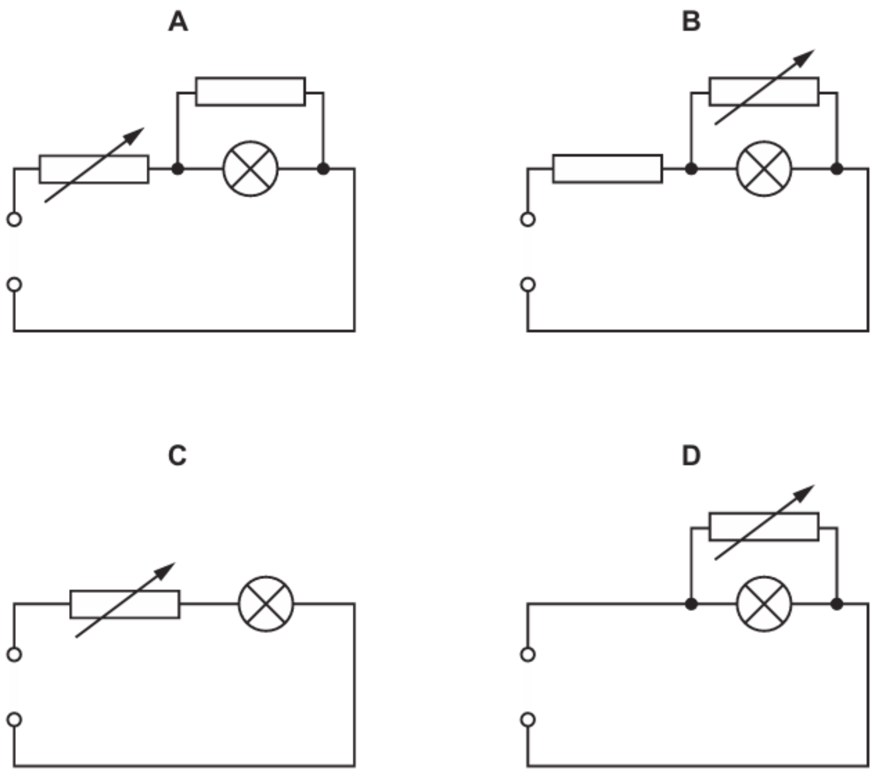
D 1.0 V to 4.0 V

218. 9702/11/O/N/20/Q35

In the circuits shown, the power supply has an electromotive force (e.m.f.) greater than the normal operating voltage of the lamp. The internal resistance of the power supply is negligible.

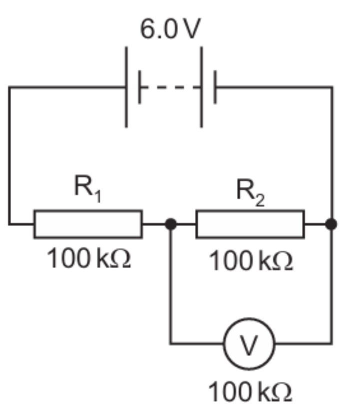
The resistance of the variable resistor is adjusted from zero to its maximum value.

In which circuit could the voltage across the lamp change from zero to its normal operating voltage and **not** exceed its normal operating voltage?



219. 9702/11/O/N/20/Q37

In the circuit shown, the 6.0 V battery has negligible internal resistance. Resistors R_1 and R_2 and the voltmeter each have a resistance of 100 kΩ.

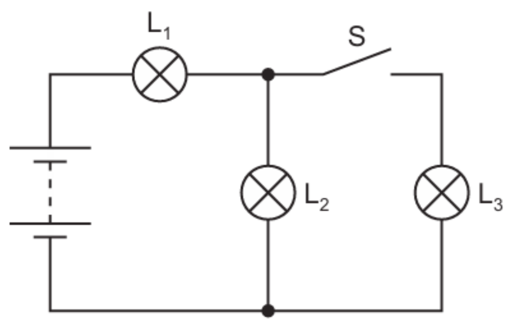


What is the current in the resistor R_2?

A 20 μA　　**B** 30 μA　　**C** 40 μA　　**D** 60 μA

220. 9702/11/O/N/20/Q36

Three identical lamps L_1, L_2 and L_3 are connected to a battery with negligible internal resistance, as shown.

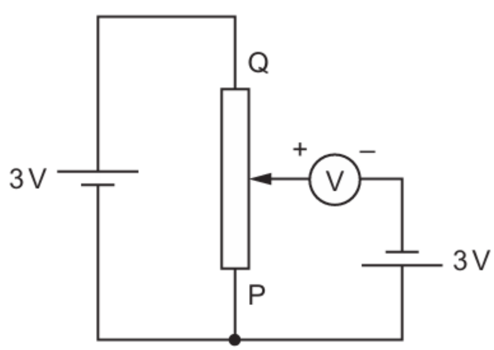


What happens to the brightness of lamps L_1 and L_2 when the switch S is closed?

	lamp L_1	lamp L_2
A	brighter	brighter
B	brighter	dimmer
C	dimmer	brighter
D	dimmer	dimmer

221. 9702/12/O/N/20/Q38

A voltmeter is connected into a circuit with the polarity shown in the diagram.



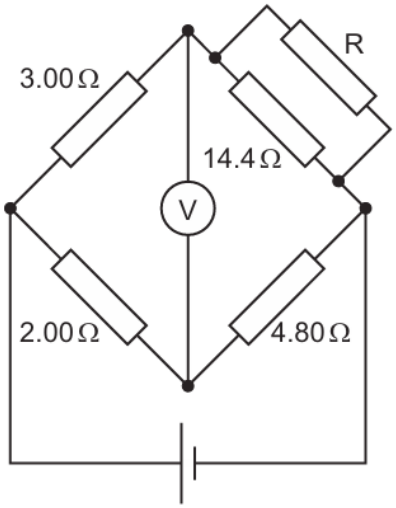
The sliding contact is moved to end P of the potentiometer and then to end Q.

What are the two readings of the voltmeter?

	sliding contact at end P	sliding contact at end Q
A	0 V	3 V
B	0 V	6 V
C	3 V	3 V
D	3 V	6 V

222. 9702/12/O/N/20/Q37

A cell of negligible internal resistance is connected to a network of resistors and a voltmeter, as shown.



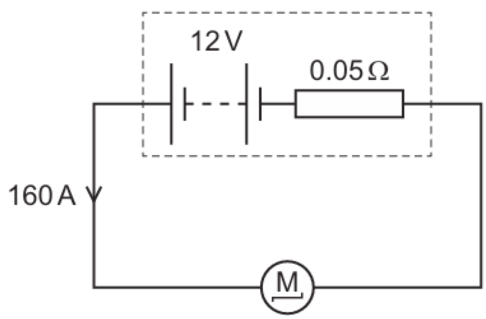
The reading on the voltmeter is zero.

What is the resistance of resistor R?

A 1.20 Ω **B** 1.80 Ω **C** 7.20 Ω **D** 14.4 Ω

223. 9702/12/O/N/20/Q36

A car battery has an electromotive force (e.m.f.) of 12 V and an internal resistance of 0.05 Ω. The battery is connected to the starter motor of a car. The current in the motor is 160 A.



What is the terminal p.d. across the battery?

A 0 V **B** 4 V **C** 8 V **D** 12 V

224. 9702/13/O/N/20/Q36

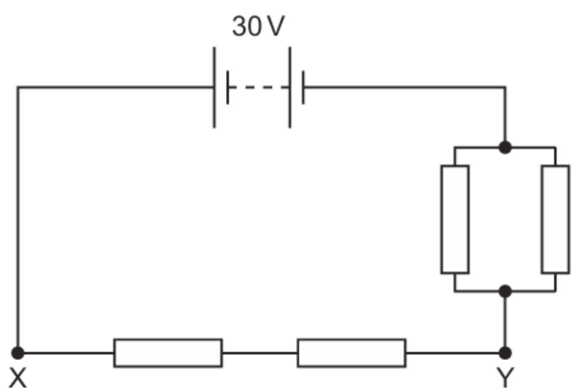
A cell is connected to a fixed resistor. Over a long period of time, the internal resistance of the cell increases.

What is the effect of the increase in internal resistance on the electromotive force (e.m.f.) of the cell and on the power dissipated by the fixed resistor?

	e.m.f.	power dissipated
A	decreases	decreases
B	decreases	no change
C	no change	decreases
D	no change	no change

225. 9702/13/O/N/20/Q37

Four identical resistors are connected in a circuit, as shown.



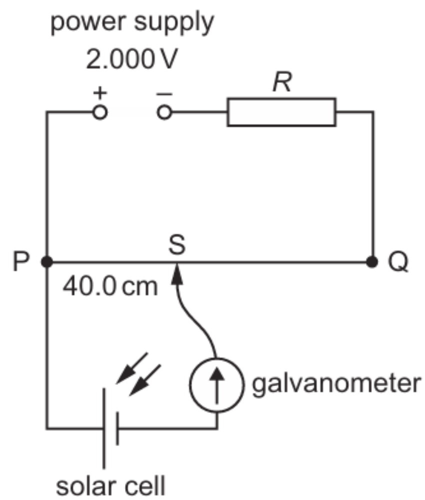
The battery has negligible internal resistance and an e.m.f. of 30 V.

What is the potential difference between the two points X and Y?

A 6.0 V **B** 15 V **C** 20 V **D** 24 V

226. 9702/13/O/N/20/Q38

A power supply and a solar cell are compared using the potentiometer circuit shown.



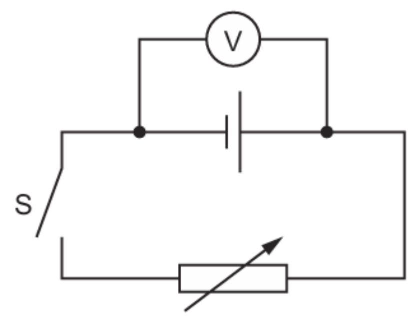
The potentiometer wire PQ is 100.0 cm long and has a resistance of 5.00 Ω. The power supply has an e.m.f. of 2.000 V and the solar cell has an e.m.f. of 5.00 mV.

Which resistance R must be used so that the galvanometer reads zero when PS = 40.0 cm?

A 395 Ω **B** 405 Ω **C** 795 Ω **D** 805 Ω

227. 9702/12/F/M/21/Q36

A cell that has internal resistance is connected to a switch S and a variable resistor. A voltmeter is connected between the terminals of the cell, as shown.



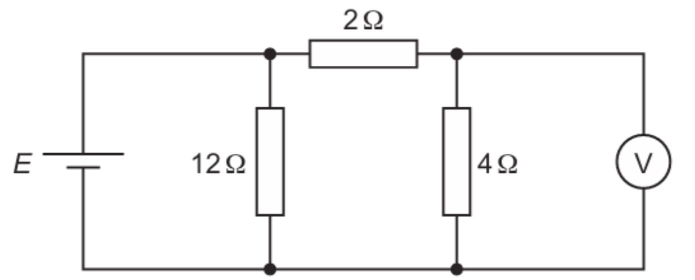
When switch S is open, the voltmeter reads 1.5 V. The switch is then closed and the variable resistor is adjusted to have a resistance of 4.0 Ω. The voltmeter now reads 0.75 V.

What is the internal resistance of the cell?

A 1.0 Ω **B** 2.0 Ω **C** 4.0 Ω **D** 8.0 Ω

228. 9702/12/F/M/21/Q37

A cell of electromotive force (e.m.f.) E and negligible internal resistance is connected into a circuit, as shown.



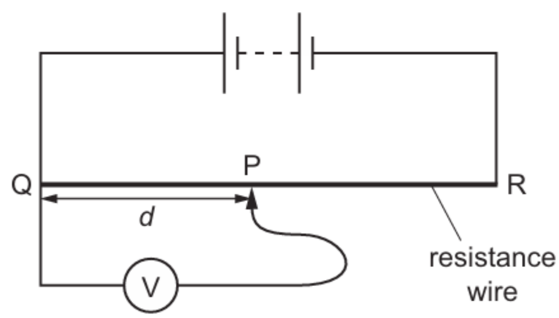
The voltmeter has a very high resistance and reads a potential difference V_{out}.

What is the ratio $\dfrac{V_{out}}{E}$?

A $\frac{1}{6}$ **B** $\frac{1}{3}$ **C** $\frac{1}{2}$ **D** $\frac{2}{3}$

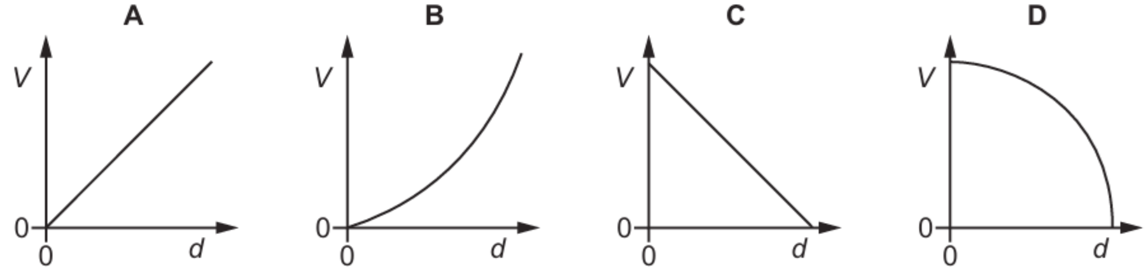
229. 9702/12/F/M/21/Q38

A battery is connected to a potentiometer. The potentiometer consists of a uniform resistance wire and a sliding contact P.



The potential difference (p.d.) V between the sliding contact P and end Q of the wire is measured using a voltmeter. The sliding contact P is moved from end Q to end R of the wire. Sliding contact P is distance d from Q.

Which graph shows the variation with distance d of the p.d. V?



230. 9702/11/M/J/21/Q35

A cell is described as having an electromotive force (e.m.f.) of 6 V.

What does this mean?

A 1 coulomb of charge always dissipates 6 J of energy in the internal resistance of the cell.

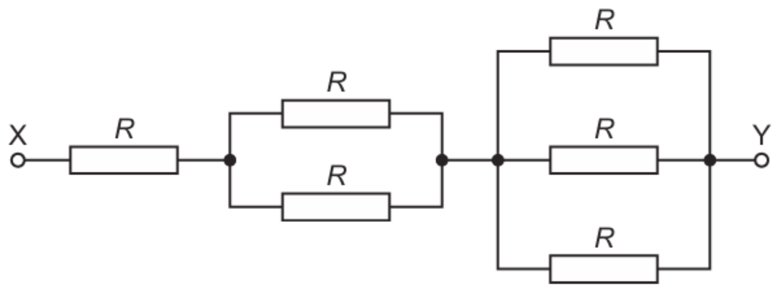
B 1 electron gains 6 J of energy when passing through the cell.

C There is a potential difference of 6 V applied across any external circuit connected to the cell.

D When 1 coulomb of charge passes through the cell, 6 J of chemical energy is transformed.

231. 9702/11/M/J/21/Q36

The diagram shows a network of resistors. Each resistor has resistance R.

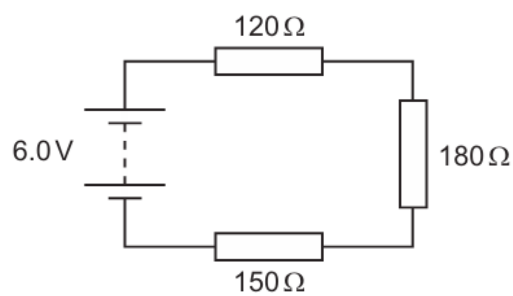


What is the total resistance of the network between points X and Y?

A $\frac{R}{6}$ **B** $\frac{6R}{11}$ **C** $\frac{11R}{6}$ **D** $6R$

232. 9702/11/M/J/21/Q37

Three resistors are connected in series with a battery, as shown. The battery has negligible internal resistance.



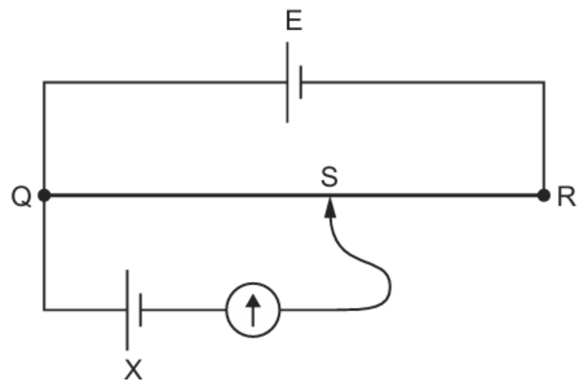
What is the potential difference across the 180 Ω resistor?

A 1.6 V **B** 2.4 V **C** 3.6 V **D** 4.0 V

233. 9702/11/M/J/21/Q38

A potentiometer circuit is used to determine the unknown electromotive force (e.m.f.) of a cell X.

In the circuit shown, E is a cell with an e.m.f. that is known accurately. QR is the potentiometer wire, which has a movable contact S. Contact S is connected to a galvanometer and to cell X.



What is **not** a necessary requirement to determine the e.m.f. of X from the circuit?

A The e.m.f. of cell X must be lower than the e.m.f. of cell E.

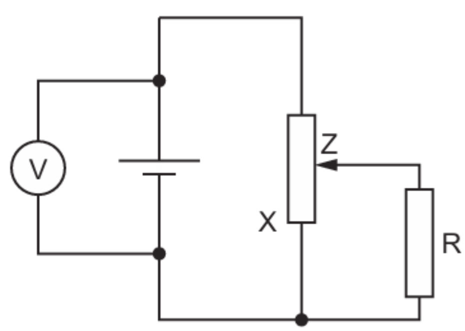
B The internal resistance of cell X must be known.

C The lengths QS and QR must be determined accurately.

D The resistance of the wire QR must be proportional to its length.

234. 9702/12/M/J/21/Q36

A cell of constant electromotive force (e.m.f.) but with internal resistance is connected to a fixed resistor R using a potentiometer. A voltmeter measures the potential difference (p.d.) between the terminals of the cell.



Which statement explains the change to the reading of the voltmeter as contact Z is moved towards end X of the potentiometer?

A The voltmeter reading decreases because the current through the cell decreases.

B The voltmeter reading decreases because the current through the cell increases.

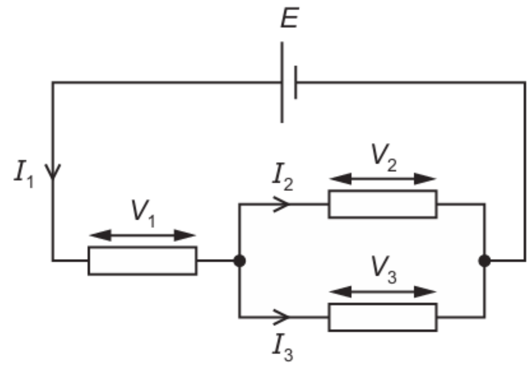
C The voltmeter reading increases because the current through the cell decreases.

D The voltmeter reading increases because the current through the cell increases.

235. 9702/12/M/J/21/Q37

A cell of electromotive force (e.m.f.) E and negligible internal resistance is connected to a circuit.

The circuit has currents I_1, I_2 and I_3, and potential differences V_1, V_2 and V_3, as shown.

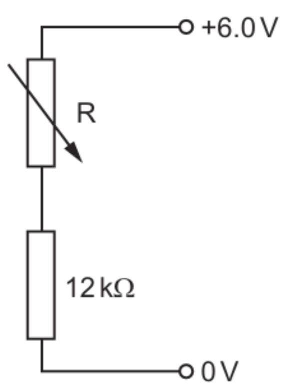


Which equation represents a statement of Kirchhoff's first law?

A $I_1 = I_2 + I_3$ **B** $I_1 = I_2 = I_3$ **C** $E = V_1 + V_2$ **D** $V_1 = V_2 = V_3$

236. 9702/12/M/J/21/Q38

Two resistors are connected in series with a 6.0 V power supply, as shown.

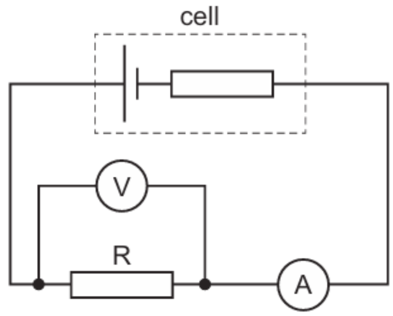


What is the resistance of the variable resistor R to give a potential difference of 1.0 V across the 12 kΩ resistor?

A 2.0 kΩ **B** 10 kΩ **C** 60 kΩ **D** 72 kΩ

237. 9702/13/M/J/21/Q36

The circuit shown includes a cell of constant internal resistance and an external resistor R.



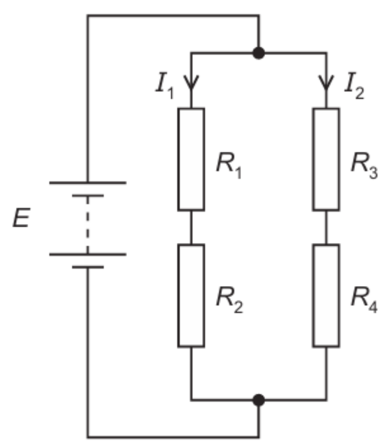
A student records the ammeter and voltmeter readings. She then connects a second identical external resistor in parallel with the first external resistor.

What happens to the ammeter reading and to the voltmeter reading?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	stays the same
C	increases	decreases
D	increases	stays the same

238. 9702/13/M/J/21/Q37

A battery of electromotive force (e.m.f.) E and negligible internal resistance is connected to four resistors of resistances R_1, R_2, R_3 and R_4.



The currents I_1 and I_2 in the resistors are as shown.

Which equation is correct?

A $0 = I_1(R_1 + R_2) + I_2(R_3 + R_4)$

B $0 = I_1(R_1 + R_2) - I_2(R_3 + R_4)$

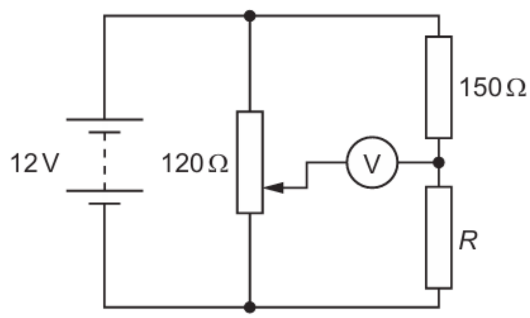
C $E = I_1(R_1 + R_2) + I_2(R_3 + R_4)$

D $E = I_1(R_1 + R_2) - I_2(R_3 + R_4)$

239. 9702/13/M/J/21/Q38

In the circuit shown, a potentiometer of total resistance 120 Ω is connected in parallel with a resistor of resistance 150 Ω and a resistor of resistance R.

The battery has electromotive force (e.m.f.) 12 V and negligible internal resistance.



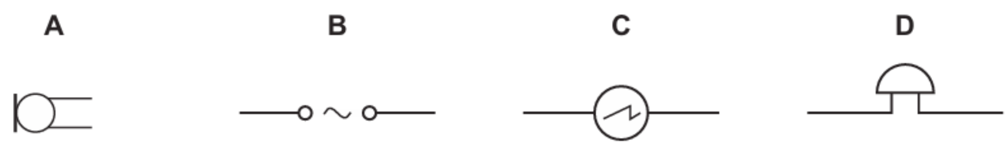
The voltmeter reads 0 V when the slider of the potentiometer is $\frac{1}{4}$ of the way from its lower end, as shown.

What is resistance R?

A 30 Ω **B** 38 Ω **C** 50 Ω **D** 450 Ω

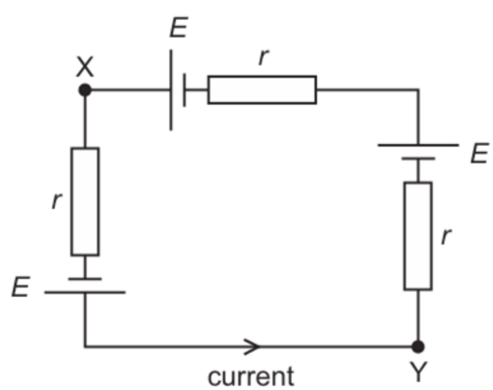
240. 9702/11/O/N/21/Q36

What is the circuit symbol for an oscilloscope?



241. 9702/11/O/N/21/Q37

Three identical cells, each of electromotive force (e.m.f.) E and internal resistance r, are connected as shown.

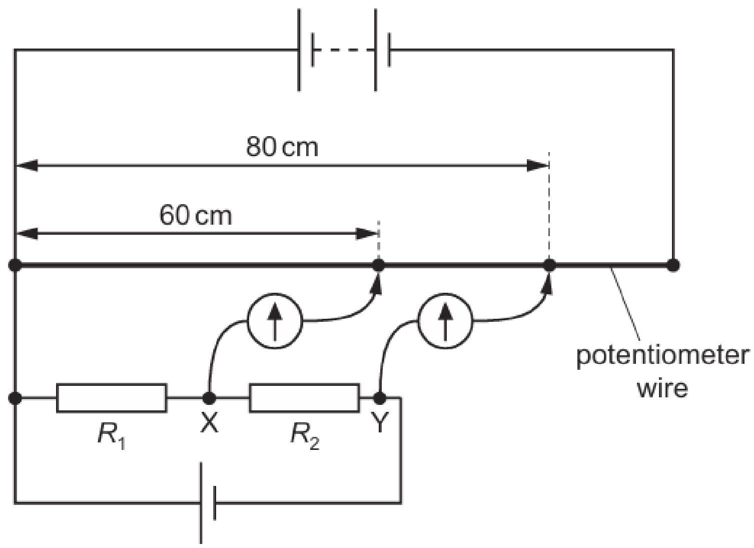


What is the potential difference between points X and Y?

A 0 **B** E **C** $2E$ **D** $3E$

242. 9702/11/O/N/21/Q38

Potential differences across two resistors of resistances R_1 and R_2 are compared using a potentiometer wire (uniform resistance wire) in the electrical circuit shown.



One terminal of a galvanometer is connected to point X. The galvanometer reads zero when its other terminal is connected to a point that is a distance of 60 cm from one end of the potentiometer wire.

One terminal of a second galvanometer is connected to point Y. This galvanometer reads zero when its other terminal is connected to a point that is a distance of 80 cm from the same end of the potentiometer wire.

What is the ratio $\frac{R_2}{R_1}$?

A $\frac{1}{3}$ **B** $\frac{3}{4}$ **C** $\frac{3}{1}$ **D** $\frac{4}{3}$

243. 9702/12/O/N/21/Q36

The electromotive force (e.m.f.) of a cell is 6.0 V. It has negligible internal resistance and is connected across a resistor. The potential difference (p.d.) across the resistor is also 6.0 V.

The e.m.f. and the p.d. have the same numerical value but represent different processes.

Which statement about the different processes is correct?

A The e.m.f. is the energy transferred from chemical energy to electrical energy in the cell and the p.d. is the energy transferred from electrical energy to thermal energy in the resistor.

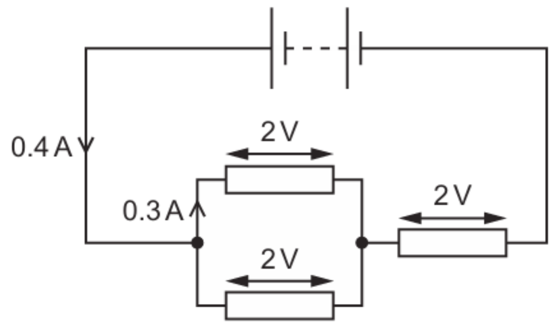
B The p.d. is the energy transferred from chemical energy to electrical energy in the cell and the e.m.f. is the energy transferred from electrical energy to thermal energy in the resistor.

C The e.m.f. is the energy transferred per unit charge from chemical energy to electrical energy in the cell and the p.d. is the energy transferred per unit charge from electrical energy to thermal energy in the resistor.

D The p.d. is the energy transferred per unit charge from chemical energy to electrical energy in the cell and the e.m.f. is the energy transferred per unit charge from electrical energy to thermal energy in the resistor.

244. 9702/12/O/N/21/Q37

A battery of negligible internal resistance is connected to three resistors, as shown.



The potential difference across each resistor is 2 V.

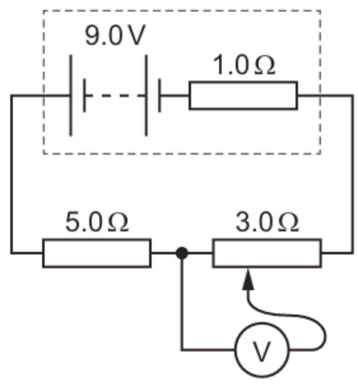
The current from the battery is 0.4 A and the current through one of the resistors connected in parallel is 0.3 A.

What is the current through the other resistor connected in parallel and what is the electromotive force (e.m.f.) of the battery?

	current / A	e.m.f. / V
A	0.1	4
B	0.3	4
C	0.1	6
D	0.3	6

245. 9702/12/O/N/21/Q38

A battery of electromotive force (e.m.f.) 9.0 V and internal resistance 1.0 Ω is connected to a fixed resistor of resistance 5.0 Ω and a potentiometer of maximum resistance 3.0 Ω, as shown.



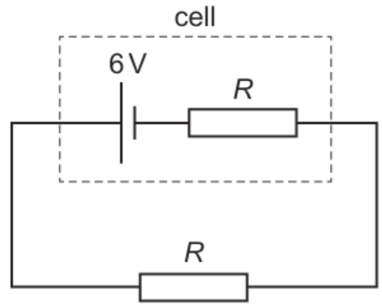
The sliding contact of the potentiometer is moved over its full range of movement.

What is the maximum value of the potential difference that is measured by the voltmeter?

A 3.0 V **B** 3.4 V **C** 4.5 V **D** 5.4 V

246. 9702/13/O/N/21/Q34

A cell has an electromotive force (e.m.f.) of 6 V and internal resistance R. An external resistor, also of resistance R, is connected across this cell, as shown.



Power P is dissipated by the external resistor.

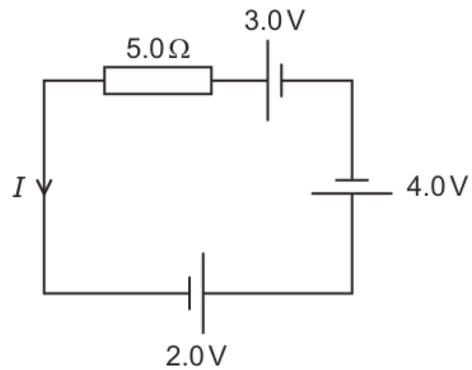
The cell is replaced by a different cell that has an e.m.f. of 6 V and negligible internal resistance.

What is the new power that is dissipated in the external resistor?

A 0.5P **B** P **C** 2P **D** 4P

247. 9702/13/O/N/21/Q37

The circuit shown contains three cells of electromotive forces 3.0 V, 2.0 V and 4.0 V, in series with a resistor of resistance 5.0 Ω. The cells have negligible internal resistance.

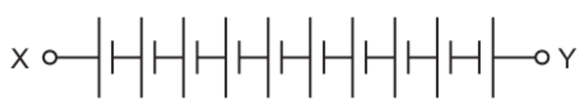


What is the current I in the circuit?

A 0.20 A **B** 0.60 A **C** 1.0 A **D** 1.8 A

248. 9702/12/F/M/22/Q33

Ten cells, each of electromotive force (e.m.f.) 1.5 V, are connected together, as shown.

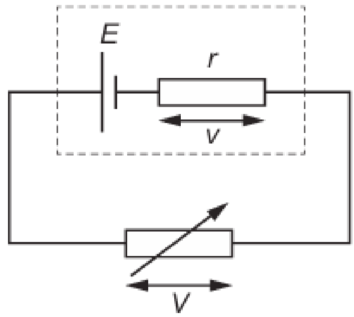


What is the combined e.m.f. between terminals X and Y?

A 8 V **B** 9 V **C** 12 V **D** 15 V

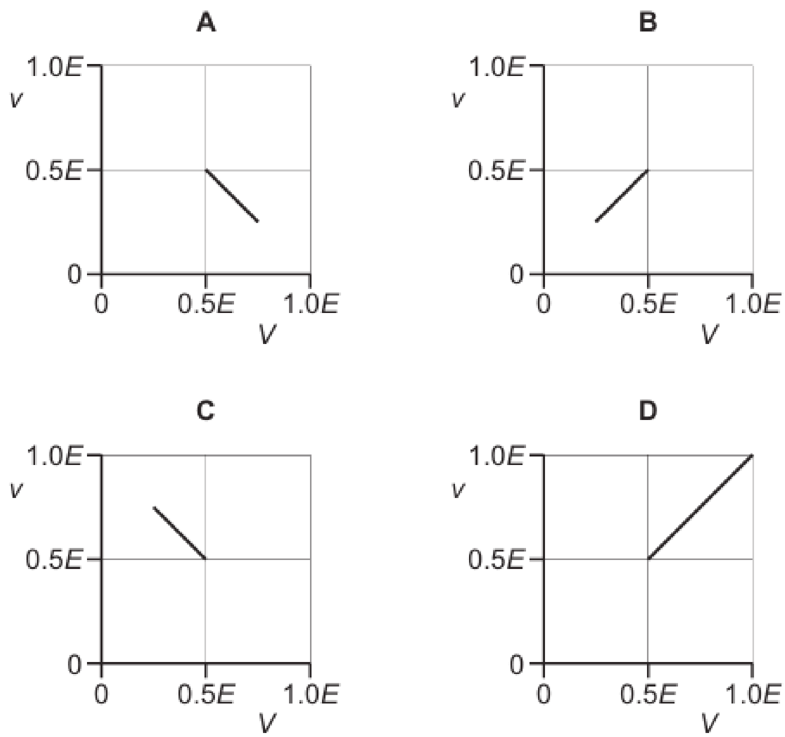
249. 9702/12/F/M/22/Q34

A cell of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor, as shown.



The resistance of the variable resistor is gradually increased from r to $3r$.

Which graph shows the variation of the potential difference (p.d.) v across the internal resistance with the p.d. V across the variable resistor?



250. 9702/12/F/M/22/Q35

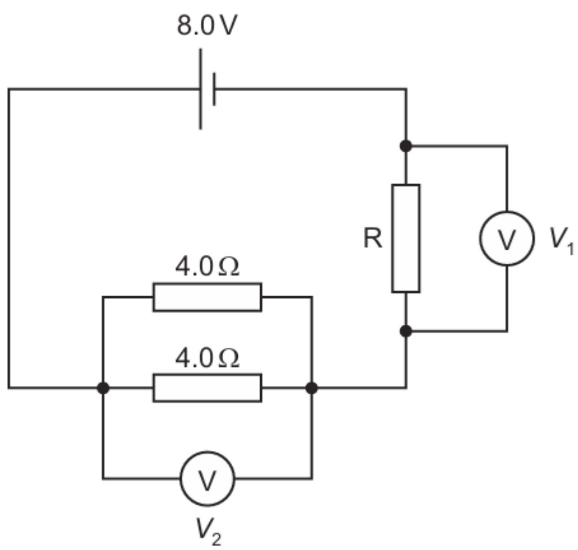
Each of Kirchhoff's two laws presumes that some quantity is conserved.

Which row states Kirchhoff's **first** law and names the quantity that is conserved?

	statement	quantity
A	the algebraic sum of currents into a junction is zero	charge
B	the algebraic sum of currents into a junction is zero	energy
C	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	charge
D	the e.m.f. in a loop is equal to the algebraic sum of the product of current and resistance round the loop	energy

251. 9702/12/F/M/22/Q36

A cell has an electromotive force (e.m.f.) of 8.0 V and negligible internal resistance. The cell forms part of a circuit, as shown.



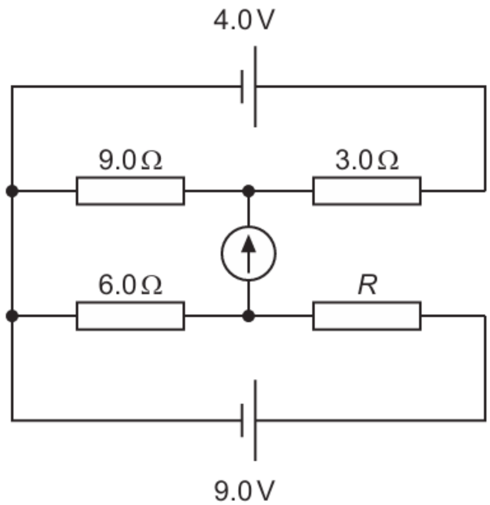
The reading V_1 is 4.0 V and the reading V_2 is also 4.0 V.

What is the resistance of resistor R?

A 0.50 Ω **B** 2.0 Ω **C** 4.0 Ω **D** 8.0 Ω

252. 9702/12/F/M/22/Q37

In the circuit shown, the cells have negligible internal resistance and the reading on the galvanometer is zero.

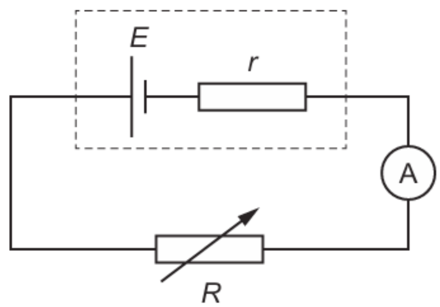


What is the value of resistor R?

A 2.0 Ω **B** 6.0 Ω **C** 12 Ω **D** 18 Ω

253. 9702/11/M/J/22/Q35

A cell has internal resistance r and electromotive force (e.m.f.) E. The cell is connected in series with an ammeter and a variable resistor of resistance R.



When R is 10 Ω the ammeter reads 0.3 A.

When R is 5 Ω the ammeter reads 0.4 A.

What is the value of E?

A 0.5 V **B** 2 V **C** 3 V **D** 6 V

254. 9702/11/M/J/22/Q36

The sum of the currents entering a junction in an electrical circuit is always equal to the sum of the currents leaving the junction.

Why is this?

A It is a consequence of the conservation of charge.

B It is a consequence of the conservation of electromotive force.

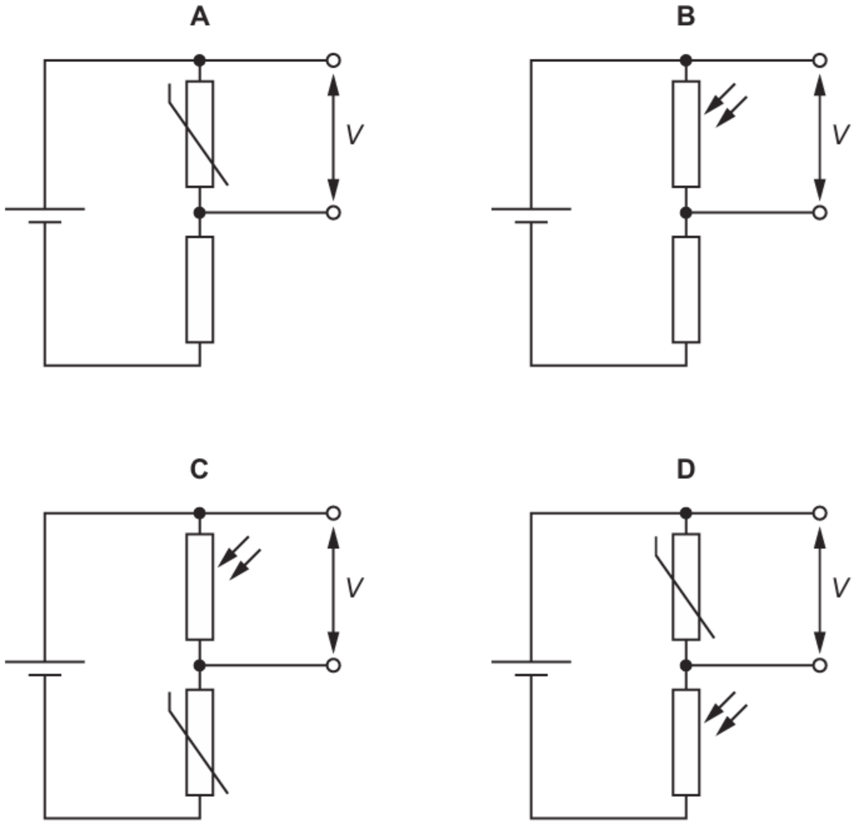
C It is a consequence of the conservation of energy.

D It is a consequence of the conservation of potential difference.

255. 9702/11/M/J/22/Q37

In the circuits shown, the temperature remains constant.

In which circuit does the potential difference (p.d.) *V* increase with increasing light intensity?



256. 9702/12/M/J/22/Q35

A cell with constant electromotive force (e.m.f.) is connected across a fixed resistor. Over time, the internal resistance of the cell increases.

Which change occurs as the internal resistance of the cell increases?

A a decrease in the charge of each charge carrier

B a decrease in the potential difference measured across the cell

C an increase in the energy dissipated per unit time in the fixed resistor

D an increase in the number of charge carriers leaving the cell per unit time

257. 9702/12/M/J/22/Q36

Kirchhoff's first and second laws are consequences of the conservation of different quantities.

What are those quantities?

	Kirchhoff's first law	Kirchhoff's second law
A	charge	energy
B	energy	current
C	current	charge
D	energy	charge

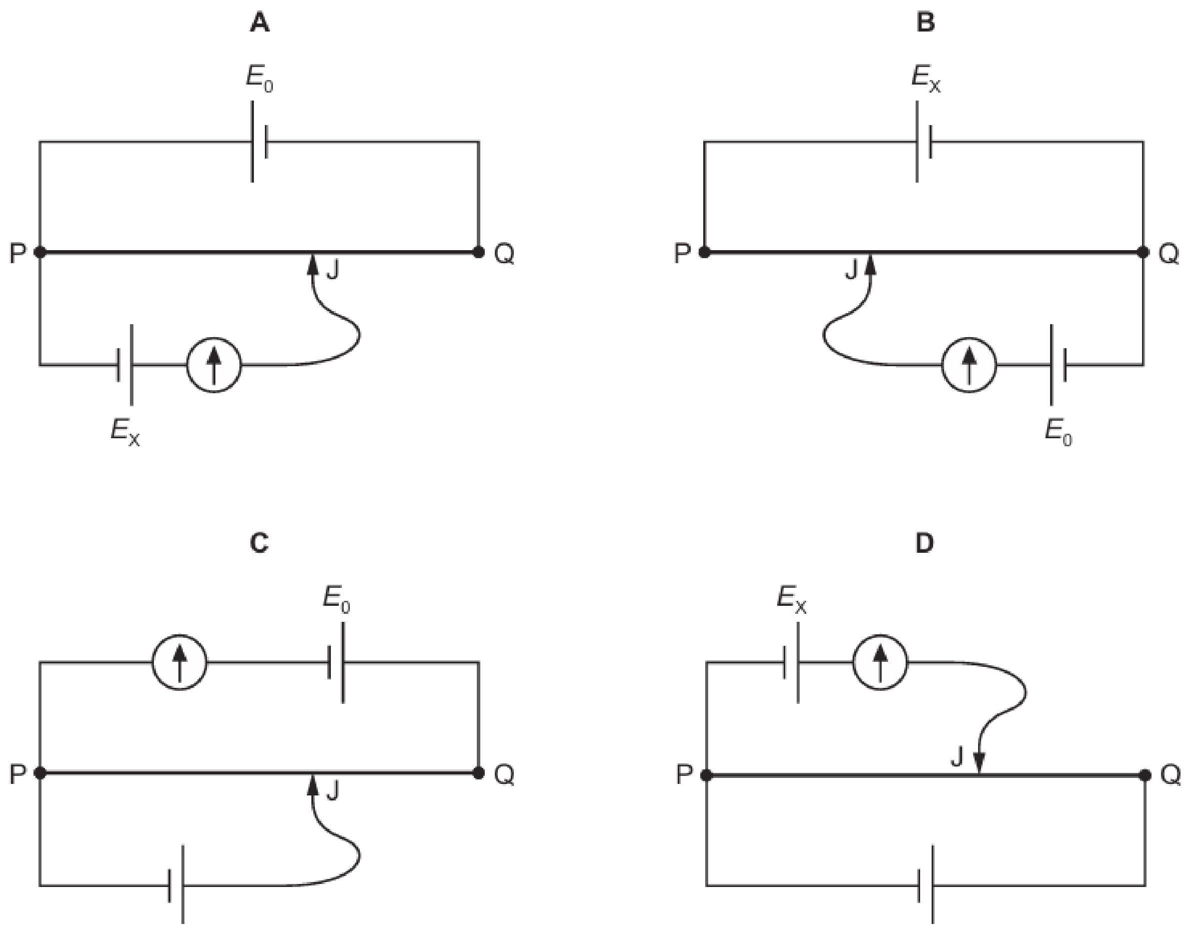
258. 9702/12/M/J/22/Q37

A potentiometer circuit is used to determine the electromotive force (e.m.f.) E_X of a cell. The circuit includes a second cell of known e.m.f. E_0 and negligible internal resistance, and a uniform resistance wire PQ of known length.

E_X is less than E_0.

The movable connection J can be positioned anywhere along the length of the resistance wire.

Which circuit is suitable for determining E_X?



259. 9702/13/M/J/22/Q36

The diagram shows the symbol for a component that may be used in an electrical circuit.

Which component is represented by this circuit symbol?

A buzzer

B electric bell

C loudspeaker

D microphone

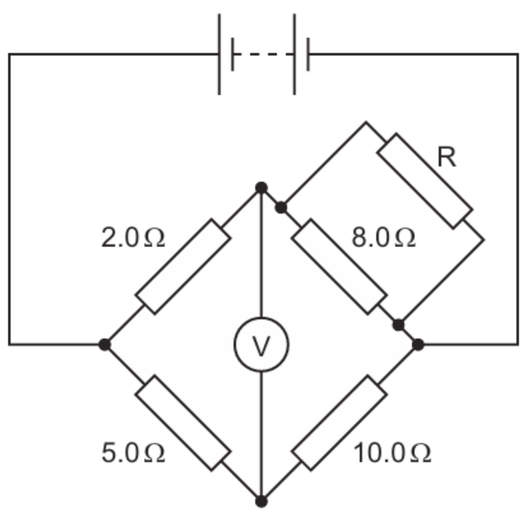
260. 9702/13/M/J/22/Q37

Which row correctly describes Kirchhoff's laws?

	Kirchhoff's first law	physics principle applied for first law	Kirchhoff's second law	physics principle applied for second law
A	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy
B	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge
C	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of energy	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of charge
D	The sum of the e.m.f.s around any closed loop in a circuit equals the sum of the p.d.s around the same loop.	conservation of charge	The sum of the currents entering a junction equals the sum of the currents leaving the junction.	conservation of energy

261. 9702/11/O/N/22/Q36

A circuit consists of a battery, a voltmeter and five fixed resistors, as shown.



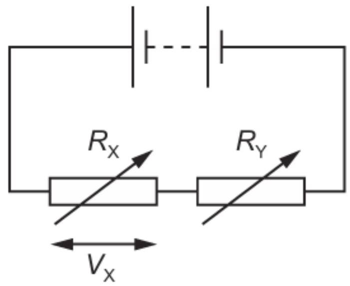
The voltmeter reading is zero.

What is the resistance of resistor R?

A 1.1Ω **B** 2.1Ω **C** 4.0Ω **D** 8.0Ω

262. 9702/13/M/J/22/Q38

A potential divider circuit is formed by connecting a battery of negligible internal resistance in series with two variable resistors, as shown.



The variable resistors have resistances R_X and R_Y.

V_X is the potential difference (p.d.) across the variable resistor with resistance R_X.

R_X and R_Y are both changed at the same time.

Which combination of changes **must** cause V_X to increase?

	R_X	R_Y
A	larger	larger
B	larger	smaller
C	smaller	larger
D	smaller	smaller

263. 9702/11/O/N/22/Q35

Which ratio has the same units as electromotive force (e.m.f.)?

A charge per unit energy transferred

B charge per unit time

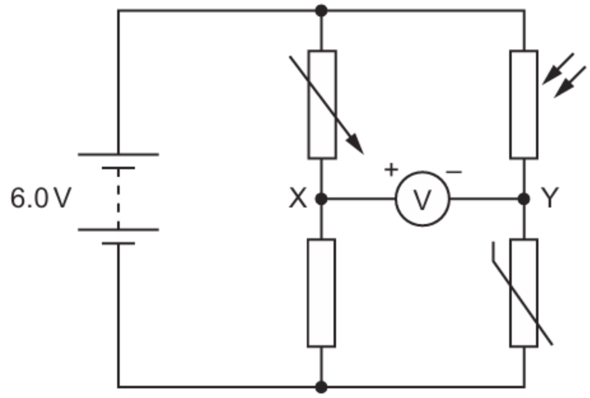
C energy transferred per unit charge

D energy transferred per unit time

264. 9702/11/O/N/22/Q37

A battery of electromotive force (e.m.f.) 6.0 V and negligible internal resistance is connected to a voltmeter and four other components, as shown.

The voltmeter is connected between points X and Y. The positive terminal of the voltmeter is connected to X and the negative terminal of the voltmeter is connected to Y.



Initially, the resistance of each of the four components is 1.0 kΩ.

Which change, on its own, will cause the voltmeter to show a positive reading?

A Decrease the temperature of the thermistor.

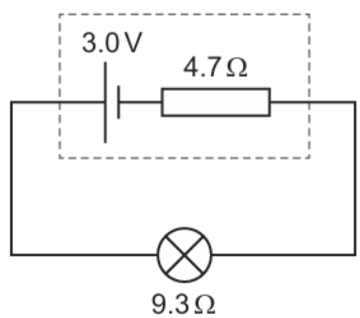
B Increase the resistance of the variable resistor.

C Reduce the intensity of light incident on the light-dependent resistor (LDR).

D Replace the fixed resistor with a 500 Ω resistor.

265. 9702/12/O/N/22/Q35

The diagram shows a cell of electromotive force (e.m.f.) 3.0 V and internal resistance 4.7 Ω connected across a lamp. The lamp has a resistance of 9.3 Ω.

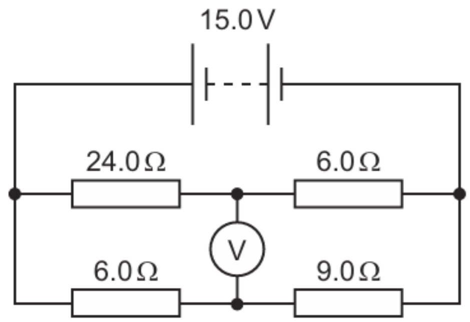


What is the power dissipated by the internal resistance of the cell?

A 0.22 W **B** 0.43 W **C** 0.64 W **D** 1.0 W

266. 9702/12/O/N/22/Q36

A circuit consists of a battery, a high-resistance voltmeter and four fixed resistors, as shown. The battery has an electromotive force (e.m.f.) of 15.0 V and negligible internal resistance.

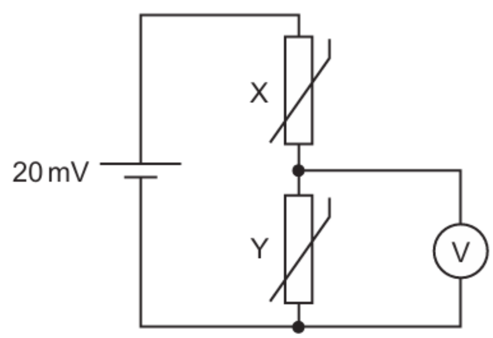


What is the reading on the voltmeter?

A 3.0 V **B** 6.0 V **C** 9.0 V **D** 12.0 V

267. 9702/12/O/N/22/Q37

A potential divider circuit is designed to detect the difference in temperature between two different places.



The cell has electromotive force (e.m.f.) 20 mV and negligible internal resistance.

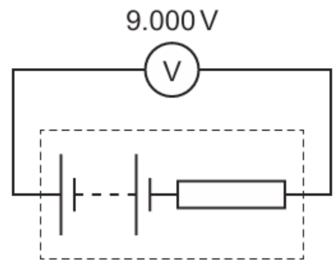
Initially, thermistors X and Y are at the same temperature and have the same resistance. The voltmeter reads 10 mV. X is then placed in a cold environment and its resistance doubles. Y is placed in a warm environment and its resistance halves.

What is the new reading on the voltmeter?

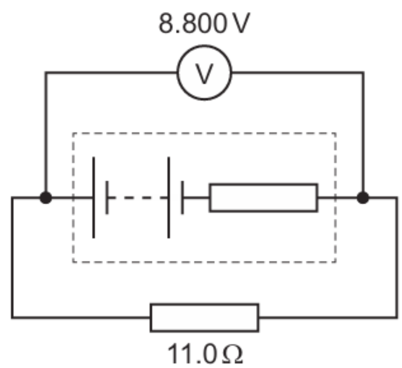
A 4 mV **B** 5 mV **C** 15 mV **D** 16 mV

268. 9702/13/O/N/22/Q34

A voltmeter reads 9.000 V when it is connected across the terminals of a battery.



When a resistor of resistance 11.0 Ω is connected in parallel with the battery, the voltmeter reading changes to 8.800 V.



What is the internal resistance of the battery?

A 0.244 Ω **B** 0.250 Ω **C** 10.8 Ω **D** 11.3 Ω

269. 9702/13/O/N/22/Q35

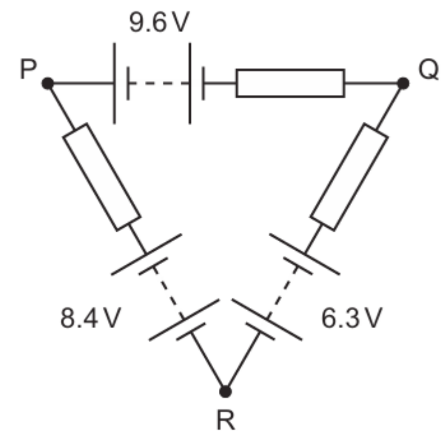
Each of Kirchhoff's laws is linked to the conservation of a physical quantity.

Which conserved physical quantities are used in the derivation of Kirchhoff's first law and of Kirchhoff's second law?

	Kirchhoff's first law	Kirchhoff's second law
A	energy	charge
B	energy	momentum
C	charge	energy
D	momentum	energy

270. 9702/13/O/N/22/Q36

Three batteries and three identical resistors are connected in a circuit PQR, as shown.



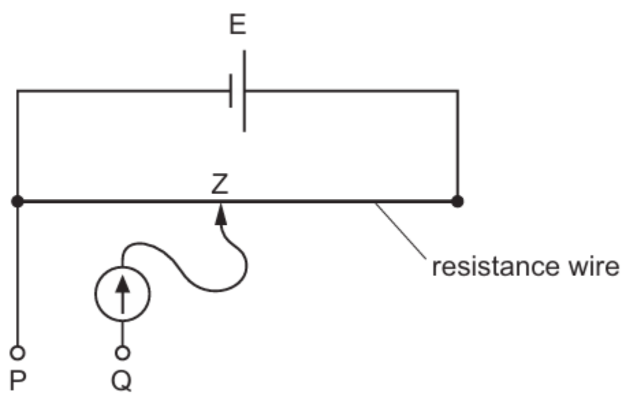
The batteries have negligible internal resistance.

What is the potential difference between points P and Q?

A 1.5 V **B** 2.1 V **C** 7.1 V **D** 12.1 V

271. 9702/13/O/N/22/Q37

A cell E, of electromotive force (e.m.f.) 2 V and negligible internal resistance, is connected to a uniform resistance wire of resistance 10 Ω and length 1.0 m.



Z is a connection that may be made at any position along the resistance wire. A galvanometer is connected between Z and a point Q.

A new source of e.m.f. of approximately 8 mV is connected between points P and Q. The e.m.f. of the new source is determined by changing the position of Z until the reading on the galvanometer is zero.

Which change to the circuit allows a much more precise value for the e.m.f. of the new source to be obtained?

A Add a resistor of resistance 0.1 Ω in series with cell E.

B Add a resistor of resistance 1000 Ω in series with cell E.

C Add a resistor of resistance 10 Ω in series with the new source.

D Add a resistor of resistance 800 Ω in series with the new source.

272. 9702/12/F/M/23/Q34

A cell has a constant electromotive force.

A variable resistor is connected between the terminals of the cell.

The resistance of the variable resistor is decreased.

Which statement about the change of the cell's terminal potential difference (p.d.) is correct?

A The terminal p.d. is decreased because more work is done moving unit charge through the internal resistance of the cell.

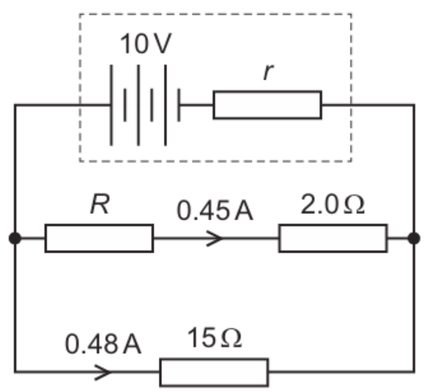
B The terminal p.d. is decreased because the current in the variable resistor is decreased.

C The terminal p.d. is increased because more work is done moving unit charge through the variable resistor.

D The terminal p.d. is increased because the current in the variable resistor is increased.

273. 9702/12/F/M/23/Q36

A battery of electromotive force (e.m.f.) 10 V and internal resistance r is connected to three resistors of resistances R, 2.0 Ω and 15 Ω, as shown. A current of 0.45 A is in the resistor of resistance 2.0 Ω and a current of 0.48 A is in the resistor of resistance 15 Ω.

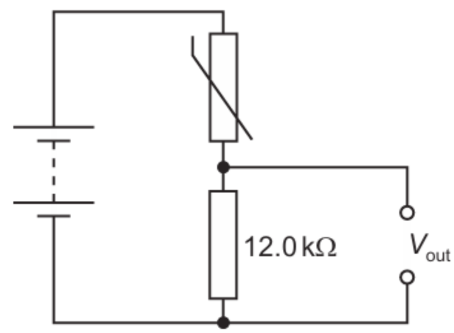


What are the values of r and R?

	r / Ω	R / Ω
A	3.0	14
B	3.0	20
C	5.8	14
D	5.8	20

274. 9702/12/F/M/23/Q37

A battery of negligible internal resistance is connected in series with a thermistor and a fixed resistor of resistance 12.0 kΩ, as shown.



The table shows the resistance of the thermistor at two different temperatures.

temperature / °C	resistance of thermistor / kΩ
20.0	12.0
50.0	5.00

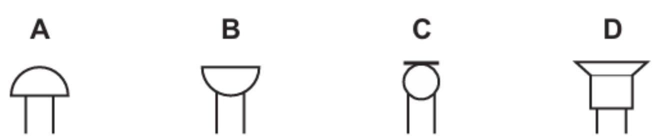
The potential difference V_{out} across the fixed resistor is 4.50 V when the thermistor is at a temperature of 20.0 °C.

What is V_{out} when the thermistor is at a temperature of 50.0 °C?

A 2.65 V **B** 3.18 V **C** 6.35 V **D** 10.8 V

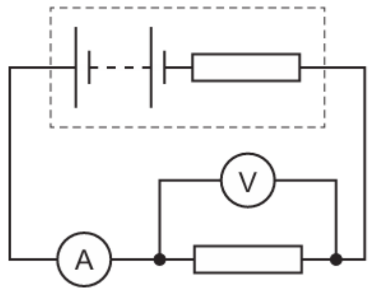
275. 9702/11/M/J/23/Q34

Which circuit symbol represents a microphone?



276. 9702/11/M/J/23/Q35

A battery with internal resistance is connected to a fixed resistor, an ammeter and a voltmeter, as shown.



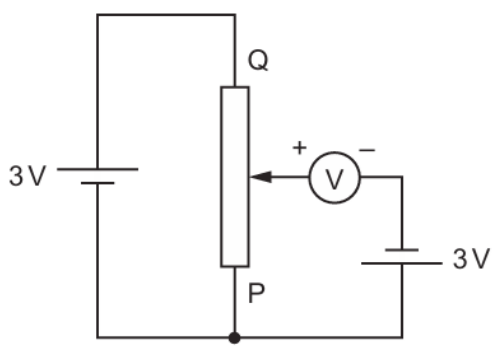
The battery is replaced by a different battery that has the same electromotive force (e.m.f.) but a greater internal resistance.

What happens to the readings on the ammeter and voltmeter?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

277. 9702/11/M/J/23/Q37

A voltmeter is connected into a circuit with the polarity shown.



The sliding contact is moved to end P of the potentiometer and then to end Q.

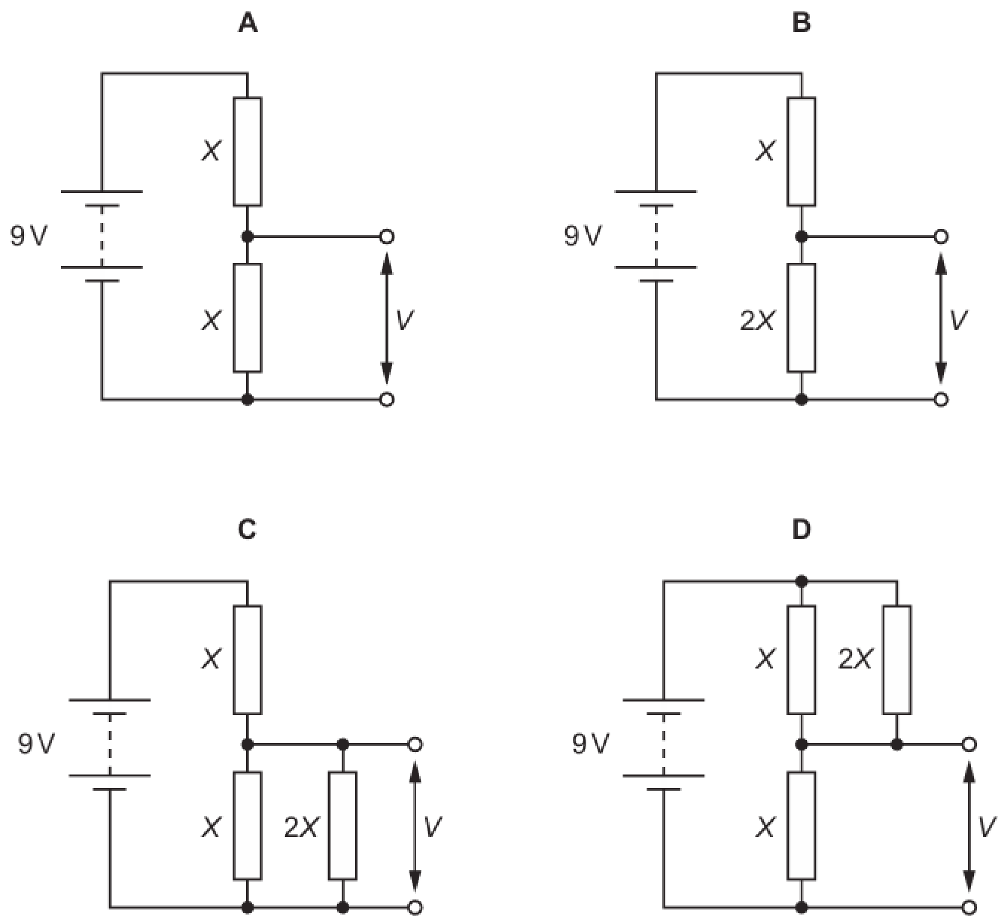
What are the two readings of the voltmeter?

	sliding contact at end P	sliding contact at end Q
A	0 V	3 V
B	0 V	6 V
C	3 V	3 V
D	3 V	6 V

278. 9702/11/M/J/23/Q36

Four potential divider circuits each consist of a battery of electromotive force (e.m.f.) 9 V and negligible internal resistance connected to a combination of resistors. Each of the resistors in the circuits has a resistance of X or $2X$.

Which circuit has the largest output voltage V?



279. 9702/12/M/J/23/Q35

Kirchhoff's first law is a consequence of the conservation of which quantity?

A charge

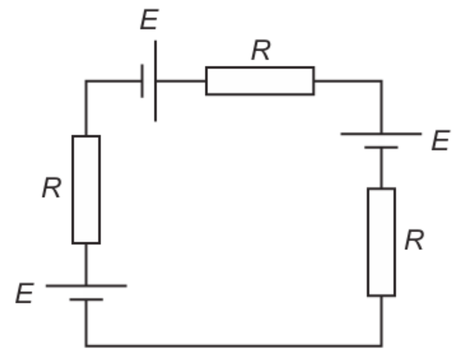
B energy

C momentum

D potential difference

280. 9702/12/M/J/23/Q36

Three identical cells each have electromotive force (e.m.f.) E and negligible internal resistance. The cells are connected to three identical resistors, each of resistance R, as shown.

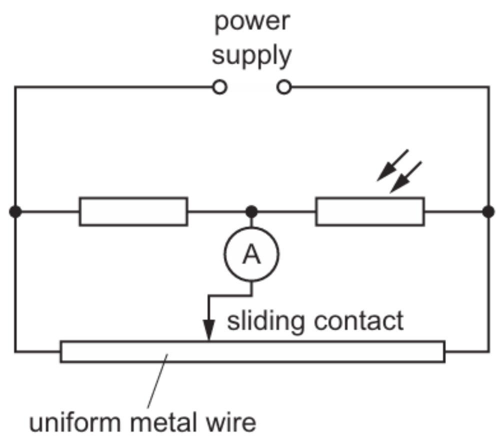


What is the potential difference across one of the resistors?

A 0 **B** $\frac{E}{3}$ **C** $\frac{2E}{3}$ **D** E

281. 9702/12/M/J/23/Q37

In the potentiometer circuit shown, the reading on the ammeter is zero.



The light-dependent resistor (LDR) is then covered and the ammeter gives a non-zero reading.

Which change could return the ammeter reading to zero?

A decreasing the supply voltage

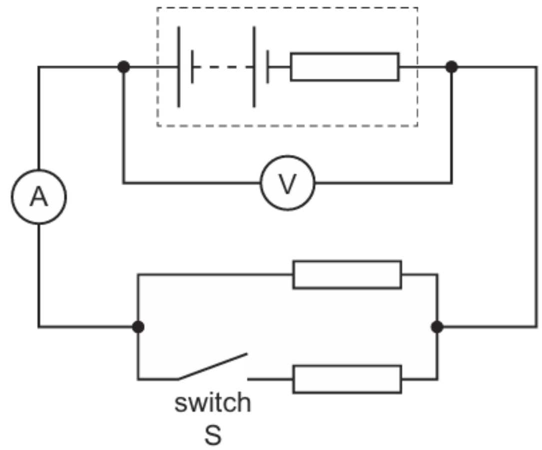
B increasing the supply voltage

C moving the sliding contact to the left

D moving the sliding contact to the right

282. 9702/13/M/J/23/Q34

A battery with internal resistance is connected to a parallel arrangement of two resistors and a switch S, as shown.



Initially, switch S is open.

What happens to the voltmeter and ammeter readings when switch S is closed?

	voltmeter reading	ammeter reading
A	decreases	increases
B	decreases	decreases
C	increases	increases
D	increases	decreases

283. 9702/13/M/J/23/Q36

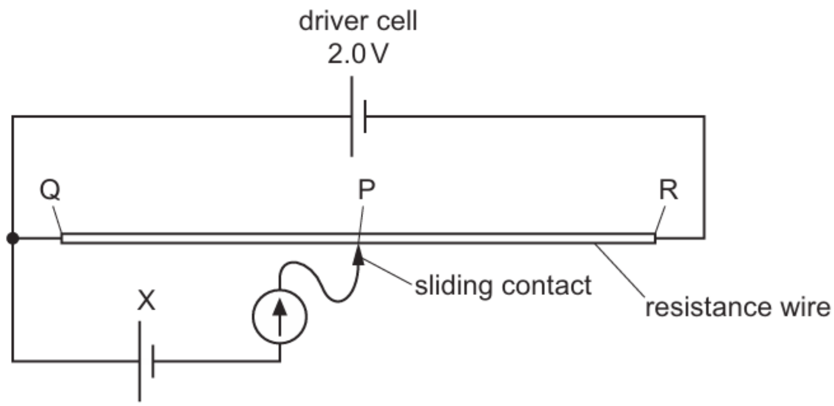
Two resistors have a combined resistance of $34\,\Omega$ when connected in series. The same resistors have a combined resistance of $7.4\,\Omega$ when connected in parallel.

What is the resistance of one of the resistors?

A $15\,\Omega$ **B** $17\,\Omega$ **C** $23\,\Omega$ **D** $27\,\Omega$

284. 9702/13/M/J/23/Q37

A potentiometer circuit is used to investigate the electromotive force (e.m.f.) of a cell X.



The e.m.f. of cell X is known to be approximately 0.50 V.

The driver cell has negligible internal resistance and an e.m.f. of 2.0 V. The sliding contact is moved along the uniform resistance wire between ends Q and R to a point P where the reading on the galvanometer is zero.

What is an expression for the approximate length QP?

A $\frac{QR}{4}$ **B** $\frac{QR}{3}$ **C** $\frac{2QR}{3}$ **D** $\frac{3QR}{4}$

285. 9702/11/O/N/23/Q34

Which statement about the electromotive force (e.m.f.) of a cell is **always** correct?

A The e.m.f. is the energy converted from electrical to other forms in the cell.

B The e.m.f. is the energy provided by the cell per unit charge passing through it.

C The e.m.f. is the potential difference across the internal resistance of the cell.

D The e.m.f. is the potential difference across the terminals of the cell.

286. 9702/11/O/N/23/Q35

Kirchhoff's first and second laws are a consequence of the conservation of which quantities?

A charge and energy

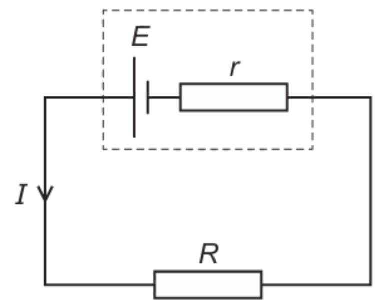
B charge and resistance

C mass and energy

D mass and resistance

287. 9702/11/O/N/23/Q36

A circuit contains a cell of electromotive force E and internal resistance r connected to a resistor of resistance R. The current in the circuit is I.

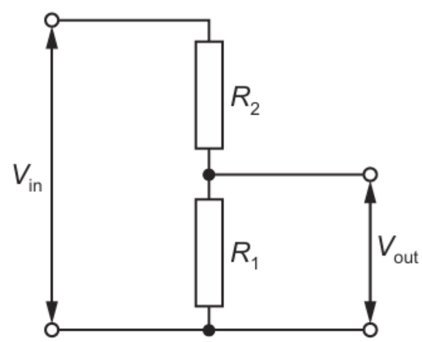


Which equation is correct?

A $E - Ir = IR$ **B** $E = Ir - IR$ **C** $E + Ir = IR$ **D** $E = IR$

288. 9702/11/O/N/23/Q37

A potential divider consists of two resistors of resistances R_1 and R_2 connected in series across a source of potential difference (p.d.) V_{in}. The p.d. across R_1 is V_{out}.

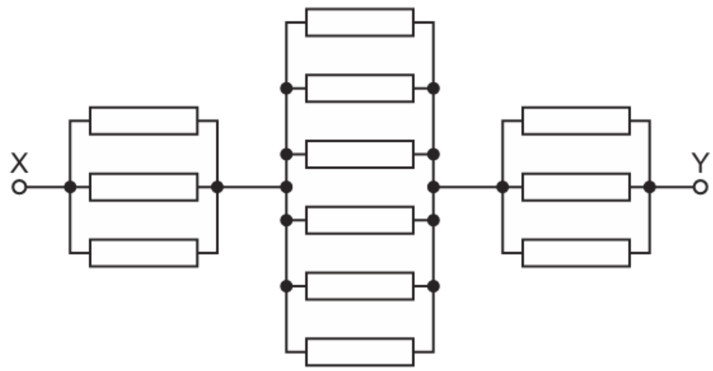


Which changes to R_1 and R_2 will increase the value of V_{out}?

	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

289. 9702/13/O/N/23/Q36

The diagram shows a network of resistors. Each resistor has a resistance of 6.0 Ω.

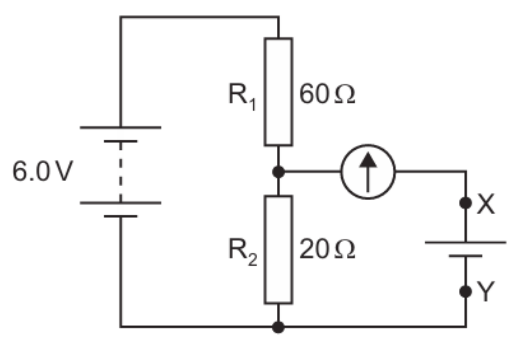


What is the total resistance of the network between points X and Y?

A 3.0 Ω **B** 5.0 Ω **C** 7.2 Ω **D** 18 Ω

290. 9702/13/O/N/23/Q37

In the circuit shown, a battery of negligible internal resistance is connected in series with a pair of fixed resistors R_1 and R_2.



The circuit is to be used to test whether the electromotive force (e.m.f.) of a particular cell is 1.5 V. The cell is connected between terminals X and Y in parallel with R_2 and in series with a galvanometer.

Which statement about the test is correct?

A Any non-zero reading on the galvanometer means the cell has an e.m.f. of 1.5 V.

B The battery does not need to have an e.m.f. of 6.0 V.

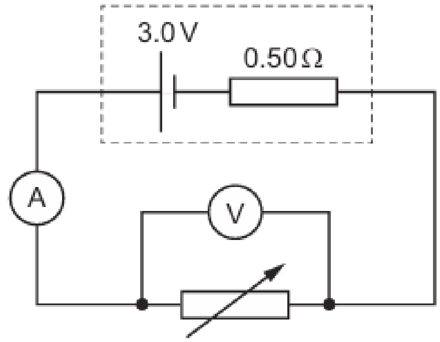
C The cell may be connected either way round between X and Y.

D The galvanometer does not need a scale calibrated in amperes.

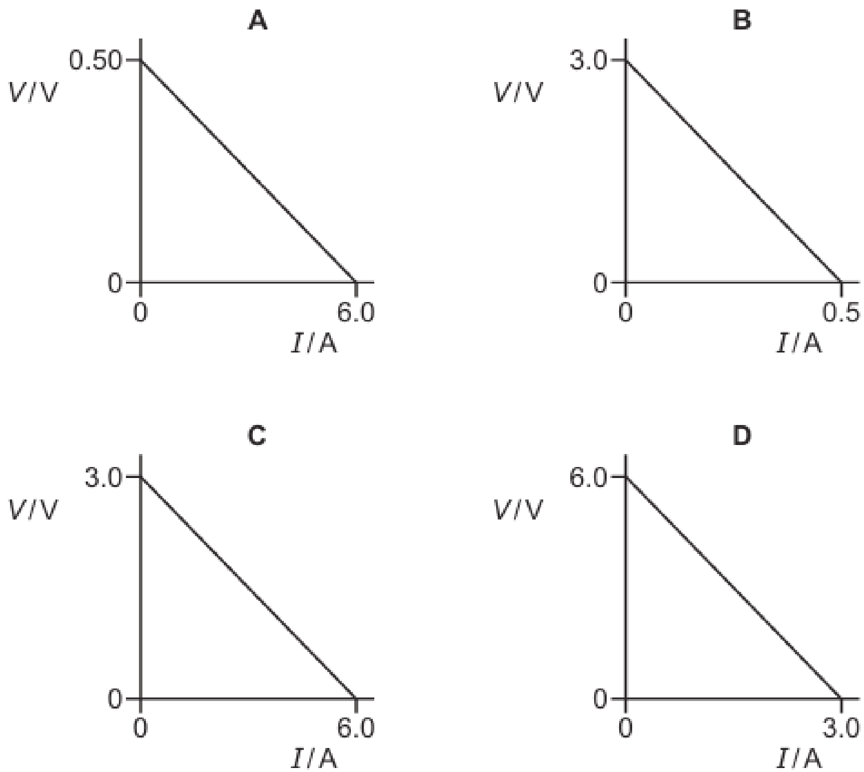
291. 9702/13/O/N/23/Q34

A cell of electromotive force (e.m.f.) 3.0 V and internal resistance 0.50 Ω is connected to a variable resistor, a voltmeter and an ammeter, as shown. The resistance of the variable resistor is varied.

The reading on the ammeter I and the reading on the voltmeter V are recorded.

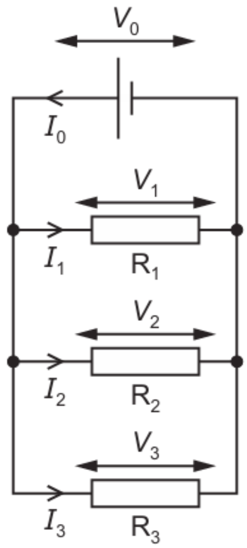


Which graph shows how V varies with I?



292. 9702/13/O/N/23/Q35

Three resistors, R_1, R_2 and R_3, are connected in parallel to a cell. The currents in the resistors are I_1, I_2 and I_3. The potential differences across the resistors are V_1, V_2 and V_3. The current in the cell is I_0. The potential difference across the cell is V_0, as shown.



Which equation can be obtained by applying Kirchhoff's second law to the circuit?

A $I_0 = I_1 = I_2 = I_3$

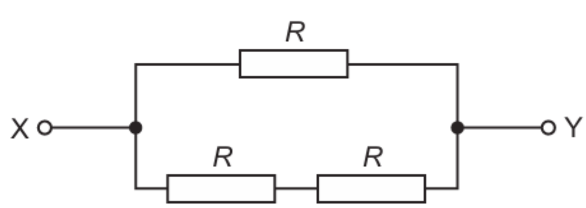
B $I_0 = I_1 + I_2 + I_3$

C $V_0 = V_1 = V_2 = V_3$

D $V_0 = V_1 + V_2 + V_3$

293. 9702/13/O/N/23/Q36

Three resistors, each of resistance R, are connected in a network, as shown.



The total resistance between points X and Y is 8.0 Ω.

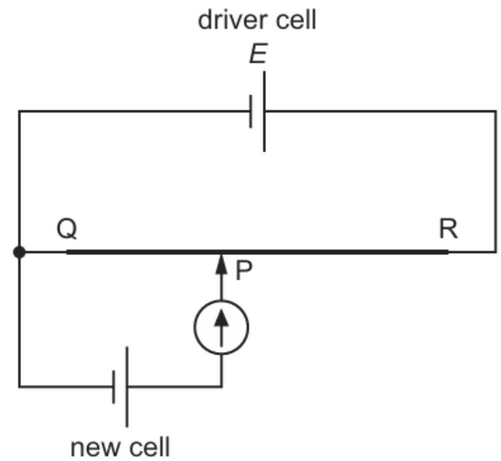
What is the value of R?

A 2.7 Ω **B** 4.0 Ω **C** 5.3 Ω **D** 12 Ω

294. 9702/13/O/N/23/Q37

A potentiometer and a driver cell of electromotive force (e.m.f.) E are used to measure the e.m.f. of a new cell.

A sliding contact at P is moved along a resistance wire QR until the reading on the galvanometer is zero.



What is an essential requirement for the e.m.f. of the new cell to be measured accurately?

A The e.m.f. of the driver cell must be less than the e.m.f. of the new cell.

B The galvanometer must have a large resistance.

C The internal resistance of the new cell must be zero.

D The resistance per unit length of the wire QR must be constant.

ANSWERS

1. C
2. A
3. C
4. C
5. C
6. D
7. A
8. A
9. B
10. B
11. B
12. A
13. C
14. B
15. D
16. B
17. D
18. D
19. B
20. A
21. B
22. D
23. B
24. B
25. B
26. C
27. B
28. C
29. B
30. D
31. A
32. B
33. B
34. A
35. A
36. C
37. C
38. C
39. [illegible]
40. D
41. D
42. A
43. C
44. C
45. B
46. D
47. B
48. B
49. B
50. D
51. D
52. A
53. A
54. B
55. C
56. D
57. A
58. A
59. A
60. C
61. B
62. D
63. B
64. B
65. B
66. C
67. C
68. D
69. C
70. B
71. C
72. B
73. D
74. B
75. B
76. C
77. B
78. A
79. D
80. D
81. D
82. C
83. B
84. C
85. C
86. A
87. C
88. B
89. D
90. D
91. A
92. B
93. B
94. D
95. D
96. D
97. C
98. C
99. B
100. C
101. A
102. D
103. D
104. C
105. C
106. A
107. D
108. A
109. C
110. A
111. D
112. B
113. B
114. B
115. A
116. A
117. C
118. A
119. C
120. D
121. A
122. B
123. A
124. D
125. A
126. C
127. A
128. C
129. B
130. A
131. A
132. B
133. B
134. C
135. A
136. B
137. B
138 A
139 D
140 D
141 B
142 C
143 A
144 B
145 A
146 C
147 B
148 C
149 D
150 A
151 C
152 A
153 B
154 A
155 B
156 A
157 A
158 C
159 B
160 D
161 A
162 C
163 D
164 D
165 D
166 D
167 C
168 D
169 C
170. A
171. A
172. B
173. A
174. B
175. B
176. C
177. B
178. A
179. D
180. C
181. A
182. C
183. C
184. A
185. C
186. A
187. A
188. D
189. A
190. C
191. B
192. B
193. B
194. D
195. D
196. C
197. D
198. A
199. C
200. C
201. A
202. C
203. D
204. A
205. A
206. B
207. B
208. B
209. C
210. D
211. B
212. D
213. C
214. A
215. A
216. D
217. A
218. B
219. A
220. B
221. D
222. D
223. B
224. C
225. D
226. C
227. C
228. D
229. A
230. D
231. C
232. B
233. B
234. C
235. A
236. C
237. C
238. B
239. C
240. C
241. A
242. A
243. C
244. A
245. A
246. D
247. A
248. C
249. A
250. A
251. B
252. C
253. D
254. A
255. D
256. B
257. A
258. D
259. A
260. A
261. D
262. B
263. C
264. C
265. A
266. B
267. A
268. B
269. C
270. C
271. B
272. A
273. A
274. C
275. C
276. A
276. A
277. D
278. B
279. A
280. B
281. C
282. A
283. C
284. A
285. B
286. A
287. A
288. B
289. B
290. D
291. C
292. C
293. D
294. D

1 **(a)** A network of resistors, each of resistance R, is shown in Fig. 7.1. **9702/22/M/J/09/Q7**

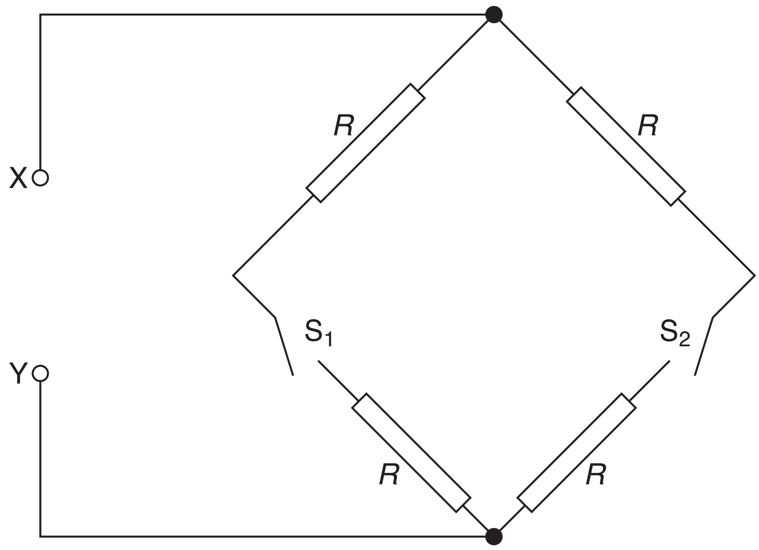


Fig. 7.1

Switches S_1 and S_2 may be 'open' or 'closed'.

Complete Fig. 7.2 by calculating the resistance, in terms of R, between points X and Y for the switches in the positions shown.

switch S_1	switch S_2	resistance between points X and Y
open	open	....
open	closed	....
closed	closed	....

Fig. 7.2

[3]

(b) Two cells of e.m.f. E_1 and E_2 and negligible internal resistance are connected into a network of resistors, as shown in Fig. 7.3.

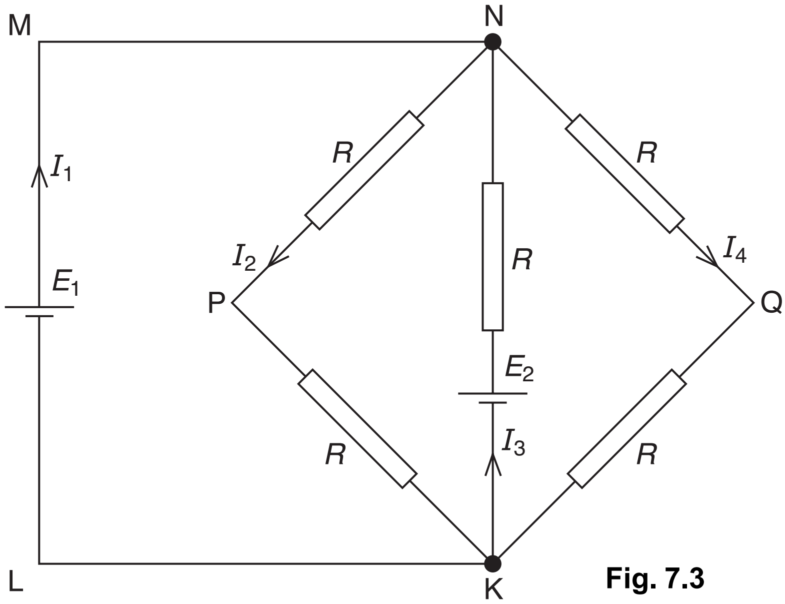


Fig. 7.3

The currents in the network are as indicated in Fig. 7.3.

Use Kirchhoff's laws to state the relation

(i) between currents I_1, I_2, I_3 and I_4,

.... [1]

(ii) between E_1, E_2, R, and I_3 in loop NKLMN,

.... [1]

(iii) between E_2, R, I_3 and I_4 in loop NKQN.

.... [1]

9702/21/O/N/09/Q6

2 A cell has electromotive force (e.m.f.) E and internal resistance r. It is connected in series with a variable resistor R, as shown in Fig. 6.1.

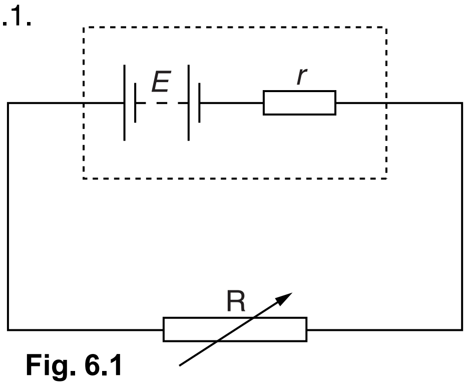


Fig. 6.1

(a) Define electromotive force (e.m.f.).

....

....

.... [2]

(b) The variable resistor R has resistance X. Show that

$$\frac{\text{power dissipated in resistor R}}{\text{power produced in cell}} = \frac{X}{X + r}.$$

[3]

(c) The variation with resistance X of the power P_R dissipated in R is shown in Fig. 6.2.

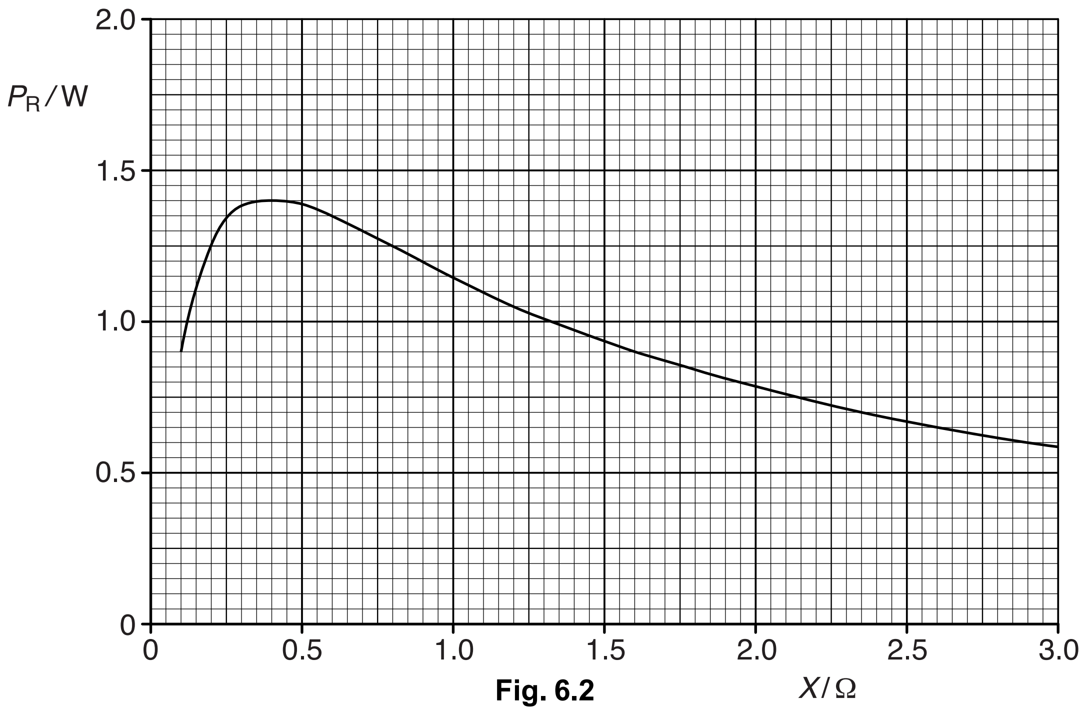


Fig. 6.2

(i) Use Fig. 6.2 to state, for maximum power dissipation in resistor R, the magnitude of this power and the resistance of R.

maximum power = ... W

resistance = ... Ω [2]

(ii) The cell has e.m.f. 1.5 V.
Use your answers in **(i)** to calculate the internal resistance of the cell.

internal resistance = ... Ω [3]

(d) In Fig. 6.2, it can be seen that, for larger values of X, the power dissipation decreases. Use the relationship in **(b)** to suggest one advantage, despite the lower power output, of using the cell in a circuit where the resistance X is larger than the internal resistance of the cell.

....

.... [1]

9702/22/O/N/09/Q6

3 **(a)** Two resistors, each of resistance R, are connected first in series and then in parallel.

Show that the ratio

$$\frac{\text{combined resistance of resistors connected in series}}{\text{combined resistance of resistors connected in parallel}}$$

is equal to 4.

[1]

(b) The variation with potential difference V of the current I in a lamp is shown in Fig. 6.1.

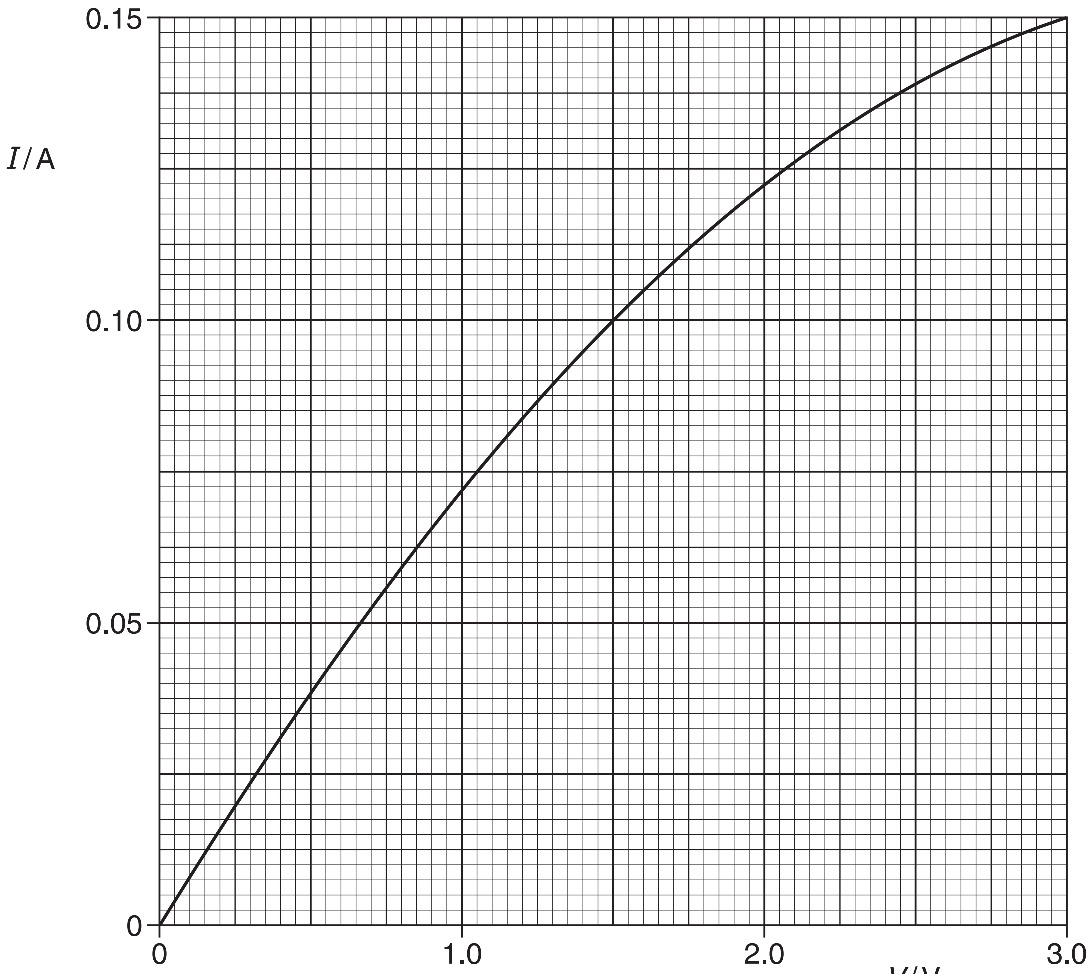


Fig. 6.1

Calculate the resistance of the lamp for a potential difference across the lamp of 1.5V.

resistance = ... Ω [2]

(c) Two lamps, each having the I-V characteristic shown in Fig. 6.1, are connected first in series and then in parallel with a battery of e.m.f. 3.0V and negligible internal resistance.

Complete the table of Fig. 6.2 for the lamps connected to the battery.

	p.d. across each lamp/V	resistance of each lamp/Ω	combined resistance of lamps/Ω
lamps connected in series	....	....	....
lamps connected in parallel	....	....	....

Fig. 6.2

[4]

(d) **(i)** Use data from the completed Fig. 6.2 to calculate the ratio

$$\frac{\text{combined resistance of lamps connected in series}}{\text{combined resistance of lamps connected in parallel}}.$$

ratio = ... [1]

(ii) The ratios in **(a)** and **(d)(i)** are not equal.

By reference to Fig. 6.1, state and explain qualitatively the change in the resistance of a lamp as the potential difference is changed.

....

....

.... [3]

9702/22/M/J/10/Q6

4 **(a)** A metal wire of constant resistance is used in an electric heater.
In order not to overload the circuit for the heater, the supply voltage to the heater is reduced from 230 V to 220 V.

Determine the percentage reduction in the power output of the heater.

reduction = ... % [2]

(b) A uniform wire AB of length 100 cm is connected between the terminals of a cell of e.m.f. 1.5 V and negligible internal resistance, as shown in Fig. 6.1.

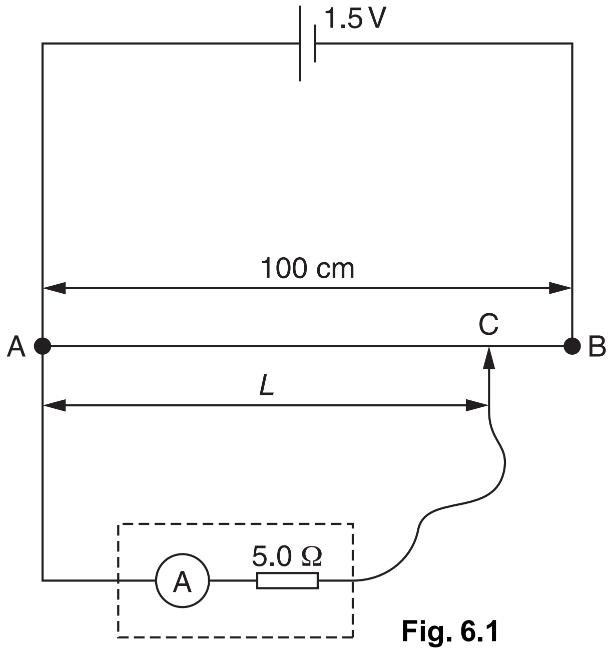


Fig. 6.1

An ammeter of internal resistance 5.0 Ω is connected to end A of the wire and to a contact C that can be moved along the wire.
Determine the reading on the ammeter for the contact C placed

(i) at A,

reading = ... A [1]

(ii) at B.

reading = ... A [1]

(c) Using the circuit in **(b)**, the ammeter reading I is recorded for different distances L of the contact C from end A of the wire. Some data points are shown on Fig. 6.2.

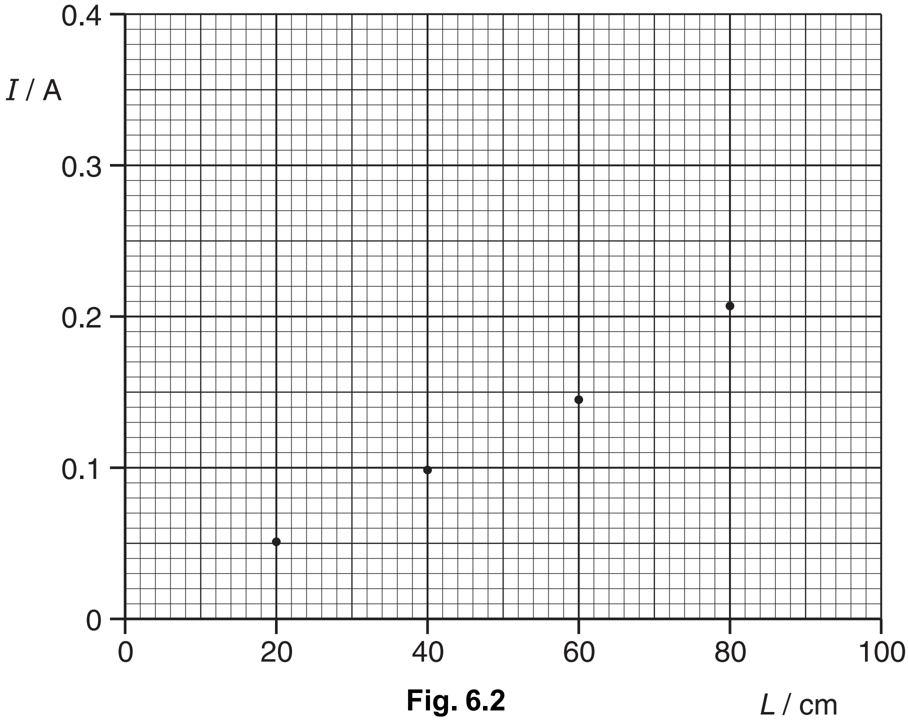


Fig. 6.2

(i) Use your answers in **(b)** to plot data points on Fig. 6.2 corresponding to the contact C placed at end A and at end B of the wire.

[1]

(ii) Draw a line of best fit for all of the data points and hence determine the ammeter reading for contact C placed at the midpoint of the wire.

reading = ... A [1]

(iii) Use your answer in **(ii)** to calculate the potential difference between A and the contact C for the contact placed at the midpoint of AB.

potential difference = ... V [2]

(d) Explain why, although the contact C is at the midpoint of wire AB, the answer in **(c)(iii)** is **not** numerically equal to one half of the e.m.f. of the cell.

....

....[2]

9702/21/O/N/10/Q6

5 **(a)** A lamp is rated as 12 V, 36 W.

(i) Calculate the resistance of the lamp at its working temperature.

resistance = ... Ω [2]

(ii) On the axes of Fig. 6.1, sketch a graph to show the current-voltage (I–V) characteristic of the lamp. Mark an appropriate scale for current on the y-axis.

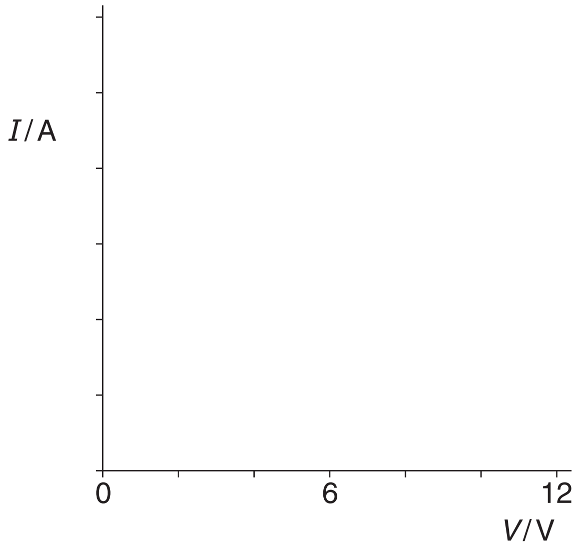


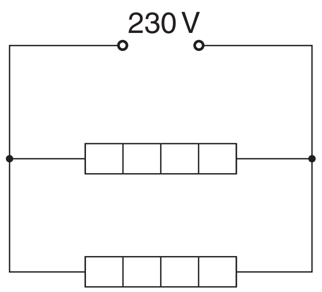
Fig. 6.1

[3]

(b) Some heaters are each labelled 230 V, 1.0 kW. The heaters have constant resistance.

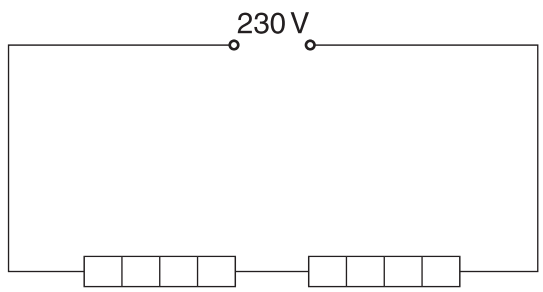
Determine the total power dissipation for the heaters connected as shown in each of the diagrams shown below.

(i)



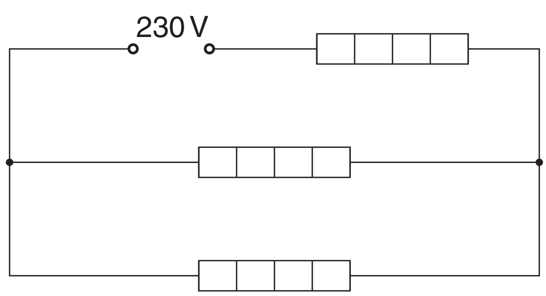
power = ... kW [1]

(ii)



power = ... kW [1]

(iii)



power = ... kW [2]

6 **(a)** A variable resistor is used to control the current in a circuit, as shown in Fig. 5.1.

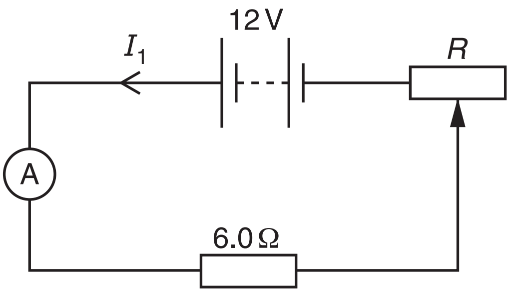


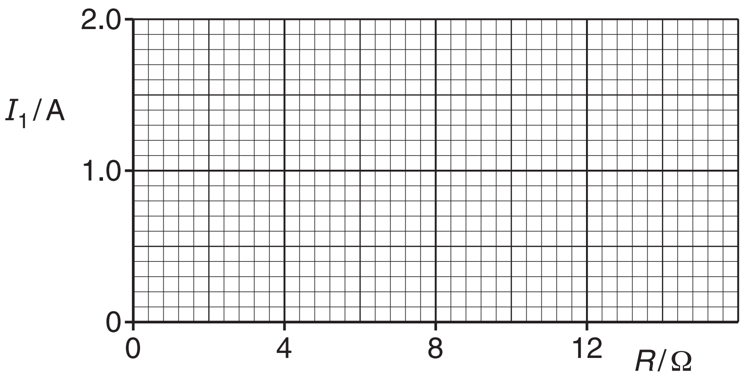
Fig. 5.1

The variable resistor is connected in series with a 12 V power supply of negligible internal resistance, an ammeter and a 6.0 Ω resistor. The resistance R of the variable resistor can be varied between 0 and 12 Ω.

(i) The maximum possible current in the circuit is 2.0 A. Calculate the minimum possible current.

minimum current = ... A [2]

(ii) On Fig. 5.2, sketch the variation with R of current I_1 in the circuit.



[2]

Fig. 5.2

(b) The variable resistor in **(a)** is now connected as a potential divider, as shown in Fig. 5.3.

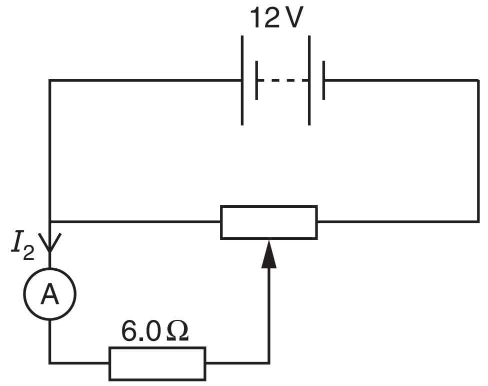


Fig. 5.3

Calculate the maximum possible and minimum possible current I_2 in the ammeter.

maximum I_2 = ... A

minimum I_2 = ... A
[2]

(c) **(i)** Sketch on Fig. 5.4 the I – V characteristic of a filament lamp.

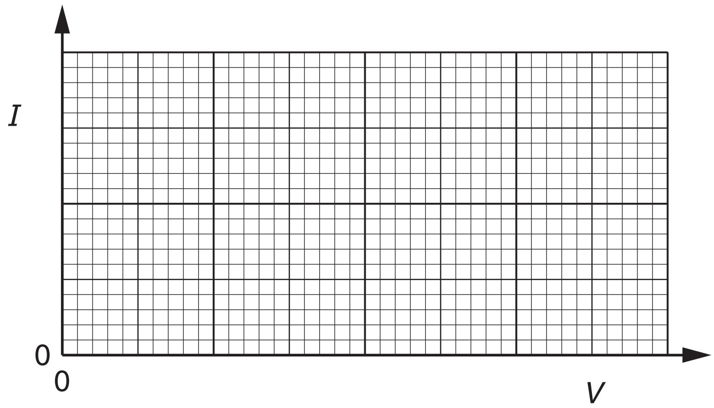


Fig. 5.4

[2]

(ii) The resistor of resistance 6.0 Ω is replaced with a filament lamp in the circuits of Fig. 5.1 and Fig. 5.3. State an advantage of using the circuit of Fig. 5.3, compared to the circuit of Fig 5.1, when using the circuits to vary the brightness of the filament lamp.

....

.... [1]

7 **(a)** For a cell, explain the terms **9702/22/M/J/11/Q5**

(i) *electromotive force* (e.m.f.),

....

.... [1]

(ii) *internal resistance*.

....

.... [1]

(b) The circuit of Fig. 5.1 shows two batteries A and B and a resistor R connected in series.

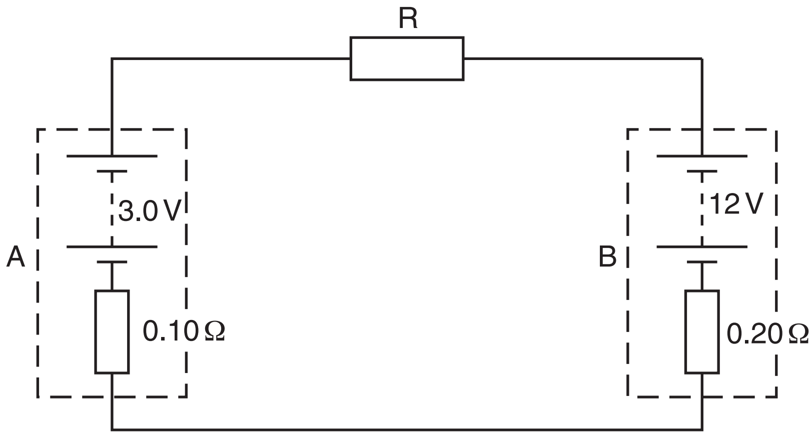


Fig. 5.1

Battery A has an e.m.f. of 3.0V and an internal resistance of 0.10Ω. Battery B has an e.m.f. of 12V and an internal resistance of 0.20Ω. Resistor R has a resistance of 3.3Ω.

(i) Apply Kirchhoff's second law to calculate the current in the circuit.

current = ... A [2]

(ii) Calculate the power transformed by battery B.

power = ... W [2]

(iii) Calculate the total energy lost per second in resistor R and the internal resistances.

energy lost per second = ... $J\,s^{-1}$ [2]

(c) The circuit of Fig. 5.1 may be used to store energy in battery A. Suggest how your answers in **(b)** support this statement.

....

.... [1]

8 A potentiometer circuit that is used as a means of comparing potential differences is shown in Fig. 5.1.

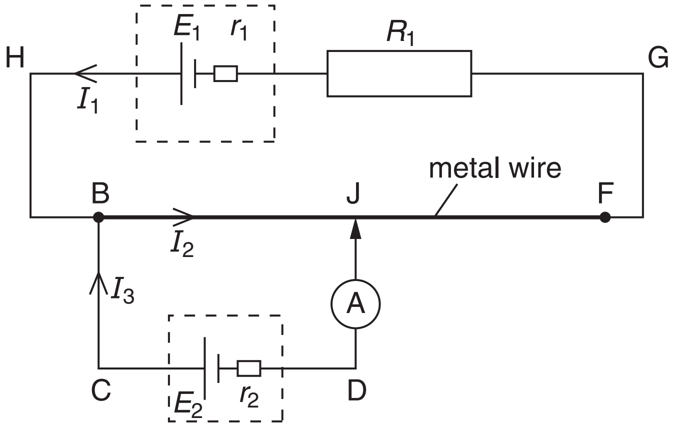


Fig. 5.1

A cell of e.m.f. E_1 and internal resistance r_1 is connected in series with a resistor of resistance R_1 and a uniform metal wire of total resistance R_2.
A second cell of e.m.f. E_2 and internal resistance r_2 is connected in series with a sensitive ammeter and is then connected across the wire at BJ. The connection at J is halfway along the wire. The current directions are shown on Fig. 5.1.

(a) Use Kirchhoff's laws to obtain the relation

(i) between the currents I_1, I_2 and I_3,

....[1]

(ii) between E_1, R_1, R_2, r_1, I_1 and I_2 in loop HBJFGH,

....[1]

(iii) between E_1, E_2, r_1, r_2, R_1, R_2, I_1 and I_3 in the loop HBCDJFGH.

....[2]

(b) The connection at J is moved along the wire. Explain why the reading on the ammeter changes.

....

....

....

....[2]

9 (a) (i) State Kirchhoff's second law. **9702/21/M/J/12/Q5**

....

.... [1]

(ii) Kirchhoff's second law is linked to the conservation of a certain quantity. State this quantity.

.... [1]

(b) The circuit shown in Fig. 5.1 is used to compare potential differences.

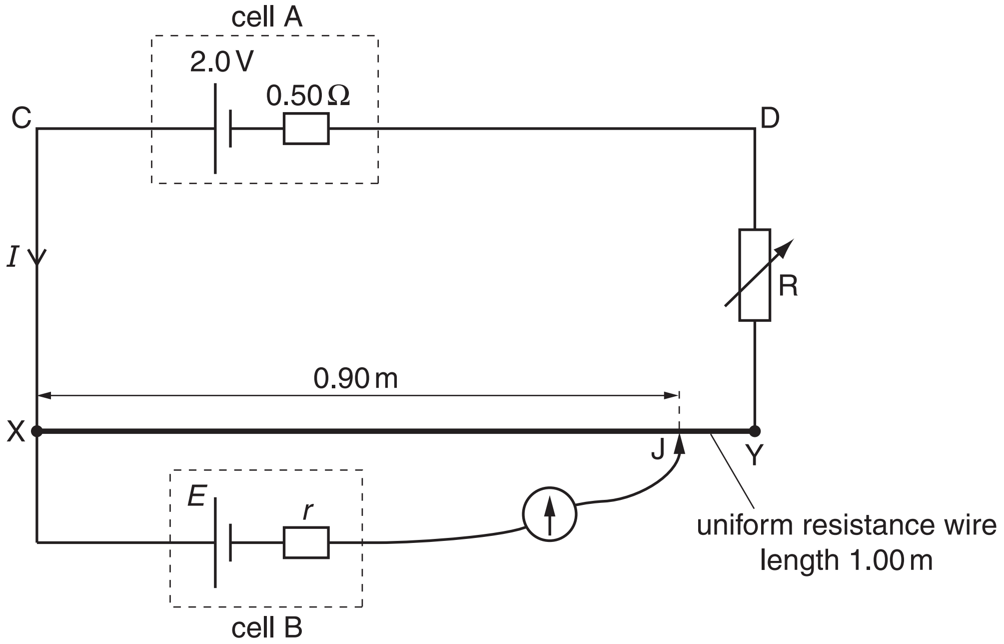


Fig. 5.1

The uniform resistance wire XY has length 1.00 m and resistance 4.0 Ω. Cell A has e.m.f. 2.0 V and internal resistance 0.50 Ω. The current through cell A is I. Cell B has e.m.f. E and internal resistance r.

The current through cell B is made zero when the movable connection J is adjusted so that the length of XJ is 0.90 m. The variable resistor R has resistance 2.5 Ω.

(i) Apply Kirchhoff's second law to the circuit CXYDC to determine the current I.

I = ... A [2]

(ii) Calculate the potential difference across the length of wire XJ.

potential difference = ... V [2]

(iii) Use your answer in **(ii)** to state the value of *E*.

E = ... V [1]

(iv) State why the value of the internal resistance of cell B is not required for the determination of *E*.

....

....[1]

10 A battery of electromotive force 12V and negligible internal resistance is connected to two resistors and a light-dependent resistor (LDR), as shown in Fig. 4.1.

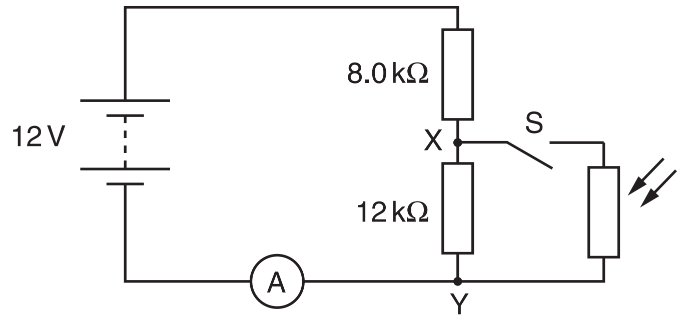


Fig. 4.1

An ammeter is connected in series with the battery. The LDR and switch S are connected across the points XY.

(a) The switch S is open. Calculate the potential difference (p.d.) across XY.

p. d. = ... V [3]

(b) The switch S is closed. The resistance of the LDR is 4.0 kΩ. Calculate the current in the ammeter.

current = ... A [3]

(c) The switch S remains closed. The intensity of the light on the LDR is increased. State and explain the change to

(i) the ammeter reading,

....

....[2]

(ii) the p.d. across XY.

....

....[2]

11 **(a)** **(i)** State Kirchhoff's first law. **9702/23/M/J/12/Q5**

....[1]

(ii) Kirchhoff's first law is linked to the conservation of a certain quantity. State this quantity.

....[1]

(b) A variable resistor of resistance R is used to control the current in a circuit, as shown in Fig. 5.1.

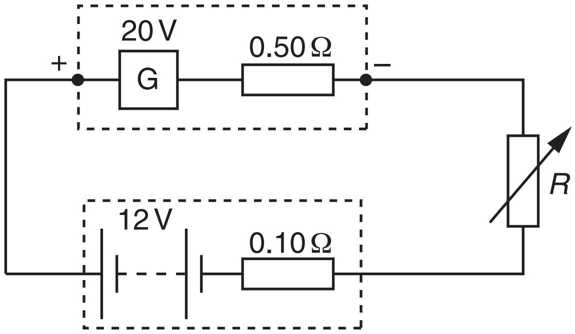


Fig. 5.1

The generator G has e.m.f. 20 V and internal resistance 0.50 Ω. The battery has e.m.f. 12 V and internal resistance 0.10 Ω. The current in the circuit is 2.0 A.

(i) Apply Kirchhoff's second law to the circuit to determine the resistance R.

R = ... Ω [2]

(ii) Calculate the total power generated by G.

power = ... W [2]

(iii) Calculate the power loss in the total resistance of the circuit.

power = ... W [2]

(iv) The circuit is used to supply energy to the battery from the generator. Determine the efficiency of the circuit.

efficiency = ... [2]

12 Fig. 5.1 shows a 12 V power supply with negligible internal resistance connected to a uniform metal wire AB. The wire has length 1.00 m and resistance 10 Ω. Two resistors of resistance 4.0 Ω and 2.0 Ω are connected in series across the wire.

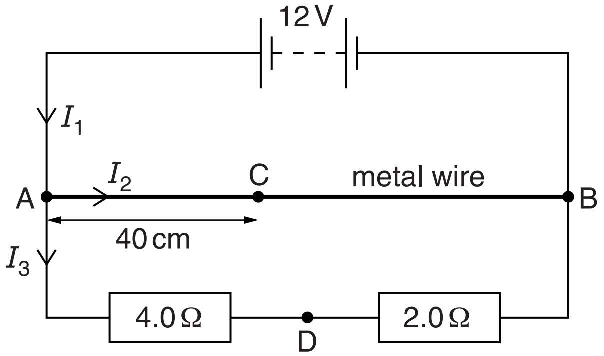


Fig. 5.1

Currents I_1, I_2 and I_3 in the circuit are as shown in Fig. 5.1.

(a) (i) Use Kirchhoff's first law to state a relationship between I_1, I_2 and I_3.

.... [1]

(ii) Calculate I_1.

I_1 = ... A [3]

(iii) Calculate the ratio *x*, where

$$x = \frac{\text{power in metal wire}}{\text{power in series resistors}}.$$

x = ... [3]

(b) Calculate the potential difference (p.d.) between the points C and D, as shown in Fig. 5.1. The distance AC is 40 cm and D is the point between the two series resistors.

p.d. = ... V [3]

13 A circuit used to measure the power transfer from a battery is shown in Fig. 4.1. The power is transferred to a variable resistor of resistance *R*.

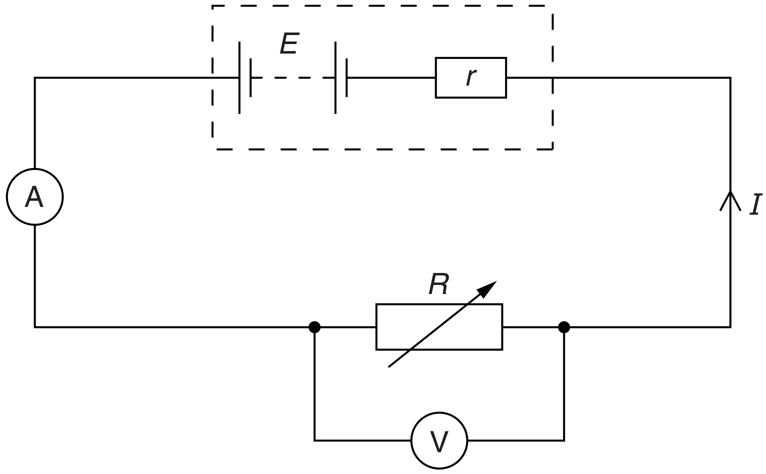


Fig. 4.1

The battery has an electromotive force (e.m.f.) *E* and an internal resistance *r*. There is a potential difference (p.d.) *V* across *R*. The current in the circuit is *I*.

(a) By reference to the circuit shown in Fig. 4.1, distinguish between the definitions of e.m.f. and p.d.

....

....

....

.... [3]

(b) Using Kirchhoff's second law, determine an expression for the current *I* in the circuit.

[1]

(c) The variation with current I of the p.d. V across R is shown in Fig. 4.2.

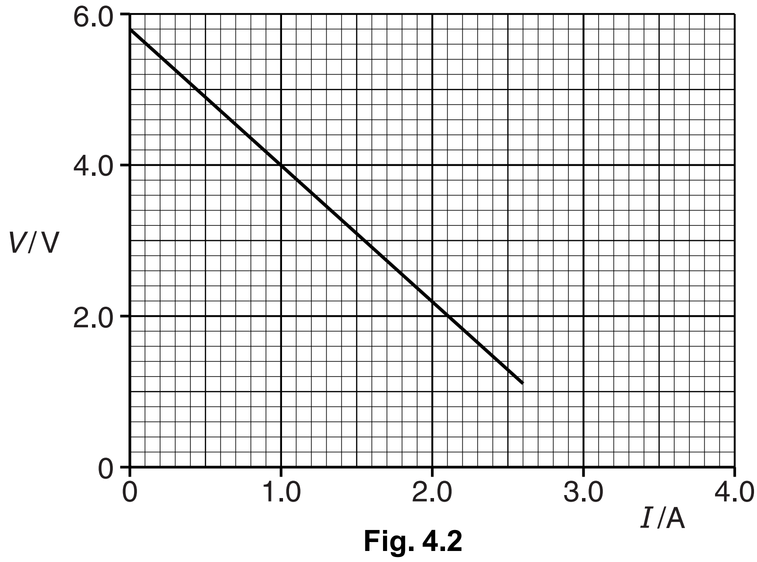


Fig. 4.2

Use Fig. 4.2 to determine

(i) the e.m.f. E,

E = ... V [1]

(ii) the internal resistance r.

r = ... Ω [2]

(d) **(i)** Using data from Fig. 4.2, calculate the power transferred to R for a current of 1.6 A.

power = ... W [2]

(ii) Use your answers from **(c)(i)** and **(d)(i)** to calculate the efficiency of the battery for a current of 1.6 A.

efficiency = ... % [2]

14 **(a)** Define *potential difference* (p.d.). **9702/23/M/J/13/Q6**

.... [1]

(b) A battery of electromotive force 20V and zero internal resistance is connected in series with two resistors R_1 and R_2, as shown in Fig. 6.1.

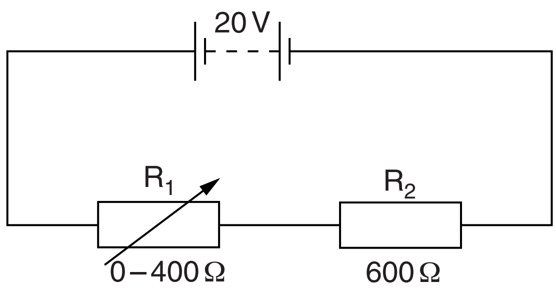


Fig. 6.1

The resistance of R_2 is 600 Ω. The resistance of R_1 is varied from 0 to 400 Ω.

Calculate

(i) the maximum p.d. across R_2,

maximum p.d. = ... V [1]

(ii) the minimum p.d. across R_2.

minimum p.d. = ... V [2]

(c) A light-dependent resistor (LDR) is connected in parallel with R_2, as shown in Fig. 6.2.

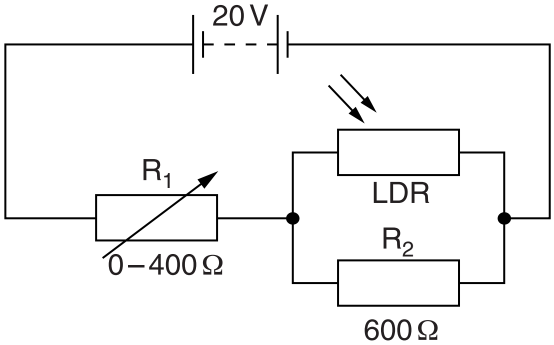


Fig. 6.2

When the light intensity is varied, the resistance of the LDR changes from 5.0 kΩ to 1.2 kΩ.

(i) For the **maximum** light intensity, calculate the total resistance of R_2 and the LDR.

total resistance = ... Ω [2]

(ii) The resistance of R_1 is varied from 0 to 400 Ω in the circuits of Fig. 6.1 and Fig. 6.2. State and explain the difference, if any, between the minimum p.d. across R_2 in each circuit. Numerical values are not required.

....

....

....[2]

15 A battery is connected in series with resistors X and Y, as shown in Fig. 6.1. **9702/22/M/J/14/Q6**

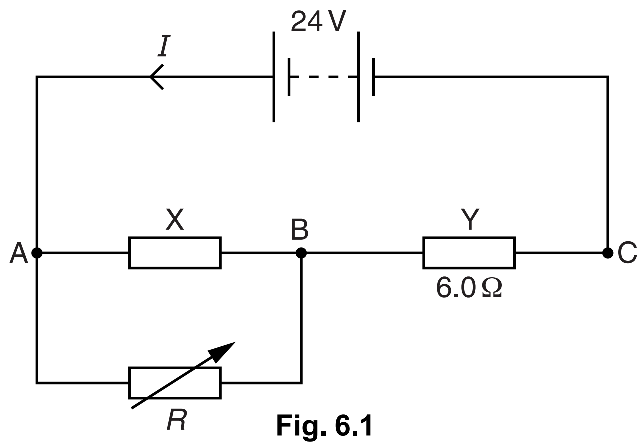


Fig. 6.1

The resistance of X is constant. The resistance of Y is 6.0 Ω. The battery has electromotive force (e.m.f.) 24 V and zero internal resistance. A variable resistor of resistance *R* is connected in parallel with X.

The current I from the battery is changed by varying *R* from 5.0 Ω to 20 Ω. The variation with *R* of I is shown in Fig. 6.2.

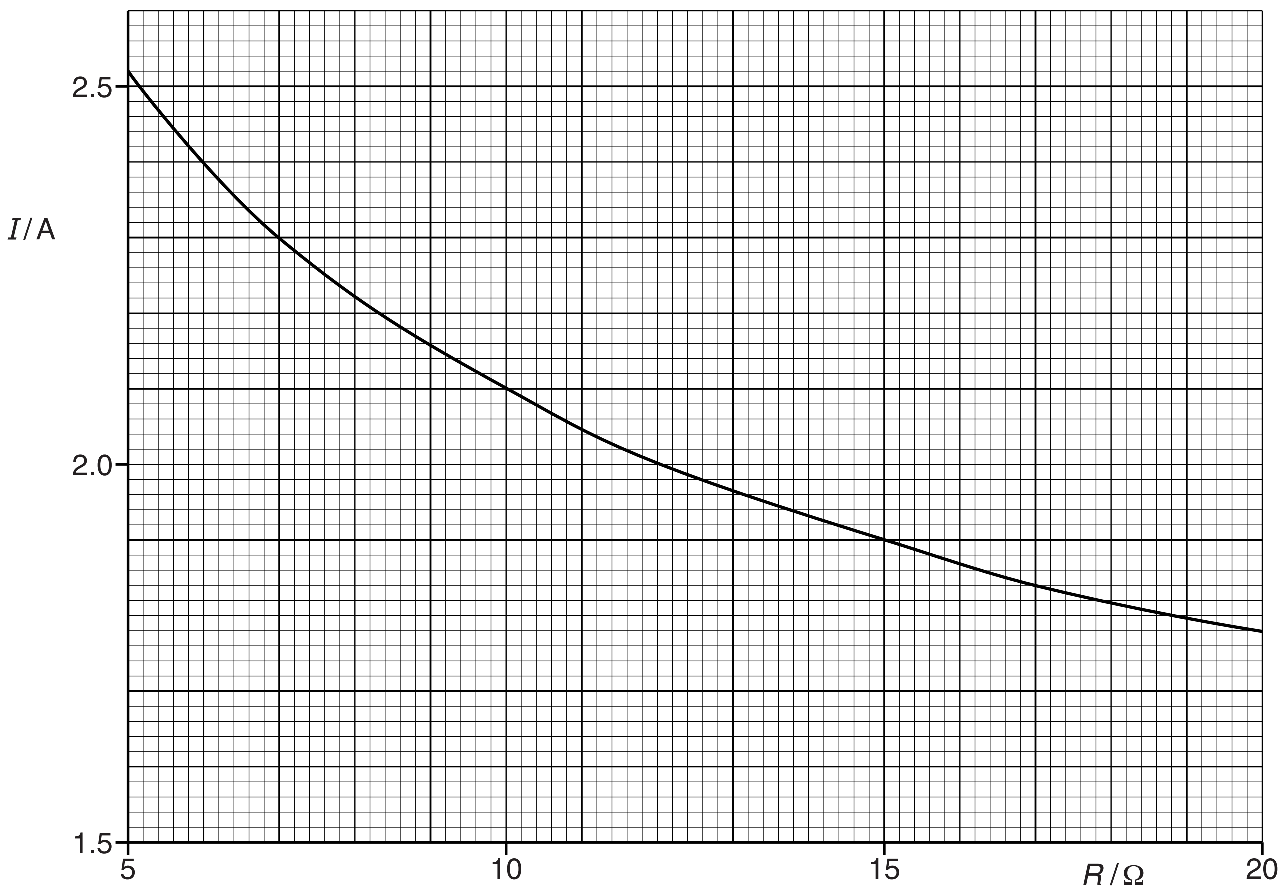


Fig. 6.2

(a) Explain why the potential difference (p.d.) between points A and C is 24 V for all values of *R*.

....

.... [1]

(b) Use Fig. 6.2 to state and explain the variation of the p.d. across resistor Y as R is increased. Numerical values are not required.

....

....

.... [2]

(c) For $R = 6.0\,\Omega$,

(i) show that the p.d. between points A and B is 9.6 V,

[2]

(ii) calculate the resistance of X,

resistance = ... Ω [3]

(iii) calculate the power provided by the battery.

power = ... W [2]

(d) State and explain qualitatively how the power provided by the battery changes as the resistance R is increased.

....

.... [1]

16 A battery of electromotive force (e.m.f.) 12 V and internal resistance r is connected in series to two resistors, each of constant resistance X, as shown in Fig. 5.1.

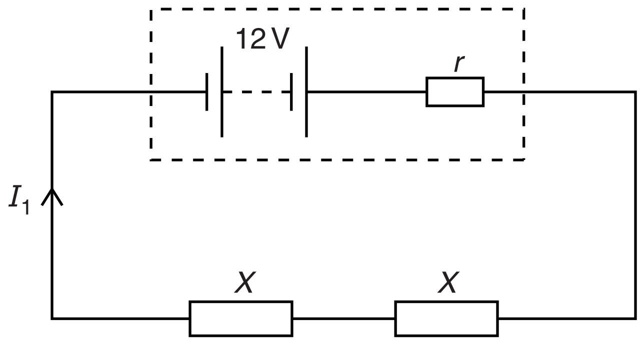


Fig. 5.1

The current I_1 supplied by the battery is 1.2 A.

The same battery is now connected to the same two resistors in parallel, as shown in Fig. 5.2.

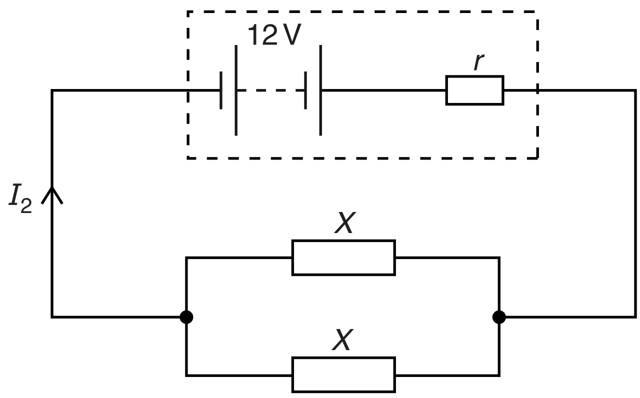


Fig. 5.2

The current I_2 supplied by the battery is 3.0 A.

(a) **(i)** Show that the combined resistance of the two resistors, each of resistance X, is four times greater in Fig. 5.1 than in Fig. 5.2.

[2]

(ii) Explain why I_2 is not four times greater than I_1.

....

....

....[2]

(iii) Using Kirchhoff's second law, state equations, in terms of e.m.f., current, *X* and *r*, for

1. the circuit of Fig. 5.1,

....

2. the circuit of Fig. 5.2.

....
[2]

(iv) Use the equations in **(iii)** to calculate the resistance *X*.

X = ... Ω [1]

(b) Calculate the ratio

$$\frac{\text{power transformed in one resistor of resistance } X \text{ in Fig. 5.1}}{\text{power transformed in one resistor of resistance } X \text{ in Fig. 5.2}}.$$

ratio = ...[2]

(c) The resistors in Fig. 5.1 and Fig. 5.2 are replaced by identical 12 V filament lamps.

Explain why the resistance of each lamp, when connected in series, is not the same as the resistance of each lamp when connected in parallel.

....

....

....

....[2]

17 (a) A wire has length 100 cm and diameter 0.38 mm. The metal of the wire has resistivity $4.5 \times 10^{-7}\,\Omega\,m$.

Show that the resistance of the wire is $4.0\,\Omega$.

[3]

(b) The ends B and D of the wire in **(a)** are connected to a cell X, as shown in Fig. 6.1.

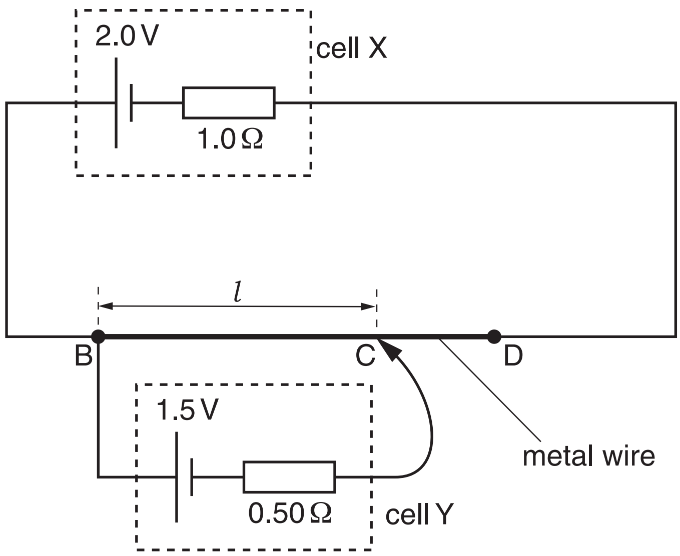


Fig. 6.1

The cell X has electromotive force (e.m.f.) 2.0 V and internal resistance $1.0\,\Omega$.

A cell Y of e.m.f. 1.5 V and internal resistance $0.50\,\Omega$ is connected to the wire at points B and C, as shown in Fig. 6.1.

The point C is distance l from point B. The current in cell Y is zero.

Calculate

(i) the current in cell X,

current = ... A [2]

(ii) the potential difference (p.d.) across the wire BD,

p.d. = ... V [1]

(iii) the distance l.

l = ... cm [2]

(c) The connection at C is moved so that l is increased. Explain why the e.m.f. of cell Y is less than its terminal p.d.

....

....

....[2]

9702/22/M/J/15/Q5

18 (a) On Fig. 5.1, sketch the temperature characteristic of a thermistor.

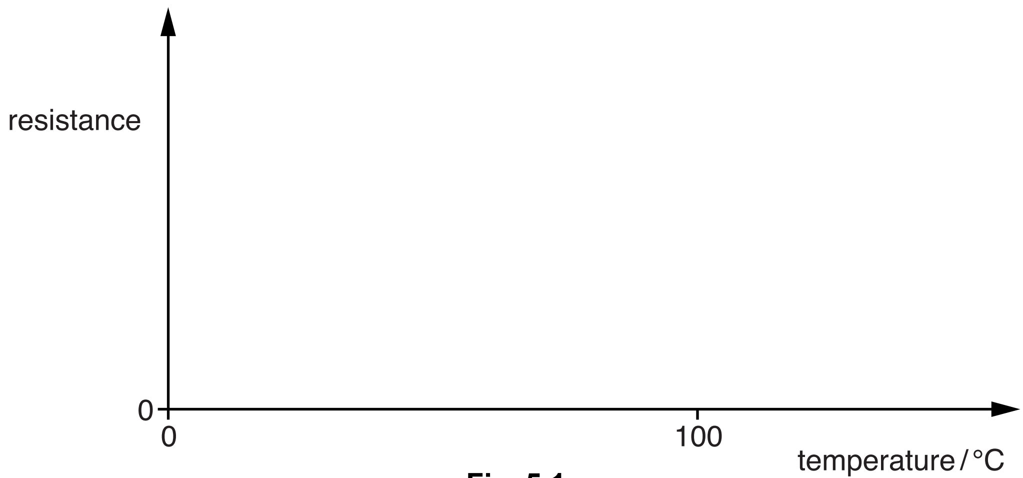


Fig. 5.1

[2]

(b) A potential divider circuit is shown in Fig. 5.2.

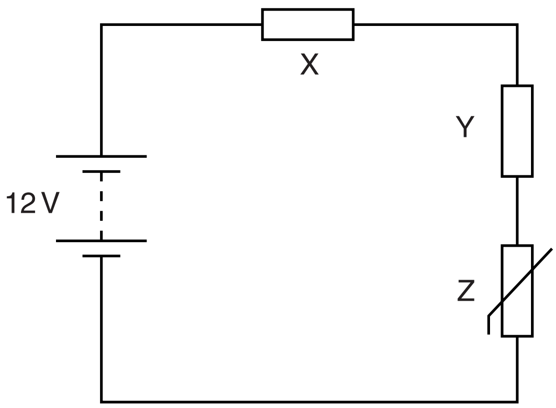


Fig. 5.2

The battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected in series with resistors X and Y and thermistor Z. The resistance of Y is 15 kΩ and the resistance of Z at a particular temperature is 3.0 kΩ. The potential difference (p.d.) across Y is 8.0 V.

(i) Explain why the power transformed in the battery equals the total power transformed in X, Y and Z.

.... [1]

(ii) Calculate the current in the circuit.

current = ... A [2]

(iii) Calculate the resistance of X.

resistance = ... Ω [3]

(iv) The temperature of Z is increased.

State and explain the effect on the potential difference across Z.

....

....

....

.... [2]

19 A uniform resistance wire AB has length 50 cm and diameter 0.36 mm. The resistivity of the metal of the wire is $5.1 \times 10^{-7}\,\Omega\,\text{m}$.

(a) Show that the resistance of the wire AB is $2.5\,\Omega$.

[2]

(b) The wire AB is connected in series with a power supply E and a resistor R as shown in Fig. 5.1.

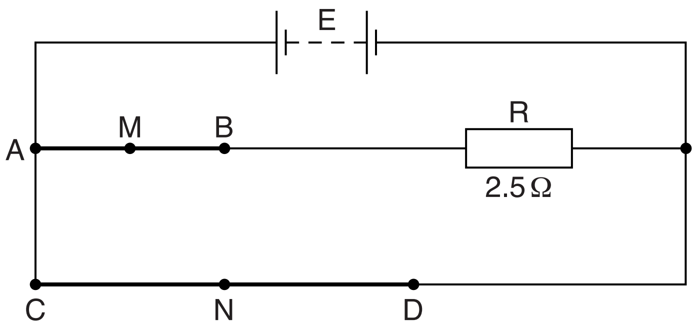


Fig. 5.1

The electromotive force (e.m.f.) of E is 6.0 V and its internal resistance is negligible. The resistance of R is $2.5\,\Omega$. A second uniform wire CD is connected across the terminals of E. The wire CD has length 100 cm, diameter 0.18 mm and is made of the same metal as wire AB.

Calculate

(i) the current supplied by E,

current = ... A [4]

(ii) the power transformed in wire AB,

power = ... W [2]

(iii) the potential difference (p.d.) between the midpoint M of wire AB and the midpoint N of wire CD.

p.d. = ... V [2]

20 **(a)** Define electric *potential difference* (*p.d.*).

....

....[1]

(b) A battery of electromotive force (e.m.f.) 14V and negligible internal resistance is connected to a resistor network, as shown in Fig. 6.1.

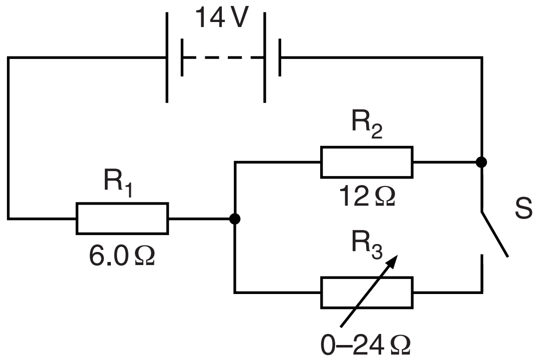


Fig. 6.1

R_1 and R_2 are fixed resistors of resistances 6.0 Ω and 12 Ω respectively. R_3 is a variable resistor.

Switch S is **closed**.

(i) Calculate the current in the battery when the resistance of R_3 is set

1. at zero,

current = ... A [2]

2. at 24 Ω.

current = ... A [2]

(ii) Use your answers in **(b)(i)** to calculate the change in the total power produced by the battery when the resistance of R_3 is changed from zero to 24 Ω.

change in power = ... W [2]

(c) Switch S in Fig. 6.1 is now **opened**.

Resistors R_1 and R_2 are made from metal wires. Some data for these resistors are shown in Fig. 6.2.

	R_1	R_2
cross-sectional area of wire	A	$1.8A$
number of free electrons per unit volume in metal	n	$0.50n$

Fig. 6.2

Determine the ratio

$$\frac{\text{average drift speed of free electrons in } R_1}{\text{average drift speed of free electrons in } R_2}.$$

ratio = ... [2]

21 (a) State Kirchhoff's second law.

....

....

....

....[2]

(b) A battery is connected in parallel with two lamps A and B, as shown in Fig. 5.1.

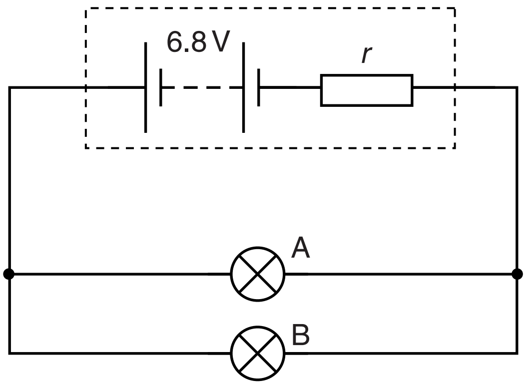


Fig. 5.1

The battery has electromotive force (e.m.f.) 6.8 V and internal resistance r.

The I–V characteristics of lamps A and B are shown in Fig. 5.2.

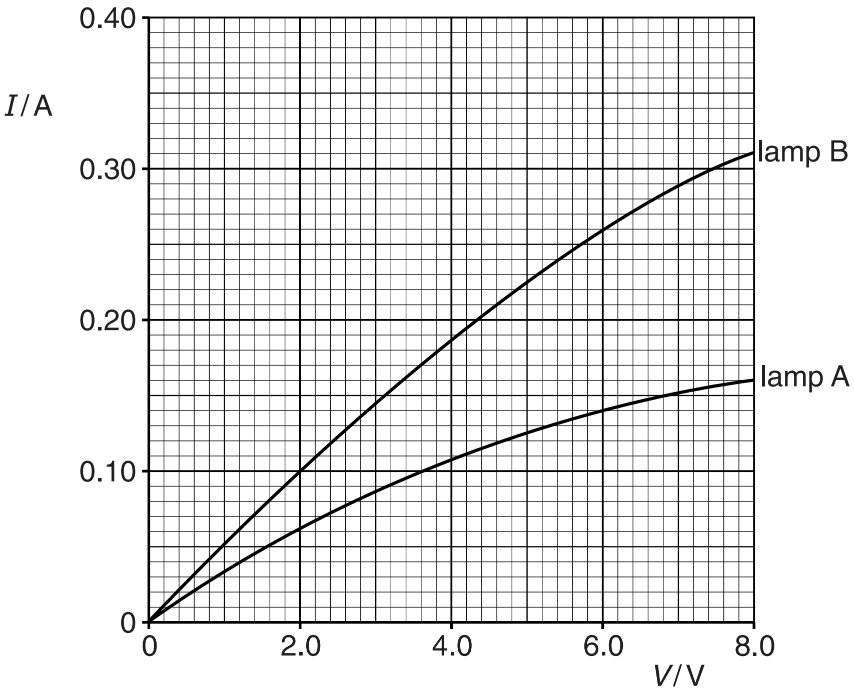


Fig. 5.2

The potential difference across the battery terminals is 6.0 V.

(i) Use Fig. 5.2 to show that the current in the battery is 0.40 A.

[2]

(ii) Calculate the internal resistance r of the battery.

r = ... Ω [2]

(iii) Determine the ratio

$$\frac{\text{resistance of lamp A}}{\text{resistance of lamp B}}.$$

ratio = ... [2]

(iv) Determine

1. the total power produced by the battery,

power = ... W [2]

2. the efficiency of the battery in the circuit.

efficiency = ... [2]

22 **(a)** Define the *ohm*. **9702/21/M/J/17/Q6**

....

....[1]

(b) A cell X of electromotive force (e.m.f.) 1.5 V and negligible internal resistance is connected in series to three resistors A, B and C, as shown in Fig. 6.1.

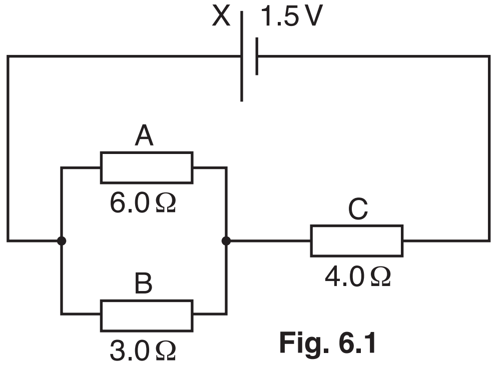


Fig. 6.1

Resistors A and B have resistances 6.0 Ω and 3.0 Ω respectively and are connected in parallel. Resistor C has resistance 4.0 Ω and is connected in series with the parallel combination.

Calculate

(i) the current in the circuit,

current = ...A [3]

(ii) the current in resistor B,

current = ...A [1]

(iii) the ratio

$$\frac{\text{power dissipated in resistor B}}{\text{power dissipated in resistor C}}.$$

ratio = ...[2]

(c) The resistors A, B and C in **(b)** are wires of the same material and have the same length.

(i) Explain how the resistors may be made with different resistance values.

....[1]

(ii) Calculate the ratio

$$\frac{\text{average drift speed of the charge carriers in resistor B}}{\text{average drift speed of the charge carriers in resistor C}}.$$

ratio = ...[2]

(d) A cell of e.m.f. 1.5 V and negligible internal resistance is connected in parallel with cell X in Fig. 6.1 with their positive terminals together.

State the change, if any, to the current in

(i) cell X,

....[1]

(ii) resistor C.

....[1]

23 **(a)** Define *electromotive force* (e.m.f.) of a cell. **9702/22/M/J/17/Q7**

....

....[1]

(b) A cell C of e.m.f. 1.50 V and internal resistance 0.200 Ω is connected in series with resistors X and Y, as shown in Fig. 7.1.

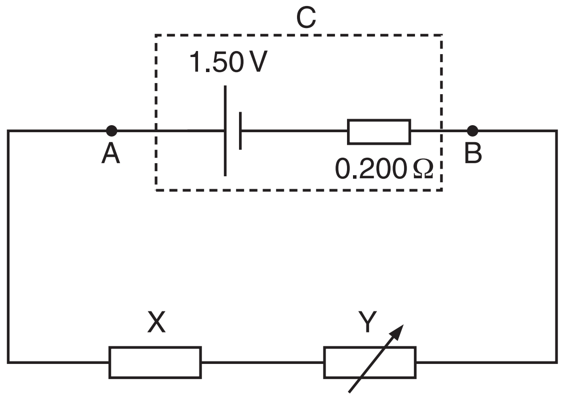


Fig. 7.1

The resistance of X is constant and the resistance of Y can be varied.

(i) The resistance of Y is varied from 0 to 8.00 Ω.

State and explain the variation in the potential difference (p.d.) between points A and B (terminal p.d. across C). Numerical values are not required.

....

....

....

....[3]

(ii) The resistance of Y is set at 6.00 Ω. The current in the circuit is 0.180 A.

Calculate

1. the resistance of X,

resistance = ... Ω [2]

2. the p.d. between points A and B,

p.d. = ... V [2]

3. the efficiency of the cell.

efficiency = ... [2]

24 Three cells of electromotive forces (e.m.f.) E_1, E_2 and E_3 are connected into a circuit, as shown in Fig. 5.1.

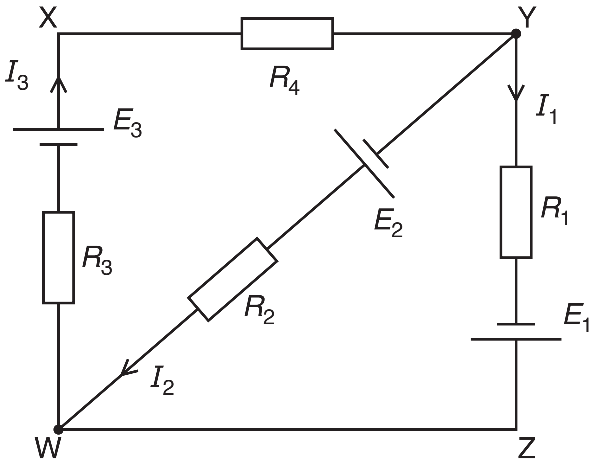


Fig. 5.1

The circuit contains resistors of resistances R_1, R_2, R_3 and R_4.
The currents in the different parts of the circuit are I_1, I_2 and I_3.
The cells have negligible internal resistance.

Use Kirchhoff's laws to state an equation relating

(a) I_1, I_2 and I_3,

....[1]

(b) E_1, E_3, R_1, R_3, R_4, I_1 and I_3 in loop WXYZW,

....

....[1]

(c) E_1, E_2, R_1, R_2, I_1 and I_2 in loop YZWY.

....

....[1]

25 (a) (i) State Kirchhoff's first law. **9702/22/M/J/18/Q6**

....

....[1]

(ii) Kirchhoff's first law is linked to the conservation of a certain quantity. State this quantity.

....[1]

(b) A battery of electromotive force (e.m.f.) 8.0 V and internal resistance 2.0 Ω is connected to a resistor X and a wire Y, as shown in Fig. 6.1.

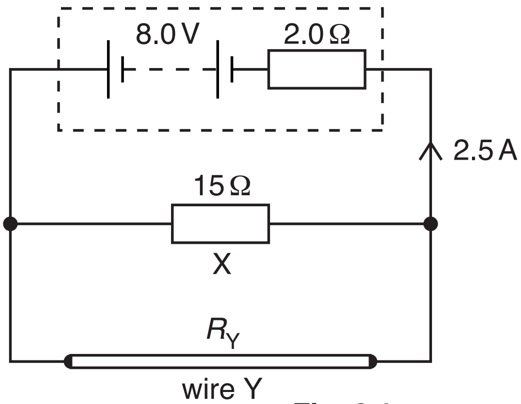


Fig. 6.1

The resistance of X is 15 Ω. The resistance of Y is R_Y. The current in the battery is 2.5 A.

(i) Calculate

1. the thermal energy dissipated in the battery in a time of 5.0 minutes,

energy = ... J [2]

2. the terminal potential difference of the battery.

terminal potential difference = ... V [1]

(ii) Determine the resistance R_Y.

R_Y = ... Ω [3]

(iii) A new wire Z has the same length but less resistance than wire Y.

1. State two possible differences between wire Z and wire Y that would separately cause wire Z to have less resistance than wire Y.

first difference: ....

....

second difference: ....

....
[2]

2. Wire Y is replaced in the circuit by wire Z. By considering the current in the battery, state and explain the effect of changing the wires on the total power produced by the battery.

....

....[2]

26. 9702/21/O/N/18/Q6

(a) State Kirchhoff's second law.

....

....

.... [2]

(b) An electric heater containing two heating wires X and Y is connected to a power supply of electromotive force (e.m.f.) 9.0 V and negligible internal resistance, as shown in Fig. 6.1.

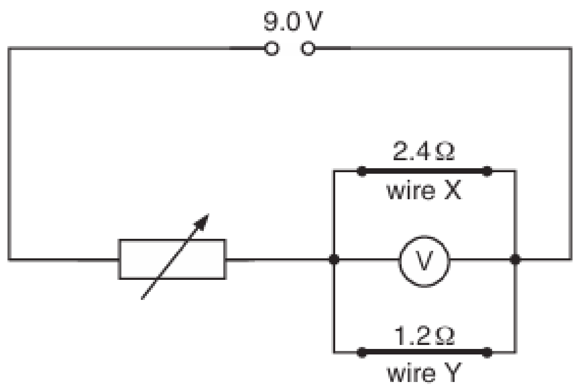


Fig. 6.1

Wire X has a resistance of 2.4 Ω and wire Y has a resistance of 1.2 Ω. A voltmeter is connected in parallel with the wires. A variable resistor is used to adjust the power dissipated in wires X and Y.

The variable resistor is adjusted so that the voltmeter reads 6.0 V.

(i) Calculate the resistance of the variable resistor.

resistance = ... Ω [3]

(ii) Calculate the power dissipated in wire X.

power = ... W [2]

(iii) The cross-sectional area of wire X is three times the cross-sectional area of wire Y. Assume that the resistivity and the number density of free electrons for the metal of both wires are the same.

Determine the ratio

1. $\dfrac{\text{length of wire X}}{\text{length of wire Y}}$,

ratio = ... [2]

2. $\dfrac{\text{average drift velocity of free electrons in wire X}}{\text{average drift velocity of free electrons in wire Y}}$.

ratio = ... [2]

[Total: 11]

27. 9702/22/O/N/18/Q6

(a) Define the *volt*.

....

....[1]

(b) A battery of electromotive force (e.m.f.) 7.0 V and negligible internal resistance is connected in series with three components, as shown in Fig. 6.1.

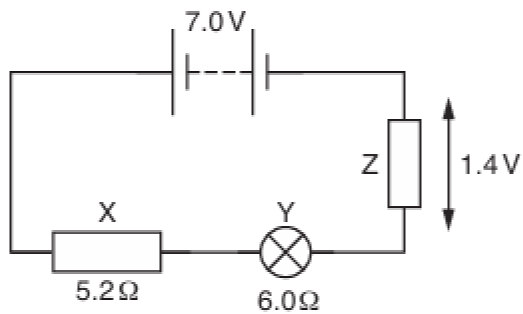


Fig. 6.1

Resistor X has a resistance of 5.2 Ω. The resistance of the filament wire of lamp Y is 6.0 Ω. The potential difference across resistor Z is 1.4 V.

(i) Calculate the current in the circuit.

current = ... A [2]

(ii) Determine the resistance of resistor Z.

resistance = ... Ω [1]

(iii) Calculate the percentage efficiency with which the battery supplies power to the lamp.

efficiency = ... % [3]

(iv) The filament wire of the lamp is made of metal of resistivity $3.7 \times 10^{-7}\,\Omega\,\text{m}$ at its operating temperature in the circuit.

Determine, for the filament wire, the value of α where

$$\alpha = \frac{\text{cross-sectional area}}{\text{length}}.$$

α = ... m [2]

[Total: 9]

28. 9702/23/O/N/18/Q7

(a) State Kirchhoff's first law.

....

....[1]

(b) A potentiometer is connected to a battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance, as shown in Fig. 7.1.

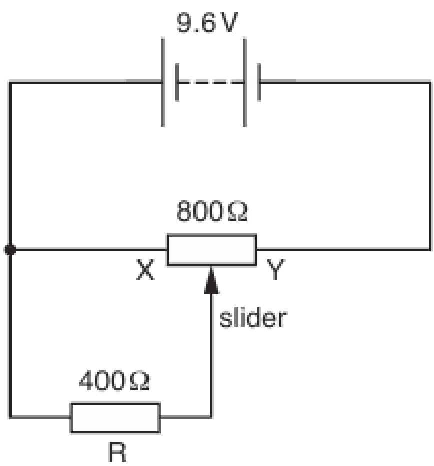


Fig. 7.1

The maximum resistance of the potentiometer is 800 Ω. A resistor R of resistance 400 Ω is connected between the slider and end X of the potentiometer.

(i) State the potential difference across resistor R when the slider is positioned

1. at end X of the potentiometer,

potential difference = ... V

2. at end Y of the potentiometer.

potential difference = ... V
[2]

(ii) Calculate the potential difference across resistor R when the slider is positioned half-way between X and Y.

potential difference = ... V [3]

[Total: 6]

29. 9702/21/M/J/19/Q6

A battery of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor of resistance R, as shown in Fig. 6.1.

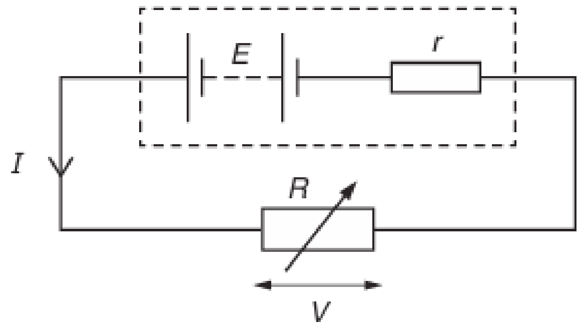


Fig. 6.1

The current in the circuit is I and the potential difference across the variable resistor is V.

(a) Explain, in terms of energy, why V is less than E.

....

....[1]

(b) State an equation relating E, I, r and V.

....[1]

(c) The resistance R of the variable resistor is varied. The variation with I of V is shown in Fig. 6.2.

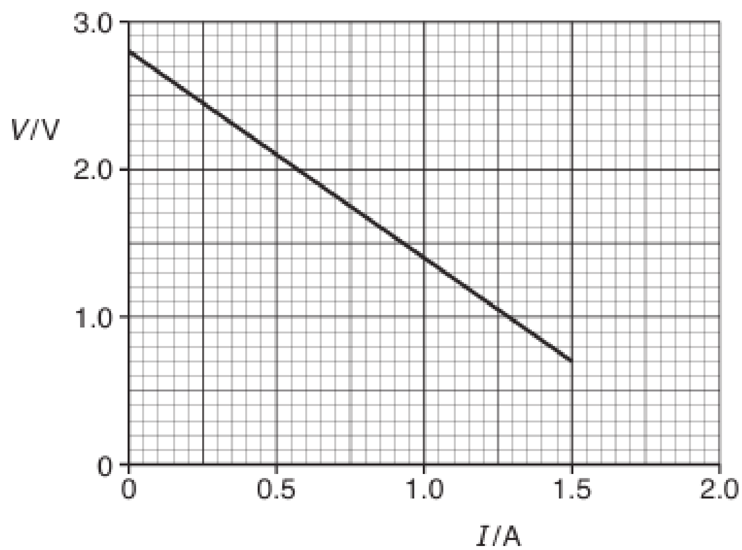


Fig. 6.2

Use Fig. 6.2 to:

(i) explain how it may be deduced that the e.m.f. of the battery is 2.8 V

....

....[1]

(ii) calculate the internal resistance *r*.

r = ... Ω [2]

(d) The battery stores 9.2 kJ of energy. The variable resistor is adjusted so that V = 2.1 V. Use Fig. 6.2 to:

(i) calculate resistance *R*

R = ... Ω [1]

(ii) calculate the number of conduction electrons moving through the battery in a time of 1.0 s

number = ... [1]

(iii) determine the time taken for the energy in the battery to become equal to 1.6 kJ. (Assume that the e.m.f. of the battery and the current in the battery remain constant.)

time taken = ... s [3]

[Total: 10]

30. 9702/22/M/J/19/Q5

(a) State Kirchhoff's second law.

....

....[2]

(b) A battery of electromotive force (e.m.f.) 5.6 V and internal resistance r is connected to two external resistors, as shown in Fig. 5.1.

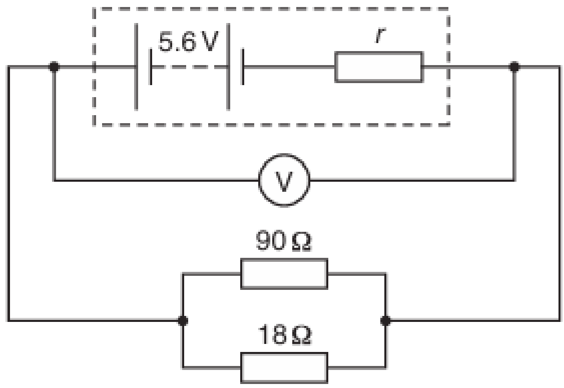


Fig. 5.1

The reading on the voltmeter is 4.8 V.

(i) Calculate:

1. the combined resistance of the two resistors connected in parallel

combined resistance = ... Ω [2]

2. the current in the battery.

current = ... A [2]

(ii) Show that the internal resistance r is 2.5 Ω.

[2]

(iii) Determine the ratio

$$\frac{\text{power dissipated by internal resistance } r}{\text{total power produced by battery}}.$$

ratio = ... [3]

(c) The battery in **(b)** is now connected to a battery of e.m.f. 7.2 V and internal resistance 3.5 Ω. The new circuit is shown in Fig. 5.2.

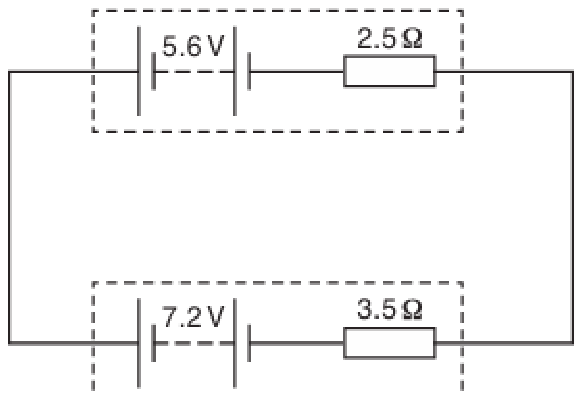


Fig. 5.2

Determine the current in the circuit.

current = ... A [2]

[Total: 13]

31. 9702/21/O/N/19/Q6

(a) Define *electric potential difference* (*p.d.*).

....

.... [1]

(b) The variation with potential difference V of the current I in a semiconductor diode is shown in Fig. 6.1.

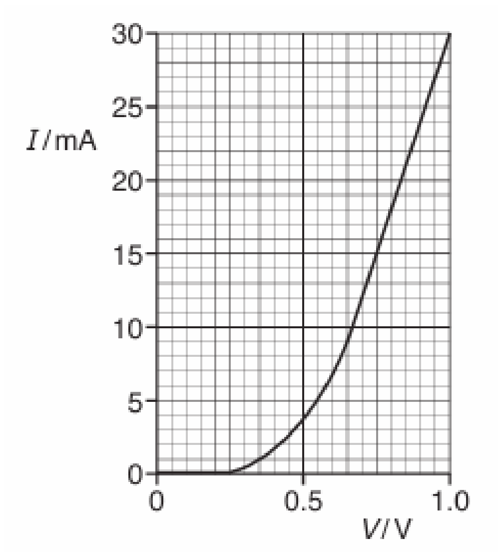


Fig. 6.1

Use Fig. 6.1 to describe qualitatively the variation of the resistance of the diode as V increases from 0 to 1.0 V.

....

....

....

.... [2]

(c) The diode in **(b)** is part of the circuit shown in Fig. 6.2.

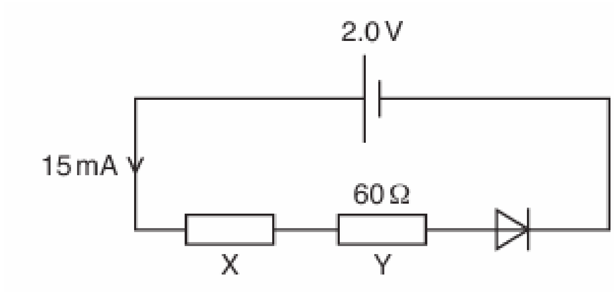


Fig. 6.2

The cell of electromotive force (e.m.f.) 2.0 V and negligible internal resistance is connected in series with the diode and resistors X and Y. The resistance of Y is 60 Ω. The current in the cell is 15 mA.

(i) Use Fig. 6.1 to determine the resistance of the diode.

resistance = ... Ω [3]

(ii) Calculate:

1. the resistance of X

resistance = ... Ω [3]

2. the ratio

$$\frac{\text{power dissipated in resistor Y}}{\text{total power produced by the cell}}.$$

ratio = ... [2]

32. 9702/22/O/N/19/Q6

(a) State Kirchhoff's first law.

....

.... [1]

(b) The variations with potential difference V of the current I for a resistor X and for a semiconductor diode are shown in Fig. 6.1.

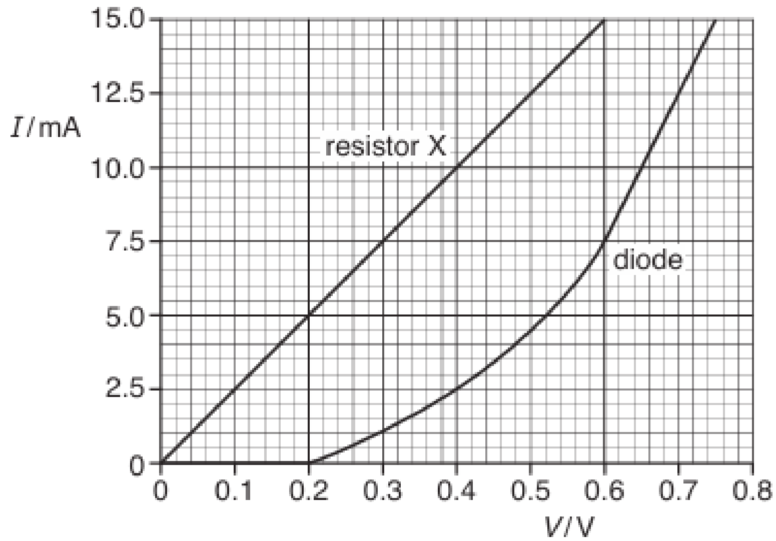


Fig. 6.1

(i) Determine the resistance of the diode for a potential difference V of 0.60 V.

resistance = ... Ω [3]

(ii) Describe, qualitatively, the variation of the resistance of the diode as V increases from 0.60 V to 0.75 V.

.... [1]

(c) The diode and the resistor X in **(b)** are connected into the circuit shown in Fig. 6.2.

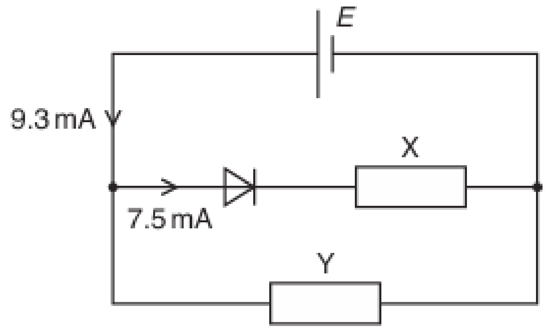


Fig. 6.2

The cell has electromotive force (e.m.f.) E and negligible internal resistance. Resistor Y is connected in parallel with resistor X and the diode. The current in the cell is 9.3 mA and the current in the diode is 7.5 mA.

(i) Use Fig. 6.1 to determine E.

E = ...V [1]

(ii) Determine the resistance of resistor Y.

resistance = ... Ω [2]

(iii) Calculate the power dissipated in the diode.

power = ...W [2]

(iv) The cell is now replaced by a new cell of e.m.f. 0.50 V and negligible internal resistance. Use Fig. 6.1 to determine the new current in the diode.

current = ...mA [1]

33. 9702/23/O/N/19/Q6

A battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected to a network of two lamps and two resistors, as shown in Fig. 6.1.

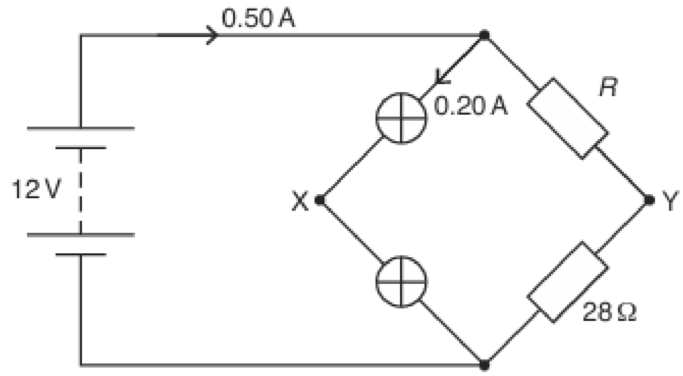


Fig. 6.1

The two lamps in the circuit have equal resistances. The two resistors have resistances R and 28 Ω. The lamps are connected at junction X and the resistors are connected at junction Y. The current in the battery is 0.50 A and the current in the lamps is 0.20 A.

(a) Calculate:

(i) the resistance of each lamp

resistance = ... Ω [2]

(ii) resistance R.

R = ... Ω [2]

(b) Determine the potential difference V_{XY} between points X and Y.

V_{XY} = ... V [3]

(c) Calculate the ratio

$$\frac{\text{total power dissipated by the lamps}}{\text{total power produced by the battery}}.$$

ratio = ... [2]

(d) The resistor of resistance *R* is now replaced by another resistor of lower resistance.

State and explain the effect, if any, of this change on the ratio in **(c)**.

....

....

....

....

.... [2]

[Total: 11]

34. 9702/21/M/J/20/Q5

(a) Metal wire is used to connect a power supply to a lamp. The wire has a total resistance of $3.4\,\Omega$ and the metal has a resistivity of $2.6 \times 10^{-8}\,\Omega\,\text{m}$. The total length of the wire is 59 m.

(i) Show that the wire has a cross-sectional area of $4.5 \times 10^{-7}\,\text{m}^2$.

[2]

(ii) The potential difference across the total length of wire is 1.8 V.

Calculate the current in the wire.

current = ... A [1]

(iii) The number density of the free electrons in the wire is $6.1 \times 10^{28}\,\text{m}^{-3}$.

Calculate the average drift speed of the free electrons in the wire.

average drift speed = ... $\text{m}\,\text{s}^{-1}$ [2]

(b) A different wire carries a current. This wire has a part that is thinner than the rest of the wire, as shown in Fig. 5.1.

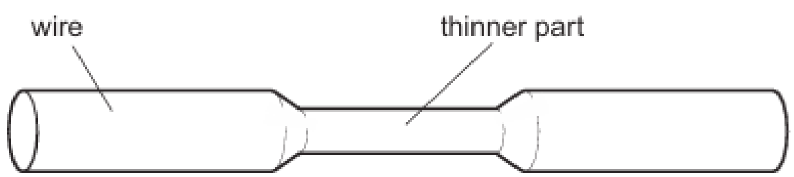


Fig. 5.1

(i) State and explain qualitatively how the average drift speed of the free electrons in the thinner part compares with that in the rest of the wire.

....

....

.... [2]

(ii) State and explain whether the power dissipated in the thinner part is the same, less or more than the power dissipated in an equal length of the rest of the wire.

....

....

.... [2]

(c) Three resistors have resistances of 180 Ω, 90 Ω and 30 Ω.

(i) Sketch a diagram showing how **two** of these three resistors may be connected together to give a combined resistance of 60 Ω between the terminals shown.
Ensure you label the values of the resistances in your diagram.

[1]

(ii) A potential divider circuit is produced by connecting the three resistors to a battery of electromotive force (e.m.f.) 12 V and negligible internal resistance. The potential divider circuit provides an output potential difference V_{OUT} of 8.0 V.
Fig. 5.2 shows the circuit diagram.

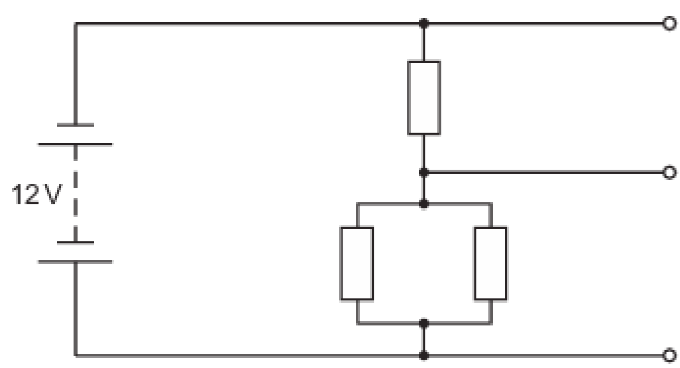


Fig. 5.2

On Fig. 5.2, label the resistances of all three resistors and the potential difference V_{OUT}. [2]

35. 9702/22/M/J/20/Q6

(a) A battery of electromotive force (e.m.f.) 7.8 V and internal resistance r is connected to a filament lamp, as shown in Fig. 6.1.

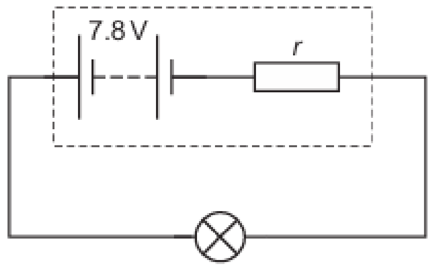


Fig. 6.1

A total charge of 750 C moves through the battery in a time interval of 1500 s. During this time the filament lamp dissipates 5.7 kJ of energy. The e.m.f. of the battery remains constant.

(i) Explain, in terms of energy and without a calculation, why the potential difference across the lamp must be less than the e.m.f. of the battery.

....

.... [1]

(ii) Calculate:

1. the current in the circuit

current = ... A [2]

2. the potential difference across the lamp

potential difference = ... V [2]

3. the internal resistance of the battery.

internal resistance = ... Ω [2]

(b) A student is provided with three resistors of resistances 90 Ω, 45 Ω and 20 Ω.

(i) Sketch a circuit diagram showing how **two** of these three resistors may be connected together to give a combined resistance of 30 Ω between the terminals shown. Label the values of the resistances on your diagram.

[1]

(ii) A potential divider circuit is produced by connecting the three resistors to a battery of e.m.f. 9.0 V and negligible internal resistance. The potential divider circuit provides an output potential difference V_{OUT} of 3.6 V. The circuit diagram is shown in Fig. 6.2.

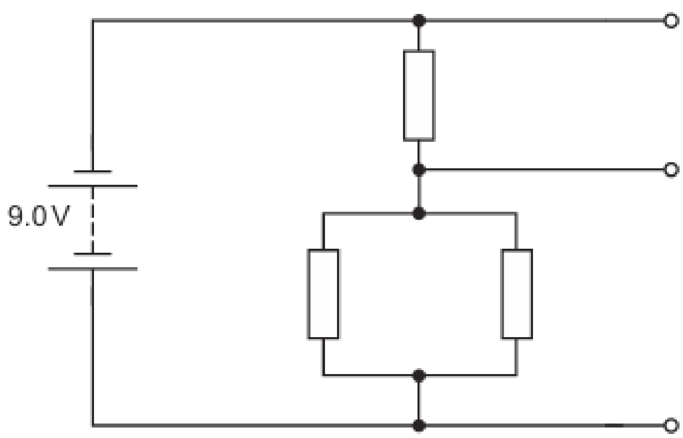


Fig. 6.2

On Fig. 6.2, label the resistances of all three resistors and the potential difference V_{OUT}.
[2]

[Total: 10]

36. 9702/21/O/N/20/Q7

(a) Define the *ohm*.

....

.... [1]

(b) A uniform wire has resistance 3.2Ω. The wire has length 2.5m and is made from metal of resistivity 460nΩm.

Calculate the cross-sectional area of the wire.

cross-sectional area = ... m^2 [3]

(c) A cell of electromotive force (e.m.f.) *E* and internal resistance *r* is connected to a variable resistor of resistance *R*, as shown in Fig. 7.1.

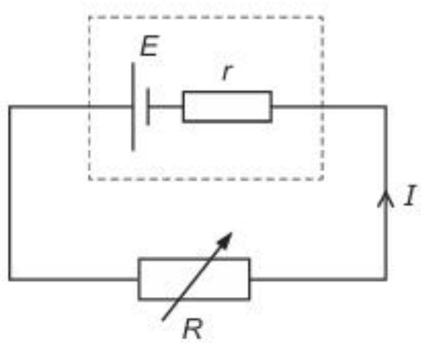


Fig. 7.1

The current in the circuit is I.

(i) State, in terms of energy, why the potential difference across the variable resistor is less than the e.m.f. of the cell.

....

.... [1]

(ii) State an expression for E in terms of I, R and r.

E = ... [1]

(iii) The resistance R of the variable resistor is changed so that it is equal to r.

Determine an expression, in terms of only E and r, for the power P dissipated in the variable resistor.

P = ... [2]

[Total: 8]

37. 9702/23/O/N/20/Q6

(a) Define *electric potential difference* (*p.d.*).

....

.... [1]

(b) A wire of cross-sectional area A is made from metal of resistivity ρ. The wire is extended. Assume that the volume V of the wire remains constant as it extends.

Show that the resistance R of the extending wire is inversely proportional to A^2.

[2]

(c) A battery of electromotive force (e.m.f.) E and internal resistance r is connected to a variable resistor of resistance R, as shown in Fig. 6.1.

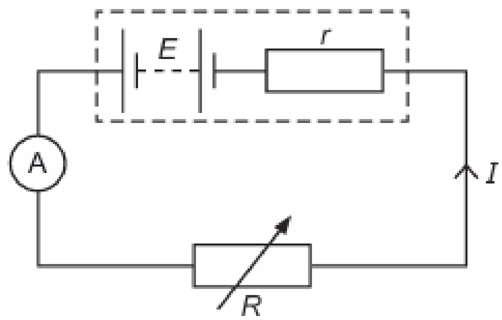


Fig. 6.1

The current in the circuit is I.

Use Kirchhoff's second law to show that

$$R = \left(\frac{E}{I}\right) - r.$$

[1]

(d) An ammeter is used in the circuit in **(c)** to measure the current I as resistance R is varied. Fig. 6.2 is a graph of R against $\frac{1}{I}$.

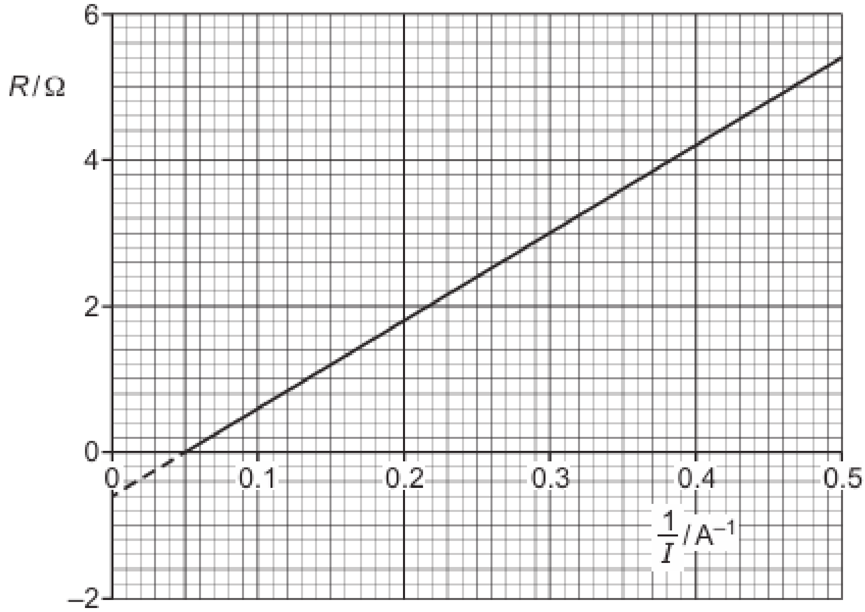


Fig. 6.2

(i) Use Fig. 6.2 to determine the power dissipated in the variable resistor when there is a current of 2.0 A in the circuit.

power = ... W [3]

(ii) Use Fig. 6.2 and the equation in **(c)** to:

1. state the internal resistance r of the battery

r = ... Ω

2. determine the e.m.f. E of the battery.

E = ... V
[3]

38. 9702/22/F/M/21/Q6

(a) State Kirchhoff's first law.

....

.... [1]

(b) A battery of electromotive force (e.m.f.) 12.0 V and internal resistance *r* is connected to a filament lamp and a resistor, as shown in Fig. 6.1.

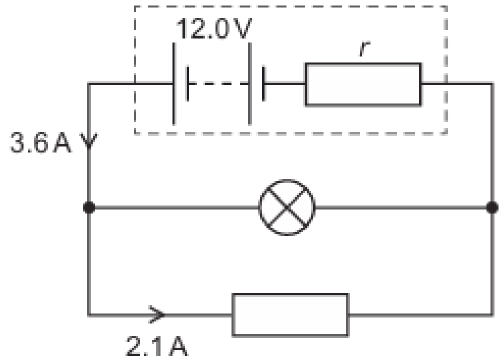


Fig. 6.1

The current in the battery is 3.6 A and the current in the resistor is 2.1 A. The I-V characteristic for the lamp is shown in Fig. 6.2.

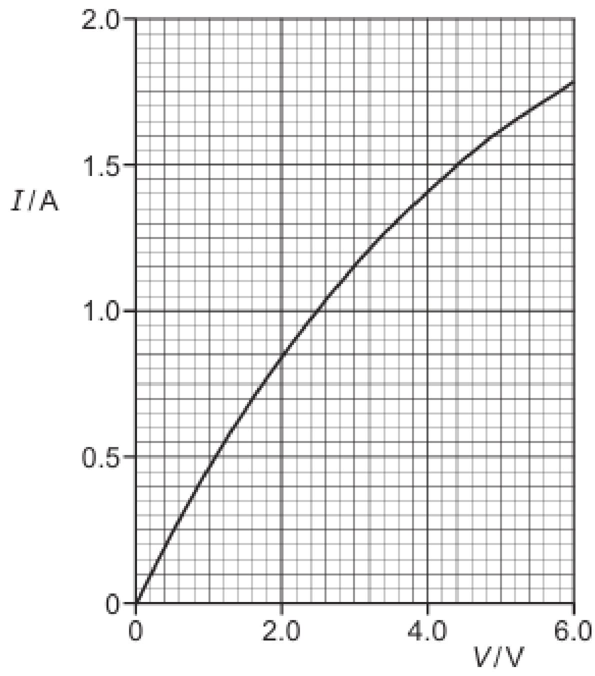


Fig. 6.2

(i) Determine the resistance of the lamp in Fig. 6.1.

resistance = ... Ω [3]

(ii) Determine the internal resistance r of the battery.

r = ... Ω [2]

(iii) The initial energy stored in the battery is 470 kJ. Assume that the e.m.f. and the current in the battery do not change with time.

Calculate the time taken for the energy stored in the battery to become 240 kJ.

time = ... s [2]

(iv) The filament wire of the lamp is connected in series with the adjacent copper connecting wire of the circuit, as illustrated in Fig. 6.3.

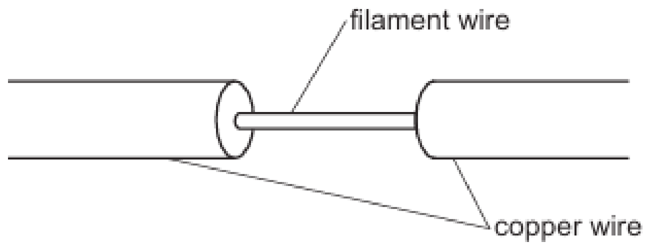


Fig. 6.3 (not to scale)

Some data for the filament wire and the adjacent copper connecting wire are given in Table 6.1.

Table 6.1

	filament wire	copper wire
cross-sectional area	A	$360A$
number density of free electrons	n	$2.5n$

Calculate the ratio

$$\frac{\text{average drift speed of free electrons in filament wire}}{\text{average drift speed of free electrons in copper wire}}.$$

ratio = ... [2]

[Total: 10]

39. 9702/21/M/J/21/Q5

(a) State Kirchhoff's second law.

....

....

.... [2]

(b) A battery has electromotive force (e.m.f.) 4.0 V and internal resistance 0.35 Ω. The battery is connected to a uniform resistance wire XY and a fixed resistor of resistance *R*, as shown in Fig. 5.1.

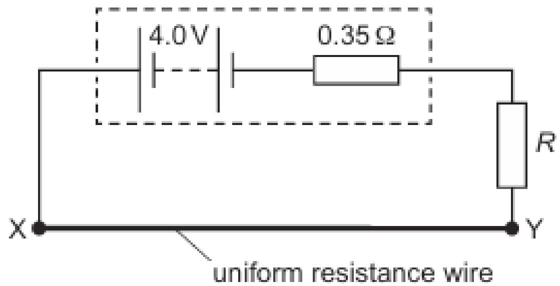


Fig. 5.1

Wire XY has resistance 0.90 Ω. The potential difference across wire XY is 1.8 V.

Calculate:

(i) the current in wire XY

current = ... A [1]

(ii) the number of free electrons that pass a point in the battery in a time of 45 s

number = ... [2]

(iii) resistance *R*.

R = ... Ω [2]

(c) A cell of e.m.f. 1.2 V is connected to the circuit in **(b)**, as shown in Fig. 5.2.

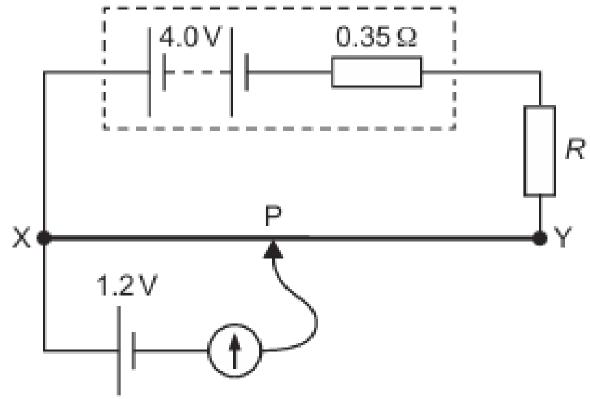


Fig. 5.2

The connection P is moved along the wire XY. The galvanometer reading is zero when distance XP is 0.30 m.

(i) Calculate the total length *L* of wire XY.

L = ... m [2]

(ii) The fixed resistor is replaced by a different fixed resistor of resistance greater than *R*.

State and explain the change, if any, that must be made to the position of P on wire XY so that the galvanometer reading is zero.

....

....

....

.... [2]

[Total: 11]

40. 9702/23/M/J/21/Q5

(a) Define the *electromotive force* (*e.m.f.*) of a source.

....

....

.... [2]

(b) The circuit shown in Fig. 5.1 contains a battery of e.m.f. E that has internal resistance r, a variable resistor, a voltmeter and an ammeter.

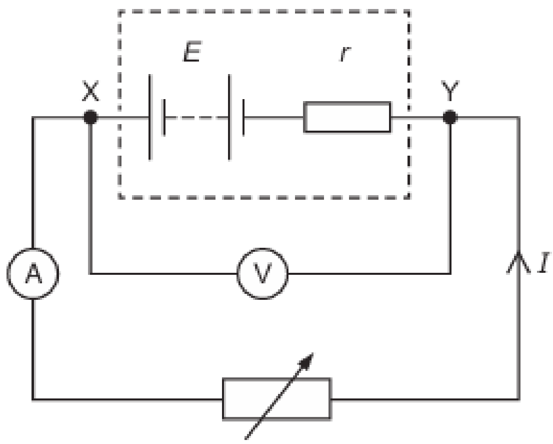


Fig. 5.1

Readings from the two meters are taken for different settings of the variable resistor. The variation with current I of the potential difference (p.d.) V across the terminals XY of the battery is shown in Fig. 5.2.

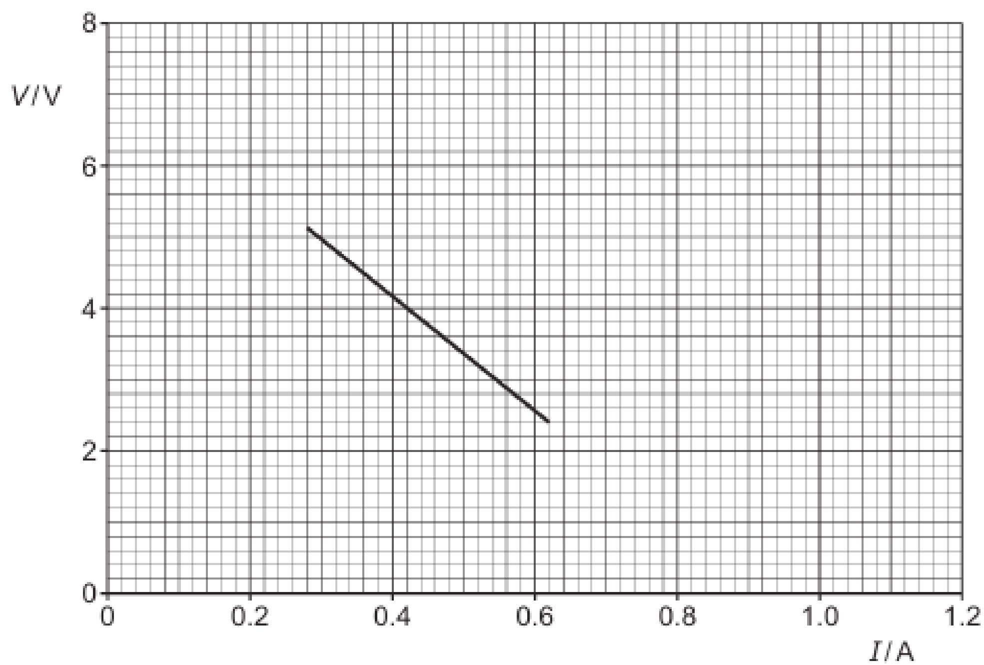


Fig. 5.2

Explain why V is not constant.

....

....

....

.... [3]

(c) For the battery in **(b)**, use Fig. 5.2 to determine:

(i) the e.m.f. E

E = ... V [1]

(ii) the maximum current that the battery can supply

maximum current = ... A [1]

(iii) the internal resistance r.

r = ... Ω [2]

(d) On Fig. 5.2, sketch a line to show a possible variation with I of V for a battery with a lower e.m.f. and a lower internal resistance than the battery in **(b)**. Your line should extend over at least the same range of currents as the original line. [2]

[Total: 11]

41. 9702/21/O/N/21/Q5

(a) State Kirchhoff's first law.

....

....

.... [2]

(b) The circuit shown in Fig. 5.1 contains a battery of electromotive force (e.m.f.) E and negligible internal resistance connected to four resistors R_1, R_2, R_3 and R_4, each of resistance R.

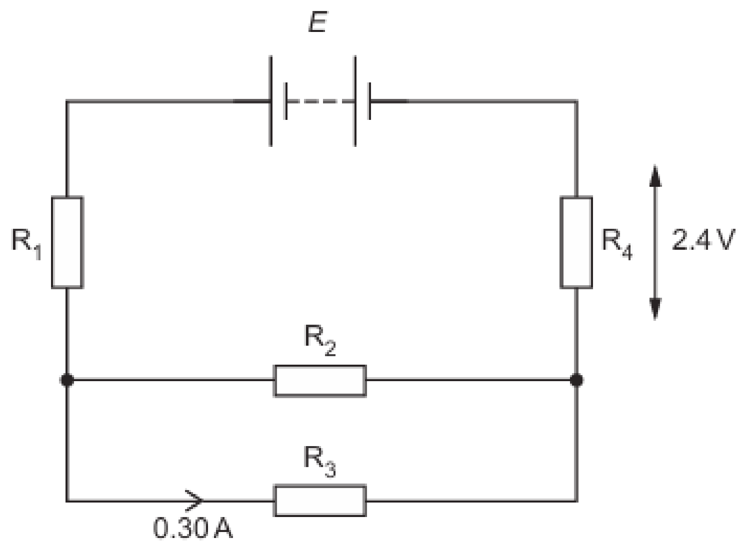


Fig. 5.1

The current in R_3 is 0.30 A and the potential difference (p.d.) across R_4 is 2.4 V.

(i) Show that R is equal to 4.0 Ω.

[2]

(ii) Determine the e.m.f. E of the battery.

E = ... V [2]

(c) The battery in **(b)** is replaced with another battery of the same e.m.f. *E* but with an internal resistance that is not negligible.

State and explain the change, if any, in the total power produced by the battery.

....

....

.... [2]

(d) The resistors in the circuit of Fig. 5.1 are made from nichrome wire of uniform radius 240 μm. The length of this wire needed to make each resistor is 0.67 m.

Calculate the resistivity of nichrome.

resistivity = ... Ω m [3]

[Total: 11]

42. 9702/23/O/N/21/Q6

(a) A resistance wire of uniform cross-sectional area $3.3 \times 10^{-7}\,m^2$ and length 2.0 m is made of metal of resistivity $5.0 \times 10^{-7}\,\Omega\,m$.

Show that the resistance of the wire is 3.0 Ω.

[2]

(b) The ends of the resistance wire in **(a)** are connected to the terminals X and Y in the circuit shown in Fig. 6.1.

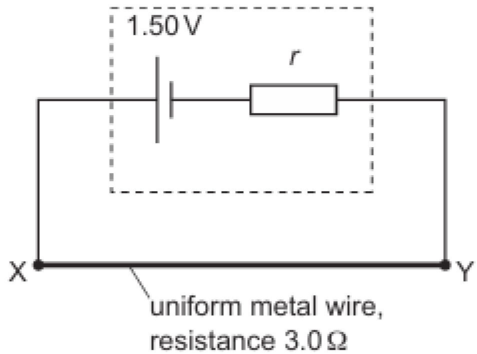


Fig. 6.1

The cell has an electromotive force (e.m.f.) of 1.50 V and internal resistance *r*. The potential difference between X and Y is 1.20 V.

Calculate:

(i) the current in the circuit

current = ... A [1]

(ii) the internal resistance *r*.

r = ... Ω [2]

(c) A galvanometer and a cell of e.m.f. E with negligible internal resistance are connected to the circuit in **(b)**, as shown in Fig. 6.2.

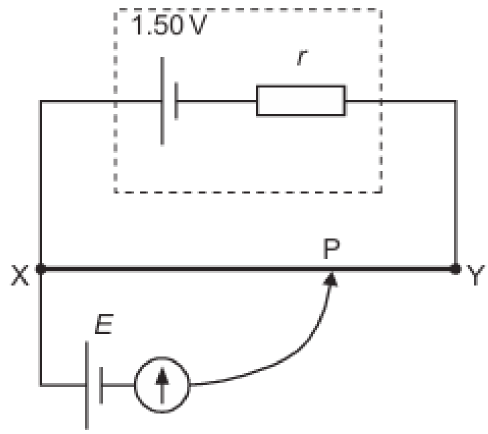


Fig. 6.2

The resistance wire between X and Y has a length of 2.0 m. The galvanometer has a reading of zero when the connection P is adjusted so that the length XP is 1.4 m.

Determine the e.m.f. E of the cell.

E = ... V [2]

(d) The circuit in Fig. 6.2 is modified by replacing the original resistance wire with a second resistance wire. The second wire has the same length as the original wire and is made of the same metal.

The second wire has a smaller cross-sectional area than the original wire.

Connection P is adjusted on the second wire so that the galvanometer has a reading of zero.

State and explain whether length XP for the second wire is shorter than, longer than or the same as length XP for the original wire when the galvanometer reading is zero.

....

....

....

....

.... [3]

43. 9702/21/M/J/22/Q6

(a) State Kirchhoff's first law.

....

.... [1]

(b) A battery is connected to two resistors X and Y, as shown in Fig. 6.1.

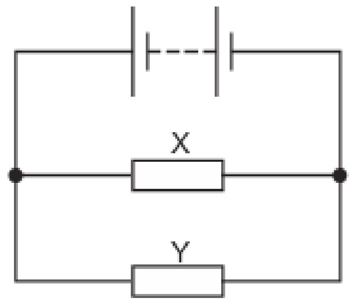


Fig. 6.1

The resistance of resistor X is greater than the resistance of resistor Y.

(i) State and explain which resistor dissipates more power.

....

....

....

....

.... [3]

(ii) The two resistors are made of wires that have the same length. Both wires are made from metal of the same resistivity.

State and explain which resistor is made of wire with the larger cross-sectional area.

....

....

.... [2]

(c) A battery of electromotive force (e.m.f.) 9.0 V and negligible internal resistance is connected in series with a light-dependent resistor (LDR) and a fixed resistor of resistance 1800 Ω, as shown in Fig. 6.2.

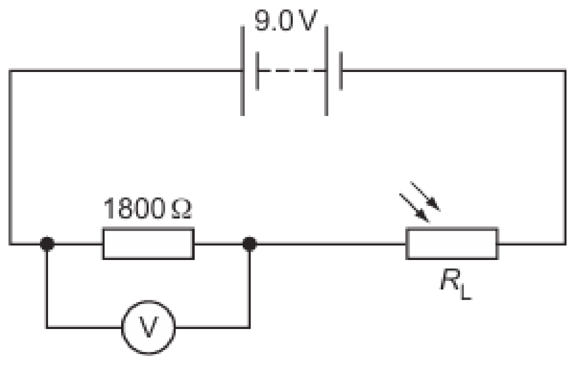


Fig. 6.2

A voltmeter is connected across the fixed resistor. The reading on the voltmeter is 5.4 V.

(i) Calculate the current in the circuit.

current = ... A [1]

(ii) Calculate the resistance R_L of the LDR.

R_L = ... Ω [2]

(iii) The intensity of the light illuminating the LDR increases.

By reference to the current in the circuit, state and explain the change, if any, to the voltmeter reading.

....

....

....

.... [2]

[Total: 11]

44. 9702/21/O/N/22/Q5

(a) State Ohm's law.

....

....

.... [2]

(b) The variation of current I with potential difference V for a filament lamp is shown in Fig. 5.1.

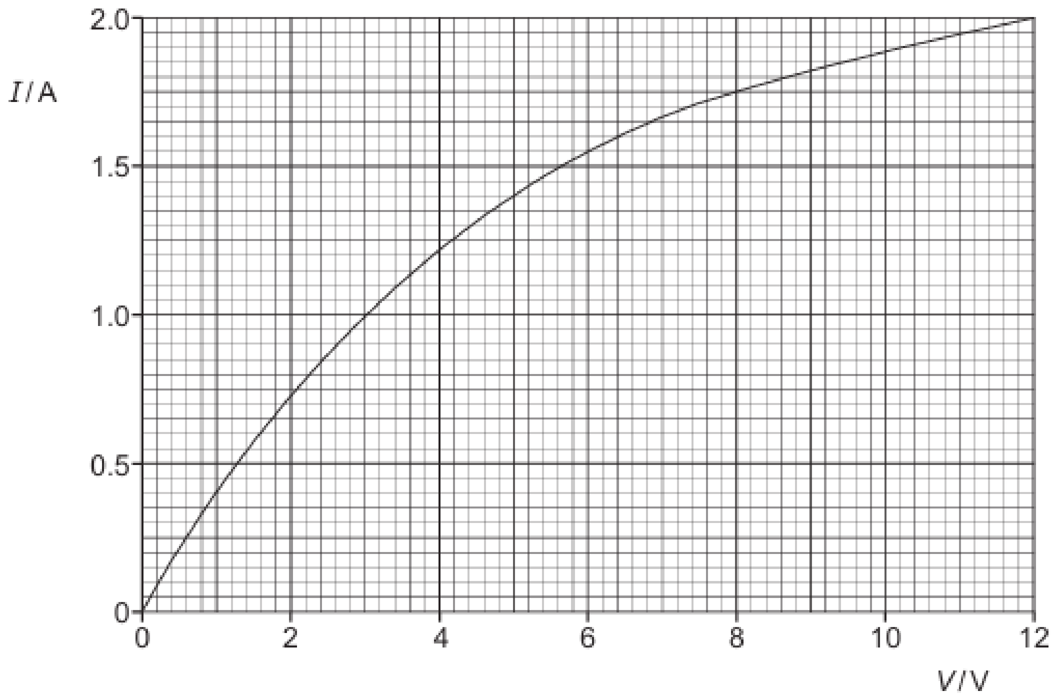


Fig. 5.1

The resistance of the filament lamp increases with potential difference.

(i) State how Fig. 5.1 shows this.

....

.... [1]

(ii) Explain why the resistance varies in this way.

....

.... [1]

(c) Fig. 5.2 shows a circuit with a battery of electromotive force (e.m.f.) 12.0V connected to a linear potentiometer AB and two identical filament lamps P and Q.

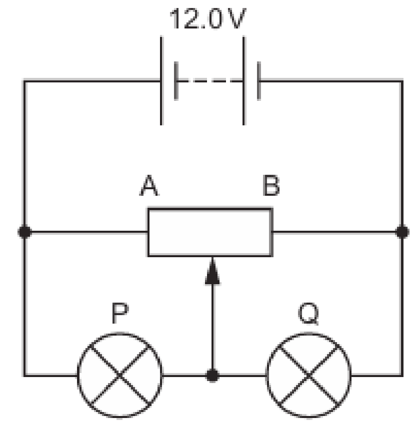


Fig. 5.2

The battery has negligible internal resistance and the lamps each have the same I–V characteristic shown in Fig. 5.1.

When the slider of the potentiometer is at its midpoint, as shown in Fig. 5.2, the current I in the battery is 1.78A.

Determine:

(i) the current in lamp P

current = ... A [1]

(ii) the total power dissipated in lamps P and Q

total power = ... W [2]

(iii) the resistance of the potentiometer between its ends A and B.

resistance = ... Ω [2]

(d) The slider of the potentiometer in **(c)** is moved to end A.

State and explain the effect on the brightness of lamps P and Q.

lamp P: ....

....

lamp Q: ....

....
[2]

[Total: 11]

45. 9702/23/O/N/22/Q5

(a) State Kirchhoff's second law.

....

....

.... [2]

(b) Three identical cells, each of electromotive force (e.m.f.) 1.5 V and internal resistance 590 mΩ, are connected in parallel across a conductor, as shown in Fig. 5.1.

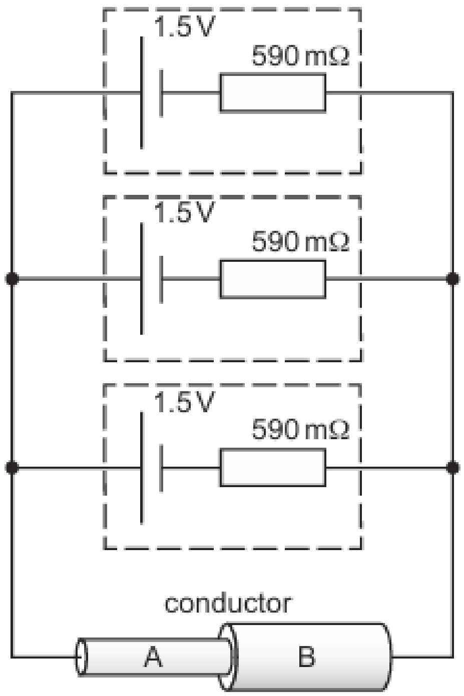


Fig. 5.1

The conductor is composed of two cylindrical sections A and B.
The total resistance of the circuit is 2.2 Ω.

(i) Show that the resistance of the conductor is 2.0 Ω.

[2]

(ii) Calculate the current in the conductor.

current = ... A [2]

(c) The two cylindrical sections A and B of the conductor in Fig. 5.1 are made from the same material and have the same length.
The diameter of section A is 4.3 mm and the diameter of section B is 7.6 mm.
The resistance of section A is R_A and the resistance of section B is R_B.

(i) Calculate the ratio $\dfrac{R_A}{R_B}$.

$\dfrac{R_A}{R_B}$ = ... [3]

(ii) Calculate the ratio

$$\frac{\text{average drift speed of free electrons in section A}}{\text{average drift speed of free electrons in section B}}.$$

Explain your reasoning.

ratio = ... [2]

(d) The circuit of Fig. 5.1 is altered by removing one of the cells.

State and explain the effect, if any, of this change on the potential difference across the conductor.

....

....

....

....

.... [3]

[Total: 14]

46. 9702/21/M/J/23/Q7

A battery of electromotive force (e.m.f.) 9.6 V and negligible internal resistance is connected in series with two fixed resistors and a thermistor, as shown in Fig. 7.1.

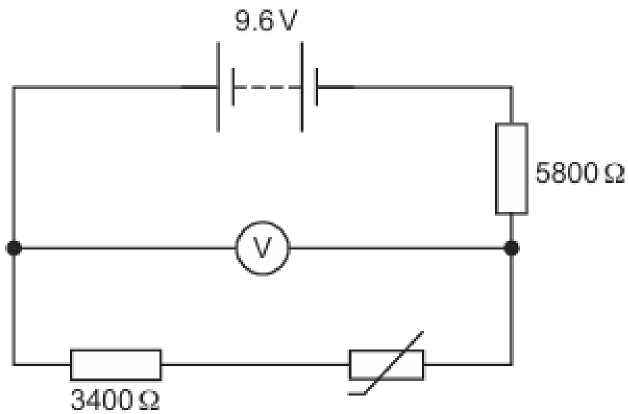


Fig. 7.1

The fixed resistors have resistances of 3400 Ω and 5800 Ω. The reading on the voltmeter in the circuit is 6.0 V.

(a) Calculate the current in the resistor of resistance 5800 Ω.

current = ... A [2]

(b) Calculate the resistance of the thermistor.

resistance = ... Ω [2]

(c) The initial energy stored in the battery is 2.6×10^4 J.

Assume that the e.m.f. of the battery is constant.

Determine the final energy stored in the battery after a charge of 330 C has moved through it.

final stored energy = ... J [2]

(d) The environmental conditions change causing an increase in the resistance of the thermistor.

State whether there is a decrease, increase or no change to:

(i) the temperature of the thermistor

.... [1]

(ii) the current in the thermistor

.... [1]

(iii) the potential difference across the thermistor.

.... [1]

[Total: 9]

47. 9702/22/M/J/23/Q7

(a) A battery of electromotive force (e.m.f.) 9.0 V and negligible internal resistance is connected to a light-dependent resistor (LDR) and a fixed resistor, as shown in Fig. 7.1.

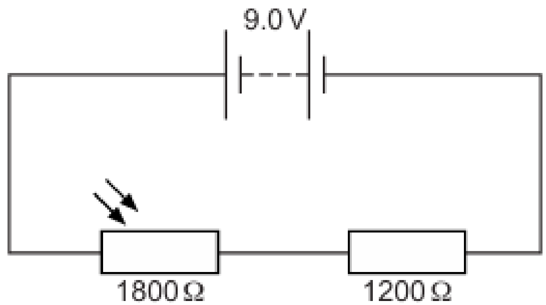


Fig. 7.1

The LDR and fixed resistor have resistances of 1800 Ω and 1200 Ω respectively.

Calculate the potential difference across the LDR.

potential difference = ... V [2]

(b) The circuit in **(a)** is now modified by adding a uniform resistance wire XY and a galvanometer, as shown in Fig. 7.2.

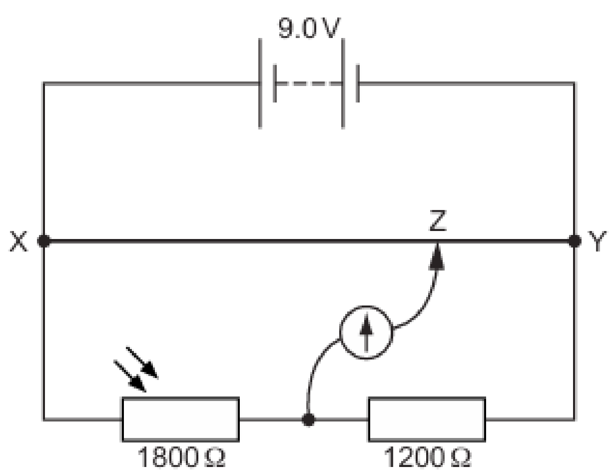


Fig. 7.2 (not to scale)

The length of the wire XY is 1.2 m. The movable connection Z is positioned on the wire XY so that the galvanometer reading is zero.

(i) Calculate the length XZ along the resistance wire.

length XZ = ... m [2]

(ii) The environmental conditions change causing a decrease in the resistance of the LDR. The temperature of the LDR remains constant.

State whether there is a decrease, increase or no change to:

- the intensity of the light illuminating the LDR

....

- the total power produced by the battery

....

- the length XZ so that the galvanometer reads zero.

....

[3]

[Total: 7]

ANSWERS

1.

(a) **(i)** R B1

(ii) $0.5R$ B1

(iii) $2.5R$ …(*allow e.c.f. from **(ii)***) B1

(b) **(i)** $I_1 + I_2 = I_3$ B1

(ii) $E_2 = I_3R + I_2R$ B1

(iii) $E_1 - E_2 = 2I_1R - I_2R$ B1

2.

(a) energy transferred from source / changed from some form to electrical M1
per unit charge (to drive charge round a complete circuit) A1

(b) <u>and</u> power in R = I^2X M1
$E = I(X + r)$ M1
power in cell = EI <u>and</u> algebra clear leading to ratio = $X / (X + r)$ A1

(c) **(i)** 1.4 W A1
0.40 Ω ………(*allow ±0.05* Ω) A1

(ii) current in circuit = $\sqrt{1.4/0.4}$ = 1.87 A C1
1.5 = 1.87 (r + 0.40) C1
r = 0.40 Ω ……….. A1

(d) *either* less power lost / energy wasted / lost
or greater efficiency (of energy transfer) B1

3.

(a) total resistance in series = $2R$
total resistance in parallel = $½R$ M1
ratio is 2R / ½R = 4 ......(*allow mark if clear numbers in the ratio*) A0

(b) at 1.5 V, current is 0.10 A C1

resistance = $V/I = \frac{1.5}{0.1}$

= 15 Ω A1

(*use of tangent or any other current scores no marks*)

(c)

	p.d. across each lamp / V	resistance of each lamp / Ω	combined resistance / Ω
series parallel	1.5 3.0	15 20	30 10

column 1 .... A1
columns 2 and 3: max 3 marks with -1 mark for each error or omission A3

(d) **(i)** ratio is 3 ....(allow e.c.f.) A1

(ii) resistance increases as potential difference increases B1
increasing p.d. increases current B1
current increases non-linearly so resistance increases B1

4.

(a) *either* $P \propto V^2$ *or* $P = V^2/R$ C1
reduction = $(230^2 - 220^2)/230^2$
= 8.5 % A1

(b) **(i)** zero A1

(ii) 0.3(0) A A1

(c) **(i)** correct plots to within ± 1 mm B1

(ii) <u>reasonable line/curve</u> through points giving current as 0.12 A
allow ± 0.005A) B1

(iii) $V = IR$ C1
$V = 0.12 \times 5.0$
= 0.6(0) V A1

(d) circuit acts as a potential divider/current divides/current in AC not the same as current in BC …………………………………………………………………… B1
resistance between A and C not equal to resistance between C and B ……. B1
or current in wire AC × R is not equal to current in wire BC × R B1
any 2 statements

5.
(a) **(i)** *either* $P = V^2 / R$ *or* $P = VI$ <u>and</u> $V = IR$ C1
$R = 4.0\ \Omega$ A1

(ii) sketch vertical axis labelled appropriately B1
(straight) line from origin then curved in correct direction B1
line passes through 12 V, 3.0 A B1

(b) **(i)** 2.0 kW A1

(ii) 0.5 kW A1

(iii) total resistance = $3R / 2$ C1
power = 0.67 kW A1

6.
(a) **(i)** $I = 12 / (6 + 12)$ C1
minimum current = 0.67 A A1

(ii) correct start and finish points M1
correct shape for curve with decreasing gradient A1

(b) maximum current = 2.0 A A1
minimum current = 0 A1

(c) **(i)** smooth curve starting at (0,0) with decreasing gradient M1
end section not horizontal A1

(ii) full range of current / p.d. possible
or currents / p.d. down to zero
or brightness ranging from off to full brightness B1

7.

(a) (i) energy converted from chemical to electrical when charge flows through cell or round <u>complete</u> circuit — B1

(ii) (resistance of the cell) causing loss of voltage or energy loss in cell — B1

(b) (i) $E_B - E_A = I\,(R + r_B + r_A)$

$12 - 3 = I\,(3.3 + 0.1 + 0.2)$ — C1

$I = 2.5$ A — A1

(ii) Power $= E \times I$

$= 12 \times 2.5$ — C1

$= 30$ W — A1

(iii) $P = I^2 \times R$ or $P = V^2 / R$ or $P = VI$

$= (2.5)^2 \times 3$ $\quad = 9^2 / 3.6$ $\quad = 9 \times 2.5$ — C1

$= 22.5\ \mathrm{J\,s^{-1}}$ — A1

(c) power supplied from cell B is greater than energy lost per second in circuit — B1

8.

(a) (i) $I_1 + I_3 = I_2$ — A1

(ii) $E_1 = I_2\dfrac{R_2}{2} + I_1\dfrac{R_2}{2} + I_1R_1 + I_1r_1$ — A1

(iii) $E_1 - E_2$ — B1

$= -I_3r_2 + I_1\,(R_1 + r_1 + R_2 / 2)$ — B1

(b) p.d. across <u>BJ</u> of wire changes / resistance of <u>BJ</u> changes — B1

there is a difference in p.d across wire and p.d. across cell E_2 — B1

9.

(a) (i) sum of e.m.f.'s = sum of p.d.'s around a loop/circuit — B1

(ii) energy — B1

(b) (i) $2.0 = I \times (4.0 + 2.5 + 0.5)$ — C1

$I = 0.286$ A (*allow 2 s.f.*) — A1

(*If total resistance is not 7Ω, 0/2 marks*)

(ii) $R = [0.90 / 1.0] \times 4\ (= 3.6)$ — C1

$V = I\,R = 0.286 \times 3.6 = 1.03$ V — A1

(*If factor of 0.9 not used, then 0/2 marks*)

(iii)	$E = 1.03\,\text{V}$	A1
(iv)	*either* no current through cell B *or* p.d. across r is zero	B1

10.

(a)	total resistance = 20 (kΩ)	C1
	current = 12 / 20 (mA) or potential divider formula	C1
	p.d. = [12 / 20] × 12 = 7.2 V	A1
(b)	parallel resistance = 3 (kΩ)	C1
	total resistance 8 + 3 = 11 (kΩ)	C1
	current = $12 / 11 \times 10^{3} = 1.09 \times 10^{-3}$ or 1.1×10^{-3} A	A1
(c) (i)	LDR resistance decreases	M1
	total resistance (of circuit) is less hence current increases	A1
(ii)	resistance across XY is less	M1
	less proportion of 12 V across XY hence p.d. is less	A1

11.

(a) (i)	sum of currents into a junction = sum of currents out of junction	B1
(ii)	charge	B1
(b) (i)	$\Sigma E = \Sigma IR$	
	$20 - 12 = 2.0(0.6 + R)$ (*not used 3 resistors 0/2)*	C1
	$R = 3.4\,\Omega$	A1
(ii)	$P = EI$	C1
	= 20 × 2	
	= 40 W	A1
(iii)	$P = I^2R$	C1
	$P = (2)^2 \times (0.1 + 0.5 + 3.4)$	
	= 16 W	A1
(iv)	efficiency = useful power / output power	C1
	24 / 40 = 0.6 or 12 × 2 / 20 × 2 or 60%	A1

12.

(a) (i)	$I_1 = I_2 + I_3$		B1
(ii)	$I = V / R$	or $I_2 = 12 / 10$ (= 1.2 A)	C1
	$R = [1/6 + 1/10]^{-1}$ [total $R = 3.75\ \Omega$]	or $I_3 = 12 / 6$ (= 2.0 A)	C1
	$I_1 = 12 / 3.75 = 3.2$ A	or $I_1 = 1.2 + 2.0 = 3.2$ A	A1
(iii)	power = VI or I^2R or V^2 / R		C1
	$x = \dfrac{\text{power in wire}}{\text{power in series resistors}} = \dfrac{I_2^2 R_w}{I_3^2 R_s}$ or $\dfrac{VI_2}{VI_3}$ or $\dfrac{V^2/R_w}{V^2/R_s}$		C1
	$x = 12 \times 1.2 / 12 \times 2.0 = 0.6(0)$ allow 3 / 5 or 3:5		A1
(b)	p.d. BC: 12 – 12 × 0.4 = 7.2 (V) / p.d. AC = 4.8 (V)		C1
	p.d. BD: 12 – 12 × 4 / 6 = 4.0 (V) / p.d. AD = 8.0 (V)		C1
	p.d. = 3.2 V		A1

13.

(a)	e.m.f. = chemical energy to electrical energy	M1
	p.d. = electrical energy to thermal energy	M1
	idea of per unit charge	A1
(b)	$E = I(R + r)$ or $I = E / (R + r)$ (any subject)	B1
(c) (i)	$E = 5.8$ V	B1
(ii)	evidence of gradient calculation or calculation with values from graph	
	e.g. $5.8 = 4 + 1.0 \times r$	C1
	$r = 1.8\ \Omega$	A1
(d) (i)	$P = VI$	C1
	$P = 2.9 \times 1.6 = 4.6$ (4.64) W	A1
(ii)	power from battery = 1.6 × 5.8 = 9.28 or efficiency = VI / EI	C1
	efficiency = (4.64 / 9.28) × 100 = 50 % or (2.9 / 5.8) × 100 = 50%	A1

14.

(a) p.d. = $\dfrac{\underline{\text{work done / energy transformed}}}{\text{charge}}$ (from electrical to other forms) B1

(b) **(i)** maximum 20 V A1

(ii) minimum = (600 / 1000) × 20 C1
= 12 V A1

(c) **(i)** use of 1.2 kΩ M1
$1/1200 + 1/600 = 1/R$, $R = 400\,\Omega$ A1

(ii) total parallel resistance (R_2 + LDR) is less than R_2 M1
(minimum) p.d. is reduced A1

15.

(a) there are no lost volts/energy lost in the battery
or there are no lost volts/energy lost in the internal resistance B1

(b) the current/I decreases (as R increases) M1
p.d. decreases (as R increases) A1

or

the parallel resistance (of X and R) increases M1
p.d. across parallel resistors increases, so p.d. (across Y) decreases A1

(c) **(i)** current = 2.4 (A) C1
p.d. across AB = $24 - 2.4 \times 6 = 9.6$ V M1

or

total resistance = 10 Ω (= 24 V/2.4 A) C1
(parallel resistance = 4 Ω), p.d. = $24 \times (4/10) = 9.6$ V M1

(ii) R (AB) = 9.6/2.4 = 4.0 Ω C1
$1/6 + 1/X = 1/4$ [must correctly substitute for R] C1
$X = 12\,\Omega$ A1

or

$I_R = 9.6/6.0 = 1.6$ (A) (C1)
$I_X = 2.4 - 1.6 = 0.8$ (A) (C1)
X (= 9.6/0.8) = 12 Ω (A1)

(iii) power $= VI$ *or* EI *or* V^2/R *or* E^2/R *or* I^2R C1
$= 24 \times 2.4$ *or* $(24)^2/10$ *or* $(2.4)^2 \times 10$
$= 57.6\,W$ (allow 2 or more s.f.) A1

(d) power decreases M0

e.m.f. constant or power $= 24 \times$ current, and current decreases
or e.m.f. constant or power $= 24^2$/resistance, and resistance increases A1

16.
(a) (i) in series $2X$ or in parallel $X/2$ M1
other relationship given and $4\times$ greater in series (than in parallel) A1

(ii) due to the internal resistance B1

total resistance for series circuit is not four times greater than resistance for parallel circuit B1

(iii) 1. $E = I_1(2X + r)$ or $12 = 1.2(2X + r)$ A1

2. $E = I_2(X/2 + r)$ or $12 = 3.0(X/2 + r)$ A1

(iv) $2X + r = 10$ and $X/2 + r = 4$
$X = 4.0\,\Omega$ A1

(b) $P = I^2R$ or V^2/R or VI C1

ratio $= [(1.2)^2 \times 4] / [(1.5)^2 \times 4]$
$= 0.64$ A1

(c) the resistance (of a lamp) changes with V or I B1

V or I is greater in parallel circuit or circuit 2
or V or I is less in series circuit or circuit 1 B1

17.
(a) $R = \rho l / A$ C1

$A = [\pi \times (0.38 \times 10^{-3})^2] / 4$ $(= 0.113 \times 10^{-6}\ m^2)$ C1

$R = (4.5 \times 10^{-7} \times 1.00) / ([\pi \times (0.38 \times 10^{-3})^2] / 4) = 4.0\ (3.97)\,\Omega$ M1

(b) **(i)** $I = V/R$ C1

$= 2.0 / 5.0 = 0.4(0)\,A$ A1

(ii) p.d. across BD $= 4 \times 0.4 = 1.6\,V$ A1

(iii) p.d. across BC (l) $= 1.5\,(V)$ C1

BC (l) $= (1.5 / 1.6) \times 100 = 94\ (93.75)\,cm$ A1

(c) p.d. across wire not balancing e.m.f. of cell OR cell Y has current B1

energy lost or lost volts due to internal resistance B1

18.

(a) curved line showing decreasing gradient with temperature rise M1

smooth line not touching temperature axis, not horizontal or vertical anywhere A1

(b) **(i)** (no energy lost in battery because) no/negligible internal resistance B1

(ii) $I = V/R$

$= 8/15 \times 10^3$ or $1.6/3.0 \times 10^3$ or $2.4/4.5 \times 10^3$ or $12/22.5 \times 10^3$ C1

$= 0.53 \times 10^{-3}\,A$ A1

(iii) p.d. across X $= 12 - 8.0 - 3.0 \times 10^3 \times 0.53 \times 10^{-3}$ ($= 2.4\,V$) C1

$R_X = 2.4/(0.53 \times 10^{-3})$ C1

or

$R_{tot} = 12/0.53 \times 10^{-3}$ ($= 22.5 \times 10^3\,\Omega$) (C1)

$R_X = (22.5 - 15.0 - 3.0) \times 10^3$ (C1)

$4.5(2) \times 10^3\,\Omega$ A1

(iv) resistance decreases hence current (in circuit) is greater M1

p.d. across X and Y is greater hence p.d across Z decreases A1

or explanation in terms of potential divider:

R_Z decreases so $R_Z/(R_X + R_Y + R_Z)$ is less (M1)

therefore p.d. across Z decreases (A1)

19.

(a) $R = \rho l / A$ C1

$= (5.1 \times 10^{-7} \times 0.50) / \pi(0.18 \times 10^{-3})^2 = 2.5\ (2.51)\,\Omega$ M1

(b) (i) resistance of CD = 8 × resistance of AB = 20 (Ω) C1

circuit resistance = $[1/5.0 + 1/20]^{-1} = 4.0\,(\Omega)$ C1

current = V/R = 6.0/4.0 C1

= 1.5 A A1

(ii) power in AB = I^2R or power = V^2/R C1

$= (1.2)^2 \times 2.5 = 3.6\,\text{W}$ $= (3.0)^2/2.5 = 3.6\,\text{W}$ A1

(iii) potential drop A to M = 1.25 × 1.2 = 1.5 V M1

potential drop C to N = 3.0 V
p.d. MN = 1.5 V A1

20.

(a) $\dfrac{\text{work done or energy (transformed) (from electrical to other forms)}}{\text{charge}}$ B1

(b) (i) 1. $V = IR$ *or* $E = IR$ C1

$I = 14/6.0$
$= 2.3\ (2.33)\,\text{A}$ A1

2. total resistance of parallel resistors = 8.0 Ω C1

current = 14/(6.0 + 8.0)
= 1.0 A A1

(ii) $P = EI$ (*allow* $P = VI$) *or* $P = V^2/R$ *or* $P = I^2R$ C1

change in power = (14 × 2.33) – (14 × 1.0)
or $(14^2/6.0) - (14^2/14)$
or $(2.33^2 \times 6.0) - (1.0^2 \times 14)$

= 19 W (18 W if 2.3 A used) A1

(c) $I = Anvq$

ratio $= (0.50n/n) \times (1.8A/A)$ or ratio $= 0.50 \times 1.8$ C1

$= 0.90$ A1

21.

(a) total/sum of electromotive forces or e.m.f.s = total/sum of potential differences or p.d.s M1
around a loop/(closed) circuit A1

(b) (i) (current in battery =) current in A + current in B *or* $I_A + I_B$ C1

(I =) 0.14 + 0.26 = 0.40 A A1

(ii) $E = V + Ir$

$6.8 = 6.0 + 0.40r$ *or* $6.8 = 0.40(15 + r)$ C1

$r = 2.0\ \Omega$ A1

(iii) $R = V/I$ C1

ratio (= R_A/R_B) = (6.0/0.14)/(6.0/0.26)
= 42.9/23.1 *or* 0.26/0.14

= 1.9 (1.86) A1

(iv) 1. $P = EI$ or VI *or* $P = I^2R$ *or* $P = V^2/R$ C1

$= 6.8 \times 0.40$ $= 0.40^2 \times 17$ $= 6.8^2/17$

= 2.7 W (2.72 W) A1

2. output power $= VI$
$= 6.0 \times 0.40$ (= 2.40 W) C1

efficiency $= (6.0 \times 0.40)/(6.8 \times 0.40) = 2.40/2.72$
= 0.88 or 88% (*allow 0.89 or 89%*) A1

22.

6(a)	volt / ampere	**B1**
6(b)(i)	R_T = [1 / 3.0 + 1 / 6.0]$^{-1}$ + 4.0 (= 6.0 Ω)	**C1**
	I = 1.5 / 6.0	**C1**
	= 0.25 A	**A1**
6(b)(ii)	V_B = 0.5 V I = 0.5 / 3.0 = 0.17 (0.167) A	**A1**
6(b)(iii)	$P = I^2R$ **or** VI **or** V^2/R	**C1**
	ratio = $(0.167^2 \times 3.0) / (0.25^2 \times 4.0)$ = 0.33	**A1**
6(c)(i)	vary/change/different radius/diameter/<u>cross-sectional</u> area (of wire)	**B1**
6(c)(ii)	$v = I / Ane$ ratio $= \dfrac{(I_B / A_B)}{(I_C / A_C)}$ **or** $\dfrac{I_B}{I_C} \times \dfrac{A_C}{A_B}$	**C1**
	($R \propto 1 / A$ so) ratio $= \dfrac{I_B}{I_C} \times \dfrac{R_B}{R_C} = \dfrac{0.167}{0.25} \times \dfrac{3.0}{4.0}$ = 0.50	**A1**
6(d)(i)	0.25 A to 0.13 (0.125) A **or** halved	**A1**
6(d)(ii)	no change	**A1**

23.

7(a)	energy transformed from <u>chemical to electrical</u> / unit charge (driven around a complete circuit)	**B1**
7(b)(i)	the current decreases (as resistance of Y increases)	**M1**
	lost volts go down (as resistance of Y increases)	**M1**
	p.d. AB increases (as resistance of Y increases)	**A1**
7(b)(ii)1.	1.50 = 0.180 × (6.00 + 0.200 + R_X)	**C1**
	R_X = 2.1(3) Ω	**A1**
7(b)(ii)2.	p.d. AB = 1.5 – (0.180 × 0.200) **or** 0.18 × (2.13 + 6.00)	**C1**
	= 1.46(4) V	**A1**
7(b)(ii)3.	efficiency = (useful) power output / (total) power input **or** IV / IE	**C1**
	(= 1.46 / 1.5) = 0.97 [0.98 if full figures used]	**A1**

24.

5(a)	$I_1 + I_2 = I_3$ [any subject]	**B1**
5(b)	$E_1 + E_3 = I_1R_1 + I_3R_3 + I_3R_4$ [any subject]	**B1**
5(c)	$E_1 – E_2 = I_1R_1 – I_2R_2$ [any subject]	**B1**

25.

6(a)(i)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out of junction **or** (algebraic) sum of current(s) at a junction is zero	**B1**
6(a)(ii)	charge	**B1**
6(b)(i)1.	$E = I^2Rt$ **or** $E = VIt$ **or** $E = (V^2 / R)t$	**C1**
	$E = 2.5^2 \times 2.0 \times 5.0 \times 60$ **or** $5.0 \times 2.5 \times 5.0 \times 60$ **or** $(5.0^2 / 2.0) \times 5.0 \times 60$ = 3800 J	**A1**
6(b)(i)2.	p.d. = 8.0 – (2.0 × 2.5) = 3.0 V	**A1**
6(b)(ii)	I_X = 3.0 / 15 = 0.20 (A)	**C1**
	I_Y = 2.5 – 0.20 = 2.3 (A)	**C1**
	R_Y = 3.0 / 2.3 = 1.3 Ω	**A1**
	or	
	R_T = 3.0 / 2.5 = 1.2 (Ω) **or** (8.0 / 2.5) – 2.0 = 1.2 (Ω)	**(C1)**
	1 / 1.2 = 1 / 15 + 1 / R_Y	**(C1)**
	R_Y = 1.3 Ω	**(A1)**

6(b)(iii)1.	Z has larger radius/diameter/(cross-sectional) area	**B1**
	Z has (material of) smaller resistivity/greater conductivity	**B1**
6(b)(iii)2.	current/I (in battery) increases	**M1**
	($P = EI$ so) power/P (produced by battery) increases	**A1**

26.

6(a)	<u>sum of</u> e.m.f.(s) equal to <u>sum of</u> p.d.(s)	**M1**
	around a loop/around a closed circuit	**A1**
6(b)(i)	current in variable resistor = (6.0 / 2.4) + (6.0 / 1.2) (= 7.5 A)	**C1**
	p.d. across variable resistor = 9.0 – 6.0 (= 3.0 V)	**C1**
	R = 3.0 / 7.5 = 0.40 Ω	**A1**
	or	
	$\frac{1}{R_T} = \frac{1}{2.4} + \frac{1}{1.2}$ R_T = 0.80 (Ω)	**(C1)**
	$\frac{3}{9} = \frac{R}{(R + 0.80)}$ **or** $\frac{3}{R} = \frac{6}{0.8}$	**(C1)**
	R = 0.40 Ω	**(A1)**
6(b)(ii)	$P = V^2 / R$ **or** $P = I^2R$ **or** $P = IV$	**C1**
	$P = 6.0^2 / 24$ **or** $2.5^2 \times 2.4$ **or** 6.0×2.5 = 15 W	**A1**

6(b)(iii)	**1.** $R = \dfrac{\rho L}{A}$	**C1**
	ratio $= (2.4/1.2) \times (3/1)$ $= 6.0$	**A1**
	2. $(I = nAvq)$ $I_X / I_Y = 2.5/5.0$ **or** $1.2/2.4$ **or** 0.5	**C1**
	ratio $= (2.5/5.0) \times (1/3)$ **or** $(1.2/2.4) \times (1/3)$ $= 0.17$	**A1**

27.

6(a)	joule / coulomb	**B1**
6(b)(i)	$7.0 = (I \times 5.2) + (I \times 6.0) + 1.4$	**C1**
	$I = 0.50\,\text{A}$	**A1**
6(b)(ii)	$R = 1.4/0.50$ $= 2.8\,\Omega$	**A1**
6(b)(iii)	$P = EI$ **or** $P = VI$ **or** $P = I^2R$ **or** $P = V^2/R$	**C1**
	efficiency = $[(0.50^2 \times 6.0)/(7.0 \times 0.50)]\ (\times 100)$ **or** efficiency = $[(0.50 \times 3.0)/(7.0 \times 0.50)]\ (\times 100)$ **or** efficiency = $[(3.0^2/6.0)/(7.0 \times 0.50)]\ (\times 100)$	**C1**
	efficiency = 43%	**A1**
6(b)(iv)	$R = \rho l / A$	**C1**
	$\alpha = \rho / R$ $= 3.7 \times 10^{-7}/6.0$ $= 6.2 \times 10^{-8}\,\text{m}$	**A1**

28.

7(a)	<u>sum of</u> current(s) in(to) junction = <u>sum of</u> current(s) out of junction **or** (algebraic) sum of current(s) at a junction is zero	**B1**
7(b)(i)	**1.** potential difference = 0	**A1**
	2. potential difference = 9.6 V	**A1**
7(b)(ii)	for resistance in parallel: $(1/R_T) = (1/400) + (1/400)$ $R_T = 200\,(\Omega)$	**C1**
	$V/9.6 = 200/600$	**C1**
	$V = 3.2\,\text{V}$	**A1**

29.

6(a)	energy is dissipated in the internal resistance	**B1**
6(b)	$E = V + Ir$	**B1**
6(c)(i)	(graph shows) maximum value of potential difference is 2.8 (V) **or** (graph shows) when current/I (from battery) is zero, V is 2.8 (V) / E	**B1**
6(c)(ii)	r = (–)gradient **or** $r = (E - V) / I$ **or** substituted values from the graph for E, V and I	**C1**
	$r = 1.4\,\Omega$	**A1**
6(d)(i)	$R = 2.1 / 0.50$ $= 4.2\,\Omega$	**A1**
6(d)(ii)	number $= 0.50 / 1.60 \times 10^{-19}$ $= 3.1 \times 10^{18}$	**A1**
6(d)(iii)	energy $= EIt$ **or** $P = EI$ **and** $P = W / t$	**C1**
	$(9.2 - 1.6) \times 10^3 = 2.8 \times 0.50 \times t$	**C1**
	$t = 5.4 \times 10^3$ s	**A1**

30.

5(a)	<u>sum of</u> e.m.f.(s) = <u>sum of</u> p.d.(s)	**M1**
	around a loop/around a closed circuit	**A1**
5(b)(i)	**1.** $1 / R = 1 / R_1 + 1 / R_2$ $1 / R = 1 / 90 + 1 / 18$	**C1**
	$R = 15\,\Omega$	**A1**
	2. $I = V / R$	**C1**
	$I = 4.8 / 15$ **or** $I = 4.8 / 90 + 4.8 / 18$ $I = 0.32$ A	**A1**
5(b)(ii)	$E = V + Ir$ **or** $E = I(R + r)$	**C1**
	$5.6 = 4.8 + 0.32\,r$ so $r = 2.5\ (\Omega)$ **or** $5.6 = 0.32 \times (15 + r)$ so $r = 2.5\ (\Omega)$	**A1**
5(b)(iii)	$P = EI$ **or** $P = VI$ **or** $P = I^2R$ **or** $P = V^2 / R$	**C1**
	ratio $= (0.32^2 \times 2.5) / (5.6 \times 0.32)$ **or** $0.256 / 1.792$	**C1**
	$= 0.14$	**A1**
5(c)	$7.2 - 5.6 - 2.5I - 3.5I = 0$	**C1**
	$I = 0.27$ A	**A1**

31.

6(a)	work done / charge **or** energy (transferred from electrical to other forms) / charge	**B1**
6(b)	for $V < 0.25$ V resistance is infinite/very high (as current is zero)	**B1**
	for $V > 0.25$ V resistance decreases (as V increases)	**B1**
6(c)(i)	$R = V / I$	**C1**
	$= 0.75 / (15 \times 10^{-3})$	**C1**
	$= 50\ \Omega$	**A1**
6(c)(ii)	**1.** $V_Y = 15 \times 10^{-3} \times 60$ (= 0.90 V)	**C1**
	$V_X = 2.0 - 0.90 - 0.75$ (= 0.35 V)	**C1**
	$R_X = 0.35 / (15 \times 10^{-3})$ $= 23\ \Omega$	**A1**
	or	
	total $R = 60 + 50 + R_X$	**(C1)**
	$60 + 50 + R_X = 2.0 / (15 \times 10^{-3})$	**(C1)**
	$R_X = 23\ \Omega$	**(A1)**
	2. $P = VI$ **or** $P = EI$ **or** $P = I^2R$ **or** $P = V^2 / R$	**C1**
	ratio $= \dfrac{(15 \times 10^{-3})^2 \times 60}{2.0 \times 15 \times 10^{-3}}$ **or** $\dfrac{0.90 \times 15 \times 10^{-3}}{2.0 \times 15 \times 10^{-3}}$ **or** $\dfrac{(0.90^2 / 60)}{2.0 \times 15 \times 10^{-3}}$ $= 0.45$	**A1**

32.

6(a)	sum of current(s) into junction = sum of current(s) out of junction **or** (algebraic) sum of current(s) at a junction is zero	**B1**
6(b)(i)	$R = V / I$	**C1**
	$= 0.60 / 7.5 \times 10^{-3}$	**C1**
	$= 80\ \Omega$	**A1**
6(b)(ii)	resistance decreases	**B1**
6(c)(i)	$E = 0.60 + 0.30$ $= 0.90$ V	**A1**
6(c)(ii)	$(I =)\ 9.3 - 7.5$	**C1**
	$I = 1.8$ (mA) **or** 1.8×10^{-3} (A) $R = 0.90 / 1.8 \times 10^{-3}$ $= 500\ \Omega$	**A1**
	or	
	total resistance $= 0.90 / 9.3 \times 10^{-3} = 96.8\ (\Omega)$ total resistance of diode and X $= 0.90 / 7.5 \times 10^{-3} = 120\ (\Omega)$ $1 / 96.8 = 1 / R + 1 / 120$	**(C1)**
	$R = 500\ \Omega$	**(A1)**

6(c)(iii)	$P = VI$ **or** I^2R **or** V^2 / R	**C1**
	$= 0.60 \times 7.5 \times 10^{-3}$ **or** $(7.5 \times 10^{-3})^2 \times 80$ **or** $0.60^2 / 80$ $= 4.5 \times 10^{-3}$ W	**A1**
6(c)(iv)	current = 2.5 mA	**A1**

33.

6(a)(i)	$R = V / I$	**C1**
	resistance = (12 / 0.20) / 2 **or** 6 / 0.20 = 30 Ω	**A1**
6(a)(ii)	I = 0.50 – 0.20 (= 0.30 A)	**C1**
	R + 28 = 12 / 0.30 (= 40 Ω) R = 12 Ω	**A1**
6(b)	p.d. across lamp = 0.20 × 30 (= 6.0 V)	**C1**
	p.d. across R = 0.30 × 12 (= 3.6 V)	**C1**
	V_{XY} = 6.0 – 3.6 = 2.4 V	**A1**
	or	
	p.d. across lamp = 0.20 × 30 (= 6.0 V)	**(C1)**
	p.d. across 28 Ω resistor = 0.30 × 28 (= 8.4 V)	**(C1)**
	V_{XY} = 8.4 – 6.0 = 2.4 V	**(A1)**
6(c)	$P = VI$ **or** $P = EI$ **or** $P = I^2R$ **or** $P = V^2 / R$	**C1**
	ratio = (6.0 × 0.20) × 2 / (12 × 0.50) **or** 0.20 / 0.50 = 0.40	**A1**
6(d)	no change to V across lamps, so power in lamps unchanged **or** current in battery/total current increases (and e.m.f. the same) so power produced by battery increases	**B1**
	both the above statements and so the ratio decreases	**B1**

34.

5(a)(i)	$R = \rho L / A$	**C1**
	$A = (2.6 \times 10^{-8} \times 59) / 3.4 = 4.5 \times 10^{-7}$ m^2	**A1**
5(a)(ii)	$I = 1.8 / 3.4$ $= 0.53$ A	**A1**
5(a)(iii)	$I = Anvq$ $v = 0.53 / (4.5 \times 10^{-7} \times 6.1 \times 10^{28} \times 1.60 \times 10^{-19})$	**C1**
	$= 1.2 \times 10^{-4}$ m s^{-1}	**A1**
5(b)(i)	(cross-sectional) area/*A* is less	**M1**
	(*I*, *n*, *e* the same so) average drift speed is greater	**A1**
5(b)(ii)	(area is less so) more resistance/*R*	**M1**
	(*I* is the same, so) more power/*P*	**A1**
	or	
	($P = I^2\rho L / A$ so) $P \propto 1 / A$	**(M1)**
	(*A* is less so) more *P*	**(A1)**
5(c)(i)	180 Ω and 90 Ω resistors shown connected in parallel	**B1**
5(c)(ii)	resistors connected in parallel labelled as 180 Ω and 90 Ω **and** the other resistor labelled as 30 Ω	**M1**
	V_{OUT} or 8.0 V labelled across the two resistors in parallel	**A1**

35.

6(a)(i)	energy is dissipated in the internal resistance/*r*	**B1**
6(a)(ii)	**1.** $I = Q / t$	**C1**
	$= 750 / 1500$ $= 0.50$ A	**A1**
	2. $V = W / Q$ **or** $V = W / It$	**C1**
	$= 5700 / 750$ **or** $5700 / (0.50 \times 1500)$ $= 7.6$ V	**A1**
	or	
	$V = P / I$ **and** $P = W / t$	**(C1)**
	$V = 3.8 / 0.50$ $= 7.6$ V	**(A1)**
	3. $r = (7.8 - 7.6) / 0.50$	**C1**
	$= 0.40$ Ω	**A1**
6(b)(i)	90 Ω and 45 Ω resistors shown connected in parallel	**B1**
6(b)(ii)	the resistors connected in parallel labelled as 90 Ω and 45 Ω with the other resistor labelled as 20 Ω	**M1**
	V_{OUT} or 3.6 V labelled across the 20 Ω resistor	**A1**

36.

7(a)	volt / ampere	**B1**
7(b)	$R = \rho L / A$	**C1**
	$A = 460 \times 10^{-9} \times 2.5 / 3.2$	**C1**
	$= 3.6 \times 10^{-7}$ m^2	**A1**
7(c)(i)	energy is dissipated in the internal resistance/r	**B1**
7(c)(ii)	$E = IR + Ir$ **or** $E = I(R + r)$	**B1**
7(c)(iii)	$P = I^2R$ **or** $P = I^2r$	**C1**
	$I = E / 2r$ (so) $P = E^2 / 4r$	**A1**

37.

6(a)	$\dfrac{\text{work (done) /energy (transferred from electrical to other forms)}}{\text{charge}}$	**B1**
6(b)	$R = \rho L / A$	**B1**
	$V = LA$ **and** (so) $R = \rho V / A^2$ (with ρ and V constant)	**B1**
6(c)	$E = IR + Ir$ **or** $E = I(R + r)$ **or** $E - Ir = IR$ **and** $R = (E / I) - r$	**A1**
6(d)(i)	$P = I^2R$ **or** $P = IV$ **or** $P = V^2 / R$	**C1**
	$R = 5.4\ (\Omega)$ **or** $V = 10.8$ (V)	**C1**
	$P = 2.0^2 \times 5.4$ $= 22$ W	**A1**
6(d)(ii)	**1.** $r = 0.60\ \Omega$	**A1**
	2. E = gradient	**C1**
	= e.g. 5.4 / 0.45 = 12 V	**A1**

38.

6(a)	<u>sum of</u> current(s) into junction = <u>sum of</u> current(s) out of junction or (algebraic) sum of current(s) at a junction is zero	**B1**
6(b)(i)	$I = 3.6 - 2.1$ $= 1.5$	**C1**
	$V = 4.4$	**C1**
	$R = 4.4 / 1.5$ $= 2.9\ \Omega$	**A1**
6(b)(ii)	$12.0 = 4.4 + 3.6r$ or $12.0 = 3.6\ (1.2 + r)$	**C1**
	$r = 2.1\ \Omega$	**A1**
6(b)(iii)	$t = (470 \times 10^3 - 240 \times 10^3) / (12 \times 3.6)$	**C1**
	$= 5300$ s	**A1**
6(b)(iv)	$I = Anvq$ ratio = $(360A / A) \times (2.5n / n)$ or 360×2.5	**C1**
	$= 900$	**A1**

39.

5(a)	<u>sum of</u> e.m.f.(s) = <u>sum of</u> p.d.(s) **or** (algebraic) sum of e.m.f.(s) and p.d.(s) is zero	**M1**
	around a loop/around a <u>closed</u> circuit	**A1**
5(b)(i)	$I = 1.8 / 0.90$ $= 2.0$ A	**A1**
5(b)(ii)	$Q = It$	**C1**
	number = $(2.0 \times 45) / 1.60 \times 10^{-19}$ $= 5.6 \times 10^{20}$	**A1**
5(b)(iii)	$4.0 = 1.8 + [2.0 \times (0.35 + R)]$ **or** $4.0 = 2.0 \times (0.90 + 0.35 + R)$	**C1**
	$R = 0.75\,\Omega$	**A1**
5(c)(i)	$1.2 / 1.8 = 0.30 / L$	**C1**
	$L = 0.45$ m	**A1**
5(c)(ii)	p.d. across XY decreases/p.d. across XP decreases	**B1**
	(so) P is moved towards Y/away from X/to the right	**B1**

40.

5(a)	energy per unit charge	**B1**
	energy transferred by source driving charge around the complete circuit **or** energy transferred from other forms to electrical energy	**B1**
5(b)	there is a p.d. across the internal resistance/r	**B1**
	change in current/I results in a change in p.d. across the internal resistance	**B1**
	$V = E$ – p.d. across internal resistance **or** change in p.d. across r causes a change in V (as e.m.f. is constant)	**B1**
5(c)(i)	$E = 7.4$ V	**A1**
5(c)(ii)	maximum current = 0.92 A	**A1**
5(c)(iii)	$r = E / I_{MAX}$ **or** (–)gradient	**C1**
	e.g. $r = 7.4 / 0.92$ $= 8.0\,\Omega$	**A1**
5(d)	straight line with negative gradient that is smaller in magnitude than the original line	**B1**
	line which would have intercept on V-axis below the original line	**B1**

41.

5(a)	sum of current(s) in = sum of current(s) out **or** (algebraic) sum of current(s) is zero	**M1**
	at a junction (in a circuit)	**A1**
5(b)(i)	(current in R_4 or R_1 =) 0.30 + 0.30 (= 0.60 A)	**B1**
	(R =) 2.4 / 0.60 = 4.0 (Ω)	**A1**
	or	
	(p.d. across R_3 or R_2 =) 2.4 / 2 (= 1.2 V)	**(B1)**
	(R =) 1.2 / 0.30 = 4.0 (Ω)	**(A1)**
5(b)(ii)	E = 2.4 + 2.4 + 1.2	**C1**
	= 6.0 V	**A1**
	or	
	total resistance = 10 (Ω)	**(C1)**
	E = 10 × 0.60 = 6.0 V	**(A1)**
5(c)	total resistance increases	**B1**
	current decreases (in battery) so total power decreases	**B1**
5(d)	resistivity = RA / L	**C1**
	$= 4.0 \times \pi \times (240 \times 10^{-6})^2 / 0.67$	**C1**
	$= 1.1 \times 10^{-6}$ Ω m	**A1**

42.

6(a)	$R = \rho L / A$	**C1**
	(R =) $5(.0) \times 10^{-7} \times 2(.0) / 3.3 \times 10^{-7} = 3.0$ Ω	**A1**
6(b)(i)	I = 1.2 / 3.0 = 0.40 A	**A1**
6(b)(ii)	r = (1.50 – 1.20) / 0.40 **or** 1.50 / 0.40 – 3.0	**C1**
	= 0.75 Ω	**A1**
6(c)	E / 1.20 = 1.4 / 2.0	**C1**
	E = 0.84 V	**A1**
	or	
	R_{XP} = (1.4 / 2.0) × 3.0 (= 2.1 Ω) E = 2.1 × 0.40	**(C1)**
	E = 0.84 V	**(A1)**
6(d)	(second wire has) larger resistance/resistance increases	**M1**
	p.d. across XY is larger/increases (for second wire) **or** p.d. across the (second) wire is larger/increases	**M1**
	(so) length XP (for second wire) is shorter	**A1**

43.

6(a)	sum of current(s) into junction = sum of current(s) out of junction **or** (algebraic) sum of current(s) at a junction is zero	**B1**
6(b)(i)	same potential difference (across X and Y as in parallel)	**B1**
	power = V^2 / R (and $R_X > R_Y$) **or** power = VI **and** $I_X < I_Y$	**M1**
	(so) Y (dissipates more power)	**A1**
6(b)(ii)	$R = \rho L / A$ (and $R_X > R_Y$)	**M1**
	(so) Y (has the larger (cross-sectional) area)	**A1**
6(c)(i)	current = 5.4 / 1800 = 3.0×10^{-3} A	**A1**
6(c)(ii)	$5.4 / 9.0 = 1800 / (1800 + R_L)$ **or** $R_L = (9.0 - 5.4) / 3.0 \times 10^{-3}$	**C1**
	$R_L = 1200\ \Omega$	**A1**
6(c)(iii)	resistance of LDR / R_L decreases	**B1**
	current (in the circuit) increases (so) voltmeter reading increases	**B1**

44.

5(a)	current (through a conductor is directly) proportional to potential difference (across the conductor)	**M1**
	(provided that) temperature (of conductor remains) constant	**A1**
5(b)(i)	(ratio of) V / I increases (as p.d. increases)	**B1**
5(b)(ii)	(as p.d. increases, current increases so) temperature increases	**B1**
5(c)(i)	I = 1.55 A	**A1**
5(c)(ii)	$P = VI$ **or** $P = I^2R$ **or** $P = V^2 / R$	**C1**
	$= 6.0 \times 1.55 \times 2$ **or** $1.55^2 \times 3.87 \times 2$ **or** $(6.0^2 / 3.87) \times 2$ = 19 W	**A1**
5(c)(iii)	$I = 1.78 - 1.55$ (= 0.23 A)	**C1**
	$R = 12.0 / 0.23$ = 52 Ω	**A1**
5(d)	lamp P: p.d. across lamp decreases to zero so goes ‘out’	**B1**
	lamp Q: p.d. across lamp increases to 12 V so gets brighter	**B1**

45.

5(a)	sum of e.m.f.(s) = sum of p.d.(s) **or** (algebraic) sum of e.m.f.(s) and p.d.(s) is zero	**M1**
	around a loop / around a <u>closed</u> circuit	**A1**
5(b)(i)	$1 / r_{(T)} = 1 / 0.59 + 1 / 0.59 + 1 / 0.59$	**B1**
	($r_{(T)}$ =) 0.197 (Ω) (R =) 2.2 – 0.197 = 2.0 Ω	**A1**
	or	
	I = 1.5 / 2.2 (= 0.68 A) **and** i = 0.68 / 3 (where I is the circuit current and i is the current from each cell)	**(B1)**
	($E = IR + ir$ =) 1.5 = 0.68R + (0.68 / 3) × 0.59 **and** R = 2.0 Ω	**(A1)**
5(b)(ii)	current = 1.5 / 2.2	**C1**
	= 0.68 A	**A1**
	or	
	p.d. across cell = p.d. across conductor 1.5 – 0.59I = 3I × 2.0 so I = 0.228 A (where I is current in cell) current = 3 × 0.228	**(C1)**
	= 0.68 A	**(A1)**
	or	
	current in conductor = 3 × current in cell V / 2.0 = 3 × (1.5 – V) / 0.59 (where V is p.d. across conductor) V = 1.37 V current = 1.37 / 2.0	**(C1)**
	= 0.68 A	**(A1)**
5(c)(i)	$R = \rho L / A$	**C1**
	$R = 4\rho L / \pi d^2$ (ρ and L are the same so) $R_A / R_B = 7.6^2 / 4.3^2$	**C1**
	= 3.1	**A1**
5(c)(ii)	$I = Anvq$ and I, n, q are same / equal / constant	**B1**
	$\frac{v_A}{v_B} = \frac{A_B}{A_A} = \frac{d_B^2}{d_A^2}$ ratio = $7.6^2 / 4.3^2$ = 3.1	**A1**
5(d)	combined internal resistance (of the cells) will be greater **or** total / circuit resistance (of circuit) greater (because a parallel resistance removed)	**B1**
	more ‘lost volts’ (inside each cell) **or** internal resistances take a greater share of total p.d. **or** conductor gets a smaller share of the total p.d. **or** current in <u>conductor</u>/total current decreases	**M1**
	(so) potential difference (across conductor) decreases	**A1**

46.

7(a)	$9.6 = 6.0 + (I \times 5800)$ **or** $3.6 = I \times 5800$	**C1**
	$I = 6.2 \times 10^{-4}$ A	**A1**
7(b)	$9.6 = 6.2 \times 10^{-4} \times (3400 + 5800 + R)$ **or** $6.0 = 6.2 \times 10^{-4} \times (3400 + R)$	**C1**
	$R = 6.3 \times 10^3\ \Omega$	**A1**
7(c)	$(\Delta E =)\ 9.6 \times 330$ (= 3170 J)	**C1**
	final stored energy = $2.6 \times 10^4 - 3170$ $= 2.3 \times 10^4$ J	**A1**
7(d)(i)	decrease	**B1**
7(d)(ii)	decrease	**B1**
7(d)(iii)	increase	**B1**

47.

7(a)	$V / 9.0 = 1800 / (1800 + 1200)$	**C1**
	$V = 5.4$ V	**A1**
	or	
	$I = 9.0 / (1800 + 1200) = 3.0 \times 10^{-3}$ (A) $V = 3.0 \times 10^{-3} \times 1800$	**(C1)**
	$= 5.4$ V	**(A1)**
7(b)(i)	$L / 1.2 = 5.4 / 9.0$ **or** $XZ / 1.2 = 5.4 / 9.0$	**C1**
	$L = 0.72$ m	**A1**
	or	
	$L / 1.2 = 1800 / (1800 + 1200)$ **or** $XZ / 1.2 = 1.8 / (1.8 + 1.2)$	**(C1)**
	$L = 0.72$ m	**(A1)**
7(b)(ii)	• (intensity) increase	**B1**
	• (power) increase	**B1**
	• (length XZ) decrease	**B1**